

The growth series of the generic right-triangle reflection group: block assembly, analytic estimates and certificates

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Abstract

This is the methods and certificates companion of [7], where the growth series U of the universal right-triangle reflection group W_{univ} and its relaxed companion V are shown to be meromorphic on $|x| < 1$ with infinitely many poles there, hence transcendental over $\mathbb{Q}(x)$, not D -finite, and with $|x| = 1$ as a natural boundary, on the transfer model (M) and the junction pairing (R-J), both theorems. The present paper contains the proofs that the headline paper cites. *The block model in full*: the travel and the bulk block are one resolvent of the semiseparable kernel $q^{\max(a,b)}$, one three-term recursion covers both, and this proves the travel telescoping, the bulk dressing, the reciprocity $t_0 = b_1$ (equivalently $\det P = 1$), the identity $t_1 = P_{12}/S_e$ and the Möbius factorisation of the gap-marked bulk resolvent; the site cost of the crossing optimisation is computed in closed form and formalised in Lean 4, with a correction of the marker clause of (M); a gap-free element has cycle count $c = 0$; and the block decomposition (M3') is proved by hand. *The analytic estimates and certificates*: an absolute rectangle bound $|T_2| \leq 0.17 \tau^{1/4}$ on an auxiliary Mellin–Barnes integral, certified by interval arithmetic over parameter boxes, which gives a second proof that the travel poles are infinite and localises them in windows of $w = \sqrt{2/\tau}$; an elementary explicit lower bound $|S_e(q_m)| \geq 0.35 \sqrt{\tau_m}$ at the travel poles; and the amplitude estimate $(\star)$, $|P_{12}| \leq 0.619 \tau^{3/2}$, which with the exact identity $S_e = \frac{qZ}{2} \mathfrak{s}(Z) - P_{12}$ closes the gate $|s| \leq 0.983$. *Further material*: the Pólya–Carlson dichotomy for the four catalytic building blocks, unconditional for the two denominators and resting on a verified, unproved coefficient-growth hypothesis for the two numerators; the spectral and sign laws at the travel poles (every travel pole is simple, and Σ_0 and S_e both have sign $(-1)^{m-1}$ at the m -th pole), resting on the inequality (T_{poles}^*) , which is proved here by a flux identity, an amplitude bound and an interval certificate, and on the shift inequality $\lambda_n \leq q^2 \lambda_{n+1}$ between consecutive eigenvalue branches of the travel operator, proved here by min-max; an orthogonal arithmetic route through Christol’s theorem; the value-transcendence question for the number β_2 , reduced to a Hahn–Exton value statement with a certified finite algebraic exclusion; and the history of the former hypotheses (R), (T) and (L). Every statement keeps the standing it has in the archival full version, except (T_{poles}^*) and the shift inequality, which are hypotheses there and are proved here.

1 Introduction

This paper is the companion of [7], and the two together are a split of the archival full version [6]. The headline theorems (the transcendence of the travel resolvent and of U and V , their non- D -finiteness, the q -difference and Mahler exclusions, the natural boundary, the poles at $-x_m$, the growth rate β_2), the transfer model (M), the meromorphic continuation, the shape of the local form, the closed

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form of the junction pairing, the q -trigonometric identification of the blocks, the travel determinant and the numerator identity are stated and proved in [7]. Everything else is here. Labels are shared with [6], and a statement of the companion is cited as [7] together with its name and label. The standing of every statement, and in particular the arithmetic model of every committed certificate, is recorded where the statement is made.

Throughout, $W_{\text{univ}} = \langle R_0, R_1, R_2 \rangle$ is the universal right-triangle reflection group of [5], $U(x) = \sum_n u_n x^n$ its growth series with respect to the three side reflections (OEIS A396406 [41]), and V its relaxed companion.

The relaxed/true dichotomy. The word length $\ell_T(g)$ is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller, analytically tamer *relaxed* length $\ell_R(g)$, with $\ell_T = \ell_R + 2c(g)$, $c(g) \in \mathbb{Z}_{\geq 0}$ a cycle count. The inequality $\ell_T \leq \ell_R + 2c$ is proved in [5]. With ℓ_R and c the site-local closed forms of that paper, the identity $\ell_T = \ell_R + 2c$ holds for every element and is formalised in Lean (`metricAll`; §7.1); that the closed form ℓ_R is also the minimum over relaxed realisations is verified there, and bears only on reading V as a relaxed count. The relaxed counting function is the companion V ; one has $v_n \geq u_n$, equality for $n \leq 7$, first strict at $n = 8$. Both are governed by a linear q -difference (dilation $t \mapsto q^2 t$) *catalytic functional equation* (§2) that is q -holonomic and not of polynomial-kernel type.

Method. The catalytic transfer is assembled from four integer q -series $\Sigma_0, \Sigma_1, S_0, S_1$ with j^2 -gapped coefficients. Pólya–Carlson gives a dichotomy for them (rational with roots-of-unity poles, or a natural boundary and transcendence over $\mathbb{Q}(q)$). For the two denominators the rational alternative is excluded outright: their level sets $\Sigma_1 = 1$ and $S_1 = 1$ in $(0, 1)$ are infinite, and a rational function that is not identically zero has finitely many zeros in $\mathbb{C}$ (Corollary 2). For the two numerators it is excluded by super-polynomial coefficient growth along a sparse set of indices, verified for the first 34 square indices for Σ_1 and out to degree 1156 for all four blocks, and not proved (§3). Both V and U acquire poles at the *travel poles* q_m , the roots of $\Sigma_1^T(q) = 1$; transcendence follows from accumulation of these poles at $q = 1$ once a certain auxiliary integral T_2 is shown to be small (§4). The travel poles are the zeros of the Hahn–Exton q -Bessel function $J_{-1/2}^{(3)}(Z(q); q^2)$, $Z = \sqrt{2(1-q)/q}$, because that function is, up to a positive factor, the Fredholm determinant $\det(I - \mathcal{T})$ of the travel transfer operator ([7, Theorem “the travel determinant”, thm:fredholm], [7, Corollary “the travel poles are Hahn–Exton zeros”, cor:qbessel]). What the theorems consume is only $|T_2| \rightarrow 0$, and that is proved self-contained in Lemma 9 by an absolute bound on a rectangle: no saddle, no path through a saddle, no cited asymptotic theorem. Its quantitative part is a verified enclosure of the contour majorant, in interval arithmetic and quantified over parameter boxes, on $\tau \in [2/101^2, 0.02]$, together with an explicit-constant estimate below that; nothing there is a scan over sample values of τ . It is the single analytic input of this kind, shared by V ([7, §sec:V]) and U ([7, §sec:U]). The sharp size of T_2 , with the constant $\sqrt{2}/36$, is recorded separately in Remark 8 as a numerically certain statement whose proof is incomplete (Remark 10); nothing here depends on it.

The U -side carries one extra ingredient. Writing the cocycle off-diagonal P_{12} in exact closed form as a Hahn–Exton q -Bessel function $J_{3/2}^{(3)}$, the transcendence gate reduces to one amplitude bound on that series at the travel poles. This bound is *proved* (Appendix B): P_{12} is the symmetric second difference of the q -cosine at its own zero (Proposition 77), so the gate is a second-derivative estimate on a window of width τ , delivered by a two-interval uniform certificate; an exact identity then converts the same certificate into the denominator lower bound, closing the gate (Corollary 84). The assembled residue at a travel pole is the square of a junction pairing $\langle \lambda, R \rangle$ ([7, Proposition “the shape of the local form”, prop:shape]), which has an exact closed form in t_1 ; the gate makes

it positive at every pole with $\tau \leq 5 \cdot 10^{-3}$, and a certificate covers the first twelve ([7, Theorem “closed form of the junction pairing”, thm:RJ]). The invariance of the travel block under the cycle marker, (T), is proved ([7, Proposition “Invariance of the travel block”, prop:travelinv]). The sharp constants behind the picture are *derived* in Remark 15: the pole–zero offset $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$ and $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$, a fourfold margin below the gate threshold. (The turning-point atom feeding that analysis, the rational envelope $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2}) \in [1, \frac{3}{2}]$ of Lemma 65, is formally verified in Lean 4.) U and V therefore rest on the same self-contained *analysis*; no special-function input separates them, and after §6 they share the junction pairing (R-J), now [7, Theorem “closed form of the junction pairing”, thm:RJ], and the transfer model (M), now [7, Theorem “the transfer model (M)”, thm:model].

Section 8 records an independent arithmetic route: by Christol’s theorem [10, 11], if $(u_n \bmod p)$ is not p -automatic for one prime p then U is transcendental over $\mathbb{Q}(x)$; the p -kernel is maximally non-automatic at $p = 3, 5$.

Appendix C treats a question separate from all of the above: whether the *number* β_2 is transcendental. That is open. What the appendix contributes is a reduction of it to a value statement for a Hahn–Exton q -Bessel function at an algebraic base, and an exact location of the obstruction, at the arithmetic ratio $\rho = \frac{1}{2}$, one notch beyond the range in which q -irrationality criteria are proved. No statement in the body depends on it.

2 The catalytic equation and its building blocks

Write $q = x^2 = e^{-\tau}$, $w = \sqrt{2/\tau}$. The relaxed length statistic satisfies a linear q -difference equation with dilation $t \mapsto q^2 t$; telescoping it produces four integer power series in q ,

$$\Sigma_1(q) = \sum_{i \geq 1} (-1)^{i-1} \frac{w^{2i}}{(2i)!} \eta_i, \quad S_e(q) = \sum_{j \geq 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q; q)_{2j}}, \quad (1)$$

and their companions Σ_0, S_0 , where $\eta_j = e^{-B_j}$ is the Gaussian-decaying form factor of §4 (equivalently $\eta_j = q^{j^2+j} [2(1-q)]^j (2j)! / [(q; q)_{2j} W^{2j}]$; the identification of this display with $1 - \Sigma_1 = \cos(Z; q^2)$ of [7, Proposition prop:qtrig] is the collapse used in Lemma 11). We write $\Sigma_1^T = \Sigma_1$ when the travel rôle is to be emphasised. The relaxed travel resolvent is $G_0^T = \Sigma_0 / (1 - \Sigma_1)$ (Proposition 31), and U inherits its denominator $1 - \Sigma_1$ ([7, Proposition “Invariance of the travel block”, prop:travelinv]); how the two blocks assemble into V and U , and how far that assembly is proved, is §6. The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \quad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at $q = 1$ (Lemma 11, proved in §4 once the collapse machinery is in place).

The invariance of the travel block, [7, Proposition “Invariance of the travel block”, prop:travelinv] (the former input (T)), is stated and proved in the companion paper.

Earlier versions of this paper recorded the proposition as a verified conjecture resting on two unproved statements, the localisation of cycles and the metric identity $\ell_T = \ell_R + 2c$. The first is now Corollary 57. The second is not needed: only the inequality $\ell_T \leq \ell_R$ is used, and it follows from the realisation exhibited by Theorem 56 together with the trivial reverse inequality, so the open lower bound of the metric identity does not enter. Statements below that invoke this proposition no longer inherit a conditional standing from it; see [7, Theorem thm:U] and [7, Remark rem:Ustatus].

3 Transcendence of the blocks

The four blocks are the k -recursion sums

$$\Sigma_k = \sum_{j \geq 0} A_{k+2j} \prod_{i < j} C_{k+2i}, \quad S_k = \sum_{j \geq 0} \alpha_{k+2j} \prod_{i < j} \gamma_{k+2i}, \quad (2)$$

$$A_k = \frac{2q}{1 - q^{k+1}}, \quad C_k = \frac{2q^{k+3}}{1 - q^{k+2}} - \frac{2q^{k+2}}{1 - q^{k+1}}, \quad \alpha_k = \frac{2q^{k+1}}{1 - q^{k+1}}, \quad \gamma_k = \alpha_{k+1} - \alpha_k,$$

Σ_0, Σ_1 carrying the travel rôle and S_0, S_1 the bulk rôle; Σ_1 and S_1 are the denominators of (1) and Σ_0, S_0 their numerators. Every factor is a Lambert series with integer coefficients. The factor C_{1+2i} starts at order $2i + 3$, so $\prod_{i < j} C_{1+2i}$ starts at order $\sum_{i < j} (2i + 3) = j^2 + 2j$ and, with A_{1+2j} starting at order 1, the j -th summand of Σ_1 starts at order exactly $(j + 1)^2$; only the finitely many $j \leq \sqrt{n} - 1$ reach $[q^n]$. Each coefficient is therefore a finite integer sum, and $\Sigma_1 = 2q + 2q^3 - 4q^4 + 6q^5 - 4q^6 + \dots$. That integrality, with convergence on $|q| < 1$, is the whole input to Pólya–Carlson.

Theorem 1. *Let Σ be any of $\Sigma_0, \Sigma_1, S_0, S_1$. Then Σ has integer coefficients and converges in $|q| < 1$, and exactly one of the following holds:*

- (a) Σ is a rational function whose poles are roots of unity; or
- (b) $|q| = 1$ is a natural boundary of Σ , and Σ is transcendental over $\mathbb{Q}(q)$.

Alternative (a) is excluded, and hence (b) holds, for any block whose coefficients grow super-polynomially along a subsequence. For Σ_1 that growth is the hypothesis

$$|[q^{(j+1)^2}] \Sigma_1| \geq 2^{j+1} \quad \text{for infinitely many } j, \quad (3)$$

which is not proved here: see Remark 5.

Proof. The coefficients are integers by the order count above: each is a finite sum of integers. Convergence in $|q| < 1$ is read off the closed forms: each block is a series whose j -th term carries the Gaussian factor $q^{j^2 + O(j)}$ against a denominator $(q; q)_{2j}$ or $(q; q)_{2j+1}$ bounded below in modulus by $|(q; q)_\infty| > 0$ on compact subsets of the punctured disc (for Σ_1 use $1 - \Sigma_1 = \cos(Z; q^2) = \sum_{j \geq 0} (-1)^j q^{j^2} (2(1 - q))^j / (q; q)_{2j}$ of [7, Proposition prop:qtrig], in which the pole of $Z^2 = 2(1 - q)/q$ at the origin is cancelled), so each series converges absolutely and locally uniformly there. Hence Pólya–Carlson [9] applies and gives the dichotomy (a)/(b): a power series with integer coefficients convergent in the unit disc is either a rational function whose poles are roots of unity, or has the unit circle as a natural boundary; a function with a natural boundary is transcendental over $\mathbb{Q}(q)$. In case (a) the coefficient sequence satisfies a linear recurrence all of whose characteristic roots are roots of unity, hence $|[q^n] \Sigma| \leq Cn^d$ for some C, d ; a subsequence growing faster than every polynomial in n is therefore incompatible with (a). Under (3), $|[q^n] \Sigma_1| \geq 2^{\sqrt{n}}$ along $n = (j + 1)^2$, which is such a subsequence. $\square$

For the two denominators the growth hypothesis is not needed at all: alternative (a) is excluded by the pole set instead.

Corollary 2 (the pole route to alternative (a)). *Alternative (a) of Theorem 1 fails for Σ_1 and fails for S_1 ; hence for each of the two denominators $|q| = 1$ is a natural boundary and the block is transcendental over $\mathbb{Q}(q)$. It also fails for at least one of the two numerators Σ_0, S_0 . Neither statement uses (3) or Hypothesis 4.*

Proof. A rational function that is not identically zero has finitely many zeros in $\mathbb{C}$. The constraint is global: unlike the identity theorem it requires no accumulation point inside a domain, so a zero set that is merely infinite already contradicts it, wherever in $\mathbb{C}$ the zeros sit and wherever they accumulate.

Suppose Σ_1 satisfies (a), so that on $|q| < 1$ it agrees with a rational function whose poles are roots of unity. Then $1 - \Sigma_1$ is rational, and it is not identically zero, because $\Sigma_1(0) = 0$: every factor A_k, C_k of (2) vanishes at $q = 0$, equivalently $1 - \Sigma_1 = \cos(Z; q^2)$ has constant term 1 ([7, Proposition prop:qtrig]). A nonzero rational function has finitely many zeros; but $\{q \in (0, 1) : \Sigma_1(q) = 1\}$ is infinite by Lemma 11. Alternative (a) is therefore impossible and Theorem 1 leaves (b). The same argument applies to S_1 , with $1 - S_1 = S_e$ and $S_1(0) = 0$ ([7, Proposition prop:qtrig]) and with Lemma 12 in place of Lemma 11.

For the numerators, suppose Σ_0 and S_0 both satisfy (a). Then $\Sigma_0 S_0$ is rational and, by [7, Proposition “the numerators do not vanish at the travel poles”, prop:numnonvanish], agrees with the rational function $2q/(1 - q)$ at every travel pole, hence at infinitely many points of $(0, 1)$ (Lemma 11); two rational functions agreeing at infinitely many points are equal. They are not: $\Sigma_0 S_0 = 4q^2 + 8q^3 + 4q^4 + \dots$ vanishes to order 2 at $q = 0$ and $2q/(1 - q)$ to order 1. This says nothing about either numerator separately, [7, (eq:numprod)] constraining only the product. $\square$

The two lemmas quoted are proved in §4 and the two identities in [7, §sec:qtrig]; of Theorem 1 they use only its first clause, that the blocks converge on $|q| < 1$, so the dependency graph is acyclic. The series data $\Sigma_1 = 2q + 2q^3 - 4q^4 + \dots$, $S_1 = 2q^2 + 2q^4 - 2q^6 + \dots$, $\Sigma_0 S_0 = 4q^2 + 8q^3 + 4q^4 + \dots$ and the two collapses $1 - \Sigma_1 = \cos(Z; q^2)$, $1 - S_1 = S_e$ are recomputed in exact integer arithmetic at two truncation degrees in `f2_pole_route.py`.

Remark 3 (what the pole route costs). Corollary 2 makes Theorem 1 unconditional for Σ_1 and S_1 , and the two routes to it are of different kinds. (3) is checked in exact integer arithmetic, with no precision to state, at 34 indices, and remains a finite verification of a statement with infinitely many instances. The pole route runs on Lemma 11, hence on Lemma 9, and that lemma is proved on all of $\tau \in (0, 0.02]$: a verified enclosure of the contour majorant, in interval arithmetic and quantified over parameter boxes rather than sampled, on $\tau \geq 2/101^2$, and an explicit-constant estimate below that. The two infinitude lemmas consume it along the infinite sequences $\tau^{(m)} = 2/(m\pi)^2$ and $\tau^{[m]}$, $m \geq 4$, all of which lie in that range. So the pole route carries no unproved arithmetic input, while (3) does; and for Σ_0 and for S_0 individually, (4) remains the only route. What the corollary buys besides the label is a shorter dependency list: the exclusion of (a) for the two denominators consumes the same single analytic input as Theorems [7, thm:travelres], [7, thm:V], [7, thm:U] and [7, Corollary cor:beta], instead of a second input of its own. [7, Proposition “infinitely many travel poles, without T_2 ”, prop:infpolesalt] gives both infinitude statements a second proof with no T_2 input, so for the two denominators, and for the joint statement on the numerators, the exclusion of (a) does not need Lemma 9 at all.

The hypothesis (3) concerns Σ_1 alone. Its counterpart for the other three blocks is the following uniform statement, which is what alternative (a) has to be weighed against for Σ_0 and S_0 , the two blocks Corollary 2 does not reach individually.

Hypothesis 4 (coefficient growth). Let Σ be any of $\Sigma_0, \Sigma_1, S_0, S_1$. Then

$$|[q^n]\Sigma| \geq 2^{\sqrt{n}-1} \quad \text{for infinitely many } n. \quad (4)$$

Statement (3) is the Σ_1 instance of (4) in the sharper form that names the witnessing indices $n = (j + 1)^2$. Either form excludes alternative (a) for the block concerned, $2^{\sqrt{n}-1}$ outgrowing every polynomial in n .

Remark 5 (why (3) is expected, and how far it is verified). (3) is verified, not proved, and the reason it is expected is exact rather than heuristic. By the order count above, the j -th summand of the telescoping $\Sigma_1 = \sum_{j \geq 0} (\prod_{i < j} C_{1+2i}) A_{1+2j}$ has lowest degree exactly $(j+1)^2$, with coefficient exactly $(-1)^j 2^{j+1}$ (the leading 2 of A_{1+2j} against the leading -2 of each of the j factors C_{1+2i}), and summands of index $> j$ start above that degree. Hence

$$[q^{(j+1)^2}] \Sigma_1 = (-1)^j 2^{j+1} + \left(\text{the degree-}(j+1)^2 \text{ tails of the } j \text{ summands of index } < j \right),$$

a *finite* correction to an exactly known leading term: (3) says precisely that those j tails do not cancel it. They do not merely fail to cancel; they reinforce. Exact integer evaluation of (2) truncated at degree 1156 gives

$$[q^{(j+1)^2}] \Sigma_1 = 2, -4, 14, -52, 178, -856, 4626, -27524, 150214, -816268, \dots$$

and confirms (3) for every $j \leq 33$, with the sign $(-1)^j$ of the leading term at every one of them and a margin that grows without bound: $|[q^{(j+1)^2}] \Sigma_1| / 2^{j+1}$ is 797 at $j = 9$, $1.2 \cdot 10^8$ at $j = 20$ and $2.7 \cdot 10^{14}$ at $j = 33$, where the value is $-4649405079364022238518260$ against the bound 2^{34} ; the successive ratios $|[q^{(j+2)^2}] \Sigma_1| / |[q^{(j+1)^2}] \Sigma_1|$ rise from 2 to 6.22 over that range. Hypothesis 4 holds for all four blocks with a witness in every window $[(j+1)^2, (j+2)^2)$ for every $j \leq 32$, the range being cut off by the truncation degree and not by a failure. The witnessing diagonals are $n = (j+1)(j+2)$ for S_1 and $n = (j+1)^2$ for S_0 , each carrying $|[q^n] \Sigma| \geq 2^{j+1}$ throughout $4 \leq j \leq 32$ and $3 \leq j \leq 32$ respectively; for Σ_0 the leading diagonal cancels: its values at $n = (j+1)^2$ are 2, 2, 2, 10, 18, 34, 266, 306, 8170, 330, which drop below 2^{j+1} at $j = 1, \dots, 5$ and again at $j = 9$, so its witnesses are off-diagonal and (4) is the only form available for it.

The computation is exact integer arithmetic in $\mathbb{Z}[[q]]/(q^{1161})$ (`blocks_growth.py`); no floating point enters, so no precision is quoted. It is checked against a second and independent derivation of the same coefficients from the q -trigonometric closed forms of [7, Proposition prop:qtrig] and Lemma 76 (agreement to degree 1160 for each of Σ_1, S_1, S_0), and against the bulk block $S_0/(1-S_1) = 2q + 2q^2 + 6q^3 + 2q^4 + 18q^5 + 6q^6 + 42q^7 + \dots$. For $j \leq 9$ the bound on Σ_1 is also machine-checked in Lean 4 (`PolyaCarlson.lean`, theorem `coeff_bound`); that proof uses `native_decide` and therefore depends, besides `propext`, on the compiler-trust axiom `PolyaCarlson.coeff_bound._native.native_decide.ax_1` which is outside the three standard axioms.

No finite verification can establish a statement quantified over infinitely many indices. (3) is numerically certain and unproved, and the transcendence conclusion (b) of Theorem 1 inherits that standing wherever it is the route used, which after Corollary 2 is Σ_0 and S_0 . What would close it is a lower bound on the degree- $(j+1)^2$ coefficient of the j preceding Lambert products, uniform in j : the statement is finite for each j and the obstruction is uniformity, not any single case.

The dichotomy of Theorem 1 uses no analytic gate. The exclusion of alternative (a) rests on (3) for Σ_0 and S_0 , and on the pole route of Corollary 2, hence on the analytic input of §4, for Σ_1 and S_1 . Neither transfers directly to V or U , which are *ratios* of blocks in which the boundary singularities can cancel; those require the pole-accumulation argument below, which is independent of (3).

4 The steepest-descent estimate

The one analytic input for both V and U is the smallness of an auxiliary integral. Let $\varphi(y) = \log \frac{\sinh(y/2)}{y/2} = \sum_{n \geq 1} \varphi_n y^{2n}$ with $\varphi_n = \frac{(-1)^{n+1} \zeta(2n)}{n(2\pi)^{2n}}$. The form factor B_s is defined by the closed form (6) of Lemma 6; the familiar expansion $B_s \sim \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s)$, $P_n(M) = \sum_{m=1}^M m^{2n}$ the

Faulhaber sum, is its Euler–Maclaurin *asymptotic* series (equality at integer $2s$, divergent otherwise; Lemma 6(v)) and is used only truncated. The block S_e decomposes as $S_e = \cos W - T_2$ with $W = we^{-\tau/2}$ and

$$T_2 = \sum_{n \geq 1} (-1)^n g_n \frac{W^{2n}}{\Gamma(2n+1)}, \quad g_s = 1 - e^{-B_s}. \quad (5)$$

What the theorems consume is only $T_2 \rightarrow 0$; its sharp size is recorded separately in Remark 8 and is not used.

Lemma 6 (amplitude regularity). *For $\operatorname{Re} s > -\frac{1}{2}$ set $M = 2s$ and*

$$B_s = \frac{\tau}{4}M(M+1) - M \log \tau - \log \Gamma(M+1) + \log(q; q)_\infty - \log(q^{M+1}; q)_\infty - s \varphi(\tau), \quad (6)$$

$\varphi(y) = \log \frac{\sinh(y/2)}{y/2}$ as above, each q -Pochhammer logarithm meaning the sum of principal logarithms $\sum_{j \geq 0} \operatorname{Log}(1 - q^{M+1+j})$: every factor has $|q^{M+1+j}| < 1$, hence $\operatorname{Re}(1 - q^{M+1+j}) > 0$, so this is the analytic branch on the half-plane (the logarithm of the product would differ by multiples of $2\pi i$). Then B_s is analytic on $\operatorname{Re} s > -\frac{1}{2}$, and at every integer $n \geq 0$

$$B_n = \sum_{i=1}^{2n} \varphi(i\tau) - n \varphi(\tau), \quad (7)$$

so (6) is the analytic interpolation of the collapse coefficients $\eta_n = e^{-B_n}$ of (1). On the strip

$$\mathcal{S} = \{-\frac{1}{2} < \operatorname{Re} s \leq 2w, \ |\Im s| \leq 2w\}, \quad \tau \leq 2/\pi^2,$$

one has the explicit bounds

$$|B_s| \leq 30.3 \sqrt{\tau}, \quad |B'_s| \leq 7.6 \tau, \quad (8)$$

and, on the smaller region $|M| \leq 2w$ which contains the saddles, the truncation bound that Remark 8 consumes:

$$|B_s - \varphi_1 \tau^2 (P_1(M) - s)| \leq 0.02 \tau^{3/2}, \quad \varphi_1 = \frac{1}{24}. \quad (9)$$

Consequently $|e^{-B_s}| \leq e^{30.3\sqrt{\tau}}$, the amplitude $A := 1 - e^{-B_s}$ satisfies $|A| \leq 1 + e^{30.3\sqrt{\tau}} \rightarrow 2$, and along any rectifiable $\Gamma \subset \mathcal{S}$ of length $\ell(\Gamma) \leq cw$,

$$V_\Gamma(A) \leq \sup_\Gamma |e^{-B_s}| \int_\Gamma |B'_s| |ds| \leq e^{30.3\sqrt{\tau}} \cdot 7.6 \tau \cdot cw = O(\sqrt{\tau}) \rightarrow 0,$$

using $\tau w = \sqrt{2\tau}$. The strip contains both saddles $s^* = -\frac{1}{4} \pm i(\frac{w}{2} + O(\sqrt{\tau}))$ of Remark 8 and the $O(w)$ -length portion of its steepest-descent contour through them (which is all the $\ell(\Gamma) \leq cw$ clause above requires, the Gaussian width being $O(\sqrt{w})$) and, being built from the larger argument $w = We^{\tau/2}$, it contains the corresponding W -strip as well.

Proof. (i) Closed form, and (7). Unwinding $\eta_n = q^{n^2+n} [2(1-q)]^n (2n)! / [(q; q)_{2n} W^{2n}]$ with $W^2 = (2/\tau)e^{-\tau}$ gives $\eta_n = e^{-\tau n^2} [(1-q)\tau]^n (2n)! / (q; q)_{2n}$. Now $1 - q^i = i\tau e^{-i\tau/2} \frac{\sinh(i\tau/2)}{i\tau/2}$, i.e. $\log(1 - q^i) = \log(i\tau) - \frac{i\tau}{2} + \varphi(i\tau)$; summing over $1 \leq i \leq M$,

$$\log(q; q)_M = \log \Gamma(M+1) + M \log \tau - \frac{\tau}{4}M(M+1) + \sum_{i=1}^M \varphi(i\tau). \quad (10)$$

Since $\log[(1-q)\tau] = -\frac{\tau}{2} + 2\log\tau + \varphi(\tau)$, substituting (10) into $-\log\eta_n$ collapses every elementary term and leaves exactly (7); and rewriting $\log(q;q)_M = \log(q;q)_\infty - \log(q^{M+1};q)_\infty$ in (10) turns (7) into (6) for all s . Every ingredient of (6) is analytic in s for $\operatorname{Re} s > -\frac{1}{2}$: there $|q^{M+1}| < 1$ keeps $(q^{M+1};q)_\infty$ zero-free and the branch above canonical, and $\operatorname{Re}(M+1) > 0$ keeps $M+1$ off the poles of Γ . (Both singular factors do in fact cancel at $M = -1$, so B_s continues past $\operatorname{Re} s = -\frac{1}{2}$; we never need that.)

(ii) *Splitting into Gamma blocks with exact cancellation.* By the Weierstrass product $\frac{\sinh(y/2)}{y/2} = \prod_{k \geq 1} (1 + \frac{y^2}{4\pi^2 k^2})$ we have $\varphi(y) = \sum_{k \geq 1} \log(1 + y^2/(4\pi^2 k^2))$, so with $R_k := 2\pi k/\tau$ and $\prod_{n=1}^M (1 + n^2 a^2) = a^{2M} \Gamma(M+1-\frac{i}{a})\Gamma(M+1+\frac{i}{a})/[\Gamma(1-\frac{i}{a})\Gamma(1+\frac{i}{a})]$ at $a = 1/R_k$, (7) becomes

$$B_s = \sum_{k \geq 1} G(M; R_k) - s\varphi(\tau), \quad G(M; R) := \int_0^M [\psi(1+u+iR) + \psi(1+u-iR) - 2\log R] du, \quad (11)$$

the integral along the segment $[0, M]$ fixing the branch (legitimate since $|M| < R_1$ puts the segment far from the poles of $\psi(1+u \pm iR)$ at $u = -1 - j \mp iR$). *The $-2\log R$ is not a normalisation: it is the exact cancellation of the two leading Stirling terms*, and it is what makes the k -sum converge; without it each block would grow like $2M \log R$.

Identity (11) is proved at integers, where the i -sum is finite and the interchange of $\sum_i$ and $\sum_k$ is trivial; it extends to all s as follows.

Domain. Both sides are analytic on the half-plane $\operatorname{Re} M > -1$: the left by (i), the right because the segment $[0, M]$ then stays in $\operatorname{Re} u > -1$ while the poles of $\psi(1+u \pm iR)$ lie in $\operatorname{Re} u \leq -1$, and because $\sum_k G$ converges locally uniformly there, all but finitely many k having $R_k > 2|M|$ and hence $|G(M; R_k)| = O(|M|^3/R_k^2)$. (The disc $|M| < R_1$ of step (iii) is where the *quantitative* majorant lives; it is not the domain of analyticity, and the recursion below would walk out of it.)

Recursion. Write $\tilde{B} = B_s + s\varphi(\tau)$ for either side with the elementary term restored. Both obey the *same* unit recursion in M : on the left $\frac{\tau}{4}[(M+1)(M+2) - M(M+1)] - \log\tau - \log(M+1) + \log(1 - q^{M+1}) = \varphi((M+1)\tau)$ term by term, and on the right $G(M+1; R) - G(M; R) = \log(1 + (M+1)^2/R^2)$, which sums over k to the same $\varphi((M+1)\tau)$ by the Weierstrass product. (For B_s itself both increments are $\varphi((M+1)\tau) - \frac{1}{2}\varphi(\tau)$, the extra term being common to the two sides.)

Conclusion. Hence the difference D is 1-periodic on $\operatorname{Re} M > -1$, so it continues to a 1-periodic *entire* function. Its Fourier coefficients satisfy $|c_m| \leq e^{2\pi m y} \sup_{0 \leq x \leq 1} |D(x+iy)|$ for every real y , and D grows at most polynomially: on the left by Stirling, on the right by splitting the k -sum at $R_k = 2|M|$: the $O(|M|\tau)$ low terms are each $O(|M|\log|M|)$ by Stirling and the remainder is $O(|M|^3 \sum_{R_k > 2|M|} R_k^{-2}) = O(|M|^2\tau)$. Letting $y \rightarrow -\infty$ for $m > 0$ and $y \rightarrow +\infty$ for $m < 0$ gives $c_m = 0$ for $m \neq 0$, so D is constant; and $D(0) = 0$ by (7) at $n = 0$, so $D \equiv 0$.

(iii) *The integrand.* For $|z| \geq 1$ with $|\arg z| \leq \frac{3\pi}{4}$ one has $|\psi(1+z) - \log z - \frac{1}{2z}| \leq |z|^{-2}$ (the constant is 0.13 on that sector; the bound fails without limit as $\arg z \rightarrow \pi$, which is why the sector, not a distance to the cut, is the right hypothesis). On $\mathcal{S}$, $|M| \leq 4\sqrt{2}w$ and $|M|/R_1 = \frac{4\sqrt{2}w\tau}{2\pi} = \frac{4}{\pi}\sqrt{\tau} \leq 0.58$, so every $R_k > |M|$; and along the segment $|\arg(u \pm iR_k)| \leq 120^\circ < 135^\circ$ with $|u \pm iR_k| \geq R_1 - |M| > 13$, so the estimate applies throughout. Adding the two conjugate terms, the logarithms combine with $-2\log R$ to $\log(1 + u^2/R^2)$ and the $\frac{1}{2z}$ terms to $u/(u^2 + R^2)$, whence for $|u| \leq |M|$

$$|\psi(1+u+iR) + \psi(1+u-iR) - 2\log R| \leq \frac{|M|^2 + |M|}{R^2 - |M|^2} + \frac{2}{(R - |M|)^2} =: \gamma(R).$$

(iv) *Summation.* Hence $|G(M; R_k)| \leq |M|\gamma(R_k)$ and, from (11),

$$|B_s| \leq |M| \sum_{k \geq 1} \gamma(R_k) + |s|\varphi(\tau), \quad |B'_s| \leq 2 \sum_{k \geq 1} \gamma(R_k) + \varphi(\tau),$$

the second because $\partial_M G = \psi(1+M+iR) + \psi(1+M-iR) - 2\log R$ and $M = 2s$. Both sums are dominated by $\sum_k R_k^{-2} = \tau^2/24$; carrying the finite corrections explicitly at $|M| = 4\sqrt{2}w$, $\tau \leq 2/\pi^2$ gives $|M|\sum_k \gamma(R_k) + |s|\varphi(\tau) \leq 30.3\sqrt{\tau}$ and $2\sum_k \gamma(R_k) + \varphi(\tau) \leq 7.6\tau$, which is (8). Both ratios are *increasing* in τ on the range (30.20, 25.14, 23.26, 22.21, 21.64 and 7.55, 6.28, 5.81, 5.55, 5.41 at $\tau = 2/\pi^2$, 0.1, 0.05, 0.02, 0.005, decreasing to the limits $\frac{64}{3}$ and $\frac{16}{3}$ as $\tau \rightarrow 0$), so each attains its maximum at the right endpoint $\tau = 2/\pi^2$, and the stated constants hold throughout.

For (9): expand the integrand of G one order further, now *keeping the third Stirling term* $-\frac{1}{12z^2}$, which is what supplies the linear part of P_1 : its integral is $\frac{M}{6(M^2+R^2)} = \frac{M}{6R^2} + O(|M|^3/R^4)$. With $\int_0^M \log(1+u^2/R^2)du = \frac{M^3}{3R^2} + O(|M|^5/R^4)$ and $\int_0^M u/(u^2+R^2)du = \frac{M^2}{2R^2} + O(|M|^4/R^4)$ this gives $R^2G(M; R) \rightarrow P_1(M) = \frac{M^3}{3} + \frac{M^2}{2} + \frac{M}{6}$, so with $\sum_k R_k^{-2} = \tau^2/24$ exactly,

$$B_s = \varphi_1 \tau^2 (P_1(M) - s) + O(|M|^5 \tau^4) + O(|s| \tau^4).$$

On $|M| \leq 2w$ every remainder is $O(\tau^{3/2})$, and the constant is 0.02 (worst 0.0177 at $\tau = 2/\pi^2$, decreasing).

(v) *What is not used.* The τ -expansion $B_s = \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s)$, P_n the Faulhaber sum, is the Euler–Maclaurin *asymptotic* expansion of (6): it is *divergent* at non-integer $2s$, since $P_n(2s)$ carries a Bernoulli part of size $(2n)! e^{2\pi i \Im 2s} / (2\pi)^{2n+1}$. Nothing above sums it; it is used only in the truncated form “leading term plus remainder”, as in Remark 8, where the $n = 1$ term at the saddle supplies the constant $\sqrt{2}/36$. $\square$

The algebraic skeleton of this proof is machine-checked in Lean 4 (`BboundedSkeleton.lean`, ten theorems, no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound`). Five carry the arithmetic content: the Gauss sum, the summation (10) built on it, the telescoping cancellation (7), the scaling identity $(\tau^2 w^3)^2 = 8\tau$ (exact for w , only $e^{-3\tau/2}$ -approximate for W), and the strip condition $|M|/R_1 < 1$. The other five (the per-factor rewriting, the collapse of $\log[(1-q)\tau]$, the cancellation $2M \log a + 2M \log R = 0$ at $aR = 1$, the w -definition, and the bounded-variation arithmetic) are *rearrangements of their hypotheses*, verified as such: they certify that the paper’s algebra is what it claims, not the transcendental facts fed into them. Those facts (the Weierstrass product, the digamma remainder, the convergence of the k -sum, and the continuation argument of step (ii)) are not formalised.

Remark 7 (sharpness). The majorant of step (iv) is lossy by about a factor 6: the true extremes on $\mathcal{S}$ are $\operatorname{Re} B_s \geq -5.03\sqrt{\tau}$ and $|B'_s| \leq 5.46\tau$, and $\sup_{\mathcal{S}} |A| = 9.02$ at $\tau = 2/\pi^2$ (`star_gate_certificate.py`, part (7)). Only the proved bounds (8) are used anywhere.

The sketch exhibits the saddles explicitly and derives the constants by Cauchy’s theorem, Stirling’s formula and elementary Gaussian integration, and it would end by invoking a second-order remainder estimate for a simple non-degenerate saddle, [34, Ch. 4, Thm. 7.1]. Its hypotheses are *not* verified here (the saddle is flat rather than non-degenerate, and the amplitude’s bounded variation is available only on a bounded piece of the path), which is one of the reasons the sketch is not a proof (Remark 10). No theorem of this paper rests on it.

Remark 8 (the sharp size of T_2 ; HEURISTIC). Numerically, $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin X - \frac{161}{1296} \tau \cos X + O(\tau^{3/2})$ uniformly in the phase, for any argument X with $\tau X^2 \in [2e^{-\tau}, 2]$; the constant and the exponent of the remainder are certain to eight digits over $m \leq 1024$. The sketch below indicates why, by steepest descent, but it does *not* constitute a proof, for the reasons set out in Remark 10. Accordingly this statement carries no weight anywhere in this paper: the only property of T_2 that the theorems consume is the crude bound $|T_2| \rightarrow 0$ of Lemma 9, which is proved, and which is what Lemma 11 cites.

Sketch, not a proof; see Remark 10. Since $\cos W = \sum_{n \geq 0} (-1)^n W^{2n} / (2n)!$, (5) gives the exact alternating sum $S_e = \cos W - T_2 = \sum_{n \geq 0} (-1)^n e^{-B_n} W^{2n} / (2n)!$ (with $B_0 = 0$). Let $b(z) = e^{-B_z} W^{2z} / \Gamma(2z + 1)$, analytic for $\operatorname{Re} z > -\frac{1}{2}$ (by (6) and Lemma 6; *not* by termwise polynomiality, the expansion being divergent off z). As $\pi \csc \pi z$ has residue $(-1)^n$ at each integer, Cauchy's theorem gives $S_e = \frac{1}{2\pi i} \oint b(z) \pi \csc(\pi z) dz$ over a contour enclosing $\{0, 1, 2, \dots\}$. Write the integrand as $e^{h(z)} \pi \csc \pi z$ with $h(z) = 2z \log W - \log \Gamma(2z + 1)$; including the $\csc$ phase, the exponent is stationary where $2 \log W - 2\psi(2z + 1) \pm i\pi = 0$, so by $\psi(w) = \log w + O(1/w)$ there are two simple, non-degenerate saddles

$$z_{\pm}^* = \pm \frac{i}{2} W (1 + O(1/W)), \quad h''(z_{\pm}^*) = \pm \frac{4i}{W} + O(W^{-2}) \neq 0.$$

Deform the contour to the steepest-descent paths through $z_{\pm}^*$; this is legitimate because b decays super-exponentially as $\operatorname{Re} z \rightarrow +\infty$ and $\csc$ suppresses $|\Im z| \rightarrow \infty$. A direct Stirling evaluation shows that at the modulation-free level $B \equiv 0$ the two saddles sum to $\cos W$ *exactly*: the divergent factors $e^{\pm \pi W/2}$ from $\csc(\pi z_{\pm}^*)$ and from $1/\Gamma(\pm iW + 1)$ cancel identically, each saddle contributing $\frac{1}{2} e^{\pm iW}$. Restoring the modulation scales each saddle by $e^{-B_{z_{\pm}^*}}$; since $P_1(2z) = \frac{2z(2z+1)(4z+1)}{6}$ and $\varphi_1 = \frac{1}{24}$,

$$B_{z_+^*} = \varphi_1 \tau^2 (P_1(iW) - \frac{iW}{2}) + O(\tau) \quad (\text{the } \tau^{3/2} \text{ part of the remainder is (9); the } O(\tau) \text{ also absorbs } B_{z_+^*} - B_{iW/2} \text{ and})$$

(using $\tau^2 W^3 = 2\sqrt{2} \sqrt{\tau} e^{-3\tau/2}$, the factor $e^{-3\tau/2}$ being absorbed in the $O(\tau)$), and $B_{z_-^*} = \overline{B_{z_+^*}}$. Hence

$$S_e = \frac{1}{2} e^{-B_{z_+^*}} e^{iW} + \frac{1}{2} e^{-B_{z_-^*}} e^{-iW} + E = \cos W - \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + E.$$

The remainder E is $O(\tau)$ by the classical second-order estimate for a simple saddle [34, Ch. 4, Thm. 7.1]: parametrising the descent path by arclength s , the integrand equals $e^{h(z^*)} e^{-s^2} (1 + O(sW^{-1/2}) + O(\tau))$, the next saddle coefficient is $\propto h'''(z^*) = O(W^{-2}) = O(\tau)$, and the bounded-variation hypothesis on the amplitude e^{-B_z} along the path is supplied by Lemma 6. This citation is the proof's one non-elementary step. The saddle prediction matches S_e to $O(\tau)$ over $\tau = 0.02, 0.01, 0.005$ numerically (`saddle_verify.py`). Thus $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + O(\tau)$, with $\sin W = \sin w + O(\sqrt{\tau})$. $\square$

Lemma 9 (absolute bound on T_2). *For $\tau \leq 0.02$, $|T_2| \leq 0.17 \tau^{1/4}$; in particular $|T_2| \leq 0.064 < 1$ there, and $|T_2| \rightarrow 0$ as $\tau \rightarrow 0$. The argument uses no saddle, no path through a saddle, and no cited asymptotic theorem: only the Mellin–Barnes residues, a rectangle contained in the strip $\mathcal{S}$ of Lemma 6, and the amplitude bounds proved there. The same holds at any argument X with $\tau X^2 \in [2e^{-\tau}, 2]$; the constant above is the one for $X = w$, which is the case Lemma 11 uses and the worse of the two.*

Proof. Since $\operatorname{Res}_{s=n} \pi / \sin \pi s = (-1)^n$ and $h(s) = X^{2s} g_s / \Gamma(2s + 1)$ is analytic on $\operatorname{Re} s > -\frac{1}{2}$ (Lemma 6(i)), the terms of (5) are the residues of $h(s) \pi / \sin \pi s$ at $s = 1, 2, \dots$. Split the series at $n = \lfloor 2w \rfloor$.

The tail. At integers, $B_n = \sum_{i \leq 2n} \varphi(i\tau) - n\varphi(\tau) \geq n\varphi(\tau) \geq 0$ by (7) and the monotonicity of φ , so $0 \leq g_n = 1 - e^{-B_n} \leq 1$ and $|\sum_{n > \sigma_*} (-1)^n g_n X^{2n} / (2n)!| \leq \sum_{n > \sigma_*} X^{2n} / (2n)!$. This is a sawtooth in τ , jumping each time $2w$ crosses an integer, with a superexponentially decreasing envelope: $7.5 \cdot 10^{-10}$ at $\tau = 0.02$ and $1.9 \cdot 10^{-11}$ at $\tau = 0.0127$. It is bounded together with the contour integral below and is nowhere the binding term.

The head. The remaining residues are enclosed by the rectangle $R = \{\frac{1}{2} \leq \operatorname{Re} s \leq \sigma_*, |\Im s| \leq X/2\}$, where σ_* is the largest half-odd-integer below $2w$ (so $|\sin \pi s| = \cosh \pi t \geq 1$ on the right edge; taking the edge at $2w$ itself would put it through a pole whenever $2w \in \mathbb{Z}$, e.g. at $\tau = 0.02$), which

lies *wholly inside* $\mathcal{S}$, so the amplitude bounds of Lemma 6 apply at every point of ∂R ; no bound on B_s outside $\mathcal{S}$ is needed anywhere. On $\operatorname{Re} s = \frac{1}{2}$ we have $|\sin \pi s| = \cosh \pi t \geq 1$, so no pole is grazed. On the horizontal edges the two exponentials cancel: $|\pi/\sin \pi s| \leq \pi/\sinh(\pi X/2) \leq 2.01\pi e^{-\pi X/2}$ (valid once $X \geq 1.69$, hence throughout $\tau \leq 0.02$, where $X \geq 9.9$), while $1/|\Gamma(2\sigma + 1 + iX)|$ carries the compensating $e^{+\pi X/2}$; the resulting kernel is $O(\sqrt{2\pi/X})$, attaining 0.7887 at $\sigma = \frac{1}{2}$ against $\sqrt{2\pi/X} = 0.7927$ when $\tau = 0.02$.

For the amplitude: where $|2s| \leq 2w$, (9) gives $|B_s| \leq \frac{\tau^2}{24} \left| \frac{M^3}{3} + \frac{M^2}{2} - \frac{M}{3} \right| + 0.02\tau^{3/2}$ with $M = 2s$, hence $|g_s| = |1 - e^{-B_s}| \leq |B_s|e^{|B_s|}$; elsewhere on ∂R , $|g_s| \leq 1 + e^{30.3\sqrt{\tau}}$ by (8). Write $\mathcal{B}(\tau, X)$ for the resulting majorant: the boundary integral of those bounds against $|X^{2s}/\Gamma(2s+1)| |\pi/\sin \pi s|$, divided by 2π , plus the tail. The prose above fixes the geometry; what remains is to bound $\mathcal{B}$ on the whole region $\tau \in (0, 0.02]$, $\tau X^2 \in [2e^{-\tau}, 2]$, and that is done in two ranges.

(1) $\tau \geq 2/101^2$: *verified enclosure*. On $w = \sqrt{2/\tau} \in [10, 101]$, $X \in [we^{-\tau/2}, w]$, the majorant is enclosed in MPFR interval arithmetic with outward directed rounding at every operation, by a branch and bound on the parameter box: a box is accepted when the enclosed supremum of $\mathcal{B}$ over the whole box does not exceed the infimum of $0.17\tau^{1/4}$ on it, and is bisected otherwise. The contour integrals are enclosed cellwise, each cell contributing its integrand enclosure times its width, so what is returned bounds the integral rather than estimating it; the cells tile each edge exactly, and the initial boxes are cut at the half-integer steps of σ_* , so no box straddles a jump of the contour. The scan terminates with 232 accepted boxes covering the range and no failures, and returns $\mathcal{B} \leq 0.16978\tau^{1/4}$, a box-level figure comparing the largest bound on a box with the smallest $\tau^{1/4}$ on it. Run at 96 and again at 160 bits it produces the same box decomposition and the same figure. Pointwise at $\tau = 0.02$, $X = w = 10$, where the cross-check below locates the maximum of the ratio, the enclosure is $\mathcal{B} \leq 0.061685 = 0.164031\tau^{1/4}$, and the scan repeated against the constant 0.166 in place of 0.17 also terminates without failures. (`tools/t2abs_iv`; the earlier adaptive-quadrature certificate `t2abs_certificate.py` shares no code with it and is kept as a cross-check, its seven tabulated ratios 0.1623, 0.0428, 0.0308, 0.0226, 0.0173, 0.0136, 0.0115 at $\tau = 2 \cdot 10^{-2}, \dots, 10^{-5}$ being reproduced from above.)

(2) $\tau < 2/101^2$: *explicit constants*. No finite scan reaches $\tau \rightarrow 0$. Here $X \geq 101e^{-1/101^2} > 100$, $w \leq Xe^{1/X^2} \leq 1.0001X$ and $\tau \leq 2/X^2$, so $0.17\tau^{1/4} = 0.17 \cdot 2^{1/4}w^{-1/2} \geq 0.2021X^{-1/2}$, and it suffices to bound $\mathcal{B}$ by that.

On the left edge $2s + 1 = 2 + 2it$, and $|\Gamma(1 + iy)|^2 = \pi y/\sinh \pi y$ makes the kernel exact: $|X^{2s}/\Gamma(2s+1)| |\pi/\sin \pi s| = X\pi\sqrt{g(\pi t)}/\sqrt{1+4t^2}$ with $g(u) = \tanh u/u \leq \min(1, 1/u)$, hence at most $X\pi$ for $t \leq 1/\pi$ and at most $\frac{1}{2}X\sqrt{\pi}t^{-3/2}$ beyond. There $|M| = \sqrt{1+4t^2} \leq 1+2t \leq 1+X < 2w$, so (9) applies along the whole edge; with $\frac{(1+2t)^3}{3} + \frac{(1+2t)^2}{2} + \frac{1+2t}{3} = \frac{8}{3}t^3 + 6t^2 + \frac{14}{3}t + \frac{7}{6}$ integrated against $t^{-3/2}$, and collecting the range $t \leq 1/\pi$ and the $0.02\tau^{3/2}$ remainder on the same power of X , the edge contributes $L \leq 0.03033X^{-1/2}$ for $X \geq 100$.

On a horizontal edge put $a = 2\sigma + 1$, $u = a/X$ and $\psi(u) = u - \arctan u - \frac{u}{2}\log(1+u^2)$, so $\psi'(u) = -\frac{1}{2}\log(1+u^2)$ and ψ is negative, decreasing and concave on $u > 0$. Stirling with Binet's remainder, $\log \Gamma(z) = (z - \frac{1}{2})\log z - z + \frac{1}{2}\log 2\pi + \mu(z)$ with $|\mu(z)| \leq \mu(\operatorname{Re} z) \leq 1/(12\operatorname{Re} z)$ (the kernel of Binet's first integral is positive on $(0, \infty)$), cancels every term of size $a \log X$ and leaves

$$\frac{X^{2\sigma}}{|\Gamma(a + iX)|} \cdot \frac{\pi}{\sinh(\pi X/2)} = \frac{\sqrt{2\pi}}{1 - e^{-\pi X}} X^{-1/2}(1+u^2)^{1/4} e^{X\psi(u)} e^{-\operatorname{Re} \mu}, \quad |\operatorname{Re} \mu| \leq \frac{1}{12a} \leq \frac{1}{24}. \quad (12)$$

With $d\sigma = \frac{X}{2}du$ and $|M| \leq X\sqrt{1+u^2}$, (9) applies for $u \leq \sqrt{3}$, where $|M| \leq 2X \leq 2w$, and gives $|g_s| \leq 1.00449[(1+u^2)^{3/2}/(18X) + 0.33502X^{-2}]$; for $u > \sqrt{3}$ only $|g_s| \leq 1 + e^{30.3\sqrt{\tau}} \leq 2.5352$ is used. Two elementary inequalities close the u -integral: $\psi(u) \leq -u^3/12$ on $[0, 1]$, since $\log(1+x) \geq x/2$

there; and $\psi(u) \leq \psi(1) + \psi'(1)(u-1)$ for $u \geq 1$, by concavity. With $\int_0^\infty e^{-Xu^3/12} du = \Gamma(\frac{4}{3})(12/X)^{1/3}$ and $(1+u^2)^{7/4} \leq 2^{7/4}$ on $[0, 1]$, one edge contributes $H \leq 0.50142 X^{-5/6} + 1.06902 X^{-11/6} + 4.5 \cdot 10^{-9} e^{-0.131(X-100)}$.

The right edge and the tail are exponentially small. On $\operatorname{Re} s = \sigma_*$ one has $|\pi/\sin \pi s| \leq \pi$ and $a = 2\sigma_* + 1 > 4X - 1$, while $\log |\Gamma(a + iY)| \geq (a - \frac{1}{2}) \log a - Y^2/a - a + \frac{1}{2} \log 2\pi - \frac{1}{12a}$ for $|Y| \leq X$; since $c \mapsto c(1 - \log c)$ decreases and $a/X \in [3.99, 4.011]$, the kernel there is at most $\pi e^{-1.2805X}$. And $n_0 = \lfloor \sigma_* \rfloor + 1 > 2X - 1$ with $(2n)! \geq (2n/e)^{2n}$ makes the tail at most $1.07 e^{-1.51X}$. Both are below 10^{-50} throughout. Collecting, for $X \geq 100$,

$$\mathcal{B} \leq 0.00966 X^{-1/2} + 0.15961 X^{-5/6} + 0.34028 X^{-11/6},$$

which is at most $0.2021 X^{-1/2}$ as soon as $0.15961 X^{-1/3} + 0.34028 X^{-4/3} \leq 0.2021 - 0.00966 = 0.19244$; the left side decreases and equals 0.03512 at $X = 100$. The constants of this range are regenerated, and the two ψ inequalities and the rearrangement (12) independently checked against `mpmath`'s complex Γ , in `t2abs_smalltau.py` at 40 and again at 60 digits.

At $\tau = \tau_4 = 0.012665$, the largest value Lemma 11 uses, the bound is 0.0144 , and what Lemma 11 consumes there is only $|T_2| < 1$. $\square$

Remark 10 (why Remark 8 is not used). The steepest-descent sketch is recorded because it produces the sharp constant, but it is not a proof, on three counts. (i) It asserts that the two saddles sum to $\cos W$ *exactly*; they do not: the leading saddle term carries a phase $e^{-i/(24W)}$, so the two sum to $\cos W + \frac{1}{24W} \sin W + O(W^{-2})$. (ii) It sizes the first saddle correction by $h'''(z^*) = O(W^{-2})$; but the saddle is *flat* ($h'' = 4i/W \rightarrow 0$), so the expansion parameter is W^{-1} and the correction is $c_1 = i/(24W) + O(W^{-2})$. Errors (i) and (ii) are each about three quarters of the quantity being computed and cancel exactly, which is why the stated constant is nevertheless right; taken literally the chain yields $\sqrt{2}/144$, four times too small. (iii) The deformation is justified by the claim that $\csc$ suppresses $|\Im z| \rightarrow \infty$, which is false for the exact B , as the numbers in the previous proof show; and the uniform remainder is attributed to a steepest-descent error estimate whose hypotheses (a parameter-independent phase and amplitude, an explicit path, an explicit truncation order) are not verified here. A complete proof needs the two-term flat-saddle expansion together with an error estimate on the curved path; that is not written, and nothing in this paper waits on it.

Lemma 11 (infinitude and accumulation of the travel poles). *The set $\{q \in (0, 1) : \Sigma_1^T(q) = 1\}$ is infinite, and together with [7, Lemma “The travel poles are locally finite”, lem:polefinite] it accumulates exactly at $q = 1$. In particular there is a travel pole with $w \in (m\pi, (m+1)\pi)$ for every large m .*

Proof. By [7, Proposition prop:qtrig], $1 - \Sigma_1^T = \cos(Z; q^2)$ with $Z^2 = z_0^2 e^\tau$. The collapse above defines the radius-free coefficients $e^{-B_n} = q^{n^2+n} [2(1-q)]^n (2n)! / [(q; q)_{2n} W^{2n}]$ through $S_e = \cos(z_0; q^2) = \sum_{n \geq 0} (-1)^n e^{-B_n} W^{2n} / (2n)!$; replacing z_0^2 by $Z^2 = z_0^2 e^\tau$ multiplies the n -th coefficient by $e^{n\tau}$, i.e. replaces W by $W e^{\tau/2} = w$ exactly:

$$1 - \Sigma_1^T = \sum_{n \geq 0} (-1)^n e^{-B_n} \frac{w^{2n}}{(2n)!} = \cos w - T_2(w),$$

with $T_2(\cdot)$ the series (5) evaluated at argument w in place of W ; the coefficients e^{-B_n} are the *same* ones, so no re-derivation of the amplitude is involved. Only the argument moves, and it moves inside the region where the two estimates are already proved: Lemma 6 is stated on a strip built from the larger argument w (and containing the W -strip), so it applies verbatim, and Lemma 9 is

stated for any argument with $\tau X^2 \in [2e^{-\tau}, 2]$, hence at $X = w$, where $\tau w^2 = 2$ identically. Thus $|T_2(w)| \leq 0.064$ once $\tau \leq 0.02$, which along the extreme-phase sequence $w = m\pi$, $\tau^{(m)} = 2/(m\pi)^2$, holds for every $m \geq 4$ (there $\tau \leq \tau_4 = 0.012665$ and the bound is 0.0144). Along that sequence

$$1 - \Sigma_1^T|_{w=m\pi} = (-1)^m - T_2(m\pi),$$

which has the sign of $(-1)^m$ for all $m \geq m_0$, with $m_0 = 4$ from Lemma 9 (only $|T_2(m\pi)| < 1$ is used, a far weaker input than the sharp size recorded in Remark 8). By continuity of $q \mapsto \Sigma_1^T(q)$ the intermediate value theorem yields a zero of $1 - \Sigma_1^T$ with $w \in (m\pi, (m+1)\pi)$ for every $m \geq m_0$: infinitely many travel poles, with $\tau \downarrow 0$, i.e. $q \uparrow 1$. Since every travel pole has $q > \frac{1}{10}$ and the pole set is locally finite in $(0, 1)$ ([7, Lemma “The travel poles are locally finite”, lem:polefinite]), its only accumulation point is $q = 1$. $\square$

The alternation is in fact immediate: $|T_2(\pi)| \leq 0.027$ by direct evaluation and $|T_2(m\pi)| = O(\tau_m)$ thereafter, so $m_0 = 1$ and there is a travel pole in every window $(m\pi, (m+1)\pi)$, $m \geq 1$. The dominant pole q_1 is the one below this ladder, in $w \in (0, \pi)$. The transport identity and the sign alternation (checked for $m \leq 40$, worst $|T_2(m\pi)| = 0.0267$) are verified in `star_gate_certificate.py`.

Alternative proof of the first sentence, without T_2 . [7, Proposition “infinitely many travel poles, without T_2 ”, prop:infpolesalt] proves that the travel poles are infinite from the Fredholm determinant of [7, Theorem “the travel determinant”, thm:fredholm] alone: $\lambda_1(q) > (1 - q^2)/(2q)$, the uniform domination $|t_k| \leq (2|y|)^k/(2k-1)!!$ with Hurwitz’s theorem, which gives $\lambda_k(q) \sim (1-q)(2k-1)^2\pi^2/8$ at fixed k , and the fact that distinct branches cross the level 1 at distinct q ; no single-crossing property is needed. With [7, Lemma “The travel poles are locally finite”, lem:polefinite] this also gives the accumulation at $q = 1$. It does not localise the poles: the window statement above, a travel pole with $w \in (m\pi, (m+1)\pi)$ for every large m , still uses Lemma 9.

Lemma 12 (infinitude of the zeros of the bulk block). *The zero set of S_e in $(0, 1)$, equivalently $\{q \in (0, 1) : S_1(q) = 1\}$, is infinite.*

Proof. Here no transport is needed: $S_e = \cos W - T_2$ at $W = we^{-\tau/2}$ is (5) itself, the series for which both estimates were set up. The map $\tau \mapsto W(\tau) = \sqrt{2/\tau} e^{-\tau/2}$ is continuous and strictly decreasing on $(0, \infty)$ with $W \rightarrow \infty$ as $\tau \downarrow 0$, so each integer $m \geq 4$ gives a unique $\tau^{[m]}$ with $W(\tau^{[m]}) = m\pi$; since $W < w$, $\tau^{[m]} < 2/(m\pi)^2 \leq \tau_4 = 0.012665$, and $\tau W^2 = 2e^{-\tau}$ identically, so Lemma 9 applies at every one of them and only $|T_2| < 1$ is used. Then $S_e = (-1)^m - T_2$ has the sign of $(-1)^m$ along the sequence, and the $\tau^{[m]}$ decrease in m , so the intermediate value theorem applied to the continuous function $q \mapsto S_e(q)$ (Theorem 1, first clause) gives a zero in each of the disjoint intervals $(e^{-\tau^{[m]}}, e^{-\tau^{[m+1]}})$, $m \geq 4$. Finally $S_e = 1 - S_1$ by [7, Proposition prop:qtrig]. $\square$

The zeros of S_e are not the travel poles: those are the zeros of $\cos(Z; q^2)$, which is the same function a half q -step away ([7, Proposition prop:qtrig]), and S_e is bounded away from 0 at every travel pole by Proposition 69. Lemma 12 is used only in Corollary 2. The alternation it asserts is checked directly for $4 \leq m \leq 40$ in `f2_pole_route.py`, at 60 and again at 100 digits: the series for S_e cancels heavily against intermediate terms far larger than its value, and 30 digits already returns wrong signs there.

Remark 13 (the Hubbard–Stratonovich integral representation). Writing $q^{j(j+1)} = q^j e^{-j^2\tau}$ and linearising the Gaussian through $e^{-j^2\tau} = (4\pi\tau)^{-1/2} \int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju} du$, the denominator block admits the *exact* real integral

$$S_e = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} e^{-u^2/(4\tau)} \Psi(u, q) du, \quad \Psi(u, q) = \sum_{j \geq 0} \frac{(-2(1-q)q e^{iu})^j}{(q; q)_{2j}}, \quad (13)$$

the interchange of sum and integral being justified by absolute convergence ($\sum_j |Z|^j/(q; q)_{2j} < \infty$, $Z = -2(1-q)qe^{iu}$, against the integrable Gaussian). Because the coefficients of Ψ in $\zeta = e^{iu}$ are real, $\Psi(-u, q) = \overline{\Psi(u, q)}$, so (13) is real and may be written over a single variable, $S_e = (4\pi\tau)^{-1/2} \int_{\mathbb{R}} e^{-u^2/(4\tau)} \operatorname{Re} \Psi du$; the exact Gaussian moment underlying it, $\int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju} du = \sqrt{4\pi\tau} e^{-\tau j^2}$, is machine-checked in Lean 4 (`GaussHS.lean`, theorem `gaussian_moment`; axioms `propext`, `Classical.choice`, `Quot.sound`, no sorry).

This representation, though exact and convenient, is *not* an analytic shortcut. The amplitude $\Psi(\cdot, q)$ is entire in ζ , but it is *not* uniformly bounded as $q \rightarrow 1^-$: on any fixed window it grows like $\cosh(w \sin(u/2))$ (numerically $|\Psi|$ already exceeds 300 at $u = 1$, $\tau = 10^{-2}$), and its termwise limit $\cos(w e^{iu/2})$ differs from Ψ by an $O(1)$ factor at the dominant scale $j \sim \tau^{-1/2}$ (where the subleading part of $(q; q)_{2j}$ is itself $O(1)$; numerically $\Psi/\cos(w e^{iu/2}) \approx 0.62$). The size of S_e is therefore governed by the balance between this exponential growth and the Gaussian; a simple *complex* saddle at $u^* \approx -\sqrt{2\tau}$, of the same order of difficulty as the direct sum, *not* a bounded-amplitude stationary-phase integral with trivial hypotheses [44, Ch. II]. In particular the bounded-variation input of Lemma 6 is *not* eliminated by passing to (13); it, together with the rectangle bound of Lemma 9, remains the analytic core. The value of (13) is representational (one real variable, an entire integrand, an exact Lean-checked moment), not a reduction of hypotheses; the numerical agreement of its saddle value with S_e is recorded in `gaussint_verify.py`.

Remark 14 (an elementary derivation of the coefficients of Remark 8). The leading term and the constant $\sqrt{2}/36$ of Remark 8 can be obtained without the steepest-descent theorem, by an exact algebraic reduction. Writing $1 - e^{-k\tau} = 2e^{-k\tau/2} \sinh(k\tau/2)$ throughout, the j -th term of $S_e = \sum_{j \geq 0} (-2(1-q))^j q^{j(j+1)}/(q; q)_{2j}$ collapses to

$$S_e = \sum_{j \geq 0} (-1)^j e^{-j\tau} \frac{\sinh^j(\tau/2)}{\prod_{k=1}^{2j} \sinh(k\tau/2)} = \sum_{j \geq 0} (-1)^j \frac{(2/\tau)^j}{(2j)!} e^{-j\tau + E_j}, \quad E_j = \frac{j\tau^2}{24} - \frac{\tau^2}{144} (2j)(2j+1)(4j+1) + O(\tau^4 j^5),$$

E_j being the exact sinh-curvature defect. Two elementary facts finish the extraction. First, the factor $e^{-j\tau}$ resums *in closed form*: $\sum_j (-1)^j (2e^{-\tau}/\tau)^j/(2j)! = \cos W$ with $W = \sqrt{2/\tau} e^{-\tau/2}$, which is exactly the frequency of Remark 8; so the phase correction needs no estimate at all. Second, the entire-function moment identity $\sum_j (-1)^j y^j j^m/(2j)! = (y d/dy)^m \cos \sqrt{y}$ evaluates the curvature term through its leading part $-\frac{\tau^2}{9} (y d/dy)^3 \cos \sqrt{y} = -\frac{\tau^2}{9} \cdot \frac{W^3}{8} \sin W = -\frac{\sqrt{2}}{36} \sqrt{\tau} \sin W$ (using $\tau^2 W^3 = 2\sqrt{2} \sqrt{\tau} e^{-3\tau/2}$, the factor $e^{-3\tau/2}$ being absorbed in the $O(\tau)$), giving

$$S_e = \cos W - \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + O(\tau), \quad W = \sqrt{2/\tau} e^{-\tau/2},$$

all constants explicit and verified to ten digits (`elemcos_verify.py`). Carried to higher moments the same reduction produces every subsequent coefficient as a finite combination of $W^k \sin W$, $W^k \cos W$, so it serves the $O(\tau^2)$ -relative demand of the U -numerator in the same elementary terms. The one input it does *not* render elementary is the remainder estimate; that the alternating expansion is asymptotic with $O(\tau)$ error. That step is a *cancellation* statement: not only does the triangle inequality on $\sum_j (2/\tau)^j/(2j)! |\cdot|$ return $\cosh W \asymp e^{\sqrt{2/\tau}}$, but the partial sums of the $\cos W$ series themselves swing up to $\asymp e^{c\sqrt{2/\tau}}$ before cancelling to $O(1)$ (numerically $\max_J |S_J| = 9, 48, 1459, 7.6 \times 10^4$ at $\tau = 0.1, 0.05, 0.02, 0.01$). Hence *every* partial-summation method (Abel, summation-by-parts, Euler transformation) fails, since each is controlled by these exponentially large partial sums; only a global resummation captures the cancellation. This global resummation is exactly the two-saddle steepest-descent computation carried out self-contained in the proof of Remark 8: the saddles $z_{\pm}^* = \pm iW/2$ supply the cancellation the partial sums cannot. Together, the elementary extraction

here and that saddle argument reduce Remark 8 to Cauchy's theorem, Stirling's formula, elementary Gaussian integration, the bounded-variation bound of Lemma 6, and *one* cited estimate: the classical remainder bound for a simple saddle [34, Ch. 4, Thm. 7.1], used once in the proof of Remark 8. It is not independent of that citation, and we do not claim otherwise; Appendix A provides the route that is.

Remark 15 (derivation of the U -numerator amplitude). The sharp picture behind Theorem 82: the asymptotic value $|\sum_k d_k| = \frac{3}{8\sqrt{2}}\tau^{5/2}(1 + O(\sqrt{\tau}))$ at the travel poles, on the one explicit series $P_{12} = \frac{2q^3}{1-q^3}\sum_k d_k$, $d_k = (-2)^k(1-q)^k q^{k^2+3k}/[(q^2; q^2)_k(q^5; q^2)_k]$, which submits to the identical reduction. Writing $1 - q^m = 2e^{-m\tau/2} \sinh(m\tau/2)$ throughout collapses the terms *exactly* to

$$d_k = (-1)^k e^{-k\tau} \frac{\sinh^{k+1}(\frac{\tau}{2}) \sinh(\frac{3\tau}{2}) \sinh((k+1)\tau)}{\prod_{m=1}^{2k+3} \sinh(\frac{m\tau}{2})},$$

and the elementary expansion (exact linear absorption into the argument, Faulhaber sums for the sinh-curvature, the moment identities $k^m \mapsto (y d/dy)^m$) yields, with $\Phi(y) = (\sin W - W \cos W)/(2W^3)$, $W = \sqrt{y}$, $D = y d/dy$,

$$\sum_k d_k = 6 \left[\Phi + c_3 D^3 \Phi + c_2 D^2 \Phi + \frac{c_2^2}{2} D^6 \Phi + (c_5 + c_2 c_3) D^5 \Phi + \frac{c_3^3}{6} D^9 \Phi \right] (y_*) + O(\tau^3),$$

$y_* = \frac{2}{\tau} e^{-\tau-23\tau^2/36}$, $c_2 = -\frac{5}{12}\tau^2$, $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$; *every constant an exact rational, derived, not fitted* (`elemY3_verify.py`; the error is verified $O(\tau^3)$ over $\tau \in [0.005, 0.08]$ and the formula reproduces the exact series at the poles $m = 7, \dots, 33$ to the displayed accuracy). At the travel poles the expansion gives $|\sum_k d_k|/\tau^{5/2} \rightarrow 3/(8\sqrt{2}) = 0.2652$, i.e. exactly the observed gate value $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$ with its fourfold margin: structurally, the pole sits at offset δw from the zero of the displayed expansion, and the constant $3/(8\sqrt{2})$ arises as $(3\tau/2) \delta w / \tau^{5/2}$.

The *phase-matching step* that fixes δw is carried out by the same machinery. The pole equation $\Sigma_1(q_m) = 1$ collapses exactly to $1 - \Sigma_1 = \sum_{j \geq 0} (-1)^j \sinh^j(\frac{\tau}{2}) / \prod_{m=1}^{2j} \sinh(\frac{m\tau}{2})$; the S_e -series stripped of its $e^{-j\tau}$ phase; so the identical expansion applies with $y_* = \frac{2}{\tau} e^{\tau^2/36}$, $c_2 = -\frac{1}{12}\tau^2$ and the *universal* $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$. Solving the two phase conditions perturbatively in the common variable $w = \sqrt{2/\tau}$ near each grid point $(M + \frac{1}{2})\pi$ gives

$$\chi_{\text{pole}} = -\frac{\sqrt{2}}{36} \sqrt{\tau} + \frac{41\sqrt{2}}{2400} \tau^{3/2} + O(\tau^2), \quad \chi_{\text{zero}} = -\frac{\sqrt{2}}{36} \sqrt{\tau} - \frac{259\sqrt{2}}{2400} \tau^{3/2} + O(\tau^2).$$

The $\sqrt{\tau}$ phases agree *identically*; both are produced by the universal cubic curvature c_3 , and the constant $\sqrt{2}/36$ is that of Remark 8; this is precisely why the travel poles track the zeros of $Y_3(1)$. The $\tau^{3/2}$ terms differ by exactly

$$\delta w = \chi_{\text{pole}} - \chi_{\text{zero}} = \frac{\sqrt{2}}{8} \tau^{3/2} + O(\tau^2), \quad \text{whence} \quad \left| \sum_k d_k \right| = \frac{3\tau}{2} \delta w (1 + O(\sqrt{\tau})) = \frac{3}{8\sqrt{2}} \tau^{5/2} (1 + O(\sqrt{\tau})),$$

and therefore $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.17678 < 1/\sqrt{2}$: *the numerator gate, derived, with its fourfold margin*. Verification: the two phase series match the exact pole and zero locations to seven digits at $m = 11, 17, 23, 29$, and $\delta w / \tau^{3/2} = 0.17699, 0.17687, 0.17683, 0.17681 \rightarrow \sqrt{2}/8 = 0.176777$ (`phase_match_verify.py`); the pole-side series independently reproduces the McMahon-type coefficients of the travel-pole expansion. The remainder discipline throughout is that of Remark 8: every coefficient extracted elementarily, the $O(\cdot)$ terms controlled by the same two-saddle argument.

5 The q -trigonometric layer: remarks and spectral consequences

The q -trigonometric identification of the blocks ([7, Proposition prop:qtrig]), the travel determinant ([7, Theorem “the travel determinant”, thm:fredholm]) and the numerator identity ([7, Proposition “the numerators do not vanish at the travel poles”, prop:numnonvanish]) are proved in the companion paper. This section records the numerical remarks attached to them and the spectral material at the travel poles, with its standing.

Remark 16. The identification is consistent with the pole data: at all 58 tabulated poles the computed value of $\cos(z_0; q^2)$ agrees with the stored S_e to at least thirty digits, and $|S_e|/\sqrt{\tau}$ lies in $[0.698, 0.708]$, approaching $1/\sqrt{2}$ (`qtrig_verify.py`). The half-step structure accounts for the size of the gate quantity: $Z - z_0 = z_0(1 - \sqrt{q})/\sqrt{q} = \tau^{3/2}/\sqrt{2} + O(\tau^{5/2})$, the pole condition annihilates $\cos(\cdot; q^2)$ at Z , and the q -derivative rule $(1 - q)D_{q,z} \sin(z; q^2) = \cos(z; q^2)$ of [23] gives the function slope $1/(1 - q) \sim 1/\tau$ near the zero, so the sampled value is of size $(\tau^{3/2}/\sqrt{2})(1/\tau) = \sqrt{\tau/2}$, the observed constant. Turning this observation into a proved lower bound on $|S_e(q_m)|$ independent of Remark 8 requires uniform control of $\cos'(\cdot; q^2)$ between z_0 and Z as $q \rightarrow 1^-$. That control is supplied by Appendix A (Lemma 73 bounds $\max |f''|$ on $[qZ, Z]$, and Proposition 69 converts it into the lower bound), independently of Remark 8.

Remark 17. The two numerators are individually of size w_m : $|\Sigma_0(q_m)|/w_m$ and $|S_0(q_m)|/w_m$ rise monotonically to 1 over the first 32 travel poles, from 0.731 and 0.893 at the dominant pole $q_1 = 0.449453\dots$ to 0.99992 and 0.99997 at the last, with Σ_0 strictly alternating in sign (`qnumerator_certificate.py`, `mpmath` at two precisions, worst deviation in [7, (eq:numprod)] $4.5 \cdot 10^{-28}$; the dominant pole is included, it lying below the extreme-phase ladder and being bracketed separately). This matches the asymptotic $\Sigma_0 \sim w \sin w$, the amplitude companion of $1 - \Sigma_1 \sim 1 - \cos w$; but [7, (eq:numprod)] is an identity and the asymptotic is not used anywhere.

5.1 Spectral consequences at the travel poles

[7, Theorem “the travel determinant”, thm:fredholm] turns questions about the travel poles into questions about the eigenvalue branches $\lambda_k(q)$. This subsection records what that gives, and at what standing. Write $\mathbf{B}(q) := 1 - \Sigma_1(q) = \det(I - \mathcal{T}(q))$.

Lemma 18 (Hellmann–Feynman). *The map $q \mapsto \mathbf{S}(q)$ is real-analytic from $(0, 1)$ into the trace-class operators, and each λ_k is real-analytic on $(0, 1)$. Let $\mathcal{T}R = \lambda_k(q)^{-1}R$ with $R \neq 0$ real ([7, Theorem “the travel determinant”, thm:fredholm](ii)), and put $u = KR$. Then*

$$\frac{\lambda'_k(q)}{\lambda_k(q)} = -\frac{\epsilon_1 + \epsilon_2}{q} - \frac{1 - 2q}{q(1 - q)}, \quad \epsilon_1 = \frac{\sum_s s q^{-s} (u_{s+1} - u_s)^2}{\sum_s q^{-s} (u_{s+1} - u_s)^2}, \quad \epsilon_2 = \frac{\sum_s s q^s u_s^2}{\sum_s q^s u_s^2}.$$

In particular $\lambda'_k(q) < 0$ if and only if $\epsilon_1 + \epsilon_2 > (2q - 1)/(1 - q)$.

Proof. With the principal branch of $q^{1/2}$, the entries $2q^{1+(s+s')/2+\max(s,s')}$ are holomorphic on $\{|q| < 1, \operatorname{Re} q > 0\}$ and the sum of their moduli converges locally uniformly there, so $\mathbf{S}$ is holomorphic in trace norm, and self-adjoint for real q . Its non-zero eigenvalues are isolated and simple ([7, Theorem “the travel determinant”, thm:fredholm](iv)), so by analytic perturbation theory for a bounded holomorphic self-adjoint family [21, Ch. II and Ch. VII] each eigenvalue $\kappa(q) = 1/\lambda_k(q)$ and its eigenvector $\phi(q)$ are real-analytic, and $\kappa' = \langle \phi, \mathbf{S}'\phi \rangle / \langle \phi, \phi \rangle$. No q -dependent domain enters: $\mathbf{S}(q)$ is a bounded family.

It remains to evaluate $\langle \phi, \mathbf{S}'\phi \rangle$. Put $R = D_e^{1/2}\phi$ and $u = KR$, so that $D_e u = \mathcal{T}R = \kappa R$ and $\phi_s^2 = \kappa^{-2} 2q^{1+s} u_s^2$; put $N = \sum_s q^s u_s^2$ and $E = \sum_s q^{-s} (u_{s+1} - u_s)^2$. Since $q\partial_q \mathbf{S}[s, s'] =$

$$(1 + \frac{s+s'}{2} + \max(s, s'))\mathbf{S}[s, s'],$$

$$q\langle\phi, \mathbf{S}'\phi\rangle = \kappa\langle\phi, \phi\rangle + \kappa \sum_s s \phi_s^2 + \sum_{s, s'} R_s R_{s'} \max(s, s') q^{\max(s, s')},$$

and the first two terms are $(2qN/\kappa)(1 + \epsilon_2)$. For the third apply $q\partial_q$ to $q^{\max(s, s')} = (1 - q) \sum_{t \geq \max(s, s')} q^t$: with $\rho_t = \sum_{s \leq t} R_s$, bounded by $\|R\|_{\ell^1}$, it equals $\sum_t (t(1 - q) - q) q^t \rho_t^2$, all sums converging absolutely. Since $u_{t+1} - u_t = \sum_{s'} (q^{\max(t+1, s')} - q^{\max(t, s')}) R_{s'} = -(1 - q) q^t \rho_t$, one has $q^t \rho_t^2 = q^{-t} (u_{t+1} - u_t)^2 / (1 - q)^2$, so the third term is $E((1 - q)\epsilon_1 - q)/(1 - q)^2$. Finally $\langle R, KR \rangle = (1 - q) \sum_t q^t \rho_t^2 = E/(1 - q)$ and $\langle R, KR \rangle = \sum_s R_s u_s = 2qN/\kappa$, so $E = 2q(1 - q)N/\kappa$. Collecting, $q\langle\phi, \mathbf{S}'\phi\rangle = (2qN/\kappa)(1 + \epsilon_1 + \epsilon_2 - q/(1 - q))$ and $\langle\phi, \phi\rangle = 2qN/\kappa^2$, so $\kappa'/\kappa = (\epsilon_1 + \epsilon_2)/q + (1 - 2q)/(q(1 - q))$, and $\lambda'_k/\lambda_k = -\kappa'/\kappa$. $\square$

At $q_1, \dots, q_6$ the right-hand side agrees to 12 digits with $\lambda'_m(q_m) = -\partial_q F / \partial_\lambda F$ computed from the explicit F of [7, Theorem “the travel determinant”, thm:fredholm]. At a travel pole the eigenvalue 1 has eigenvector the travel null vector ([7, Corollary “the travel poles are Hahn–Exton zeros”, cor:qbessel]), so the following is a statement about R alone.

Theorem 19 ((T_{poles}^*)). *Let $\frac{1}{2} < q < 1$, and let $R \neq 0$ be real with $R \in \ell^1$ and $\mathcal{T}(q)R = R$. With $u = KR$ and ϵ_1, ϵ_2 as in Lemma 18,*

$$\epsilon_1 + \epsilon_2 \geq \frac{5}{4} \cdot \frac{2q - 1}{1 - q} > \frac{2q - 1}{1 - q}.$$

In particular the strict inequality holds for the travel null vector at every travel pole $q_m > \frac{1}{2}$. One step of the proof, the inequality (15) on $[\frac{1}{2}, 1 - 10^{-6}]$, is computer-assisted (interval arithmetic).

Proof. Put $\alpha = 2q(1 - q)$. Everything below is for this fixed q .

Step 0: the recursion. From $R = \mathcal{T}R = D_e u$, $R_s = 2q^{1+s} u_s$, so $(1 - q)R_s = \alpha q^s u_s$. Define $p_s = -\alpha \sum_{t \leq s} q^t u_t$ for $s \geq -1$, so $p_{-1} = 0$. Since $q^{\max(s+1, t)} - q^{\max(s, t)}$ equals $-(1 - q)q^s$ for $t \leq s$ and 0 for $t > s$,

$$u_{s+1} = u_s + q^s p_s, \quad p_s = p_{s-1} - \alpha q^s u_s \quad (s \geq 0). \quad (14)$$

Also $|u_s| \leq q^s \sum_{t \leq s} |R_t| + \sum_{t > s} |R_t| \rightarrow 0$, and $|p_s| \leq \alpha \sup |u|/(1 - q)$. The map $(u_s, p_s) \mapsto (u_{s+1}, p_{s+1})$ given by (14) is linear and invertible, and $(u_0, p_0) = (u_0, -\alpha u_0)$. If $u_0 = 0$ then $u \equiv 0$ and $R = D_e u = 0$. So $u_0 \neq 0$ and $(u_s, p_s) \neq (0, 0)$ for every $s \geq 0$.

Step 1: the flux identity. Summation by parts in (14) gives, for every N ,

$$\sum_{s=0}^N q^s p_s^2 = p_N u_{N+1} + \alpha \sum_{s=0}^N q^s u_s^2.$$

Here p is bounded and $u_{N+1} \rightarrow 0$, so $\sum_s q^s p_s^2 = \alpha \sum_s q^s u_s^2$. Since $q^{-s}(u_{s+1} - u_s)^2 = q^s p_s^2$, the two quotients of Lemma 18 have denominators $\alpha \mathcal{N}$ and $\mathcal{N}$, where $\mathcal{N} = \sum_s q^s u_s^2 > 0$. Put $F_s = u_s^2 + p_s^2/\alpha$ and $f_s = q^s F_s$, so that $\sum_s f_s = 2\mathcal{N}$. Then

$$\epsilon_1 + \epsilon_2 = \frac{\sum_s s q^s p_s^2}{\alpha \mathcal{N}} + \frac{\sum_s s q^s u_s^2}{\mathcal{N}} = 2 \langle s \rangle_f, \quad \langle s \rangle_f := \frac{\sum_s s f_s}{\sum_s f_s}.$$

If F were constant, $\langle s \rangle_f$ would be $q/(1 - q)$. The rest of the proof shows that F is constant up to a bounded factor.

Step 2: the amplitude. Put $c = q\sqrt{(1-q)/2}$, so $c^2 = \alpha q/4$ and $0 < c \leq \sqrt{6}/9 < 1$ on $(\frac{1}{2}, 1)$. Put $F'_s = u_s^2 + p_s^2/(\alpha q)$ and $G_s = F'_s + q^s u_s p_s$. By the arithmetic–geometric mean inequality $|u_s p_s| \leq c F'_s$, so

$$(1-c)F'_s \leq G_s \leq (1+c)F'_s, \quad qF'_s \leq F_s \leq F'_s,$$

and $G_s > 0$ by Step 0. A direct computation from (14) gives the drift identity

$$G_{s+1} - G_s = -(1-q)q^s u_{s+1} p_{s+1}.$$

Hence $|G_s - G_{s+1}| \leq (1-q)q^s c F'_{s+1} \leq \bar{d} q^s G_{s+1}$ with $\bar{d} = (1-q)c/(1-c) < 1$, so $|\log G_s - \log G_{s+1}| \leq \bar{d} q^s/(1-\bar{d})$. Summing over $s \geq 0$, for all $s, t \geq 0$,

$$|\log G_s - \log G_t| \leq \Lambda(q) := \frac{\bar{d}}{(1-\bar{d})(1-q)} = \frac{c}{(1-c)(1-\bar{d})},$$

and therefore

$$\frac{F_s}{F_t} \leq \frac{G_s/(1-c)}{qG_t/(1+c)} \leq \mathcal{R}(q) := \frac{e^{\Lambda(q)}(1+c)}{q(1-c)}.$$

Step 3: the mean. Let $m = \inf_s F_s$ and $M = \sup_s F_s$. By Step 2, $M \leq \mathcal{R}m$, and $M \geq F_0 > 0$, so $m > 0$. For $s \geq 1$ put $A = \sum_{t \geq s} f_t \geq m q^s/(1-q)$ and $B = \sum_{t < s} f_t \leq M(1-q^s)/(1-q)$. Since $A/(A+B)$ increases in A and decreases in B ,

$$\frac{\sum_{t \geq s} f_t}{\sum_{t \geq 0} f_t} \geq g(s) := \frac{q^s}{\mathcal{R} - (\mathcal{R} - 1)q^s}.$$

Summing over $s \geq 1$ gives $\langle s \rangle_f \geq \sum_{s \geq 1} g(s)$. The function g decreases in s , so $\sum_{s > K} g(s) \geq \int_{K+1}^{\infty} g$, and for every $K \geq 0$

$$\mathfrak{e}_1 + \mathfrak{e}_2 \geq \Phi_K(q) := 2 \sum_{s=1}^K g(s) + \frac{2}{(\mathcal{R} - 1) \log(1/q)} \log \frac{\mathcal{R}}{\mathcal{R} - (\mathcal{R} - 1)q^{K+1}}.$$

Here $\mathcal{R} = \mathcal{R}(q) > 1$.

Step 4: the certificate. It remains to show $\Phi_K(q) \geq \frac{5}{4}(2q-1)/(1-q)$. On $[\frac{1}{2}, 1 - 10^{-6}]$ this is checked with $K = 30$:

$$\Phi_{30}(q) - \frac{5}{4} \cdot \frac{2q-1}{1-q} > 0 \quad \left(\frac{1}{2} \leq q \leq 1 - 10^{-6}\right). \quad (15)$$

The check is done by interval arithmetic (`mpmath.iv`, 30 digits, outward rounding) on 770 consecutive subintervals covering $[\frac{1}{2}, 0.9999990026]$. Each subinterval passes on the first evaluation or after at most eight bisections, and none fails (`tstar_certificate.py`). The smallest ratio of Φ_{30} to the right-hand side is about 1.069, near $q \approx 0.781$. On $[1 - 10^{-6}, 1)$ use $g(s) \geq q^s/\mathcal{R}$ instead, valid because $\mathcal{R} \geq 1$. This gives $\mathfrak{e}_1 + \mathfrak{e}_2 \geq 2q/(\mathcal{R}(1-q))$. There $\mathcal{R} \leq 1.0021251$, so $2q/\mathcal{R} - \frac{5}{4}(2q-1) \geq 0.7457 > 0$, again checked in interval arithmetic on the closed interval $[1 - 10^{-6}, 1]$ (`tstar_endpoint.py`). Finally $\frac{5}{4} \frac{2q-1}{1-q} > \frac{2q-1}{1-q}$ because $q > \frac{1}{2}$.

For the last sentence of the theorem: at a travel pole the eigenvector of $\mathcal{T}$ for the eigenvalue 1 is the travel null vector ([7, Corollary “the travel poles are Hahn–Exton zeros”, cor:qbessel]). It lies in ℓ^1 and can be taken real ([7, Theorem “the travel determinant”, thm:fredholm](ii), $\mathbf{S}$ being real symmetric). $\square$

Standing and checks. Theorem 19 is proved, with one computer-assisted step: the interval certificate for (15) and the endpoint check. Both are in `code/zeta_probe/tstar_certificate/`, and their trust base is `mpmath.iv`. They are not formally verified. The proof uses no Sturm oscillation theorem, no analyticity in q and no information on the index of the eigenvalue: only $\mathcal{T}R = R$ with $R \in \ell^1$. The algebraic and amplitude steps are machine-checked in Lean 4 with Mathlib [28]. Each theorem listed depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`, with no `sorry` and no `native_decide`. The files are: `TstarCore.lean` (`sbp`, the finite summation by parts of Step 1, and `drift`, the one-step drift identity); `TstarAmplitude.lean` (`amgm`, `G_bounds`, `drift` for the recursion (14), the per-step bound `step`, and the finite-horizon bound `log_ratio` on $|\log G_m - \log G_n|$, which assumes $G > 0$); and `RoomB34.lean` (`leapfrog_ne`, `G_F'_compare`, and `G_pos`, which discharges that positivity hypothesis). The following are *not* formalised: Step 0, the passage $N \rightarrow \infty$ in Steps 1 and 2, Step 3, and Step 4. As a check, at $q_2 = 0.913486638731\dots$ the left-hand side is $\epsilon_1 + \epsilon_2 = 20.507$ and $(2q_2 - 1)/(1 - q_2) = 9.559$, a ratio of 2.145. Numerically the ratio exceeds 2 at every pole $2 \leq m \leq 50$ and decreases towards 2 (2.014 at $m = 5$). The constant $\frac{5}{4}$ is therefore not sharp.

At a pole $q_m \leq \frac{1}{2}$ (only $q_1 = 0.4494\dots$) the inequality holds automatically: the right-hand side is ≤ 0 and $\epsilon_2 > 0$, since u is not supported at $s = 0$ (if it were, $R = \kappa^{-1}D_e u$ would be a multiple of δ_0 and $(KR)_1 = qR_0 \neq 0$).

Proposition 20 (simple poles and the sign of $\mathbf{B}'$). *The following hold.*

- (i) *each λ_k takes the value 1 at exactly one point $c_k \in (0, 1)$, and $c_1 < c_2 < \dots$; the m -th travel pole is $q_m = c_m$, and $\lambda_j(q_m) < 1$ for $j < m$, $\lambda_m(q_m) = 1$, $\lambda_j(q_m) > 1$ for $j > m$;*
- (ii) *$\lambda'_m(q_m) < 0$, and $\mathbf{B}$ has a simple zero at q_m : $\mathbf{B}'(q_m) = \lambda'_m(q_m) \prod_{j \neq m} (1 - 1/\lambda_j(q_m)) \neq 0$. Hence in [7, Proposition “the shape of the local form”, prop:shape] $p = \pi = 1$: every travel pole is a simple pole of $(I - \mathcal{T})^{-1}$;*
- (iii) *$\text{sign } \mathbf{B}'(q_m) = (-1)^m$.*

Proof. (i) At a point c with $\lambda_k(c) = 1$, c is a travel pole with eigenvector R ([7, Corollary “the travel poles are Hahn–Exton zeros”, cor:qbessel]), so Theorem 19 (if $c > \frac{1}{2}$) or the note after it (if $c \leq \frac{1}{2}$), with Lemma 18, gives $\lambda'_k(c) < 0$. A real-analytic function that is strictly decreasing at every point where it equals 1 takes the value 1 at most once: between two consecutive such points it would be < 1 just after the first and > 1 just before the second, and the intermediate value theorem would give a third point between them. With [7, Proposition “infinitely many travel poles, without T_2 ”, prop:infpolesalt], λ_k takes the value 1 exactly once, at c_k , and $\lambda_k < 1$ exactly on $(c_k, 1)$. Since $\lambda_k(c_{k+1}) < \lambda_{k+1}(c_{k+1}) = 1$, $c_k < c_{k+1}$. The travel poles are the points where some λ_k equals 1 ([7, Corollary “the travel poles are Hahn–Exton zeros”, cor:qbessel]), so their increasing enumeration is $q_m = c_m$, and $\lambda_j(q_m) < 1$ exactly for $j < m$.

(ii) Near q_m let $P(q)$ be the analytic rank-one eigenprojection of $\kappa_m = 1/\lambda_m$ [21, Ch. VII]. Since $\mathbf{S}$ commutes with P , $\mathbf{B} = \det(I - \mathbf{S}) = (1 - \kappa_m) \det(I - \mathbf{S}(I - P))$, and the second factor is real-analytic near q_m , equal to $\prod_{j \neq m} (1 - 1/\lambda_j(q_m)) \neq 0$ at q_m . So $\mathbf{B}'(q_m) = -\kappa'_m(q_m) \prod_{j \neq m} (1 - 1/\lambda_j(q_m))$ with $-\kappa'_m(q_m) = \lambda'_m(q_m) < 0$, and $p = 1$. For π , on ℓ^1

$$(I - \mathcal{T})^{-1} = I + D_e^{1/2}(I - \mathbf{S})^{-1}D_e^{1/2}K,$$

as follows from $(I - D_e K)D_e^{1/2} = D_e^{1/2}(I - \mathbf{S})$ and $D_e^{1/2}K D_e K = \mathbf{S}D_e^{1/2}K$; here $D_e^{1/2}K : \ell^1 \rightarrow \ell^2$ and $D_e^{1/2} : \ell^2 \rightarrow \ell^1$ are bounded and analytic in q . Near q_m , $(I - \mathbf{S})^{-1} = P/(1 - \kappa_m)$ plus the reduced resolvent, which is holomorphic there because the other eigenvalues stay away from 1; and

$1 - \kappa_m$ has a simple zero. So $(I - \mathcal{T})^{-1}$ has a pole of order at most 1, and $\pi \geq p \geq 1$ ([7, Proposition “the shape of the local form”, prop:shape]) gives $\pi = p = 1$.

(iii) By (i) the product in (ii) has exactly $m - 1$ negative factors, and $\lambda'_m(q_m) < 0$. $\square$

The alternation sign $\mathbf{B}'(q_m) = (-1)^m$ is checked numerically for $m \leq 10$ ($\mathbf{B}'(q_1) = -2.630$, $\mathbf{B}'(q_2) = 28.71$, $\dots$, $\mathbf{B}'(q_{10}) = 6661$).

The variational setting. Fix $q \in (0, 1)$; all sequences are real. For a sequence $u = (u_s)_{s \geq 0}$ put

$$E(u) = \sum_{s \geq 0} q^{-s} (u_{s+1} - u_s)^2, \quad M(u) = \sum_{s \geq 0} 2(1 - q) q^{s+1} u_s^2,$$

both in $[0, \infty]$; E is the energy of Lemma 18, and $M = 2q(1 - q)N$ in the notation of its proof. The quotient E/M is the Rayleigh quotient of the q -string of [7, Remark “a q -string”, rem:qstring] in its Friedrichs realisation, free at $s = 0$ and fixed at $s = \infty$ ($u_s \rightarrow 0$). We do not identify the closed form domain of that realisation. We work instead on the explicit subspace

$$\mathcal{D} := KD_e^{1/2} \ell^2 = \left\{ KX : \sum_{s \geq 0} X_s^2 q^{-s} < \infty \right\}, \quad J := KD_e^{1/2} : \ell^2 \rightarrow \mathcal{D},$$

which contains every eigenfunction $u = KR$, and we derive the variational principle on $\mathcal{D}$ directly from the positive trace-class operator $\mathbf{S}$ of [7, Theorem “the travel determinant”, thm:fredholm](i). On $\mathcal{D}$ every quantity below is finite and no closure is taken, so nothing depends on the choice of form domain. Let $\phi_1, \phi_2, \dots$ be the orthonormal eigenvectors of $\mathbf{S}$, $\mathbf{S}\phi_k = \kappa_k \phi_k$ with $\kappa_k = 1/\lambda_k(q)$ ([7, Theorem “the travel determinant”, thm:fredholm](iv)). They form a complete orthonormal system, $\mathbf{S}$ being compact, self-adjoint and injective, and they can be taken real, $\mathbf{S}$ being real symmetric.

Lemma 21 (the shift inequality). *For every $q \in (0, 1)$ and every $n \geq 1$,*

$$\lambda_n(q) \leq q^2 \lambda_{n+1}(q).$$

Proof. Step 1: the quotient on $\mathcal{D}$. Let $x \in \ell^2$, $X = D_e^{1/2} x$, which lies in ℓ^1 by Cauchy–Schwarz, and $u = Jx = KX$. Then

$$E(u) = (1 - q) \langle x, \mathbf{S}x \rangle, \quad M(u) = (1 - q) \|\mathbf{S}x\|^2. \quad (16)$$

Indeed, with $\rho_t = \sum_{s \leq t} X_s$ one has $u_{t+1} - u_t = -(1 - q)q^t \rho_t$, as in the proof of Lemma 18, so $E(u) = (1 - q)^2 \sum_t q^t \rho_t^2 = (1 - q) \langle x, \mathbf{S}x \rangle$ by the proof of [7, Theorem “the travel determinant”, thm:fredholm](i); and $M(u) = (1 - q) \|D_e^{1/2} u\|^2$ with $D_e^{1/2} u = D_e^{1/2} K D_e^{1/2} x = \mathbf{S}x$. In particular J is injective: $Jx = 0$ gives $\mathbf{S}x = 0$, hence $x = 0$. So $M(u) > 0$ for every $u \in \mathcal{D} \setminus \{0\}$.

Step 2: the variational principle. Let $V \subset \mathcal{D}$ be a subspace of dimension n . Then

$$\max_{u \in V \setminus \{0\}} \frac{E(u)}{M(u)} \geq \lambda_n, \quad (17)$$

and on $V_n := J \text{span}\{\phi_1, \dots, \phi_n\}$ one has $E(u)/M(u) \leq \lambda_n$ for every $u \neq 0$. (The maximum exists: E/M is a quotient of positive definite quadratic forms on the finite-dimensional V .) For the proof write $V = J V'$ with $\dim V' = n$ and put $Y = \mathbf{S}^{1/2} V'$, again of dimension n because $\mathbf{S}^{1/2}$ is injective. For $u = Jx$ and $y = \mathbf{S}^{1/2} x$, (16) gives $E(u)/M(u) = \|y\|^2 / \langle y, \mathbf{S}y \rangle$. Since $\dim Y = n$, some $y \in Y \setminus \{0\}$ is orthogonal to $\phi_1, \dots, \phi_{n-1}$, and for it $\langle y, \mathbf{S}y \rangle = \sum_{k \geq n} \kappa_k \langle y, \phi_k \rangle^2 \leq \kappa_n \|y\|^2$, by completeness of the ϕ_k and $\kappa_k \leq \kappa_n$ for $k \geq n$. So $E/M \geq 1/\kappa_n = \lambda_n$ at the corresponding u , which

is (17). If $V' = \text{span}\{\phi_1, \dots, \phi_n\}$, then $Y = V'$, and $\langle y, \mathbf{S}y \rangle = \sum_{k \leq n} \kappa_k \langle y, \phi_k \rangle^2 \geq \kappa_n \|y\|^2$ for $y \in Y$, so $E/M \leq \lambda_n$ on $V_n \setminus \{0\}$.

Step 3: the shift. For a sequence u put $(\sigma u)_s = u_{s+1}$. If $u = KX \in \mathcal{D}$, then $\sigma u = KX'$ with $X'_0 = q(X_0 + X_1)$ and $X'_t = qX_{t+1}$ for $t \geq 1$. Indeed $q^{\max(s+1, t)} = q^{s+1}$ for $t \in \{0, 1\}$ and $q^{\max(s+1, r+1)} = q q^{\max(s, r)}$, so

$$u_{s+1} = \sum_{t \geq 0} q^{\max(s+1, t)} X_t = q^{\max(s, 0)} q(X_0 + X_1) + \sum_{r \geq 1} q^{\max(s, r)} qX_{r+1} = (KX')_s,$$

and $\sum_t X_t^2 q^{-t} \leq 2q^2(X_0^2 + X_1^2) + q^3 \sum_{r \geq 2} X_r^2 q^{-r} < \infty$. So $\sigma\mathcal{D} \subset \mathcal{D}$. Reindexing $t = s + 1$ in the two series gives

$$E(\sigma u) = q(E(u) - (u_1 - u_0)^2), \quad M(\sigma u) = \frac{M(u) - 2(1-q)q u_0^2}{q}, \quad (18)$$

all four series converging by Step 1. Hence, if $u \in \mathcal{D} \setminus \{0\}$ and $u_0 = 0$,

$$\frac{E(\sigma u)}{M(\sigma u)} = q^2 \frac{E(u) - u_1^2}{M(u)} \leq q^2 \frac{E(u)}{M(u)}.$$

On $\{u_0 = 0\}$ the map σ is injective: $\sigma u = 0$ and $u_0 = 0$ give $u = 0$.

Step 4: conclusion. The space V_{n+1} has dimension $n + 1$ (Step 1, J injective), so its subspace $\{u \in V_{n+1} : u_0 = 0\}$, the kernel of one linear functional, has dimension at least n . Choose an n -dimensional subspace W' of it and put $W = \sigma W'$. By Step 3, $W \subset \mathcal{D}$ and $\dim W = n$. For $w = \sigma u \in W \setminus \{0\}$, $u \in W' \setminus \{0\} \subset V_{n+1}$, Steps 3 and 2 give $E(w)/M(w) \leq q^2 E(u)/M(u) \leq q^2 \lambda_{n+1}$. By (17) applied to W , $\lambda_n \leq q^2 \lambda_{n+1}$. $\square$

Standing and checks. Lemma 21 is proved, with no computer-assisted step. It uses only [7, Theorem “the travel determinant”, thm:fredholm](i),(iv); it uses neither Theorem 19, nor a Sturm oscillation theorem, nor any information on the index of an eigenvalue. The finite-horizon form of Step 3 is machine-checked in Lean 4 with Mathlib [28], in RoomB34.lean: with $E_N(u) = \sum_{s < N} q^{-s} (u_{s+1} - u_s)^2$ and $M_N(u) = \sum_{s < N} 2(1-q)q^{s+1} u_s^2$, `E_shift_eq` is $E_N(\sigma u) = q(E_{N+1}(u) - (u_1 - u_0)^2)$, `E_shift_le` is $E_N(\sigma u) \leq q E_{N+1}(u)$, `M_shift_eq_general` and `M_shift_eq` are the identity for M_N without and with $u_0 = 0$, and `rayleigh_shift` is $E_N(\sigma u)/M_N(\sigma u) \leq q^2 E_{N+1}(u)/M_{N+1}(u)$ for $u_0 = 0$ and $M_{N+1}(u) > 0$. Each depends only on `propext`, `Classical.choice` and `Quot.sound`. Letting $N \rightarrow \infty$ in these identities of partial sums of non-negative series gives (18). Steps 1, 2 and 4, the description of σ on X , and the passage $N \rightarrow \infty$ are not formalised. The inequality is sharp as $n \rightarrow \infty$ at fixed q : at $q = 0.6$ (40 digits, truncation 90) the ratios $\lambda_n/(q^2 \lambda_{n+1})$ for $n = 1, \dots, 6$ are 0.3292, 0.9187, 0.99869, 0.9999979, $1 - 3.8 \cdot 10^{-10}$ and $1 - 8 \cdot 10^{-15}$.

Proposition 22 (the sign of S_e , and half-interlacing). *Put $\mathbf{v} = D_e^{1/2} \mathbf{1} = (\sqrt{2q^{1+s}})_{s \geq 0} \in \ell^2$.*

(a) *Off the travel poles in $(0, 1)$, $\Sigma_0/\mathbf{B} = \langle \mathbf{v}, (I - \mathbf{S})^{-1} \mathbf{v} \rangle$. At a travel pole q_m , with $\lambda_k(q_m) = 1$ and ϕ the normalised eigenvector,*

$$\Sigma_0(q_m) = w_m \prod_{j \neq k} (1 - 1/\lambda_j(q_m)), \quad w_m = \langle \mathbf{v}, \phi \rangle^2 > 0.$$

(b) *sign $\Sigma_0(q_m) = (-1)^{m-1}$; hence Σ_0 has at least one zero in each interval (q_m, q_{m+1}) .*

(c) *$\lambda_{m-1}(q_m) \leq q_m^2 < q_m$ for $m \geq 2$, and sign $S_e(q_m) = (-1)^{m-1}$; hence S_e has at least one zero in each interval (q_m, q_{m+1}) .*

Proof. (a) By Proposition 31, $\Sigma_0/\mathbf{B} = \mathbf{1}^\top (I - \mathcal{T})^{-1} D_e \mathbf{1}$ off the poles, and by the resolvent identity in the proof of Proposition 20, $(I - \mathcal{T})^{-1} D_e = D_e + D_e^{1/2} (I - \mathbf{S})^{-1} \mathbf{S} D_e^{1/2} = D_e^{1/2} (I - \mathbf{S})^{-1} D_e^{1/2}$; and $\mathbf{1}^\top D_e^{1/2} g = \langle \mathbf{v}, g \rangle$ for $g \in \ell^2$. For the value at a pole fix q and put $N_q(\lambda) = \det(I - \lambda \mathbf{S} + \mathbf{v} \mathbf{v}^\top) - \det(I - \lambda \mathbf{S})$, entire in λ . Where $I - \lambda \mathbf{S}$ is invertible, the rank-one identity $\det(A + \mathbf{v} \mathbf{v}^\top) = \det A (1 + \langle \mathbf{v}, A^{-1} \mathbf{v} \rangle)$ and the expansion of $\mathbf{S}$ in its orthonormal eigenbasis (ϕ_j) (complete, $\mathbf{S}$ being compact, self-adjoint and injective) give

$$N_q(\lambda) = \det(I - \lambda \mathbf{S}) \langle \mathbf{v}, (I - \lambda \mathbf{S})^{-1} \mathbf{v} \rangle = \sum_j w_j \prod_{i \neq j} (1 - \lambda / \lambda_i(q)), \quad w_j = \langle \mathbf{v}, \phi_j \rangle^2,$$

and since $|\prod_{i \neq j} (1 - \lambda / \lambda_i)| \leq e^{|\lambda| \operatorname{tr} \mathbf{S}}$ and $\sum_j w_j = \|\mathbf{v}\|^2$, the series converges locally uniformly and the identity holds for all λ . At $\lambda = 1$ and q off the poles, $N_q(1) = \Sigma_0(q)$; both sides are continuous in q , $N_q(1)$ as a difference of Fredholm determinants of trace-norm continuous families, so $N_q(1) = \Sigma_0(q)$ on all of $(0, 1)$. At $q = q_m$ every term $j \neq k$ of the series contains the factor $1 - 1/\lambda_k(q_m) = 0$. This gives the formula, and $w_m > 0$ because $\Sigma_0(q_m) \neq 0$ ([7, Proposition “the numerators do not vanish at the travel poles”, prop:numnonvanish]).

(b) By Proposition 20(i), $k = m$ and exactly $m - 1$ factors are negative. So Σ_0 alternates in sign along the poles, and, being continuous, vanishes between consecutive ones.

(c) Let $\lambda_k(q_m) = 1$. By Lemma 21 with $n = k - 1$, if $k \geq 2$ then

$$\lambda_{k-1}(q_m) \leq q_m^2 \lambda_k(q_m) = q_m^2 < q_m.$$

By [7, Theorem “the travel determinant”, thm:fredholm](iv),(v) at $\lambda = q$, $S_e(q_m) = \prod_j (1 - q_m / \lambda_j(q_m))$. For $j > k$, $\lambda_j > 1 > q_m$ and the factor is positive; for $j = k$ the factor is $1 - q_m > 0$; for $j < k$, $\lambda_j \leq \lambda_{k-1} < q_m$ and the factor is negative. So $\operatorname{sign} S_e(q_m) = (-1)^{k-1}$, for whichever k has $\lambda_k(q_m) = 1$. By Proposition 20(i), $k = m$, which gives both statements of (c). Finally $S_e(q) = F(q, q)$ is continuous in q on $(0, 1)$, so it vanishes between consecutive poles. $\square$

The first half of the proof of (c) does not use the index: without Proposition 20, and so without Theorem 19, it gives $\operatorname{sign} S_e(q_m) = (-1)^{k-1}$ where $\lambda_k(q_m) = 1$. The index identification $k = m$ is Proposition 20(i); it is the only place where (c) uses Theorem 19. Numerically, at $m = 2, \dots, 6$, $\lambda_{m-1}(q_m) = 0.112, 0.360, 0.510, 0.605, 0.670$, well below q_m^2 , and $S_e(q_m) = 0.520, -0.210, 0.127, -0.091, 0.071, -0.058$ for $m = 1, \dots, 6$ (50 digits). The sign alternation of Σ_0 is the one recorded for the first 32 poles in Remark 17.

Remark 23 (standing of Propositions 20 and 22). Earlier versions of this paper carried (T_{poles}^*) as a hypothesis. Our working notes also carried three inputs beside it for the same purpose: (I1) a Sturm oscillation theorem for the Friedrichs realisation of the eigenvalue problem; (I2) Hellmann–Feynman on a q -dependent form domain; (I3) [7, Proposition prop:qtrig] with Lemma 11; and Proposition 22(c) carried a shift inequality as an extra hypothesis. Their standing is now as follows.

- (T_{poles}^*) is Theorem 19. It is proved, with one computer-assisted step (the interval certificate (15) and its endpoint check). Its proof uses none of (I1)–(I3).
- (I2) is Lemma 18. It is proved: $\mathbf{S}(q)$ is a bounded norm-analytic family, so Kato’s theory applies and no domain question arises.
- (I3) is covered by [7, Proposition prop:qtrig], [7, Theorem “the travel determinant”, thm:fredholm], [7, Corollary “the travel poles are Hahn–Exton zeros”, cor:qbessel] and [7, Proposition “infinitely many travel poles, without T_2 ”, prop:infpolesalt], all proved in the companion. Lemma 11 is not used.

- (I1) is not used. It would turn the index statement of Proposition 20(i) into the count of $m - 1$ sign changes of R_s , which nothing here needs; that count remains unproved here (it is checked numerically for $m \leq 50$). The Friedrichs-domain question behind (I1) and (I2) does not arise either: $u_s = (KR)_s \rightarrow 0$ follows from $R \in \ell^1$ (Step 0 of the proof of Theorem 19).
- The shift inequality $\lambda_n \leq q^2 \lambda_{n+1}$, formerly an extra hypothesis of Proposition 22(c), is Lemma 21. It is proved, by min-max on the explicit subspace $\mathcal{D} = KD_e^{1/2} \ell^2$, with no computer-assisted step and none of (I1)–(I3) or Theorem 19; its finite-horizon shift identities are formalised in `RoomB34.lean`.

So Proposition 20 and Proposition 22 are proved, all three parts of the latter without extra hypotheses, at the standing of Theorem 19. For Proposition 22(c) that standing enters only through the index identification $\lambda_m(q_m) = 1$ of Proposition 20(i); the sign $(-1)^{k-1}$ at a pole where $\lambda_k = 1$ follows from Lemma 21 alone. No other theorem of this paper depends on these propositions: [7, Proposition “the shape of the local form”, prop:shape] needs only $\pi \geq p \geq 1$, and [7, Theorem “closed form of the junction pairing”, thm:RJ] gives the non-vanishing of the residue directly. Proposition 20(ii) adds that the orders there are $\pi = p = 1$ at every travel pole.

The pole-location law. We now locate the m -th travel pole to relative accuracy $O(1/m)$. The input is a two-sided bound on every eigenvalue branch, obtained by comparing the quotient E/M of the variational setting with the Rayleigh quotient of the string $-g'' = k^2 g$ on $[0, 1]$, $g(0) = 0$, $g'(1) = 0$. That string has eigenvalues k_n^2 , where

$$k_n := \frac{(2n-1)\pi}{2} \quad (n \geq 1).$$

Throughout, $\varepsilon = 1 - q$ and $\lambda_n(q)$ is the n -th smallest zero of $F(\cdot, q)$, that is, $1/\kappa_n$ with $\kappa_1 > \kappa_2 > \dots$ the eigenvalues of **S** ([7, Theorem “the travel determinant”, thm:fredholm](iii),(iv)). This is the labelling used in (17) and in Proposition 20. All the eigenvalues are simple, so the ordered labelling and the analytic branches coincide.

Lemma 24 (the eigenvalue sandwich). *For every $q \in (0, 1)$ and every $n \geq 1$,*

$$\lambda_n(q) \geq \frac{\varepsilon k_n^2}{2q(1 + 2\varepsilon k_n)},$$

and, if $2\varepsilon k_n < 1$,

$$\lambda_n(q) \leq \frac{\varepsilon k_n^2}{2q(1 - 2\varepsilon k_n)}.$$

Proof. Step 1: the grid, and two transfer relations. Put $z_s = q^s$ and $I_s = [z_{s+1}, z_s]$ ($s \geq 0$), so $|I_s| = \varepsilon q^s \leq \varepsilon$. The I_s cover $(0, 1]$ and overlap only at endpoints. With this grid

$$E(u) = \sum_{s \geq 0} \frac{(u_s - u_{s+1})^2}{q^s}, \quad M(u) = 2q \sum_{s \geq 0} |I_s| u_s^2,$$

the second because $2(1 - q)q^{s+1} = 2q|I_s|$. So M is a Riemann sum for $2q \int_0^1 g^2$, with no factor ε , while E carries the factor ε below. Let $\mathcal{H}$ be the space of absolutely continuous $g : [0, 1] \rightarrow \mathbb{R}$ with $g(0) = 0$ and $g' \in L^2(0, 1)$, and write $\|\cdot\|$ for the norm of $L^2(0, 1)$. We use two relations.

(a) *Energy.* Let $g \in \mathcal{H}$ and $u_s = g(z_s)$. By Cauchy–Schwarz on each I_s , $(u_s - u_{s+1})^2 = (\int_{I_s} g')^2 \leq |I_s| \int_{I_s} g'^2$, so $E(u) \leq \varepsilon \|g'\|^2$. If g is linear on every I_s , this is an equality term by term, so $E(u) = \varepsilon \|g'\|^2$.

(b) *Riemann error.* Let $g \in \mathcal{H}$ and $u_s = g(z_s)$. For $z \in I_s$, $|g(z_s)^2 - g(z)^2| \leq \int_{I_s} 2|gg'|$. Integrating over I_s gives $|\int_{I_s} u_s^2 - \int_{I_s} g^2| \leq \varepsilon \int_{I_s} 2|gg'|$. Summing over s and applying Cauchy–Schwarz,

$$|M(u) - 2q\|g\|^2| \leq 4q\varepsilon \|g\| \|g'\|,$$

the series defining $M(u)$ converging because its partial sums are bounded by $2q(\|g\|^2 + 2\varepsilon\|g\|\|g'\|)$.

Step 2: the continuum quotient. Put $\psi_j(z) = \sqrt{2}\sin(k_j z)$ and $\chi_j(z) = \sqrt{2}\cos(k_j z)$. Each of $(\psi_j)_{j \geq 1}$ and $(\chi_j)_{j \geq 1}$ is a complete orthonormal system in $L^2(0, 1)$. (They are the eigenfunctions of $-d^2/dz^2$ on $[0, 1]$ with the boundary conditions $g(0) = g'(1) = 0$, respectively $g'(0) = g(1) = 0$.) For $g \in \mathcal{H}$, integration by parts gives $\langle g', \chi_j \rangle = [g\chi_j]_0^1 + k_j \langle g, \psi_j \rangle = k_j \langle g, \psi_j \rangle$, since $g(0) = 0$ and $\cos k_j = 0$. Parseval in both systems then gives

$$\|g'\|^2 = \sum_j k_j^2 \langle g, \psi_j \rangle^2, \quad \|g\|^2 = \sum_j \langle g, \psi_j \rangle^2.$$

Two consequences. (a) If $W \subset \mathcal{H}$ has dimension n , some $g \in W \setminus \{0\}$ is orthogonal to $\psi_1, \dots, \psi_{n-1}$, and for it $\|g'\| \geq k_n \|g\|$. (b) If $g \in G_n := \text{span}\{\psi_1, \dots, \psi_n\}$, then $\|g'\| \leq k_n \|g\|$.

Step 3: the lower bound. Let $V_n = J \text{span}\{\phi_1, \dots, \phi_n\}$. By Step 2 of the proof of Lemma 21, $V_n \subset \mathcal{D}$ has dimension n , and $E(u) \leq \lambda_n M(u)$ on V_n . Every $u = KX \in \mathcal{D}$ has $u_s \rightarrow 0$, because $|u_s| \leq q^s \sum_{t \leq s} |X_t| + \sum_{t > s} |X_t|$ and $X \in \ell^1$. So the function g_u that is linear on each I_s , with $g_u(z_s) = u_s$ and $g_u(0) = 0$, is continuous on $[0, 1]$. By Step 1(a), $\|g'_u\|^2 = E(u)/\varepsilon < \infty$, so $g_u \in \mathcal{H}$. The map $u \mapsto g_u$ is linear and injective, so $W = \{g_u : u \in V_n\}$ has dimension n . By Step 2(a) there is $u \in V_n \setminus \{0\}$ with $a := \|g'_u\| \geq k_n b$, where $b := \|g_u\| > 0$. Step 1 gives $E(u) = \varepsilon a^2$ and $M(u) \leq 2q(b^2 + 2\varepsilon ab)$, and $M(u) > 0$ by Step 1 of the proof of Lemma 21. Hence, with $t = a/b \geq k_n$,

$$\lambda_n \geq \frac{E(u)}{M(u)} \geq \frac{\varepsilon t^2}{2q(1 + 2\varepsilon t)} \geq \frac{\varepsilon k_n^2}{2q(1 + 2\varepsilon k_n)},$$

because $t \mapsto t^2/(1 + 2\varepsilon t)$ increases on $(0, \infty)$.

Step 4: the upper bound. Assume $2\varepsilon k_n < 1$. For $g \in G_n$ put $u_s^g = g(z_s)$. First, $u^g \in \mathcal{D}$. Put $\rho_t = (u_t^g - u_{t+1}^g)/(\varepsilon q^t)$, the mean of g' over I_t , and $X_0 = \rho_0$, $X_t = \rho_t - \rho_{t-1}$ for $t \geq 1$. Since ρ_t and ρ_{t-1} are values of g' on $[z_{t+1}, z_{t-1}]$, $|X_t| \leq \|g''\|_\infty (z_{t-1} - z_{t+1}) \leq 2\|g''\|_\infty q^{t-1}$. So $X \in \ell^1$ and $\sum_t X_t^2 q^{-t} < \infty$. As in the proof of Lemma 18, $(KX)_s - (KX)_{s+1} = (1 - q)q^s \sum_{t \leq s} X_t = \varepsilon q^s \rho_s = u_s^g - u_{s+1}^g$. Both $(KX)_s$ and u_s^g tend to 0, the latter because $g(0) = 0$. Telescoping gives $u^g = KX \in \mathcal{D}$.

Now let $g \in G_n \setminus \{0\}$, $b = \|g\|$ and $a = \|g'\| \leq k_n b$ (Step 2(b)). By Step 1, $E(u^g) \leq \varepsilon a^2 \leq \varepsilon k_n^2 b^2$ and $M(u^g) \geq 2q(b^2 - 2\varepsilon ab) \geq 2qb^2(1 - 2\varepsilon k_n) > 0$. In particular $g \mapsto u^g$ is injective on G_n , so $V = \{u^g : g \in G_n\} \subset \mathcal{D}$ has dimension n , and $E/M \leq \varepsilon k_n^2/(2q(1 - 2\varepsilon k_n))$ on $V \setminus \{0\}$. By (17), $\lambda_n \leq \max_{V \setminus \{0\}} E/M$, which is the upper bound. $\square$

Standing and checks. Lemma 24 is proved, with no computer-assisted step. It uses only [7, Theorem “the travel determinant”, thm:fredholm](i),(iii),(iv), through the variational principle (17) on the explicit subspace $\mathcal{D}$. It uses neither Theorem 19 nor a Sturm oscillation theorem. Three points of the statement are deliberate. First, λ_n is the min–max value on $\mathcal{D}$, which by (17) is the n -th zero of $F(\cdot, q)$; no Friedrichs form domain is identified. Second, the upper bound is claimed only when $2\varepsilon k_n < 1$, which is where the Riemann error keeps $M(u^g) > 0$ and makes the sampling injective. Third, M is normalised with the factor $1 - q$, so $M \approx 2q\|g\|^2$ while $E = \varepsilon\|g'\|^2$; the quotient is $\varepsilon\|g'\|^2/(2q\|g\|^2)$, not $\|g'\|^2/(2q\|g\|^2)$. As a check, at $q = q_m$ for $m \in \{1, 2, 3, 4, 5, 8, 12, 20, 30, 40\}$ we computed all the zeros of $F(\cdot, q_m)$ below $1.08^2(k_{m+3}/k_m)^2$ by a sign-change scan in $\sqrt{\lambda}$ (120 digits). That is at least the first $m + 2$ zeros. Every one satisfies both bounds of the lemma wherever they apply, and the zero at $\lambda = 1$ has index exactly m , as Proposition 20(i) requires (`code/zeta_probe/rj_certificates/polelaw_check.py`).

Theorem 25 (the pole-location law). For $m \geq 1$ put $\varepsilon_m = 1 - q_m$ and

$$x_m := \frac{\varepsilon_m k_m^2}{2q_m}, \quad y_m := \frac{\varepsilon_m k_m^2}{2} = \varepsilon_m / \frac{8}{\pi^2(2m-1)^2} = q_m x_m.$$

(i) For every $m \geq 1$, $|x_m - 1| \leq 2\varepsilon_m k_m$.

(ii) For every $m \geq 1$, $\varepsilon^-(k_m) \leq \varepsilon_m \leq \varepsilon^+(k_m)$, where

$$\varepsilon^+(k) = \frac{\sqrt{(k^2 - 4k + 2)^2 + 32k} - (k^2 - 4k + 2)}{8k}, \quad \varepsilon^-(k) = \frac{k^2 + 4k + 2 - \sqrt{(k^2 + 4k + 2)^2 - 32k}}{8k}.$$

(iii) For every $m \geq 2$,

$$\varepsilon_m < \frac{2}{k_m(k_m - 4)}, \quad |x_m - 1| \leq 2\varepsilon_m k_m < \delta_m := \frac{8}{(2m-1)\pi - 8},$$

and $\delta_m < 1$ exactly when $m \geq 4$.

(iv) For every $m \geq 4$, $|y_m - 1| < \eta_m := 2(2k_m + 1)/(k_m(k_m - 4))$. The sequence $m\eta_m$ decreases, from 2.3912 at $m = 4$ to the limit $4/\pi$. For every $m \geq 1$, $|y_m - 1| \leq 3/m$.

In particular

$$1 - q_m = \frac{2q_m}{k_m^2} (1 + O(1/m)) = \frac{8}{\pi^2(2m-1)^2} (1 + O(1/m)),$$

with the explicit constants of (iii) and (iv). Parts (i)–(iii) hold, with k_m replaced by k_n , at every $q \in (0, 1)$ where $\lambda_n(q) = 1$, whatever n is.

Proof. By Proposition 20(i), $\lambda_m(q_m) = 1$. Write $q = q_m$, $\varepsilon = \varepsilon_m$, $k = k_m$, $x = x_m$, $y = y_m$.

(i) The lower bound of Lemma 24 at $n = m$ reads $1 \geq x/(1 + 2\varepsilon k)$, that is $x \leq 1 + 2\varepsilon k$. If $2\varepsilon k < 1$, its upper bound reads $1 \leq x/(1 - 2\varepsilon k)$, that is $x \geq 1 - 2\varepsilon k$. If $2\varepsilon k \geq 1$, then $x > 0 \geq 1 - 2\varepsilon k$ anyway. So $|x - 1| \leq 2\varepsilon k$ for every m , with no condition.

(ii) By (i), $\varepsilon k^2 \leq 2q(1 + 2\varepsilon k)$ and $\varepsilon k^2 \geq 2q(1 - 2\varepsilon k)$. Substitute $q = 1 - \varepsilon$. The first inequality becomes $4k\varepsilon^2 + (k^2 - 4k + 2)\varepsilon - 2 \leq 0$. This quadratic is -2 at $\varepsilon = 0$ and has a positive leading coefficient, so $\varepsilon \leq \varepsilon^+(k)$, its positive root. The second becomes $4k\varepsilon^2 - (k^2 + 4k + 2)\varepsilon + 2 \leq 0$. Its discriminant exceeds $(4k + 2)^2 - 32k = 4(2k - 1)^2 \geq 0$, so it has two real roots, and ε lies between them. In particular $\varepsilon \geq \varepsilon^-(k)$, the smaller root.

(iii) From (i) and $q < 1$, $\varepsilon k^2 < 2(1 + 2\varepsilon k)$, that is $\varepsilon k(k - 4) < 2$. For $m \geq 2$, $k \geq 3\pi/2 > 4$, so $\varepsilon < 2/(k(k - 4))$ and $2\varepsilon k < 4/(k - 4) = 8/((2m - 1)\pi - 8) = \delta_m$. Finally, $\delta_m < 1$ if and only if $(2m - 1)\pi > 16$, that is, $m \geq 4$ (at $m = 3$, $\delta_3 = 1.038$).

(iv) Let $m \geq 4$. Then $2\varepsilon k < \delta_m < 1$ by (iii). Upward, $y = qx \leq x \leq 1 + 2\varepsilon k < 1 + 4/(k - 4) = 1 + 4k/(k(k - 4)) < 1 + \eta_m$. Downward, $y = (1 - \varepsilon)x \geq (1 - \varepsilon)(1 - 2\varepsilon k) \geq 1 - \varepsilon(1 + 2k) > 1 - \eta_m$, by the bound on ε in (iii). For the monotonicity, $m = k/\pi + \frac{1}{2}$ gives

$$m\eta_m = \frac{(2k + \pi)(2k + 1)}{\pi k(k - 4)} = \frac{1}{\pi} \left(4 + \frac{(18 + 2\pi)k + \pi}{k^2 - 4k} \right).$$

For $a, b > 0$ the function $h(k) = (ak + b)/(k^2 - 4k)$ satisfies $(k^2 - 4k)^2 h'(k) = -ak^2 - 2bk + 4b < 0$ for $k \geq 2$, so $m\eta_m$ decreases on $k > 4$. Its value at $m = 4$ ($k = 7\pi/2$) is 2.39116..., and its limit is $4/\pi$. So $|y_m - 1| < \eta_m < 3/m$ for $m \geq 4$. For $m = 1, 2, 3$, part (ii) gives $y_m = \varepsilon_m k_m^2/2 \in [0.2620, 0.8968]$, $[0.5266, 2.3682]$ and $[0.6531, 1.8084]$ respectively. Hence $m|y_m - 1| \leq 0.738, 2.737$ and 2.426 , all below 3.

The final displays restate (iii) and (iv), using $2q_m/k_m^2 = \frac{8q_m}{\pi^2(2m-1)^2}$ and $q_m = 1 - \varepsilon_m = 1 + O(m^{-2})$. The index-free statement follows from the same proof, with Lemma 24 applied at n in place of m . Proposition 20(i) is used only to identify that index as m . $\square$

Standing and checks. Parts (i)–(iii) in their index-free form rest only on Lemma 24. They are unconditional and have no computer-assisted step. The identification $n = m$, and with it the statements indexed by m , is Proposition 20(i). So Theorem 25 as stated is proved at the standing of Theorem 19, that is, with the interval certificate (15) as its one computer-assisted input. The constants $0.2620, \dots, 1.8084$ at $m \leq 3$ and 2.39116 are evaluations of the closed forms of (ii) and (iv), rounded outward.

Since $\tau_m = -\log q_m$ satisfies $\varepsilon_m \leq \tau_m \leq \varepsilon_m/(1 - \varepsilon_m)$, the parameter $w_m = \sqrt{2/\tau_m}$ satisfies $(1 - \varepsilon_m)/y_m \leq w_m^2/k_m^2 \leq 1/y_m$. So $w_m = (m - \frac{1}{2})\pi(1 + O(1/m))$ as well.

We checked the theorem against the 41 poles $q_1, \dots, q_{41}$ of `code/zeta_probe/rj_certificates/models/poles_40.json` (200 digits). Every pole lies in the interval of (ii) and satisfies (i), and (iii) for $m \geq 2$ and (iv) for $m \geq 4$. The measured $m|y_m - 1|$ falls from 0.321 at $m = 1$ to 0.0023 at $m = 41$ (`polelaw_check.py`). So the measured relative error behaves like m^{-2} , not merely $1/m$: $1 - y_m = 8/(9k_m^2) + O(k_m^{-4})$ reproduces the measured $1 - y_{40} = 5.77 \cdot 10^{-5}$, in line with the second-order law of Remark 86. The theorem proves only the first order. The $O(1/m)$ loss comes from the Riemann error of Step 1(b). For $g = \psi_n$, a first-order expansion of the Riemann sum gives $M/(2q) - \|g\|^2 \approx \frac{\varepsilon}{2}(g(1)^2 - \|g\|^2)$, which is $O(\varepsilon)$ when $\varepsilon k_n^2 = O(1)$; this is a heuristic and is not used. The bound of Step 1(b) takes absolute values before integrating, and so charges $O(\varepsilon k_n)$.

Remark 26 (what the known theory does not give). Two statements about the travel poles lie outside what [7, Theorem “the travel determinant”, thm:fredholm] and the literature on Hahn–Exton zeros deliver. The first is now proved here; the second is open.

The accumulation law (now proved). The zero theory of [22, 42] concerns the zeros in x at fixed q ; the travel poles sit on the diagonal $x = Z(q)$. With Hurwitz’s theorem it gives each branch at fixed k , $\lambda_k(q) \sim \varepsilon(2k - 1)^2\pi^2/8$ with $\varepsilon = 1 - q$ ([7, Proposition “infinitely many travel poles, without T_2 ”, prop:infpolesalt](b)). The m -th pole is where $\lambda_m = 1$, that is in the regime $k = m \asymp \varepsilon^{-1/2}$, and no fixed- k statement reaches it. So the law $\varepsilon_m = 1 - q_m \sim 8/(\pi^2(2m - 1)^2)$, equivalently $w_m \sim (m - \frac{1}{2})\pi$, does not follow from that theory. Earlier versions of this paper recorded it only numerically, and said it needed a count lemma: an asymptotic for $\#\{k : \lambda_k(q) < 1\}$, uniform as $q \rightarrow 1$. It is now Theorem 25, with explicit error bounds for every $m \geq 1$: $\varepsilon_m = \frac{8}{\pi^2(2m-1)^2}(1 + \theta_m)$ with $|\theta_m| \leq 3/m$, and $|\theta_m| < \eta_m \sim 4/(\pi m)$ for $m \geq 4$. No count lemma is needed. Two inputs replace it. The uniform-in- n eigenvalue sandwich of Lemma 24 is valid in the regime $n \asymp \varepsilon^{-1/2}$ that the fixed- k theory misses. The index identification $\lambda_m(q_m) = 1$ of Proposition 20(i) supplies the count at $q = q_m$, namely exactly $m - 1$ branches below 1. The law is therefore proved at the standing of Theorem 19. Its index-free form is unconditional: at any q with $\lambda_n(q) = 1$, $|\varepsilon k_n^2/(2q) - 1| \leq 2\varepsilon k_n$. Only the first order is proved. The rational second-order term $-8/(9\mu_j)$ of Remark 86 is derived there by a formal crossing calculus and verified numerically, but it is not proved.

Complex q . The identity $\det(I - \mathcal{T}) = \cos(Z; q^2)$ continues analytically to complex q , but positivity and self-adjointness do not, and the zero theory cited is for $0 < q < 1$. We know of no result on zeros of q -Bessel functions at complex base that would locate the non-real zeros of [7, Remark “where on the circle the poles accumulate”, rem:polesdensity]; their density on the circle remains a conjecture.

6 The transfer model and the block assembly

The travel block and the bulk block are one object seen twice. Both are the resolvent of the semiseparable kernel $q^{\max(a,b)}$ against a geometric diagonal weight, and a single three-term recursion in the section index k covers both: its coefficients are α_k, γ_k in the bulk case and A_k, C_k in the travel case, which is why the four series of (2) are four ladders on one recursion and not four separate

constructions. This section develops that into the block assembly the rest of the paper consumes.

Sections 6.1–6.5 are unconditional statements about explicitly given operators, and §6.6 is an unconditional statement about the crossing optimisation that those operators encode. [7, Section sec:modelassembly] states the strand-walk input (M) that identifies the one with the other and proves it ([7, Theorem “the transfer model (M)”, thm:model], with the assembly step proved in Appendix D). [7, Section sec:RJ] evaluates the junction pairing at the travel poles in closed form.

Throughout, $|q| < 1$ and $x = q^{1/2}$. Besides the four blocks of (2) we use the third ladder on the same recursion,

$$T_k = \sum_{j \geq 0} \alpha_{k+2j+1} \prod_{i < j} \gamma_{k+2i} \quad (k \geq 0). \quad (19)$$

Since $\gamma_k = -2q^{k+1}(1-q)/[(1-q^{k+1})(1-q^{k+2})]$ and $C_k = q\gamma_k$, on every compact subset of $|q| < 1$ there is a Ξ with $|\gamma_k|, |C_k| \leq \Xi|q|^{k+1}$, so $|\prod_{i < j} \gamma_{k+2i}| \leq \Xi^j |q|^{j^2+jk}$ and likewise for C ; as α_k and A_k are bounded there, S_k, Σ_k, T_k converge absolutely and locally uniformly and are holomorphic on $|q| < 1$ for every $k \geq 0$.

6.1 The section identity

Definition 27 (section identity). Fix $s_0 \in \{0, 1\}$ and $\vartheta \in \{0, 1\}$ with $s_0 + \vartheta = 1$, and put $e_s = 2q^{\vartheta+s}$ for $s \geq s_0$. Given a bounded *source* $\mathbf{c} = (c_s)_{s \geq s_0}$, a family $\Phi = (\Phi_s)_{s \geq s_0}$ satisfies the *section identity* with data $(s_0, \vartheta, \mathbf{c})$ if

$$\Phi_s = e_s \left(c_s + \sum_{s' \geq s_0} q^{\max(s, s')} \Phi_{s'} \right), \quad s \geq s_0. \quad (20)$$

Its *sections* are $G_k = \sum_{s \geq s_0} q^{ks} \Phi_s$, $k \geq 0$. In operator form $(I - M)\Phi = E$ with

$$M[s, s'] = e_s q^{\max(s, s')}, \quad E_s = e_s c_s. \quad (21)$$

The *bulk* instance is $(s_0, \vartheta) = (1, 0)$ and the *travel* instance is $(s_0, \vartheta) = (0, 1)$; we write M_0 for the bulk operator and $\mathcal{T}$ for the travel one.

Lemma 28 (existence, uniqueness, analyticity). For $|q| < \frac{1}{3}$ and any bounded $\mathbf{c}$, equation (20) has a unique solution in ℓ^1 , namely $\Phi = \sum_{n \geq 0} M^n E$, the series converging in ℓ^1 and coefficientwise in q , one extra power of q per term. One has $|\Phi_s| \leq C|q|^{\vartheta+s}$ with C independent of s , and each section G_k is holomorphic at $q = 0$.

Proof. $\|M\|_{\ell^1 \rightarrow \ell^1} = \sup_{s'} \sum_s |M[s, s']| \leq \sup_{s'} \sum_{s \geq s_0} 2|q|^{\vartheta+s} = 2|q|/(1-|q|) < 1$ for $|q| < \frac{1}{3}$, using $\vartheta + s_0 = 1$ and $|q|^{\max(s, s')} \leq 1$. Hence $I - M$ is invertible on ℓ^1 and the Neumann series converges. Every entry of M is divisible by $q^{\vartheta+s} q^{\max(s, s')}$, hence by q , so $M^n E$ is divisible by q^{n+1} and the series converges coefficientwise. Reading the s -th row of $\Phi = E + M\Phi$ gives $|\Phi_s| \leq 2|q|^{\vartheta+s} (\|\mathbf{c}\|_\infty + \|\Phi\|_{\ell^1})$. Since $|\Phi_s| = O(|q|^s)$, each G_k is a locally uniformly convergent sum of holomorphic functions. $\square$

6.2 One telescoping lemma for both blocks

Lemma 29 (catalytic telescoping). Let $0 < |q| < 1$ and let $\Phi \in \ell^1$ satisfy (20) with data $(s_0, \vartheta, \mathbf{c})$. Put

$$a_k = \frac{2q^{\vartheta+(k+1)s_0}}{1-q^{k+1}}, \quad b_k = \frac{2q^{\vartheta+k+2}}{1-q^{k+2}} - \frac{2q^{\vartheta+k+1}}{1-q^{k+1}}, \quad c_k = \sum_{s \geq s_0} e_s c_s q^{ks},$$

and $\mathbf{a}_k = \sum_{j \geq 0} a_{k+2j} \prod_{i < j} b_{k+2i}$, $\mathbf{c}_k = \sum_{j \geq 0} c_{k+2j} \prod_{i < j} b_{k+2i}$. Then for every $k \geq 0$

$$G_k = C_k + a_k G_1 + b_k G_{k+2}, \quad (22)$$

the series $\mathbf{a}_k, \mathbf{c}_k$ converge, $G_k = \mathbf{c}_k + \mathbf{a}_k G_1$, and if $\mathbf{a}_1 \neq 1$ then

$$G_1 = \frac{\mathbf{c}_1}{1 - \mathbf{a}_1}, \quad G_k = \mathbf{c}_k + \mathbf{a}_k \frac{\mathbf{c}_1}{1 - \mathbf{a}_1} \quad (k \geq 0). \quad (23)$$

In the bulk instance $a_k = \alpha_k$, $b_k = \gamma_k$ and $\mathbf{a}_k = S_k$; in the travel instance $a_k = A_k$, $b_k = C_k$ and $\mathbf{a}_k = \Sigma_k$.

Proof. Split the inner sum of (20) at $s' = s$, where $\max(s, s')$ switches:

$$\Phi_s = e_s c_s + e_s \sum_{s' \geq s} q^{s'} \Phi_{s'} + e_s q^s \sum_{s_0 \leq s' < s} \Phi_{s'}.$$

Multiply by q^{ks} and sum over $s \geq s_0$. All three double sums converge absolutely, since $\Phi \in \ell^1$ and $|q| < 1$, so the orders of summation may be exchanged. The first gives $\mathcal{C}_k$. In the second, with $e_s = 2q^{\vartheta+s}$,

$$\sum_{s \geq s_0} 2q^{\vartheta+(k+1)s} \sum_{s' \geq s} q^{s'} \Phi_{s'} = 2q^{\vartheta} \sum_{s' \geq s_0} q^{s'} \Phi_{s'} \sum_{s=s_0}^{s'} q^{(k+1)s} = \frac{2q^{\vartheta}}{1 - q^{k+1}} \sum_{s' \geq s_0} q^{s'} \Phi_{s'} (q^{(k+1)s_0} - q^{(k+1)(s'+1)}),$$

which is $a_k G_1 - \frac{2q^{\vartheta+k+1}}{1 - q^{k+1}} G_{k+2}$, because $\sum_{s'} q^{s'} \Phi_{s'} = G_1$ and $\sum_{s'} q^{(k+2)s'} \Phi_{s'} = G_{k+2}$. In the third,

$$\sum_{s \geq s_0} 2q^{\vartheta+(k+2)s} \sum_{s_0 \leq s' < s} \Phi_{s'} = 2q^{\vartheta} \sum_{s' \geq s_0} \Phi_{s'} \sum_{s > s'} q^{(k+2)s} = \frac{2q^{\vartheta+k+2}}{1 - q^{k+2}} G_{k+2}.$$

Adding the three gives (22). Iterating it J times,

$$G_k = \sum_{j < J} (\mathcal{C}_{k+2j} + a_{k+2j} G_1) \prod_{i < j} b_{k+2i} + G_{k+2J} \prod_{i < J} b_{k+2i}.$$

For fixed q one has $|b_m| \leq \Xi |q|^{m+1}$ with $\Xi = 4/(1 - |q|)$, so $|\prod_{i < J} b_{k+2i}| \leq \Xi^J |q|^{J^2+Jk} \rightarrow 0$; and $|G_m| \leq \|\Phi\|_{\ell^1}$ uniformly in m , so the remainder vanishes as $J \rightarrow \infty$. The same bound, against $\sup_k |a_k| < \infty$ and $\sup_k |\mathcal{C}_k| \leq 2\|\mathbf{c}\|_{\infty}/(1 - |q|)$, makes $\mathbf{a}_k$ and $\mathbf{c}_k$ converge. This gives $G_k = \mathbf{c}_k + \mathbf{a}_k G_1$; taking $k = 1$ gives $G_1(1 - \mathbf{a}_1) = \mathbf{c}_1$, whence (23). The identifications of a_k, b_k are read off the displays: $(s_0, \vartheta) = (1, 0)$ gives $a_k = 2q^{k+1}/(1 - q^{k+1}) = \alpha_k$ and $b_k = \alpha_{k+1} - \alpha_k = \gamma_k$, while $(s_0, \vartheta) = (0, 1)$ gives $a_k = 2q/(1 - q^{k+1}) = A_k$ and $b_k = 2q^{k+3}/(1 - q^{k+2}) - 2q^{k+2}/(1 - q^{k+1}) = C_k$. $\square$

Remark 30 (where each feature of the model is used). Three features enter (22), each exactly once. That the coupling is $q^{\max(s, s')}$ is what makes the split at $s' = s$ produce two *geometric* sums; a symmetric kernel not of the form $\mu^{\max}$ destroys the three-term relation. That e_s is geometric in s is what makes those sums summable in closed form; a non-geometric edge weight leaves G_k coupled to a continuum of sections. That the source is geometric is used only to evaluate $\mathcal{C}_k$; both (22) and (23) hold for any bounded source. The step $k \mapsto k + 2$, and with it the two-ladder structure of S_k and Σ_k , comes from the two occurrences of $q^{(k+2)s'}$ above.

Proposition 31 (the travel k -sum telescopes). *Let Φ solve the travel section identity, $(s_0, \vartheta, \mathbf{c}) = (0, 1, \mathbf{1})$. Then, as meromorphic functions on $|q| < 1$,*

$$G_k^T = \frac{\Sigma_k}{1 - \Sigma_1} \quad (k \geq 0), \quad \text{in particular} \quad G_0^T = \sum_{s \geq 0} \Phi_s = \frac{\Sigma_0}{1 - \Sigma_1}.$$

Proof. Lemma 29 with $(s_0, \vartheta) = (0, 1)$ and $c_s \equiv 1$ gives $\mathcal{C}_k = \sum_{s \geq 0} 2q^{1+s} q^{ks} = 2q/(1 - q^{k+1}) = A_k = a_k$, so $\mathbf{c}_k = \mathbf{a}_k = \Sigma_k$ and (23) reads $G_k^T = \Sigma_k + \Sigma_k \Sigma_1 / (1 - \Sigma_1) = \Sigma_k / (1 - \Sigma_1)$. Both sides are holomorphic near $q = 0$ and the right-hand side is meromorphic on $|q| < 1$, so the identity extends by analytic continuation. Combinatorially this is the Neumann series of Lemma 28: $\sum_s \Phi_s = \sum_{n \geq 0} \mathbf{1}^\top \mathcal{T}^n E$, whose n -th term is the generating function of the travel runs with exactly $n + 1$ edges. $\square$

Proposition 32 (the bulk dressing). *Let Φ solve the bulk section identity with $\mathbf{c} = \mathbf{1}$ and let Ψ solve it with $c_s = q^s$. Set*

$$b_0 = \sum_{s \geq 1} \Phi_s, \quad t_0 = \sum_{s \geq 1} q^s \Phi_s, \quad b_1 = \sum_{s \geq 1} \Psi_s, \quad t_1 = \sum_{s \geq 1} q^s \Psi_s.$$

Then, as meromorphic functions on $|q| < 1$,

$$b_0 = \frac{S_0}{1 - S_1}, \quad t_0 = \frac{S_1}{1 - S_1}, \quad t_1 = \frac{T_1}{1 - S_1}, \quad b_1 = T_0 + \frac{S_0 T_1}{1 - S_1}.$$

Proof. Lemma 29 with $(s_0, \vartheta) = (1, 0)$. For $\mathbf{c} = \mathbf{1}$, $\mathcal{C}_k = \sum_{s \geq 1} 2q^{s(k+1)} = \alpha_k = a_k$, so $\mathbf{c}_k = \mathbf{a}_k = S_k$ and $G_k = S_k / (1 - S_1)$; take $k = 0, 1$. For $c_s = q^s$, $\mathcal{C}_k = \sum_{s \geq 1} 2q^{s(k+2)} = \alpha_{k+1}$, so $\mathbf{c}_k = \sum_j \alpha_{k+2j+1} \prod_{i < j} \gamma_{k+2i} = T_k$ by (19), and (23) gives $t_1 = G_1 = T_1 / (1 - S_1)$ and $b_1 = G_0 = T_0 + S_0 T_1 / (1 - S_1)$. Analytic continuation as before. $\square$

With $1 - S_1 = S_e$ ([7, Proposition prop:qtrig]) the first of these is $b_0 = S_0 / S_e$, the third line of the dictionary (32); and b_0 is holomorphic and non-zero at every travel pole, by Propositions 69 and [7, prop:numnonvanish] respectively.

6.3 The cocycle, the reciprocity, and the dictionary

The bulk section identity has a second, equivalent, description as a two-point problem for a first-order system. That system is the cocycle of §7.4, and comparing the two descriptions identifies t_1 with P_{12} / S_e .

Corollary 33 (the cocycle is the same object). *Let Φ solve the bulk section identity with source $\mathbf{c}$, and put $L_b = \sum_{a \leq b} \Phi_a$ and $\mathcal{E}_b = -\sum_{a \geq b} q^a \Phi_a$. Then $\Phi_b = 2q^b c_b + 2q^{2b} L_{b-1} - 2q^b \mathcal{E}_b$ and*

$$\begin{pmatrix} L_b \\ \mathcal{E}_{b+1} \end{pmatrix} = \mathcal{M}_b \begin{pmatrix} L_{b-1} \\ \mathcal{E}_b \end{pmatrix} + \begin{pmatrix} 2q^b c_b \\ 2q^{2b} c_b \end{pmatrix}, \quad \mathcal{M}_b = \begin{pmatrix} 1 + 2q^{2b} & -2q^b \\ 2q^{3b} & 1 - 2q^{2b} \end{pmatrix}, \quad (24)$$

with $L_0 = 0$, $\mathcal{E}_1 = -t_0$ or $-t_1$ according to the source, and $\mathcal{E}_b \rightarrow 0$. Let (ℓ, r) and (ℓ', r') be the solutions of the homogeneous recursion ($\mathbf{c} = 0$) with data $(\ell_0, r_1) = (1, 0)$ and $(\ell'_0, r'_1) = (0, 1)$, and put $(P_{11}, P_{12}) = \lim_b (\ell_b, r_{b+1})$ and $(P_{21}, P_{22}) = \lim_b (\ell'_b, r'_{b+1})$, the limits existing because $\sum_b \|\mathcal{M}_b - I\| < \infty$. Eliminating L between the two lines of (24) at $\mathbf{c} = 0$ gives

$$\mathcal{E}_{b+2} = (1 + q^3 - 2(1 - q)q^{2b+2})\mathcal{E}_{b+1} - q^3 \mathcal{E}_b,$$

which is the scalar three-term recursion of §7.4; so r and r' are its two standard solutions and P_{12}, P_{22} their limits, P_{12} being the entry whose closed form is (34). Then, identically in $0 < q < 1$,

$$P_{11} = 1 + T_0, \quad P_{12} = T_1, \quad P_{21} = -S_0, \quad P_{22} = S_e = 1 - S_1.$$

In particular $t_1 = P_{12} / S_e$ is an identity between independently defined objects, not a definition.

Proof. Substituting $L_b = L_{b-1} + \Phi_b$ and $\mathcal{E}_{b+1} = \mathcal{E}_b + q^b \Phi_b$ into $\Phi_b = 2q^b(c_b + q^b L_b - \mathcal{E}_{b+1})$, which is (20) after the split at $s' = b$, the two terms $\pm 2q^{2b} \Phi_b$ cancel and leave $\Phi_b = 2q^b c_b + 2q^{2b} L_{b-1} - 2q^b \mathcal{E}_b$; feeding this back gives (24). The homogeneous solutions form a two-dimensional space, since (24) determines $(L_b, \mathcal{E}_{b+1})$ from $(L_{b-1}, \mathcal{E}_b)$.

Take $\mathbf{c} = \mathbf{1}$, so $L_0 = 0$, $L_b \rightarrow b_0$, $\mathcal{E}_1 = -t_0$ and $\mathcal{E}_{b+1} \rightarrow 0$. Then $(L, \mathcal{E} - 1)$ solves the homogeneous recursion: in the first line the source $2q^b c_b = 2q^b$ is cancelled by $-2q^b \cdot (-1)$, and in the second the source $2q^{2b} c_b = 2q^{2b}$ is cancelled by $(1 - 2q^{2b})(-1) - (-1)$. Its data are $(0, -1 - t_0)$, so $(L, \mathcal{E} - 1) = -(1 + t_0)(\ell', r')$, and letting $b \rightarrow \infty$,

$$b_0 = -(1 + t_0)P_{21}, \quad -1 = -(1 + t_0)P_{22}.$$

By Proposition 32 and $S_e = 1 - S_1$ one has $1 + t_0 = 1 + S_1/S_e = 1/S_e$, whence $P_{22} = S_e$ and $P_{21} = -b_0 S_e = -S_0$.

Take now $c_s = q^s$, with data $L'_0 = 0$, $L'_b \rightarrow b_1$, $\mathcal{E}'_1 = -t_1$, $\mathcal{E}'_{b+1} \rightarrow 0$. Here $(L' + 1, \mathcal{E}')$ solves the homogeneous recursion: in the first line $(1 + 2q^{2b}) \cdot 1$ exceeds 1 by $2q^{2b}$, which is the source $2q^b c_b = 2q^{2b}$, and in the second $2q^{3b} \cdot 1$ is the source $2q^{2b} c_b = 2q^{3b}$. Its data are $(1, -t_1)$, so $(L' + 1, \mathcal{E}') = (\ell, r) - t_1(\ell', r')$, and letting $b \rightarrow \infty$,

$$b_1 + 1 = P_{11} - t_1 P_{21} = P_{11} + t_1 S_0, \quad 0 = P_{12} - t_1 P_{22} = P_{12} - t_1 S_e.$$

The second gives $P_{12} = t_1 S_e = T_1$ by Proposition 32, and the first then gives $P_{11} = b_1 + 1 - t_1 S_0 = T_0 + S_0 T_1 / S_e + 1 - (T_1 / S_e) S_0 = 1 + T_0$. All four limits are taken for $|q| < \frac{1}{3}$, where Lemma 28 applies, and extend to $0 < q < 1$ by analytic continuation, both sides being meromorphic there. $\square$

Proposition 34 (reciprocity). $t_0 = b_1$ identically in q . Equivalently $T_0 S_e + T_1 S_0 = S_1$, equivalently $P_{11} P_{22} - P_{12} P_{21} = 1$.

Proof. Two independent proofs.

Kernel symmetry. Write $D = \text{diag}(2q^b)_{b \geq 1}$ and $K[b, a] = q^{\max(a, b)}$, so that $M_0 = DK$, $E = D\mathbf{1}$, and $u := (2q^{2b})_b = Dv$ with $v = (q^a)_a$. The two sources of Proposition 32 are E and u , so $\Phi = (I - M_0)^{-1}E$, $\Psi = (I - M_0)^{-1}u$, and

$$t_0 = v^\top (I - M_0)^{-1}E, \quad b_1 = \mathbf{1}^\top (I - M_0)^{-1}u.$$

For $|q| < \frac{1}{3}$, $X := (I - M_0)^{-1}D = \sum_{n \geq 0} (DK)^n D$, and the n -th summand is $DKD \cdots KD$, a palindromic product of the symmetric matrices D and K , hence symmetric; the series converges entrywise. Therefore $t_0 = v^\top X \mathbf{1} = \sum_{a, b} v_a X[a, b] = \sum_{a, b} X[b, a] v_a = \mathbf{1}^\top X v = b_1$, every sum being absolutely convergent. Both sides are meromorphic on $|q| < 1$ by Proposition 32, so the identity extends.

Casoratian. $\det \mathcal{M}_b = (1 + 2q^{2b})(1 - 2q^{2b}) + 4q^{3b}q^b = 1$, so $\ell_b r'_{b+1} - \ell'_b r_{b+1}$ does not depend on b ; at $b = 0$ it is 1, whence $P_{11} P_{22} - P_{21} P_{12} = 1$. Substituting the four entries of Corollary 33 gives $(1 + T_0)S_e + T_1 S_0 = 1$, that is $T_0 S_e + T_1 S_0 = 1 - S_e = S_1$; dividing by S_e and using Proposition 32 turns this into $b_1 = T_0 + S_0 T_1 / S_e = S_1 / S_e = t_0$. $\square$

The symmetry of K is used and is necessary: replacing $q^{\max(a, b)}$ by the asymmetric $q^{\max(a, b)} + q^{a+2b}$ makes $t_0 - b_1 = 4q^8 + O(q^9)$ (`assembly_certificate.py`, Part 2). The determinant identity $P_{11} P_{22} - P_{12} P_{21} = 1$ is the Casoratian of the cocycle and is automatic; what carries content is the identification of its four entries in Corollary 33, which turns it into the reciprocity.

Corollary 35 (the dictionary). *Identically in $0 < q < 1$,*

$$S_e = 1 - S_1, \quad S_o = \frac{1-q}{2q} S_0, \quad b_0 = \frac{S_0}{S_e}, \quad t_0 = b_1 = \frac{S_1}{S_e}, \quad t_1 = \frac{T_1}{S_e} = \frac{P_{12}}{S_e},$$

and consequently $\kappa := t_0 b_1 - b_0 t_1 = (S_1^2 - S_0 P_{12})/S_e^2$. This is (32) together with its two remaining entries.

Proof. The first line is [7, Proposition prop:qtrig] and [7, Lemma “the numerators in closed form”, lem:qsinenum]; the rest is Proposition 32 with Proposition 34 and Corollary 33, and κ is substitution. $\square$

6.4 Compactness, the null space, and the singularities

Lemma 36 (compactness and meromorphy). *For $|q| < 1$ the operator M of (21), and its gap-marked version with kernel (25) below, are compact on ℓ^1 . Hence $q \mapsto (I - M)^{-1}$ is meromorphic on $|q| < 1$, with poles of finite order, and at each pole $\ker(I - M) \neq 0$.*

Proof. Let M_N agree with M on rows $s \leq N$ and vanish above. Then $\|M - M_N\|_{\ell^1 \rightarrow \ell^1} \leq \sup_{s'} \sum_{s > N} |e_s| |q|^{\max(s, s')} \leq \sum_{s > N} 2|q|^{\vartheta+s} \rightarrow 0$, so M is a norm limit of finite-rank operators and is compact; the rank-one gap term is compact a fortiori. The family is holomorphic in q on the connected domain $|q| < 1$ and $I - M$ is invertible near $q = 0$ (Lemma 28), so the analytic Fredholm theorem applies. $\square$

Lemma 37 (the null space is at most one-dimensional). *Let $0 < |q| < 1$ and let $R \in \ell^1$ satisfy $R_s = e_s \sum_{s'} q^{\max(s, s')} R_{s'}$ for all $s \geq s_0$. Then its sections satisfy $G_k^R = G_1^R \mathbf{a}_k$ for all $k \geq 0$, and $G_1^R = 0$ forces $R = 0$. In particular $\dim \ker(I - M) \leq 1$, and a non-zero kernel element exists only where $\mathbf{a}_1 = 1$.*

Proof. Lemma 29 with $\mathbf{c} = 0$ gives $\mathcal{C}_k = 0$, hence $\mathbf{c}_k = 0$ and $G_k^R = \mathbf{a}_k G_1^R$; its derivation used only $R \in \ell^1$ and $|q| < 1$. Taking $k = 1$ gives $G_1^R = \mathbf{a}_1 G_1^R$, so $\mathbf{a}_1 = 1$ unless $G_1^R = 0$. If $G_1^R = 0$ then $G_k^R = 0$ for every k . The function $F(t) = \sum_{s \geq s_0} R_s t^s$ is holomorphic on $|t| < 1$ because $R \in \ell^1$, and $F(q^k) = G_k^R = 0$ for all $k \geq 0$; the points q^k accumulate at the interior point $t = 0$, so $F \equiv 0$ and $R = 0$. Hence the kernel is spanned by the class normalised by $G_1^R = 1$. $\square$

Corollary 38 (singularities of the two gapless resolvents). *On $|q| < 1$, the poles of $(I - M_0)^{-1}$ lie in the zero set of S_e and the poles of $(I - \mathcal{T})^{-1}$ lie in the zero set of $1 - \Sigma_1$; conversely every travel pole is a pole of $(I - \mathcal{T})^{-1}$. In particular b_0, b_1, t_0, t_1 are holomorphic at every travel pole q_m , where $S_e(q_m) \neq 0$ by Proposition 69, and $b_0(q_m) = S_0(q_m)/S_e(q_m) \neq 0$ by [7, Proposition “the numerators do not vanish at the travel poles”, prop:numnonvanish].*

Proof. By Lemma 36 a pole forces a non-zero kernel element, and by Lemma 37 that forces $\mathbf{a}_1 = 1$, which is $S_1 = 1$, i.e. $S_e = 0$, in the bulk instance and $\Sigma_1 = 1$ in the travel one. Conversely, at a travel pole $\Sigma_0/(1 - \Sigma_1)$ is unbounded because $\Sigma_0(q_m) \neq 0$ ([7, Proposition “the numerators do not vanish at the travel poles”, prop:numnonvanish]), so by Proposition 31 the travel resolvent is singular there. $\square$

6.5 The gap term is rank one: the Möbius factorisation

Marking the gap bridges changes the bulk kernel by one term. With $\mathfrak{g} = \mathfrak{g}(q, y) = \sum_{L \geq 1} q^L y^{L-1} = q/(1 - qy)$, the gap-marked bulk kernel and system are

$$K_{\mathfrak{g}}(a, b) = q^{\max(a, b)} + q^{a+b} \mathfrak{g}, \quad P_b = 2q^b \left(q^b + \sum_{a \geq 1} K_{\mathfrak{g}}(a, b) P_a \right), \quad B(q, y) = \sum_{b \geq 1} P_b, \quad (25)$$

the source q^b being the cost of the outer span-end site of a bulk run (Lemma 119). In operator form $(I - M_0 - \mathfrak{g}uv^\top)P = u$ with

$$M_0[b, a] = 2q^b q^{\max(a, b)}, \quad E_b = 2q^b, \quad u_b = 2q^{2b}, \quad v_a = q^a, \quad (26)$$

since $2q^b \cdot q^{a+b} = (2q^{2b})(q^a)$: the gap term is exactly rank one. With the four scalars of Proposition 32,

$$b_0 = \mathbf{1}^\top (I - M_0)^{-1} E, \quad b_1 = \mathbf{1}^\top (I - M_0)^{-1} u, \quad t_0 = v^\top (I - M_0)^{-1} E, \quad t_1 = v^\top (I - M_0)^{-1} u.$$

Proposition 39 (Möbius factorisation). *Let q be such that $I - M_0$ is invertible on ℓ^1 and $1 - \mathfrak{g}t_1 \neq 0$. Then $I - M_0 - \mathfrak{g}uv^\top$ is invertible on ℓ^1 ,*

$$(I - M_0 - \mathfrak{g}uv^\top)^{-1} = R_0 + \frac{\mathfrak{g}}{1 - \mathfrak{g}t_1} R_0 uv^\top R_0, \quad R_0 = (I - M_0)^{-1}, \quad (27)$$

and the E -sourced sum $B^E(q, y) := \mathbf{1}^\top (I - M_0 - \mathfrak{g}uv^\top)^{-1} E$ is $(b_0 + \mathfrak{g}\kappa)/(1 - \mathfrak{g}t_1)$ with $\kappa = t_0 b_1 - b_0 t_1$, which is (30). (The bulk sum of the model has source u ; see Remark 40.) In particular B^E is a Möbius function of $\mathfrak{g}$, with a single zero, at $\mathfrak{g}_0 = -b_0/\kappa$, and a single pole, at $\mathfrak{g} = 1/t_1$.

Proof. Sherman–Morrison. Put $\delta = 1 - \mathfrak{g}v^\top R_0 u = 1 - \mathfrak{g}t_1 \neq 0$. Then the right-hand side of (27) is a two-sided inverse of $I - M_0 - \mathfrak{g}uv^\top$:

$$(I - M_0 - \mathfrak{g}uv^\top)(R_0 + \mathfrak{g}\delta^{-1}R_0 uv^\top R_0) = I + \mathfrak{g}(\delta^{-1} - 1 - \mathfrak{g}\delta^{-1}t_1)uv^\top R_0 = I,$$

since $\delta^{-1}(1 - \delta - \mathfrak{g}t_1) = 0$; the other side is identical. All pairings converge absolutely: $u, E \in \ell^1$, $v, \mathbf{1} \in \ell^\infty$, and R_0 is bounded on ℓ^1 . Applying $\mathbf{1}^\top(\cdot)E$,

$$B^E = \mathbf{1}^\top R_0 E + \mathfrak{g}\delta^{-1}(\mathbf{1}^\top R_0 u)(v^\top R_0 E) = b_0 + \frac{\mathfrak{g}b_1 t_0}{1 - \mathfrak{g}t_1} = \frac{b_0 + \mathfrak{g}(t_0 b_1 - b_0 t_1)}{1 - \mathfrak{g}t_1}. \quad \square$$

Remark 40 (the bulk sum of the model). The system (25) has right-hand side u , not E . By (27),

$$B(q, y) = \mathbf{1}^\top (I - M_0 - \mathfrak{g}uv^\top)^{-1} u = \frac{b_1}{1 - \mathfrak{g}t_1}, \quad B_0(q, y) := 1 + \mathfrak{g} \sum_{b \geq 1} q^b P_b = \frac{1}{1 - \mathfrak{g}t_1},$$

where B_0 is the weight of the junction-adjacent bulk chains that are empty or begin with a gap run. The Möbius form $(b_0 + \mathfrak{g}\kappa)/(1 - \mathfrak{g}t_1)$ of Proposition 39 is the resolvent of E . It enters below only through b_0 , t_1 and the gate, and not as the bulk sum.

Remark 41 (the two hypotheses are the two the paper already carries). Invertibility of $I - M_0$ is $S_e \neq 0$ (Corollary 38), which at the travel poles is Proposition 69. And $1 - \mathfrak{g}v^\top t_1 \neq 0$ is $s \neq 1$, the gate half $|s| < 1$ of (31), which is Theorem 82 with Lemma 83; $1 - \mathfrak{g}u^\top t_1 \neq 0$ follows, since $\mathfrak{g}_U = \mathfrak{g}_V/(1 + q) < \mathfrak{g}_V$ and $t_1 > 0$ at the poles. So Proposition 39 consumes no hypothesis that §7.3 was not already carrying.

Machine verification of the telescoping, the factorisation and the reciprocity

Proposition 39 is formally verified in Lean 4 against Mathlib [28] in `Mobius.lean` and `MobiusL1.lean`. Its hypotheses are its Lean hypotheses: ‘ $I - M_0$ is invertible’ is an operator R_0 with $R_0(I - M_0) = (I - M_0)R_0 = I$, and $1 - \mathfrak{g}t_1 \neq 0$ is carried as stated. With those in hand nothing analytic remains. `Mobius.lean` proves the factorisation for an arbitrary module over an arbitrary field: `opN_opS` and `opS_opN` are the two halves of (27), `solution_unique` is the unique solvability of (25), `B_mobius` is (30), and `B_eq_zero_iff` and `B_pole` are the single zero at $\mathfrak{g}_0 = -b_0/\kappa$ and the single pole at $\mathfrak{g} = 1/t_1$.

`MobiusL1.lean` instantiates that statement at the case the proposition is about, $K = \mathbb{C}$ and $E = \ell^1$ with the concrete data of (26). For $|q| < 1$ a kernel with rows dominated by a summable sequence defines a $\mathbb{C}$ -linear operator on ℓ^1 (`opOfKer`) and a bounded sequence defines a functional (`funcOfSeq`); `M0`, `uElt`, `EElt`, `vFun` and `oneFun` are M_0 , u , E , $v^\top$ and $\mathbf{1}^\top$ of (26), with the index n standing for site $n + 1$. `rank_one` is the identity $2q^b q^{a+b} = (2q^{2b})(q^a)$, `gapSystem_iff` identifies the operator equation $(I - M_0 - \mathfrak{g}uv^\top)P = E$ with the display (25) coordinatewise, `l1_solution_unique` is the unique ℓ^1 solvability of (25), `l1_smresolvent_left` and `l1_smresolvent_right` are (27) for those operators, and `l1_B_of_solution` is the conclusion: an ℓ^1 solution of (25) has $B = \sum_b P_b = (b_0 + \mathfrak{g}\kappa)/(1 - \mathfrak{g}t_1)$.

The first sentence of Proposition 34, $t_0 = b_1$, is verified in `Reciprocity.lean` in the operator form its first proof uses, with E and F kept apart: $D: F \rightarrow E$ and $K: E \rightarrow F$ symmetric for the pairing, $M_0 = DK$, and the two sources $D\mathbf{1}$ and Dv . `pushThrough` is $(I - DK)^{-1}D = D(I - KD)^{-1}$, which replaces the palindromic Neumann series of the paper’s proof by an algebraic identity; `X_symm` is the symmetry of X and `t0_eq_b1` the conclusion. The necessity of the symmetry of K is recorded as `K_symm_needed`.

Lemma 29 is verified in `Telescope.lean`. The site $s = n + s_0$ is indexed by $n \in \mathbb{Z}_{\geq 0}$, so e_s is `eCo` and $s_0 + \vartheta = 1$ is the hypothesis `hs`; the source is bounded and $\Phi \in \ell^1$ is carried as summability of $\|\Phi_n\|$, which is what `MobiusL1.summable_norm` supplies for a Φ given as an element of the ℓ^1 space of `MobiusL1.lean` (`l1_three_term`, `l1_G_eq`). `row_sum` is the split of the inner sum at $s' = s$ together with the finite and the infinite geometric sum it produces; `three_term` is (22), the interchange of the two summations being Fubini against the majorant $2|q|^{n+1}\|\Phi_n\|$, which is summable on $\mathbb{Z}_{\geq 0}^2$ as a product; `norm_bTel_le` is the bound $|b_k| \leq \Xi|q|^{k+1}$ with $\Xi = 4/(1 - |q|)$ and `summable_bProd_norm` is the paper’s $\Xi^J|q|^{J^2+Jk} \rightarrow 0$, obtained by the ratio test; `summable_aFrak` and `summable_cFrak` are the convergence of $\mathfrak{a}_k$ and $\mathfrak{c}_k$; `G_eq` is $G_k = \mathfrak{c}_k + \mathfrak{a}_k G_1$, proved by telescoping $R_j = G_{k+2j} \prod_{i < j} b_{k+2i}$ and letting $R_J \rightarrow 0$; `G_one_solved` and `G_solved` are (23). The last sentence of the lemma, the identification of the two instances, is `aTel_bulk`, `bTel_bulk`, `aTel_travel` and `bTel_travel`.

Proposition 31 and Proposition 32 are verified in `SectionLadders.lean`, which evaluates $\mathcal{C}_k$ at the three geometric sources: `CC_travel` is $\mathcal{C}_k = 2q/(1 - q^{k+1}) = A_k$, `CC_bulk` is $\mathcal{C}_k = \alpha_k$ and `CC_bulk_geo` is $\mathcal{C}_k = \alpha_{k+1}$, whence `cFrak_travel`, `cFrak_bulk` and `cFrak_bulk_geo` identify $\mathfrak{c}_k$ with Σ_k , with S_k and with T_k . `travelsum` is Proposition 31 and `bulkdress_b0_t0`, `bulkdress_t1_b1` are the four scalars of Proposition 32. `Sladder_eq`, `SigmaLadder_eq` and `Tladder_eq` are the three ladders written out as in (2) and (19). What is verified is the implication, not the existence of the solution: the ℓ^1 solution is a hypothesis, which is where the paper’s $|q| < \frac{1}{3}$ lives, and no analytic continuation step is needed, `G_eq` holding on the whole of $|q| < 1$ for any ℓ^1 solution. Lemma 28 is not formalised.

The first of the two clauses of Proposition 34 that begin “equivalently” is verified as `SectionLadders.recip_ladder` given the two bulk solutions, $t_0 = b_1$ and $T_0 S_e + T_1 S_0 = S_1$ are the same statement, the equivalence being `recip_ladder_iff`. The reciprocity itself enters there as a hypothesis; it is proved separately in `Reciprocity.lean`, in the operator form recorded above, and the two are not joined, that joining

being the instantiation of `Reciprocity.lean` at the concrete ℓ^1 resolvent.

Not formalised in this section: Corollary 33, and with it the second “equivalently” clause of Proposition 34, the one in terms of P_{ij} . The obstruction is the existence of the four limits $\lim_b(\ell_b, r_{b+1})$ and $\lim_b(\ell'_b, r'_{b+1})$: Mathlib carries no convergence theorem for an infinite product of matrices $\prod_b \mathcal{M}_b$ with $\sum_b \|\mathcal{M}_b - I\| < \infty$, and supplying one is a statement about products, not about this cocycle. The rest of the corollary, the passage from (20) to (24) and the identification of the four entries once the limits exist, needs only the split already carried out in `Telescope.row_sum`.

Lemma 42 (a positivity segment). *For $0 < q \leq 0.45$ one has $S_e(q) \geq 0.32$ and $|s(q)| = \frac{q}{1-q} |P_{12}|/S_e \leq 0.78$. Since $q_1 = 0.4494536 \dots < 0.45$, both hold on $(0, q_1]$.*

Proof. By [7, Proposition prop:qtrig], $S_e = \sum_{j \geq 0} (-1)^j [2(1-q)]^j q^{j^2+j} / (q; q)_{2j}$, and by (37), $P_{12} = \sum_{k \geq 1} (-1)^{k-1} [2(1-q)]^k q^{k^2+k+1} (1-q^{2k}) / (q; q)_{2k+1}$. For every n , $(q; q)_n \geq (q; q)_\infty \geq 1 - q - q^2$, the last step by truncating Euler’s pentagonal series $1 - q - q^2 + q^5 + q^7 - \dots$, whose tail is positive for $q \leq \frac{1}{2}$; on $q \leq 0.45$ this is ≥ 0.3475 . Each term $[2(1-q)]^j q^{j^2+j}$ with $j \geq 1$ increases on $(0, 0.45]$, since $(j^2+j)/q - j/(1-q) > 0$ there, so it is at most its value at $q = 0.45$: successively 0.222750, 0.010048, 0.000092, the remaining terms summing to less than 10^{-5} . Hence $S_e \geq 1 - 0.23290/0.3475 \geq 0.32$. The same estimate applied to $[2(1-q)]^k q^{k^2+k+1}$, likewise increasing, and to $(1 - q^{2k}) \leq 1$, gives 0.100238, 0.004521, 0.000042 and a tail below 10^{-6} , so $|P_{12}| \leq 0.104801/0.3475 \leq 0.3016$ and $|s| \leq \frac{0.45}{0.55} \cdot \frac{0.3016}{0.32} \leq 0.78$. $\square$

6.6 The site cost

The subsections above concern the operators of Definition 27. This one concerns the combinatorial quantity those operators are meant to encode: the per-site cost of the crossing optimisation of [5]. Its value is computed here in closed form, which settles the first half of the model input (M) below and corrects the marker clause of the form in which that input was previously stated.

Definition 43 (the pairing optimisation). Fix marker data $(\varepsilon^*, \delta^*) \in \{\pm 1\} \times \{0, 1\}$, a displacement $k^* \in \mathbb{Z}$, and deposits $d_j \in \mathbb{Z}$ on the integer edges j , edge j joining site j to site $j + 1$. The travel indicator is $f_j = 1$ for $0 \leq j < k^*$, $f_j = -1$ for $k^* \leq j < 0$ and $f_j = 0$ otherwise, and $d_j \equiv f_j \pmod{2}$. A *realisation* consists of

- crossing counts $m_j \geq \max(|d_j|, |f_j|)$ with $m_j \equiv d_j \pmod{2}$, whose support is an interval containing 0; edge j is crossed $u_j = (m_j + f_j)/2$ times upward and $dn_j = (m_j - f_j)/2$ times downward;
- a sign in $\{\pm\}$ on each crossing, with p_j^u of the up-crossings and p_j^d of the down-crossings carrying $+$, subject to $d_j = 2p_j^d - dn_j + u_j - 2p_j^u$;
- at every site s , a bijection between the arrivals and the departures at s .

The arrivals at s are the up-crossings of edge $s - 1$, of class $(L, \pm)$ by their sign, and the down-crossings of edge s , of class $(R, \pm)$; the departures are the down-crossings of edge $s - 1$, of class $(L, \pm)$, and the up-crossings of edge s , of class $(R, \pm)$. In addition there is a virtual arrival of class $(L, +)$ at $s = 0$, and a virtual departure at $s = k^*$, of class (R, ε^*) if $\delta^* = 1$ and (L, ε^*) if $\delta^* = 0$. A matched pair costs 0 on the same side with equal signs, 2 on the same side with opposite signs, and 1 on opposite sides. The cost of a realisation is $\sum_j m_j$ plus the sum of the pairing costs; ℓ_R is the minimum cost, and c is the least number of components other than the one carrying the two virtual events, over the realisations of minimum cost.

Remark 44 (standing of Definition 43). This is the optimisation of [5, Metric formula, as verified], restated. That it computes the relaxed word length is verified there and is not proved. Everything in this subsection is a statement about the optimisation itself and is proved outright. The growth-series statements of this paper do not consume it: they go through the site-local closed forms ℓ_R, c and the Lean identity `metricAll` ([7, Theorem “the transfer model (M)”, thm:model]), and the optimisation enters only the interpretation of V as a relaxed count.

Lemma 45 (the transportation value). *Let $A^\pm, B^\pm, C^\pm, D^\pm$ be non-negative integers with $A^+ + A^- + B^+ + B^- = C^+ + C^- + D^+ + D^-$, and consider the transportation problem with supplies A^+, A^-, B^+, B^- and demands C^+, C^-, D^+, D^- in the four classes $(L, +), (L, -), (R, +), (R, -)$ under the cost of Definition 43. Put*

$$\alpha = (C^+ - C^-) - (A^+ - A^-), \quad \beta = (B^+ - B^-) - (D^+ - D^-), \quad \Phi = (A^+ + A^-) - (C^+ + C^-).$$

Then the minimum cost equals $\max(|\alpha|, |\beta|, |\Phi|)$.

Proof. Write $N_L^a = A^+ + A^-$, $N_L^d = C^+ + C^-$, $N_R^a = B^+ + B^-$, $N_R^d = D^+ + D^-$, so $\Phi = N_L^a - N_L^d = N_R^d - N_R^a$, and $\sigma_L^a = A^+ - A^-$, $\sigma_L^d = C^+ - C^-$, and similarly on the right, so that $\alpha = \sigma_L^d - \sigma_L^a$ and $\beta = \sigma_R^a - \sigma_R^d$.

Let π be the mass sent from left supplies to right demands and π' that from right supplies to left demands. The mass staying on the left is $g = N_L^a - \pi = N_L^d - \pi'$, so $\pi' = \pi - \Phi$; put $P = \pi + \pi' = 2\pi - \Phi$. Cross transports cost 1 each and their cost does not depend on the signs, so for fixed π, π' the cost is $P + 2\text{mis}_L + 2\text{mis}_R$, where mis_L is the least number of opposite-sign pairs among the g same-side pairs on the left, minimised over which supplies stay. If a^+ of the staying supplies and c^+ of the served demands carry $+$, that number is $|a^+ - c^+|$, since $\min(a^+, c^+) + \min(g - a^+, g - c^+) = g - |a^+ - c^+|$ pairs can be matched with equal signs and no more. Here a^+ ranges over the integer interval $[\max(0, g - A^-), \min(A^+, g)]$, which is non-empty because $0 \leq g \leq A^+ + A^-$, and c^+ over $[\max(0, g - C^-), \min(C^+, g)]$, likewise non-empty. The least $|a^+ - c^+|$ over two non-empty integer intervals $[l_1, r_1], [l_2, r_2]$ is $\max(0, l_1 - r_2, l_2 - r_1)$. Here $l_1 - r_2 = \max(0, g - A^-) - \min(C^+, g)$, which is $g - A^- - C^+$ when $g \geq A^-$ and $C^+ \leq g$, and is at most 0 otherwise, so $\max(0, l_1 - r_2) = \max(0, g - A^- - C^+)$; symmetrically $\max(0, l_2 - r_1) = \max(0, g - C^- - A^+)$, and

$$\text{mis}_L = \max(0, g - A^- - C^+, g - C^- - A^+).$$

Substituting $A^\pm = (N_L^a \pm \sigma_L^a)/2$ and $C^\pm = (N_L^d \pm \sigma_L^d)/2$ gives $g - A^- - C^+ = \frac{\Phi - \alpha}{2} - \pi$ and $g - C^- - A^+ = \frac{\Phi + \alpha}{2} - \pi$, hence $2\text{mis}_L = \max(0, |\alpha| + \Phi - 2\pi) = \max(0, |\alpha| - P)$. The same computation on the right gives $2\text{mis}_R = \max(0, |\beta| - \Phi - 2\pi') = \max(0, |\beta| - P)$. Therefore the cost is

$$\phi(P) = P + (|\alpha| - P)^+ + (|\beta| - P)^+,$$

which for $P \geq 0$ is at least $\max(|\alpha|, |\beta|)$, with equality exactly on $\min(|\alpha|, |\beta|) \leq P \leq \max(|\alpha|, |\beta|)$, and which equals P above that range.

It remains to identify the admissible P . From $\pi \geq 0$ and $\pi' \geq 0$ one gets $P \geq |\Phi|$; from $\pi \leq \min(N_L^a, N_R^d)$ and $\pi' \leq \min(N_R^a, N_L^d)$ one gets, using $\Phi = N_L^a - N_L^d = N_R^d - N_R^a$, the single constraint $P \leq \min(N_L^a + N_L^d, N_R^a + N_R^d)$. Finally $P \equiv \Phi \pmod{2}$, and $\alpha \equiv \beta \equiv \Phi \pmod{2}$ because $\sigma \equiv N \pmod{2}$ on each side. Now $|\alpha| \leq |\sigma_L^d| + |\sigma_L^a| \leq N_L^d + N_L^a$ and $|\Phi| \leq N_L^a + N_L^d$, and symmetrically $|\beta|, |\Phi| \leq N_R^a + N_R^d$. Hence $P^* := \max(|\Phi|, \min(|\alpha|, |\beta|))$ is admissible: it has the right parity, it is at least $|\Phi|$, and it is at most $\max(|\Phi|, |\alpha|) \leq N_L^a + N_L^d$ and at most $\max(|\Phi|, |\beta|) \leq N_R^a + N_R^d$. If $|\Phi| \leq \max(|\alpha|, |\beta|)$ then P^* lies in the equality range and $\phi(P^*) = \max(|\alpha|, |\beta|)$, which is optimal

since $\phi \geq \max(|\alpha|, |\beta|)$ throughout. If $|\Phi| > \max(|\alpha|, |\beta|)$ then every admissible P satisfies $P \geq |\Phi|$ and $\phi(P) = P$, so the minimum is $|\Phi|$, attained at $P^* = |\Phi|$. In both cases the minimum is $\max(|\alpha|, |\beta|, |\Phi|)$. $\square$

Corollary 46 (the local cost law). *In Definition 43, let $\alpha_s, \beta_s, \Phi_s$ be the three quantities of Lemma 45 formed from the arrivals and departures at site s . Then*

$$\begin{aligned}\alpha_s &= d_{s-1} - [s = 0] + \varepsilon^*[s = k^*][\delta^* = 0], & \beta_s &= d_s - \varepsilon^*[s = k^*][\delta^* = 1], \\ \Phi_s &= f_{s-1} + [s = 0] - [s = k^*][\delta^* = 0],\end{aligned}$$

one has $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$ in every case, and the minimal pairing cost at s is

$$\text{Site}(s) = \max(|\alpha_s|, |\beta_s|),$$

independently of the crossing counts m_j , of the sign splits p_j^u, p_j^d , and of the travel indicators. In particular at a site carrying no virtual event $\text{Site}(s) = \max(|d_{s-1}|, |d_s|)$.

Proof. The displays are read off Definition 43: the up-crossings of edge $s-1$ contribute σ_L^a , the down-crossings contribute σ_L^d , and $d_{s-1} = 2p_{s-1}^d - \text{dn}_{s-1} + u_{s-1} - 2p_{s-1}^u = \sigma_L^d - \sigma_L^a$, which is α_s before the virtual events; the virtual arrival adds one to A^+ and the virtual departure adds one to C^{ε^*} or to D^{ε^*} . Likewise $d_s = \sigma_R^a - \sigma_R^d$ and $\Phi_s = u_{s-1} - \text{dn}_{s-1} = f_{s-1}$ before the virtual events. Since the sign splits enter only through σ^a, σ^d , and those enter $\alpha_s, \beta_s, \Phi_s$ only through the fixed combinations d_{s-1}, d_s, f_{s-1} , none of the three depends on m_j or on p_j^u, p_j^d .

For $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$: at a site with no virtual event $\Phi_s = f_{s-1} = f_s$, since arrivals and departures balance, and $|d_j| \geq |f_j|$, since a bulk deposit is even with $f = 0$ and a travel deposit is odd with $|f| = 1$. The virtual events sit at $s = 0$ and $s = k^*$, which for $k^* \neq 0$ are exactly the two sites where f jumps, and there are eight remaining cases. For $s = 0 < k^*$: $\Phi_0 = f_{-1} + 1 = 1$, while $\alpha_0 = d_{-1} - 1$ is odd and $\beta_0 = d_0$ is odd. For $s = 0$ and $k^* < 0$: $\Phi_0 = -1 + 1 = 0$. For $s = k^* > 0$: $\Phi = 1 - 1 = 0$ if $\delta^* = 0$, and $\Phi = 1$ with $\alpha = d_{k^*-1}$ odd, $\beta = d_{k^*} - \varepsilon^*$ odd, if $\delta^* = 1$. For $s = k^* < 0$: $\Phi = -1$ with $\alpha = d_{k^*-1} + \varepsilon^*$ odd and $\beta = d_{k^*}$ odd if $\delta^* = 0$, and $\Phi = 0$ if $\delta^* = 1$. For $k^* = 0$, where the two virtual events share the site: $\Phi_0 = 0$ if $\delta^* = 0$, and $\Phi_0 = 1$ with $\alpha_0 = d_{-1} - 1$ and $\beta_0 = d_0 - \varepsilon^*$ both odd if $\delta^* = 1$. In each case $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$, so Lemma 45 gives $\text{Site}(s) = \max(|\alpha_s|, |\beta_s|)$. $\square$

Corollary 47 (the closed form for ℓ_R , and rigidity of the crossing counts). *Let the span be the least interval of edges containing 0, all j with $d_j \neq 0$ and all j with $f_j \neq 0$. Every realisation of minimum cost has support exactly the span and crossing counts*

$$m_j = \max(|d_j|, |f_j|), \quad \text{forced to 2 when that value is 0,}$$

and

$$\ell_R = \sum_{j \in \text{span}} m_j + \sum_s \max(|\alpha_s|, |\beta_s|),$$

the second sum over the sites of the span.

Proof. By Corollary 46 the pairing cost at every site is independent of the m_j , so the total cost is $\sum_j m_j$ plus a quantity that does not depend on the m_j ; hence a minimum-cost realisation minimises $\sum_j m_j$ alone, subject to $m_j \geq \max(|d_j|, |f_j|)$, $m_j \equiv d_j$, and the support being an interval containing 0. Since $d_j \equiv f_j$, the value $\max(|d_j|, |f_j|)$ has the parity of d_j and is the least admissible value on the span except where it vanishes, where the support condition forces $m_j \geq 2$. Enlarging the support beyond the span adds an edge with $m \geq 2$ and creates one further site, of cost $\max(0, 0) = 0$, without changing any other site cost, so it adds 2 and never helps. $\square$

Corollary 46 is the first clause of (M1) below, proved. The marker clause needs the sign, and the form previously used does not carry it.

Corollary 48 (the marker junctions). *Let $k^* > 0$, let $d_L = d_{-1}$ be the last bulk deposit (even) and $d_R = d_0$ the first travel deposit (odd). Then the cost of the junction at site 0 is*

$$\text{Site}_0(d_L, d_R) = \max(|d_L - 1|, |d_R|),$$

for all four marker data. At the far junction, site k^ , with $d_L = d_{k^*-1}$ odd and $d_R = d_{k^*}$ even, the cost is $\max(|d_L + \varepsilon^*|, |d_R|)$ if $\delta^* = 0$ and $\max(|d_L|, |d_R - \varepsilon^*|)$ if $\delta^* = 1$; it is not independent of $(\varepsilon^*, \delta^*)$ and it is not the mirror image of Site_0 .*

Proof. Corollary 46. At site 0 neither ε^* nor δ^* occurs. $\square$

Remark 49 (the earlier marker form is false). Earlier versions of this paper, and [8], carried $\text{Site}_0(d_L, d_R) = \max(|d_L| - 1, |d_R|)$. That agrees with Corollary 48 exactly on $d_L \geq 0$ and fails on every cell with $d_L \leq -2$ and $|d_R| \leq |d_L|$: the smallest is $d_L = -2$, $d_R = 1$, where the true cost is 3 and the earlier form gives 1. The verification on which the earlier form rested covered $0 \leq d_L \leq 12$ only. The junction cost therefore depends on the *sign* of the last bulk deposit, not on its magnitude alone.

The correction is local. Every bulk edge cost and every interior bulk site cost depends on the deposit only through its magnitude (Corollary 46), so reversing the sign of the junction-adjacent deposit of a bulk run is a weight-preserving involution on bulk runs; writing B_σ for the generating function of the bulk runs whose junction-adjacent deposit has magnitude σ (so $B_{2b} = P_b$ of (25), and $B_0 = 1/(1 - \mathfrak{g}t_1) \neq 0$ of Remark 40 collects the empty chain and the chains opening with a gap run), the two signs carry $B_\sigma/2$ each, and the near-junction vector of (M3) is

$$\mu_s^{(y)} = B_0 x^{2s+1} + \sum_{b \geq 1} \frac{P_b}{2} \left(x^{\max(2b-1, 2s+1)} + x^{\max(2b+1, 2s+1)} \right), \quad (28)$$

in place of $\sum_\sigma B_\sigma x^{\max(\sigma-1, 2s+1)}$. The far-junction vector is

$$E_s^{(y)} = B_0^{[s]} x^{2s} + B_0 x^{2s+2} + \sum_{b \geq 1} P_b \left(x^{\max(2s, 2b)} + x^{\max(2s+2, 2b)} \right), \quad B_0^{[s]} = \begin{cases} B_{0z} := 1 + y \mathfrak{g} \sum_b q^b P_b, & s = 0, \\ B_0, & s \geq 1, \end{cases} \quad (29)$$

and $\lambda^{(y)} := E^{(y)} + 2\mu^{(y)}$. The slot B_{0z} carries the junction cut: at $y = 1$ it equals B_0 , and at $y = q$ it equals $B_0(1 - t_1 q / (1 + q))$. This changes no rank statement below: Proposition 60 concludes that the junction is not of rank one, and the symmetrised matrix (28) has rank 6 on the block $\sigma \in \{0, 2, 4, 6, 8, 10\}$, $2s + 1 \in \{1, 3, 5, 7, 9, 11\}$, by exact elimination over $\mathbb{Q}$ at $x = \frac{1}{2}$, against rank 5 for the earlier form on the same block.

Proposition 50 (the cut criterion and the shield lower bound). *Call a site s of the span cut if $\alpha_s = \beta_s = \Phi_s = 0$, and let Z be the set of cut sites interior to the span. At a cut site every minimum-cost pairing matches each arrival with a departure on its own side, so no strand crosses s ; at any other interior site s some minimum-cost pairing lets a strand cross. Consequently every minimum-cost realisation has at least $|Z| + 1$ components and*

$$c \geq |Z|.$$

A maximal run of L gap edges of the bulk ($d_j = f_j = 0$) flanked by bulk deposits and containing no virtual event contributes exactly its $L - 1$ interior sites to Z .

Proof. At a cut site $\phi(P) = P$ in the notation of Lemma 45, so the unique minimiser is $P = 0$, which is the assertion. At any other interior site $\max(|\alpha_s|, |\beta_s|) > 0$, since $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$; the equality range of ϕ is $[\min(|\alpha_s|, |\beta_s|), \max(|\alpha_s|, |\beta_s|)]$ and contains a value $P > 0$ admissible for the caps of Lemma 45: namely $\min(|\alpha_s|, |\beta_s|)$ when that is positive, which is the choice P^* of Lemma 45; and 2 otherwise, since $\min(|\alpha_s|, |\beta_s|) = 0$ forces $\Phi_s = 0$, hence $\alpha_s \equiv \beta_s \equiv 0 \pmod{2}$ and $\max(|\alpha_s|, |\beta_s|) \geq 2$. The two caps of Lemma 45 are m_{s-1} and m_s , each augmented by 1 for a virtual event of s on that side, and both are at least 2 when $\Phi_s = 0$: an edge with $f = 0$ has $m \geq 2$ on the span by Corollary 47, while $\Phi_s = f_{s-1} + [s = 0] - [s = k^*][\delta^* = 0] = f_s + [s = k^*][\delta^* = 1]$ shows that an adjacent edge with $|f| = 1$ forces a virtual event of s on its side, which supplies the missing unit. Removing the $|Z|$ cut sites disconnects the crossings of the span into at least $|Z| + 1$ classes, no two of which can lie in the same component; one component carries the virtual events, so at least $|Z|$ are isolated cycles. For the last sentence, an interior site of such a run has $d_{s-1} = d_s = 0$ and $f_{s-1} = 0$ and no virtual event, hence is cut, while the two sites at the ends of the run have $|\alpha|$ resp. $|\beta|$ equal to the flanking bulk deposit, which is even and non-zero. $\square$

Remark 51 (what the shield law still owes). Proposition 50 gives $c \geq |Z|$. The reverse inequality $c \leq |Z|$ asks for one minimum-cost realisation whose components are exactly the $|Z| + 1$ classes cut out by Z , that is, for pairings that connect all crossings within each class. It is not proved here. It is verified by exhaustive enumeration, over every sign split and every site bijection, on all bulk configurations with $k^* = 0$, up to 8 edges, even deposits of magnitude at most 4 and both signs, and all four marker data: 1 048 544 configurations, gap runs up to length 6, 0 exceptions to both $c = |Z|$ and the closed form of Corollary 47 (`code/zeta_probe/tools/sitecost`, mode `shield`). The closed form $|Z|$ also recovers, by inspection, the boundary shield $\text{sh}_R = \mathbf{1}[\varepsilon^* = -1 \text{ or } \delta^* = 1]$ that [5, the explicit transducer] records at $k^* = 0$: by Corollary 46 the marker site is cut exactly when $d_{-1} = 0$, $\delta^* = 0$ and $\varepsilon^* = 1$, which is exactly where that indicator vanishes.

Remark 52 (the boundary term in the cut count, and the check at $k^* \neq 0$). The set Z of Proposition 50 consists of cut sites *interior* to the span. That count is short by one whenever both markers sit at an endpoint of the span on the side opposite the edges, since cutting there isolates the marker component and leaves every crossing in closed walks. The configuration is exactly

$$k^* = 0, \quad \delta^* = 0, \quad \text{lo} = 0 < \text{hi},$$

and it is the boundary shield of Remark 51: the marker site is cut precisely where sh_R vanishes. The clause $\text{lo} = 0 < \text{hi}$ is needed as well, since without it the identity is miscounted, having $c = 0$ and one cut site.

Writing Z^+ for Z together with that site when the displayed condition holds, the shield law $c = |Z^+|$ has no exceptions on the 3 336 511 elements of word length at most 31, over all k^* , with $c = (\ell_T - \ell_R)/2$ computed from breadth-first word length and the closed form of Corollary 47 (`code/zeta_probe/tools/nogap`). This extends the check of Remark 51, which covers the bulk $k^* = 0$ only. Enlarging Z by every marker site instead, rather than by the boundary configuration above, produces 8555 violations of Proposition 50 at word length 19, which is how the condition was located.

The count $|Z^+|$ is the site-local cut count `cTrue` of `CorrectedSpan.lean`, and $\ell_T = \ell_R + 2|Z^+|$, with ℓ_R the closed form of Corollary 47, is now the Lean theorem `metricAll` ([7, Theorem “the transfer model (M)”, `thm:model`]). What remains unproved is only the reverse inequality $c \leq |Z^+|$ for the component count c of the optimisation of Definition 43; the growth-series results do not use it.

6.7 The gap-free case: the reverse inequality holds when Z is empty

Call an edge j of the span a *gap edge* when $d_j = 0$ and $f_j = 0$; by Corollary 47 such an edge is forced to $m_j = 2$ by reachability. The reverse inequality is out of reach in general (Remark 51), but it holds, with a short proof, on the elements that carry no gap edge at all.

Lemma 53 (the sign split is forced at minimum multiplicity). *Let j be a span edge with no gap, so $d_j \neq 0$. Then $m_j^* = |d_j|$, and the split $p_j^d = p_j^u + (d_j - f_j)/2$ subject to $0 \leq p_j^u \leq u_j$ and $0 \leq p_j^d \leq dn_j$ has exactly one solution: $p_j^u = 0$ when $d_j > 0$ and $p_j^u = u_j$ when $d_j < 0$. Consequently every up-crossing of edge j carries the sign $-\text{sgn}(d_j)$ and every down-crossing the sign $+\text{sgn}(d_j)$.*

Proof. $d_j \equiv f_j \pmod{2}$ and $d_j \neq 0$ give $|d_j| \geq |f_j|$ in both parities, so $m_j^* = |d_j|$ and $2dn_j = d_j - f_j$ when $d_j > 0$. Then $p_j^d = p_j^u + dn_j \leq dn_j$ forces $p_j^u \leq 0$, hence $p_j^u = 0$. The case $d_j < 0$ is the mirror image. $\square$

Lemma 54 (the pair cost sees only the side pattern). *At a site both of whose edges are gap-free, the left arrivals all carry one sign and the left departures the opposite sign, and likewise on the right. Hence $\text{pc} = 2$ for a bounce and 1 for a pass, independently of which strands are matched.*

Proof. Left arrivals are the up-crossings of the left edge and left departures its down-crossings, so Lemma 53 gives them opposite signs; same on the right. $\square$

Lemma 55 (free swaps). *Under Lemma 54, the 2-swap exchanging the departures of two pairs changes the site cost by 0 exactly when the two arrivals lie on the same side or the two departures do; the only cost-lowering patterns are two bounces on opposite sides. In particular a pair that is a bounce never blocks.*

Proof. Sixteen side patterns, checked exhaustively; formalised as `NoGapMerge.swap_free_iff`, `swap_neg_iff` and `bounce_never_blocks`. $\square$

Theorem 56 (no gap edge forces $c = 0$). *If g has no gap edge then $c(g) = 0$.*

Proof. Since the site cost of Lemma 45 depends on neither the crossing counts nor the sign splits, every relaxed-optimal realisation has $m_j = m_j^*$, so Lemmas 53–55 apply at every site. Take a relaxed-optimal realisation with the fewest components and suppose there are at least two. If two components hold strands of the leftmost span edge, that site is shared and both have $(R \rightarrow R)$ bounces there, so by Lemma 55 they merge at no cost. Otherwise one component owns that edge; let B be another component whose leftmost site r is minimal, so r exceeds the left end of the span. Edge $r - 1$ then lies in the span with $m_{r-1} \geq 1$, and its strands are not B 's, so site r is shared; B 's ends at r come only from edge r , so B bounces there and again the merge is free. A 2-swap across two components merges them, so the component count drops at no cost, contradicting minimality. Hence there is one component and no isolated cycle.

One configuration escapes the last step: if r carries the virtual arrival, B may have a single $(L \rightarrow R)$ pass and no bounce, and a neighbour with an opposing $(R \rightarrow L)$ pass and $d_{r-1} < 0$ would block. That neighbour then has no left arrival at r , so its strands on edge $r - 1$ are all down-crossings, and any up-crossing there would belong to a third component which merges with B freely; so the blocker also needs $u_{r-1} = 0$, that is $d_{r-1} = f_{r-1} = -1$ and a single strand. But $f_{r-1} = -1$ forces $k^* < 0$, and $r = 0$ since r carries the virtual arrival, so the marker component runs from site 0 to site $k^* < 0$ and must traverse edge $r - 1$; its one strand is therefore the marker component's, contradicting the assumption that it belongs to another. The configuration does not occur. $\square$

Corollary 57. *If the lamp support of g lies inside its travel interval then $c(g) = 0$.*

Proof. Support inside I_k makes the span equal to I_k , on which $f_j = \pm 1$, so no edge is a gap edge; apply Theorem 56. $\square$

Corollary 57 is the statement used in the lifting argument for the true series, where it had been verified but not proved. Verification of Theorem 56 independent of the proof: 2 505 271 elements of word length at most 31, no exceptions (`code/zeta_probe/tools/nogap`).

Remark 58 (the argument does not extend past $Z = \emptyset$). Theorem 56 rests on Lemma 53, which needs $d_j \neq 0$. A gap edge has $m = 2$ and $d = 0$, so $p^d = p^u$ with $u = dn = 1$ admits two splits and the homogeneity of Lemma 54 fails. Enumerating all 256 sign-and-side configurations of a 2-swap, 52 block, of which 36 share an arrival or a departure side, and a cost-0 bounce is blocked in 28; both criteria of Lemma 55 are therefore false in general. The reverse shield inequality is not reachable by this route.

Remark 59 (the relaxed length decomposes at every k^*). Corollary 47 is verified by direct enumeration only for $k^* = 0$ (Remark 51), but it holds at every k^* . The site cost depends on neither m_j nor the sign splits, so $\sum_j m_j$ is minimised separately at $m_j = m_j^*$; edge j presents a departure at site j and an arrival at site $j + 1$, so the matchings at distinct sites act on disjoint sets of ends and are chosen independently; and in the relaxed model every family of per-site matchings is admissible, isolated cycles being free. No term falls below its local minimum and all local minima are attained together. The true metric behaves differently precisely because the single-walk requirement couples the sites, which is what Theorem 56 has to handle.

Machine verification of the site cost

Lemma 45, Corollary 46, Corollary 47, Corollary 48, the counterexample of Remark 49 and all of Proposition 50 except its component count are formally verified in Lean 4 against Mathlib [28], in six files: `SiteCost.lean` (the transportation problem at one site), `MarkedSite.lean` (the virtual events and the local cost law), `PairingMatrix.lean` (bijections against multiplicity matrices), `Realisation.lean` (the whole edge path), `CutCross.lean` (the middle sentence of Proposition 50) and `CutComponents.lean` (the counting step of the component bound, over an abstract graph).

A pairing at a site is the structure `Plan`: sixteen non-negative integers with the eight matching equations of Definition 43; `Plan.cost` is its cost and `Plan.cross` the number of pairs on opposite sides. The minimum of Lemma 45 is formalised as `IsLeast`, so both halves are proved separately: `cost_lower_bound` holds for every pairing and `exists_plan_cost_eq` exhibits one of exactly that cost. Definition 43 asks for a *bijection* between the arrivals and the departures rather than for a matrix, and the two are reconciled: `planOfBij` sends a bijection to its multiplicity matrix without changing the cost, and `bijOfMatrix` realises every matrix with the right row and column sums by a bijection, so `bij_min` is Lemma 45 stated over bijections and `MarkedSite.site_cost_bij` is Corollary 46 stated over bijections. A site is the structure `MarkedSite`: the displacement k^* , the site s , the marker data, and the crossing data of the two adjacent edges, constrained to realise the travel indicator `travel`. A realisation of a whole path is the structure `Realisation` over a `PathData`, which carries k^* , the marker data, the deposits and the span $[A, B]$; `PathData.emptyPath` is a witness that the hypotheses are consistent.

Paper step	Lean declaration	axioms
$\text{cost} \geq \max(\alpha , \beta , \Phi)$, six duals	<code>cost_lower_bound</code>	standard
a pairing attaining it	<code>exists_plan_cost_eq</code>	standard
Lemma 45, the minimum	<code>transport_min</code>	standard
a bijection gives a matrix of equal cost	<code>planOfBij_cost</code>	standard
a matrix gives a bijection of equal cost	<code>bijOfMatrix_cost</code>	standard
Lemma 45 over bijections	<code>bij_min</code>	standard
collapse to $\max(\alpha , \beta)$	<code>siteCost_eq</code>	standard
$\alpha_s = d_{s-1} - [s=0] + \varepsilon^*[s=k^*][\delta^*=0]$	<code>MarkedSite.alpha_eq</code>	standard
$\beta_s = d_s - \varepsilon^*[s=k^*][\delta^*=1]$	<code>MarkedSite.beta_eq</code>	standard
$\Phi_s = f_{s-1} + [s=0] - [s=k^*][\delta^*=0]$	<code>MarkedSite.Phi_eq</code>	standard
$ \Phi_s \leq \min(\alpha_s , \beta_s)$, all site types	<code>MarkedSite.Phi_le_min</code>	standard
Corollary 46, the minimum at any site	<code>MarkedSite.site_cost</code>	standard
Corollary 46 over bijections	<code>MarkedSite.site_cost_bij</code>	standard
independence of m_j and of p_j^u, p_j^d	<code>MarkedSite.cost_indep</code>	propext
interior site cost $\max(d_{s-1} , d_s)$	<code>MarkedSite.cost_interior</code>	propext
Corollary 48, near junction	<code>PathData.marker_near</code>	propext
Corollary 48, far junction, $\delta^*=0$	<code>PathData.marker_far_left</code>	propext
Corollary 48, far junction, $\delta^*=1$	<code>PathData.marker_far_right</code>	propext
and it is neither $(\varepsilon^*, \delta^*)$ -free nor the mirror of Site_0	<code>PathData.marker_far_depends</code> (the same declaration)	none
off the span every site is empty	<code>Realisation.pair_cost_zero_outside</code>	standard
Corollary 47, the minimum	<code>lR_closed</code>	standard
Corollary 47, rigidity of m_j	<code>Realisation.rigidity</code>	standard
Proposition 50, cut site: no crossing	<code>cut_forces_no_cross</code>	standard
the same over a realisation	<code>Realisation.cut_no_cross</code>	standard
the transportation value at a prescribed cross mass	<code>exists_plan_of_cross</code>	standard
the two caps $m_{s-1}, m_s \geq 2$ where $\Phi_s = 0$	<code>Realisation.capL_two</code>	standard
	<code>Realisation.capR_two</code>	standard
Proposition 50, a non-cut site: a crossing	<code>Realisation.exists_cross_of_not_cut</code>	standard
Proposition 50, a gap run's cut sites	<code>PathData.gap_run_cut</code>	standard
Z interior to the span	<code>PathData.cutSet_lt, cutSet_le</code>	standard
every edge of the span carries a crossing	<code>Realisation.m_pos_of_min</code>	standard
blk constant on components	<code>CutComponents.blk_reachable</code>	propext
blk attains $0, \dots, Z $ on the span	<code>CutComponents.exists_pos_with_gz</code>	standard
at least $ Z + 1$ components, given the graph	<code>CutComponents.exists_injective_components</code>	standard
and $c \geq Z $ from it	<code>..._avoiding</code>	standard
the two marker forms agree on $d_L \geq 0$	<code>marker_forms_agree</code>	standard
and differ at $(d_L, d_R) = (-2, 1)$	<code>marker_forms_differ</code>	none

“standard” means `propext`, `Classical.choice`, `Quot.sound`, or a subset of them; several of the rows above in fact report only `propext` and `Quot.sound`. The last row is closed by `decide` and depends on no axiom at all, as do `PathData.marker_far_depends` and `Realisation.site_cost_eq`. The cost of a realisation is summed over the sites of its support interval, which is legitimate because both virtual events lie there (`Realisation.virtual_in_window`) and every site outside it is empty.

The middle sentence of Proposition 50, that at an interior site which is not cut some minimum-cost pairing lets a strand cross, is verified in the form the proof uses. `exists_plan_cost_eq` settles every case except $\min(|\alpha_s|, |\beta_s|) = 0$ with $\Phi_s = 0$; there `exists_plan_of_cross` runs the construction of Lemma 45 at the prescribed cross mass $P = 2$, which is admissible because $\Phi_s = 0$ forces both caps to be at least 2, and that is a path-level fact, proved as `Realisation.capL_two` and `Realisation.capR_two` from $m_j \geq \max(|d_j|, |f_j|)$ on the span together with $\Phi_s = f_{s-1} + [s=0] - [s=k^*][\delta^*=0] = f_s + [s=k^*][\delta^*=1]$.

The counting step of the component bound is verified in `CutComponents.lean`, over an abstract

graph. Write $\text{blk}(v)$ for the number of cut sites at or below the edge carrying the crossing v . `blk_adj` and `blk_reachable` are that `blk` is constant on connected components, given the locality hypothesis **Local**: an edge of the graph joins crossings on the two edges adjacent to one site, and joins crossings on *different* edges only at a site that is not cut. `exists_pos_with_gz` is the discrete intermediate value argument, that `blk` takes every value in $0, \dots, |Z|$ on the span, and `exists_injective_components` concludes with an injection of $\{0, \dots, |Z|\}$ into the connected components, which is “at least $|Z| + 1$ components” with no finiteness hypothesis. `exists_injective_components_avoiding` is the passage to $c \geq |Z|$ once the two virtual events are known to share a component. Two of the three inputs are supplied against the structures of `Realisation.lean`: `PathData.cutSet` is Z with its two bounds (`cutSet_lt`, `cutSet_le`), and `Realisation.m_pos_of_min` is the occupancy, every edge of the span of a minimum-cost realisation carrying a crossing.

Two things in this subsection are *not* formalised. First, that the optimisation of Definition 43 computes the relaxed word length (Remark 44, an input of [5], verified there and not proved). Second, the strand graph itself, and with it the component count $c \geq |Z|$. `Realisation.pair` stores the multiplicity matrix of the site pairing rather than the bijection, so the vertex set and the incidence relation that `CutComponents.lean` quantifies over are not definable against the present structure and would need a further definition. `Realisation.cut_no_cross` already carries the mathematical content of the locality hypothesis at the level of the matrix; what is missing is the passage from a matrix with vanishing cross mass to an incidence relation on individual crossings.

6.8 The junction matrix and the bulk dictionary

The transfer model (M), [7, Theorem “the transfer model (M)”, thm:model], the meromorphic continuation ([7, Lemma “meromorphic continuation”, lem:mero]), the shape of the local form ([7, Proposition “the shape of the local form”, prop:shape]) and the closed form of the junction pairing ([7, Theorem “closed form of the junction pairing”, thm:RJ]) are stated and proved in the companion paper; their proofs cite this section and Appendix D. The following statements complete the block assembly.

Proposition 60 (the marker junction is not rank one). *The junction matrix $J[\sigma, s]$ of (28) has rank at least 3. Already the one-sign matrix $[x^{\max(\sigma-1, 2s+1)}]$ has rank at least 3: on $\sigma \in \{0, 4, 8\}$, $2s + 1 \in \{1, 5, 9\}$ it is*

$$\begin{pmatrix} x & x^5 & x^9 \\ x^3 & x^5 & x^9 \\ x^7 & x^7 & x^9 \end{pmatrix}, \quad \det = x^9 (x - x^3)(x^5 - x^7) \neq 0 \quad (0 < x < 1),$$

its rank on $\sigma \in \{0, 2, 4, 6, 8\}$, $2s + 1 \in \{1, 3, 5, 7, 9\}$ is 4, and the symmetrised matrix (28) has rank 6 on $\sigma \in \{0, 2, 4, 6, 8, 10\}$, $2s + 1 \in \{1, 3, 5, 7, 9, 11\}$. By contrast the gap-bridge term q^{a+b} of (25) is rank one, which is why (30) factorises. Consequently $\mu^{(y)}$ is not a function of the total bulk sum $B(q, y)$ alone, and $\Pi_y(q_m) \neq 0$ does not follow from $\Sigma_0(q_m) \neq 0$ together with $B(q_m, y) \neq 0$. At a travel pole, however, the pairing against the null vector R collapses to a function of t_1 alone ([7, Theorem “closed form of the junction pairing”, thm:RJ]).

Proof. The displayed block is the case $\sigma - 1 \in \{-1, 3, 7\}$, $2s + 1 \in \{1, 5, 9\}$, which interleaves as $-1 < 1 < 3 < 5 < 7 < 9$; subtracting the second row from the first gives $(x - x^3, 0, 0)$, and expanding along it leaves $(x - x^3)(x^5x^9 - x^9x^7) = x^9(x - x^3)(x^5 - x^7)$. The two rank statements are exact Gaussian elimination over $\mathbb{Q}$, at $q = \frac{1}{3}$ and at $x = \frac{1}{2}$ respectively (`assembly_certificate.py`, Part 4). For the second-last sentence: by (M3), $\mu^{(y)}$ is the image of $(B_\sigma)_\sigma$ under the junction matrix, whose rank is at least 3. $\square$

Theorem 61 (the bulk dictionary at the travel poles). *At every travel pole q_m the factorisation (30) holds with $\kappa = t_0 b_1 - b_0 t_1$ and $t_0 = b_1$, and the dictionary (32) holds identically in $0 < q < 1$, in the completed form of Corollary 35. Consequently the bulk quantities $B_0 = 1/(1 - \mathfrak{g}t_1)$ and $B = b_1/(1 - \mathfrak{g}t_1)$ of Remark 40 are finite at every travel pole, for $y = 1$ and $y = q$.*

Proof. Proposition 39, whose two hypotheses are supplied by Corollary 38 with Proposition 69 and by Remark 41; Proposition 34; Corollary 35. Finiteness is $|\mathfrak{g}_V t_1| \leq 0.983$ (Corollary 84) and $\mathfrak{g}_U < \mathfrak{g}_V$. $\square$

Remark 62 (what (L) consumed). That the bulk transfer is the operator of (26), that is, (M1) and (M2). Of these, (M1) is now Corollary 46, the exponent $q^{\max(s, s')}$ of the bulk kernel being $x^{\max(|d|, |d'|)}$ at even deposits; and of (M2) the cost q^L of a gap run and the inequality $c \geq L - 1$ are Corollary 47 and Proposition 50, and $c \leq L - 1$ follows from the metric identity `metricAll` with the site-local cut count ([7, Theorem “the transfer model (M)”, thm:model]), which no longer consumes the optimisation of Definition 43. Everything the paper previously listed as unproved *downstream* of that, namely the factorisation (30), the reciprocity $t_0 = b_1$, the entry $t_1 = P_{12}/S_e$, and the resolvent lines of (32), is proved above. The residue at q_m is governed not by the bulk sum but by the pairing $\Pi_y(q_m)$ ([7, Proposition “the shape of the local form”, prop:shape]), which [7, Theorem “closed form of the junction pairing”, thm:RJ] evaluates.

Earlier versions carried (M) as a hypothesis, with three named weak links: the reverse shield inequality $c \leq L - 1$ (Remark 51), the metric formula, and (M3). The first two are no longer separate inputs. The cut count c that enters $\mathcal{W}$ is the site-local count `cTrue`, and `metricAll` proves `wordLength =  $\ell_R + 2c$` for it directly, so neither the reverse inequality nor the optimisation of Definition 43 is consumed. One caveat: ℓ_R in $\mathcal{W}$ is the closed-form relaxed length of [5]. That it also equals the minimum over relaxed realisations ([5, rem:metric-formula-status]) affects only the *interpretation* of V as a relaxed count, not U and not the series identities. The third, (M3), is proved in Appendix D, in the corrected form stated above. Nothing in §§2–[7, sec:numerators], §4, §§6.1–6.6 above, or Appendices A–B uses it: those speak only about the explicit series of (2), about P_{12} , about the operators just defined, and about the optimisation of Definition 43.

6.9 Formalisation of the junction pairing and of the model

The following files in `lean/with_mathlib` are checked by Lean 4 against Mathlib [28]. Each carries `#print axioms` checks on its main statements, and the recorded output lists only the standard axioms; none uses `sorry` or `native_decide`.

- `RJMain.lean` and `RJShift.lean`: the X and Y shift identities (`X_shift`, `Y_shift`) and their solutions (`X_closed`, `Y_closed`), from the eigen-recursion of R , the bulk recursion of Ψ and the normalisation, including the interchange of the double sums from ℓ^1 summability (`summable_swap`).
- `RJClosedForm.lean`: the algebra of [7, (eq:RJclosed1)]–[7, (eq:RJclosedq)] (`closed_form`, `closed_form_yq`) and the positivity step from the gate bound (`gate_bound`, `bracket_pos`, `bracket_yq_pos`).
- `RJIdentities.lean`: auxiliary finite identities for the travel recursion.
- `RJLemmaJ.lean`: the cut criterion at the junction sites used in Appendix D.
- `RJPhi.lean`: the faithfulness of the model, i.e. the affine realisation intertwines the three generators and is injective.

- `RJMetricAll.lean`: `metricAll`, `wordLength =  $\ell_R + 2c$` for every element, with ℓ_R, c the closed forms `lRTrue`, `cTrue`.

What is *not* formalised: the assembly (M3) of Appendix D; the gate (Appendix B) and every analytic estimate behind it; the identification of the Lean eigen-recursion hypotheses with the operators $\mathcal{T}$ and M_0 at an actual travel pole; and the interval certificates. The formal statements take these as hypotheses.

6.10 Certificate

Remark 63 (the computational certificate). `code/zeta_probe/assembly_certificate.py` re-derives the statements of this section in two independent arithmetic models and exits 0 with every check passing. Numerical agreement is evidence and is not a proof; the proofs above are what carry the statements. What the certificate adds is an independent arithmetic model and the hypothesis deletions.

Part 1, exact arithmetic in $\mathbb{Z}[[q]]/(q^{N+1})$ at the two truncation orders $N = 28$ and $N = 40$, no floating point: Propositions 31, 32 and 34 and Corollaries 33 and 35, coefficient by coefficient; the bulk block b_0 has coefficients 2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, 190, 706, 594 and the travel block the travel-run counts 2, 6, 10, 26, 54, 114, 274, 582, 1298; $P_{11} = 1 + T_0$, $P_{12} = T_1$, $P_{21} = -S_0$, $P_{22} = S_e = 1 - S_1$, $P_{11}S_e + P_{12}S_0 = 1$, the Hahn–Exton form (34), and $S_o = \frac{1-q}{2q}S_0$.

Part 2, exact arithmetic in $\mathbb{Q}[\mathfrak{g}][[q]]/(q^{19})$: the factorisation of Proposition 39 holds identically in the free parameter $\mathfrak{g}$, and three hypothesis deletions each break it, with the first surviving order recorded. A rank-two gap term fails at q^6 ; a gap term q^{a+2b} in place of q^{a+b} fails at q^4 ; and the asymmetric gapless kernel $q^{\max(a,b)} + q^{a+2b}$ breaks $t_0 = b_1$ at q^8 .

Part 3, two precisions, 40 and 70 significant decimal digits: at the first ten travel poles, $q_1 = 0.4494536306$ through $q_{10} = 0.997756895$, the dictionary $b_0 = S_0/S_e$, $t_0 = b_1 = S_1/S_e$, $t_1 = P_{12}/S_e$ holds to relative error $2.5 \cdot 10^{-26}$ at 40 digits and $8.8 \cdot 10^{-57}$ at 70, the loss of about 14 digits being the cancellation in the alternating ladders at the highest pole; the two precisions agree on the pole locations to $1.6 \cdot 10^{-31}$. The resolvent is evaluated there by the $O(N)$ two-point solve of Corollary 33, that is by the analytic continuation, the defining Neumann series not converging at those q . The Möbius value is cross-checked against an independent dense solve of the truncated system at $q = 0.5, 0.7, q_1$ and $y = 1, q$, agreeing to $6.2 \cdot 10^{-28}$. The gate signs $b_0 > 0$ and $s < 1$ hold at all ten, with $b_0\tau \rightarrow 2$ and $s \rightarrow \frac{1}{4}$. (Earlier versions also checked a “lifting increment” $(1-s)B_V + qb_0$; it refers to the E -sourced Möbius form and not to the model’s bulk sum, and is withdrawn.)

Part 4, exact rational arithmetic: the one-sign junction matrix $[x^{\max(\sigma-1, 2s+1)}]$ has rank 4 over $\mathbb{Q}$ at $q = \frac{1}{3}$ on the 5×5 block, its interleaved 3×3 minor equals $9/524288$ at $x = \frac{1}{2}$, the symmetrised junction matrix (28) has rank 6 on the 6×6 block at $x = \frac{1}{2}$, and the gap-bridge matrix $[q^{a+b}]$ has rank 1.

Part 5, exact integer arithmetic, `code/zeta_probe/tools/sitecost`: the site cost is computed by two independent exact solvers, a min-cost flow and a transportation dynamic program, which agree on all 1 012 664 supply-demand pairs with entries at most 7; on the same range the closed form $\max(|\alpha|, |\beta|, |\Phi|)$ of Lemma 45 is exact, with 0 misses. Corollary 46 is confirmed on 4 532 157 site configurations, each evaluated by both solvers, namely every crossing count from the minimum to the minimum plus 10, every sign split, both signs of both deposits with $|d| \leq 24$, every travel indicator, all four marker data and all eight site types (bulk interior, travel interior in both directions, the near and far junctions for $k^* > 0$ and $k^* < 0$, and the coincident junction at $k^* = 0$): 0 exceptions, with the cost independent of the crossing counts and of the sign splits in every case. Deleting any one of the three pairing costs is witnessed necessary: replacing the sign-flip bounce cost 2

by 0, 1, 3, 4, the pass cost 1 by 0, 2, 3, or the bounce cost 0 by 1, 2 each breaks Corollary 46, with 2544 to 7392 exceptions out of 7524 configurations and the smallest counterexample recorded. The marker closed forms of Corollary 48 hold with 0 exceptions on the full signed range, and the form $\max(|d_L| - 1, |d_R|)$ fails on 42 of 156 cells at the near junction and on 36 of 156 at its $k^* < 0$ counterpart, all with $d_L < 0$. Corollaries 47 and the shield count are confirmed as recorded in Remark 51.

7 The inputs of V and U , and their history

The transcendence theorems for the travel resolvent, for V and for U (Theorems [7, thm:travelres], [7, thm:V], [7, thm:U]) and the growth rate ([7, Corollary cor:beta]) are proved in the companion paper. This section records the bivariate bridge, the former inputs (R), (T) and (L) with their present status, the closed form of the gate block, and the turning-point atom.

Remark 64 (what became of (R)). *What (R) was.* Route B decomposes a relaxed element into two bulk runs, one travel interval of length $|k|$ and four marker junctions; (R) was that decomposition read as a local form for V at a travel pole, *together with* a product form for its numerator, $\mathcal{N}_V = \Sigma_0 \cdot S_0 / (1 - S_1) \cdot J$ with J a junction factor.

What is now proved. The k -sum telescoping (Proposition 31), the bulk dressing (Proposition 32), and the shape of the local form, with a rank-one numerator paired against the null vector of the travel transfer ([7, Proposition “the shape of the local form”, prop:shape]). These are the whole of (R) except the product form.

What is refuted. The product form. The marker junction cost is $\max(|d_L - 1|, |d_R|)$ (Corollary 48), a max-coupling of rank at least 3 with an explicit non-zero 3×3 minor (Proposition 60), so the numerator is not a multiple of the total bulk sum and the step “ $\mathcal{N}_V(q_m) \neq 0$ because $\Sigma_0(q_m) \neq 0$ and $S_0(q_m) \neq 0$ ” is not available. [7, Proposition “the numerators do not vanish at the travel poles”, prop:numnonvanish] is unaffected and is used, in [7, Theorem “the travel resolvent”, thm:travelres] and inside [7, Proposition “the shape of the local form”, prop:shape]; what fails is the deduction that leant on it here.

What replaces it. The junction pairing (R-J), that Π_1 is non-zero at infinitely many travel poles, which is strictly weaker than (R) and is [7, Theorem “closed form of the junction pairing”, thm:RJ]: at a travel pole Π_1 collapses to an explicit function of t_1 , and the gate makes it positive.

Where it is used. Only in [7, Theorem thm:V] and, at $y = q$, in [7, Theorem thm:U]. [7, Theorem “the travel resolvent”, thm:travelres], Propositions [7, prop:numnonvanish] and 69, §4 and Appendices A–B do not touch it. All of §6 rests on the transfer model (M), [7, Theorem “the transfer model (M)”, thm:model].

Evidence. The blocks reproduce their own combinatorial series exactly: $S_0/(1 - S_1)$ against the bulk-run counts 2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, 190, 706, 594 and $\Sigma_0/(1 - \Sigma_1)$ against the travel-run counts 2, 6, 10, 26, 54, 114, 274, 582, 1298 (`blocks_growth.py`, and `assembly_certificate.py` Part 1 in exact integer arithmetic at two truncation orders). V itself is computed to 131 exact terms by an independent catalytic transfer, not from the blocks.

Transcendence of U needs two things that the analysis of §4 and Appendix B does not supply: that U has the travel poles at all, and that the residue of U at each of them is non-zero. The first was the combinatorial hypothesis (T); it is now [7, Proposition “Invariance of the travel block”, prop:travelinv] and is proved (§7.2). The second was previously the hypothesis (L); its algebraic content is now a theorem (§6.5, Theorem 61), and the residue itself is the junction pairing (R-J) at $y = q$, [7, Theorem “closed form of the junction pairing”, thm:RJ], the same statement that [7, Theorem thm:V] consumes at $y = 1$. This section states (T) in full, with its dependencies, with

every place it is used, and with the proof that closes it, and records what the bulk dictionary gives, before turning to the analytic content. A reader should not have to open [8] to see what [7, Theorem thm:U] rests on.

7.1 The bivariate bridge

With $c(g) \in \mathbb{Z}_{\geq 0}$ the cycle count of §1, set

$$\mathcal{W}(x, y) = \sum_{n, c \geq 0} w_{n, c} x^n y^c, \quad w_{n, c} = \#\{g : \ell_R(g) = n, c(g) = c\},$$

the grading being by *relaxed* length. Then $V(x) = \mathcal{W}(x, 1)$ and, since $\ell_T = \ell_R + 2c$ and $q = x^2$, $U(x) = \mathcal{W}(x, x^2) = \mathcal{W}(x, q)$: the two series are two evaluations of one bivariate object, at $y = 1$ and at $y = q$, and everything separating them is carried by the marker y . The second identity is the one place where the metric identity $\ell_T = \ell_R + 2c$ is used. With ℓ_R, c the site-local closed forms of [5], $\text{wordLength} = \ell_R + 2c$ holds for every element (Lean, `metricAll` in `RJMetricAll.lean`, resting on `wordLength_eq_lRTrue_add_two_cTrue` and `reaches_of_phiZ` in `PhiLipschitz.lean`), and the word length of the model is the Cayley length in W_{univ} (Lean, `RJPhi.lean`); so $U = \mathcal{W}(x, q)$ unconditionally. That the closed form ℓ_R is also the minimum over relaxed realisations is [5]’s metric formula (its `rem:metric-formula-status`); it bears on reading V as a relaxed count, and on nothing proved here. The corrected assembly [7, (eq:assembly)] agrees with a breadth-first enumeration of W_{univ} as exact series: at $y = q$ to x^{26} in all three sectors; at $y = q^2$ to x^{22} ; and at symbolic y in all 816 coefficients with $\ell_T \leq 31$ ($5.03 \cdot 10^6$ elements), where a deliberately corrupted model is detected. $v_0, \dots, v_{19}$ are certified by the same enumeration together with the bound $\ell_R \geq 2c + 6$. Earlier versions cited agreement to v_{14} ; that check (`cycle_weighted_gf.py`) was a sweep of the bridge $\ell_T = \ell_R + 2c$ and did not assemble the blocks, and the model as then printed fails it.

7.2 The former input (T): the travel block carries no marker

(T) is [7, Proposition “Invariance of the travel block”, `prop:travelinv`]: the denominator $1 - \Sigma_1$ is y -free, hence literally the same series in $\mathcal{W}(x, 1)$ and in $\mathcal{W}(x, q)$, so U has the travel poles q_m of §2.

Where it is used. In the proof of [7, Theorem thm:U], to transport the pole set of Lemma 11 from V to U , and in the same way in [7, Corollary cor:beta] for the growth rate of u_n . Nothing in §4, Appendix A or Appendix B touches it.

What it rests on. Nothing further: it is proved. The localisation of cycles, that a pure-travel element has $c = 0$, is Corollary 57, a case of Theorem 56. The metric identity is not needed. Theorem 56 exhibits a relaxed-optimal realisation with no isolated cycle; that realisation is a single open walk, so it is admissible for ℓ_T and gives $\ell_T \leq \ell_R$, while $\ell_T \geq \ell_R$ holds because the relaxed minimum ranges over a superset of the realisations. Hence $\ell_T = \ell_R$ on pure-travel elements, the two gradings agree termwise, and the travel recursion (2), assembled from pure-travel runs alone, is computed inside a $c \equiv 0$ sector, so the marker cannot enter it.

In particular the reverse inequality of $\ell_T = \ell_R + 2c$ – open when earlier versions of this section listed it as a dependency, and since proved in [5] – plays no role here regardless: only the direction that follows from an exhibited realisation is used.

Evidence. The pure-travel generating function computed in both gradings to radius 18 is identical coefficient by coefficient, 1, 3, 5, 8, 12, 17, 24, 35, 52, 78, 116, 173, 256, 383, 572, 858, 1276, 1907, 2832, with ratio tending to β_2 .

7.3 The former input (L): the bulk dictionary and the gate

The marker enters the block assembly through exactly one factor, the gap-bridge kernel $\mathfrak{g}(q, y) = q/(1 - qy)$ of (M2), summing a maximal run of L forced gap edges, each of weight q and each an isolated cycle except the one shielded by the inner block's turn-around. Its two evaluations, $\mathfrak{g}_V = \mathfrak{g}(q, 1) = \frac{q}{1-q}$ and $\mathfrak{g}_U = \mathfrak{g}(q, q) = \frac{q}{1-q^2} = \frac{\mathfrak{g}_V}{1+q}$, are holomorphic and non-zero on $(0, 1)$, hence at every travel pole.

The bulk resolvent is semiseparable, $M_0[b, a] = 2q^b q^{\max(a, b)}$ plus a rank-one gap term of weight $\mathfrak{g}$, and the resolvent of E is a Möbius function of $\mathfrak{g}$:

$$B^E(q, y) = \frac{b_0 + \mathfrak{g} \kappa}{1 - \mathfrak{g} t_1}, \quad \kappa = t_0 b_1 - b_0 t_1, \quad (30)$$

with b_0, b_1, t_0, t_1 the y -free gapless-resolvent sums of Proposition 32, which satisfy the reciprocity $t_0 = b_1$ (Proposition 34). The factorisation (30) is proved as Proposition 39; its two hypotheses are $S_e \neq 0$ and $s \neq 1$, which the paper was carrying already (Remark 41). Write $s = \mathfrak{g}_V t_1$, so that $q t_1 = s(1 - q)$. What the assembly consumes is the *gate*

$$b_0 > 0 \quad \text{and} \quad s := \frac{q}{1-q} t_1 < 1, \quad t_1 = \frac{P_{12}}{S_e}, \quad (31)$$

on the off-diagonal cocycle entry P_{12} , together with the dictionary identifying the entries of (30) with the blocks of (1),

$$S_e = 1 - S_1, \quad S_o = \frac{1-q}{2q} S_0, \quad b_0 = \frac{S_0}{1 - S_1} = \frac{2q}{1 - q} \cdot \frac{S_o}{S_e}, \quad (32)$$

S_o being identified in Lemma 76 and the two resolvent lines in Corollary 35. The remaining entry $t_1 = P_{12}/S_e$ is an identity between independently defined objects (Corollary 33), not a definition, so all four lines of (32) now stand on the same footing.

What is proved. The factorisation (30) and the dictionary (32) are Theorem 61. Three ingredients previously listed as verified rather than proved are now proved: the reciprocity $t_0 = b_1$ (Proposition 34), the entry $t_1 = P_{12}/S_e$ (Corollary 33), and the identification of the resolvent quantities b_0, b_1, t_0 with the blocks (Proposition 32). That the bulk transfer is M_0 with the rank-one gap term and source u is part of (M), now [7, Theorem “the transfer model (M)”, thm:model]. On the numerical side, the kernel $K(a, b) = q^{\max(a, b)}$ on active half-sizes reproduces $S_0/(1 - S_1) = 2q + 2q^2 + 6q^3 + 2q^4 + 18q^5 + 6q^6 + 42q^7 + \dots$ (`blocks_growth.py`), and the whole dictionary is re-derived in exact integer arithmetic and at two precisions in `assembly_certificate.py`.

Where it is used. In [7, Theorem “closed form of the junction pairing”, thm:RJ], through t_1 and $1 - \mathfrak{g} t_1$; and inside Corollary 84, where the half $b_0 > 0$ is proved from a sign chain running through the closed form in (32). The other half of the gate, $|s| < 1$, uses neither: it is Theorem 82 with Lemma 83, and it is unconditional.

7.4 The gate block in closed form

Both columns of the cocycle satisfy the scalar three-term recursion $y_{n+1} = (1 + q^3 - 2(1 - q)q^{2n+2})y_n - q^3 y_{n-1}$; setting $x = q^n$ turns it into the linear q -difference equation

$$Y(qx) + q^3 Y(x/q) = (1 + q^3 - 2(1 - q)q^2 x^2) Y(x), \quad (33)$$

whose regular solution is, in closed form, a Hahn–Exton q -Bessel function [18, 15, 23],

$$Y_3(x) = \sum_{k \geq 0} d_k x^{2k+3}, \quad d_k = \frac{(-2)^k (1 - q)^k q^{k^2+3k}}{(q^2; q^2)_k (q^5; q^2)_k} = J_{3/2}^{(3)} \text{ normalised}, \quad (34)$$

and $P_{12} = \frac{2q^3}{1-q^3}Y_3(1)$. The denominator S_e is the block of (1), controlled by Lemma 83. It remains to bound the numerator residual against its confluent limit; this is where the two-Bessel identity enters.

7.5 The amplitude atom, formally verified

As $q \rightarrow 1$, equation (33) degenerates to $\tau Y'' - (2\tau/x)Y' + 2Y = 0$, solved by $x^{3/2}J_{3/2}(Wx)$. The q -corrected connection coefficient across the turning point $X = \nu = \frac{3}{2}$ is governed by the following identity, whose point is that the confluence is *entire*; it passes *through* the turning point, bypassing the Liouville–Green singularity where asymptotic methods stall.

Lemma 65 (two-Bessel confluence; amplitude normalisation). *The regular solution of (33) has the confluent form*

$$Y_3(x) = \gamma_{\text{cl}} x^{3/2} \left[J_{3/2}(X) + \frac{\tau X}{2} J_{5/2}(X) \right] + O(\tau^2), \quad X = wx, \quad \gamma_{\text{cl}} = \frac{3\sqrt{\pi}}{4} \left(\frac{w}{2} \right)^{-3/2},$$

and its single-solution envelope satisfies $A/A_{\text{cl}} = 1 + \tau h(X) + O(\tau^2)$ with the explicit rational

$$h(X) = \frac{1 + 3/(2X^2)}{1 + 1/X^2} \in [1, \frac{3}{2}] \quad (X > 0).$$

Consequently $\gamma/\gamma_{\text{cl}} = 1 + O(\tau)$ with explicit constant $\frac{3}{2}$.

Proof. From the q -Pochhammer confluence $d_k = \tau^{-k} D_k(1 - k\tau + O(\tau^2))$ with $D_k = (-1)^k / [2^k k! (\frac{5}{2})_k]$, the leading sum is $\gamma_{\text{cl}} x^{3/2} J_{3/2}(X)$ by the Bessel representation of ${}_0F_1$, and the $-k\tau$ correction contributes $\frac{\tau X}{2} \gamma_{\text{cl}} x^{3/2} J_{5/2}(X)$ via ${}_0F_1(\cdot; \frac{7}{2}; \cdot)$. Both series are entire in X . The envelope of $J_{3/2} + \frac{\tau X}{2} J_{5/2}$ follows from $A^2 = M_{3/2}^2 + (\frac{\tau X}{2})^2 M_{5/2}^2 + \tau X (J_{3/2} J_{5/2} + Y_{3/2} Y_{5/2})$ and the closed half-integer cross-term identity

$$J_{3/2}(X) J_{5/2}(X) + Y_{3/2}(X) Y_{5/2}(X) = \frac{4}{\pi X^2} \left(1 + \frac{3}{2X^2} \right),$$

which with $M_{3/2}^2 = \frac{2}{\pi X} (1 + \frac{1}{X^2})$ gives $h(X) = (1 + \frac{3}{2X^2}) / (1 + \frac{1}{X^2})$; this is monotone decreasing from $h(0^+) = \frac{3}{2}$ to $h(\infty) = 1$, so $1 \leq h \leq \frac{3}{2}$. $\square$

Remark 66 (machine verification). The cross-term identity and the bound $1 \leq h(X) \leq \frac{3}{2}$ have been formalised and checked in Lean 4 against Mathlib (file `AtomN.lean`); the proof terms depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`. The identity is proved by a single `linear_combination` off $\sin^2 + \cos^2 = 1$, and the bound by `nlinarith`.

Remark 67 (the assembly is machine-checked). The composition just performed (the gate arithmetic, the two halves of (31), the passage to the pole set, and the step from infinitely many poles to non-algebraicity) is verified in Lean 4 (`UAssembly.lean`, seven theorems, no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound`), each analytic input entering as an explicit hypothesis: Theorem 82, Lemma 83, Corollary 84, Lemma 11, the reduction (L) (the Lean hypothesis `hred`), and the finiteness of an algebraic function’s pole set in the disc. The file formalises *no* analysis: no q -series, no steepest descent, no numerics; its content is exactly that those hypotheses *compose*, so its hypothesis list is an audit of what [7, Theorem thm:U] rests on, not a verification of any of them. In particular `hred`, the hypothesis “the gate implies a non-zero residue”, is assumed there and not derived; its content is [7, Theorem “closed form of the junction pairing”, thm:RJ], whose identities and positivity step are checked in `RJClosedForm.lean` and `RJMain.lean` with the eigen-relations

and the gate as hypotheses (§6.9). The quantitative step is isolated as `gate_ratio_lt_one`, whose hypothesis $K < c$ (numerator constant strictly below denominator constant) is what the gate needs and what a numerator bound alone cannot supply: with $K = 0.619$ from Theorem 82 and the constant 0.35 of Proposition 69 it is *false*, which is precisely why Lemma 83’s 0.63 is required.

Remark 68 (what the turning-point atom does and does not give). Lemma 65 discharges the amplitude *through the turning point* $X = \frac{3}{2}$, which is the obstruction to any Liouville–Green treatment. It does *not* supply the amplitude at the *pole* $X = w$ to the $O(\tau^2)$ -relative accuracy the numerator gate requires. There the two-Bessel form carries a relative defect whose $O(1)$ part is exactly $\frac{1}{36}$; i.e. the classical turning-point-to-pole connection factor $\frac{36}{35}$, which is itself derivable and, once applied, matches $Y_3(1/q)$ to $O(\tau)$ -relative; and whose *residual* $O(\tau)$ part (the amplitude drift from the turning point to the pole) is not supplied by this route. Because $P_{12} = \frac{2q^3}{3(1-q^3)}Y_3(1/q) - \frac{2}{3}S_e$ is an $O(\sqrt{\tau})$ cancellation, that residual is amplified to the size of P_{12} itself. This residual is closed in Appendix B, not by resumming the amplitude at all, but by the second-difference identity of Proposition 77: P_{12} is the symmetric second difference of the q -cosine at its own zero, so the gate needs only $\max |\psi''| \leq 3.5w$ on a window of width τ , which Theorem 82 proves; Lemma 83 and Corollary 84 then close the remaining halves of the gate ($|S_e| \geq 0.63\sqrt{\tau}$ and $b_0 > 0$). The two off-diagonal cocycle entries obey the Abel relation $x_N Y_N - X_N y_N = -1$ (the discrete q -Wronskian, conserved because $\det M_n \equiv 1$), whence

$$P_{12} = \frac{X_N S_e - 1}{x_N} \quad \text{exactly,}$$

x_N, X_N being the leading components of the same three-term recursion. Hence P_{12} is fixed by the *same* family of Hahn–Exton blocks that controls S_e : one analytic class; that of Remark 8; not two. In this precise sense U sits on V ’s footing. What U additionally requires is the pole asymptotics of x_N, X_N ; again Hahn–Exton, again governed by the exact q -Wronskian of Olde Daalhuis [33], $W_q(J_\nu, Y_\nu) = q^{\nu(\nu-1)}(1-q^2)/\Gamma_{q^2}(\frac{1}{2})^2$ for the numerically satisfactory pair, whose $q \rightarrow 1$ expansion is explicit via Moak’s q -Stirling formula [29] (e.g. $\Gamma_{q^2}(\frac{1}{2}) = \sqrt{\pi}(1 - \frac{3}{8}\tau + O(\tau^2))$). The block-asymptotic assembly is carried out *elementarily* in Remark 15: the numerator amplitude has a derived closed-form expansion with exact rational constants, and the phase-matching of the travel-pole equation against the amplitude’s zero (both collapsing to the same sinh-product family) derives the gate value $1/(4\sqrt{2})$. U ’s analytic inputs thereby coincide with V ’s: the further pieces (($\star$) and the closure of the gate) are proved in Appendix B. (The orthogonal arithmetic route of §8 would settle U without any of this analysis, but is itself open.)

8 An orthogonal arithmetic route

For each prime p , $U \bmod p \in \mathbb{F}_p[[q]]$ is algebraic over $\mathbb{F}_p(q)$ iff its coefficient sequence is p -automatic (Christol [10, 11]), iff its p -kernel is finite. We find the p -kernel to be *maximally* non-automatic: through the accessible depth its size tracks the full p -ary tree with *no* deficit. (An apparent deficit of 1 at level 3; 1, 4, 13, 39 vs. the full 1, 4, 13, 40 at $p = 3$; is a *comparison-length artifact*: a level- k decimation subsequence retains only $\sim N/p^k$ terms, so at a short prefix two distinct decimations can coincide; the deficit disappears once the comparison length reaches 9.) Two instances are machine-checked in Lean 4; both proofs use `native_decide`, so besides the three standard axioms they depend on the compiler-trust axiom `_native.native_decide.ax_1_1` of the declaration concerned. For the true series itself, the 3-kernel of $(u_n \bmod 3)$ through level 3 is the full ternary tree, deficit 0: all $1 + 3 + 9 + 27 = 40$ decimation words of length ≤ 3 are pairwise distinct: at comparison lengths 9 and 10 alike; on the 271-term prefix $n \leq 270$ (`UKernel.lean`, from

the validated `tools/u_modp_rust` output, whose u_n are self-checked against the 43 known terms and byte-identical to the earlier validated $n \leq 180$ run on the overlap). For the sibling block Σ_1 , transcendental over $\mathbb{Q}(q)$ by Corollary 2, the same level-3 full tree is certified in `SigmaKernel.lean` (built from the A/C recursion over $\mathbb{F}_3$), with the short-prefix deficit-1 confirmed spurious. The Berlekamp–Massey [27] linear complexity of $(u_n \bmod 3)$ is $\approx N/2$. This is the strongest finite-data evidence for non-automaticity, and hence for transcendence over $\mathbb{Q}(x)$, independent of all of §§4–[7, sec:U]. A proof for U would require a non-automaticity criterion valid for *dense*-support series (the coefficients carry no lacunary gaps), which the standard gap/Mahler methods do not provide.

9 Conclusion

The arithmetic of A396406 separates cleanly. The Pólya–Carlson dichotomy for the catalytic building blocks is unconditional (Theorem 1), and so is the lower bound β_2 for the growth rate; the matching upper bound, and with it the identification of β_2 as the rate, consumes the transfer model (M), [7, Theorem “the transfer model (M)”, thm:model], and nothing else ([7, Corollary cor:beta]). For the two denominators Σ_1, S_1 the exclusion of the rational alternative needs no growth hypothesis: it follows from the infinitude of their level sets in $(0, 1)$, hence from the same Lemma 9 that the rest of the paper consumes (Corollary 2). For the two numerators it rests on Hypothesis 4, verified out to degree 1156 and unproved, except for the joint statement that Σ_0 and S_0 are not both rational, which the pole route supplies. **The travel resolvent $\Sigma_0/(1 - \Sigma_1)$ is transcendental over $\mathbb{Q}(x)$, unconditionally** ([7, Theorem “the travel resolvent”, thm:travelres]), and it needs no growth hypothesis. Its non-cancellation inputs are two and are separate. The denominator input is Proposition 69, proved in Appendix A by an elementary chain with explicit numerical constants: an annulus representation, an Euler bound, a Poisson theta envelope, and a q -Wronskian invariant. No asymptotic theorem is cited on that path. The numerator input is [7, Proposition “the numerators do not vanish at the travel poles”, prop:numnonvanish], the identity $\Sigma_0(q_m)S_0(q_m) = 2q_m/(1 - q_m)$, which follows from the closed forms $\Sigma_0 = \frac{2q}{Z}\mathfrak{s}(Z)$, $S_0 = \frac{2q}{z_0}\mathfrak{s}(z_0)$ and the q -Pythagorean identity at a zero of the q -cosine; it is exact, it holds at every pole with no exceptional set, and it consumes no asymptotics. Remark 8 gives the denominator conclusion by a different and shorter route (the saddles $z_\pm^* = \pm iW/2$ with all constants derived explicitly, the two saddles supplying the global cancellation that no partial-summation method can), but that route is a sketch only (Remark 10), and no theorem uses it. Both V and U take their pole set from Lemma 11, which consumes only the rectangle bound $|T_2| \rightarrow 0$ of Lemma 9. The block assembly is developed in §6, on the observation that the travel and the bulk block are the same object, the resolvent of $q^{\max(a,b)}$ against a geometric weight: this proves the travel telescoping, the bulk dressing, the reciprocity $t_0 = b_1$ with its cocycle form $\det P = 1$, the identity $t_1 = P_{12}/S_e$, the Möbius factorisation (30), and the shape of the local form at a travel pole, whose leading numerator is the square of the junction pairing $\langle \lambda, R \rangle$. It also refutes the product form of the numerator: the marker junction has rank at least 3 (Proposition 60); at a travel pole the pairing nevertheless collapses to an exact closed form in t_1 , positive at every pole ([7, Theorem “closed form of the junction pairing”, thm:RJ]). **V is transcendental over $\mathbb{Q}(x)$** ([7, Theorem thm:V]). The true series U stands on the same *analytic* footing: its further analytic inputs are proved in Appendix B: the gate block is the symmetric second difference of the q -cosine at its own zero (Proposition 77), the two-interval uniform certificate of Theorem 82 gives $|P_{12}| \leq 0.619 \tau^{3/2} < \tau^{3/2}/\sqrt{2}$ for $\tau \leq 5 \cdot 10^{-3}$ with the six poles above that threshold checked directly, and the exact identity $S_e = \frac{qZ}{2}\mathfrak{s}(Z) - P_{12}$ turns the same certificate into the denominator bound that closes the gate ($|s| \leq 0.983$; Corollary 84), consistent with the phase-matched leading values $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$ and $s \rightarrow \frac{1}{4}$. **U is transcendental over $\mathbb{Q}(x)$** ([7, Theorem thm:U]).

The input (T) that earlier versions of this theorem carried is gone: [7, Proposition “Invariance of the travel block”, prop:travelinv] is now proved (§7.2), via Theorem 56 and Corollary 57, and does not depend on the metric identity. (R-J) is [7, Theorem “closed form of the junction pairing”, thm:RJ], and the hypothesis (L) of earlier versions is Theorem 61. Neither series is D -finite, and $|x| = 1$ is a natural boundary of both (Theorems [7, thm:nonDfinite] and [7, thm:natboundary], by the Pólya–Bertrandias theorem); for the travel resolvent the natural boundary $|q| = 1$ is unconditional. Two questions remain open: whether the poles are dense on the circle, and whether U and V are D -algebraic. Every statement in §6 beyond the operator algebra rests on the transfer model (M), now [7, Theorem “the transfer model (M)”, thm:model], whose assembly step is a hand proof not yet formalised. No input separates U from V . Two finite ranges are computer-assisted: the six gate poles above $\tau = 5 \cdot 10^{-3}$ and the positivity of the pairing at the first twelve poles; neither is needed for transcendence, only for the statements “at every travel pole”. The series in question is the growth series of the universal group W_{univ} ; for a particular rational triangle it agrees with that of W_T only to the effective universality depth of [5]. The turning-point amplitude atom on the U -side is the elementary rational bound of Lemma 65, formally verified in Lean 4. The independent mod- p route of §8 would settle U by a completely different mechanism, if a dense-support non-automaticity criterion can be found. Whether the *number* β_2 is transcendental is untouched by all of this and remains open; Appendix C reduces it to a value statement for a Hahn–Exton q -Bessel function and shows the obstruction to be the arithmetic ratio $\rho = \frac{1}{2}$.

A An explicit lower bound for S_e at the travel poles

Throughout $z_0 = \sqrt{2(1-q)}$, $Z = z_0/\sqrt{q}$, $w = \sqrt{2/\tau}$, $f(z) = \cos(z; q^2) = C(z^2)$, $C(y) = \sum_{j \geq 0} (-1)^j q^{j^2+j} y^j / (q; q)_{2j}$, and $s(z) = \sin(z; q^2)$; by [7, Proposition prop:qtrig], $S_e = f(z_0)$ and the travel poles are the solutions of $f(Z) = 0$.

Proposition 69. *At every travel pole q_m with $\tau_m \leq 1.29 \cdot 10^{-4}$ one has $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$. The finitely many ([7, Lemma “The travel poles are locally finite”, lem:polefinite]) poles with $\tau_m > 1.29 \cdot 10^{-4}$ (the dominant pole $q_1 = 0.449453\dots$ ($|S_e|/\sqrt{\tau_1} = 0.581$; `star_gate_certificate.py`, Corollary 84) and the thirty-nine tabulated below it, $\tau_{41} = 1.2354 \cdot 10^{-4}$ being the first below the threshold) satisfy the same bound by direct high-precision computation.*

The proof occupies the rest of the appendix: an annulus representation turns all derivative bounds into cancellation-free integrals, and the pole condition supplies the value-level input.

Lemma 70. *For $0 < r < 1$, $C(y) = \frac{1}{2\pi} \int_0^{2\pi} \Theta(y/x^2)(x; q)_\infty^{-1} d\theta$ with $x = re^{i\theta}$ and $\Theta(t) = \sum_{j \in \mathbb{Z}} (-1)^j q^{j^2+j} t^j = (q^2; q^2)_\infty (q^2 t; q^2)_\infty (1/t; q^2)_\infty$.*

Proof. Euler’s identity gives $1/(q; q)_{2j} = [x^{2j}](x; q)_\infty^{-1}$; extract the coefficient by Cauchy’s formula on $|x| = r$. For $j \leq -1$ the integrand $x^{-2j}(x; q)_\infty^{-1}$ is analytic in $|x| < 1$ (the zeros of $(x; q)_\infty$ lie at $x = q^{-k}$, $|q^{-k}| \geq 1$), so the negative-index terms vanish and the sum completes to the bilateral one, which is the triple product. Equivalently this is the $p = q^2$ case of the q -Borel- q -Laplace inversion of [45] and [30, Lemma 1], the q^2 -Borel transform of C being the even part of $(x; q)_\infty^{-1}$. $\square$

Lemma 71. *For $0 < r \leq \frac{1}{2}$: $(x; q)_\infty^{-1} \leq \exp(w_r \cos \theta + K_r)$ in modulus, with $w_r = r/(1-q)$ and $K_r = r^2/(2(1-r)(1-q^2))$.*

Proof. $-\log|1-u| = \operatorname{Re} u + \operatorname{Re} \sum_{n \geq 2} u^n/n \leq \operatorname{Re} u + \frac{1}{2}|u|^2/(1-|u|)$ for $|u| < 1$; summing over $u = xq^j$ gives $\sum_j \operatorname{Re}(xq^j) = w_r \cos \theta$ and $\sum_j r^2 q^{2j}/(2(1-rq^j)) \leq r^2/(2(1-r)(1-q^2)) = K_r$. $\square$

Lemma 72. Let $p = q^2$, $t = \rho e^{i\varphi}$, $a = (\log \rho - \tau) + i\tilde{\varphi}$ with $\tilde{\varphi} \in (-\pi, \pi]$ representing $\varphi + \pi$. For $n \in \{0, 1, 2\}$,

$$|\Theta^{(n)}(t)| \leq \sqrt{\pi/\tau} e^{(\log \rho - \tau)^2/(4\tau)} \rho^{-n} (|a|/(2\tau) + \tau^{-1/2})^n G(\tilde{\varphi}), \quad G(\tilde{\varphi}) = \sum_{k \in \mathbb{Z}} e^{-\pi^2(k - \tilde{\varphi}/2\pi)^2/\tau}.$$

Proof. $\Theta(t) = \theta_p(-pt)$ with $\theta_p(y) = \sum_n p^{n(n-1)/2} y^n$. Poisson summation gives the *exact* identity

$$S(a) := \sum_{j \in \mathbb{Z}} e^{-\tau j^2 + aj} = \sqrt{\pi/\tau} \sum_{k \in \mathbb{Z}} e^{a_k^2/(4\tau)}, \quad a_k := a - 2\pi i k. \quad (35)$$

Since $\operatorname{Re}(a_k^2) = (\log \rho - \tau)^2 - (\tilde{\varphi} - 2\pi k)^2$, the imaginary part of a contributes *decay*, and $\sum_k e^{\operatorname{Re}(a_k^2)/(4\tau)} = e^{(\log \rho - \tau)^2/(4\tau)} G(\tilde{\varphi})$; no cancellation is involved, the term $-(\tilde{\varphi} - 2\pi k)^2$ being precisely what constitutes G . Differentiating (35) in a , the j - and $j(j-1)$ -weighted sums are S' and $S'' - S'$, and $\partial_a e^{a_k^2/(4\tau)} = (a_k/2\tau) e^{a_k^2/(4\tau)}$, so the k -th image carries the weight

$$W_1(a_k) = \frac{|a_k|}{2\tau}, \quad W_2(a_k) = \frac{|a_k|^2}{4\tau^2} + \frac{1}{2\tau} + \frac{|a_k|}{2\tau}.$$

At the radius $\rho = e^\tau$ of Lemma 73 one has $a_k = i(\tilde{\varphi} - 2\pi k)$ exactly, so $|a_k| = |\tilde{\varphi} - 2\pi k|$. To pull the weight out of the k -sum, note that for $k \neq 0$ the excess $|a_k| - |a_0|$ is paid for by the Gaussian factor, which is smaller by $e^{-((\tilde{\varphi} - 2\pi k)^2 - \tilde{\varphi}^2)/(4\tau)}$; the aggregate excess is maximised at the nearest image and is at most $\max_{\delta > 0} 2\delta e^{-\pi\delta/\tau} = 2\tau/(\pi e) \leq 0.2342\tau$, which is below $\tau^{-1/2}$ for $\tau \leq 0.0198$. Hence $\sum_k W_n(a_k) e^{\operatorname{Re}(a_k^2)/(4\tau)} \leq (|a|/(2\tau) + \tau^{-1/2})^n \sum_k e^{\operatorname{Re}(a_k^2)/(4\tau)}$, which is the stated bound. $\square$

Lemma 73. For $\tau \leq 0.0198$ and every $\xi \in [qZ, Z]$, with $r^2 = \xi^2 e^{-\tau}$: $M_2 := \max_{[qZ, Z]} |f''| \leq 7w^4$.

Proof. $f''(z) = 2C'(z^2) + 4z^2 C''(z^2)$, and differentiating Lemma 70 under the integral, $|C^{(n)}(y)| \leq (2\pi r^{2n})^{-1} \int_0^{2\pi} |\Theta^{(n)}(t)| (x; q)_\infty^{-1} d\theta$ at $\rho = y/r^2 = e^\tau$, which zeroes $(\log \rho - \tau)^2$. The key step is the exact unfolding: as θ runs over $[0, 2\pi)$ the variable $u = \pi - 2\theta$ runs over two periods, consecutive branches carrying opposite signs of $\sin(u/2)$, so

$$\frac{1}{2\pi} \int_0^{2\pi} e^{w_r \cos \theta} G(\pi - 2\theta) d\theta = \frac{1}{2\pi} \int_{\mathbb{R}} \cosh(w_r \sin \frac{u}{2}) e^{-u^2/(4\tau)} du,$$

an identity; the derivative weights unfold with it, becoming polynomials in u , and $|\sin(u/2)| \leq |u|/2$. The even weight gives an exact Gaussian moment, $\int v^2 e^{-(v-s)^2/(4\tau)} dv = \sqrt{4\pi\tau} (2\tau + s^2)$ with $s = \tau w_r \leq \sqrt{2\tau} e^{\tau/2}$, and $\sqrt{\pi/\tau} \cdot \sqrt{4\pi\tau} = 2\pi$ cancels the prefactors; the odd weight does *not*, giving instead $\int |v| e^{-(v-s)^2/(4\tau)} dv = \sqrt{4\pi\tau} s \operatorname{erf}(s/(2\sqrt{\tau})) + 4\tau e^{-s^2/(4\tau)}$, and the error function and the exponential are kept, not discarded. Writing $s = \tau w_r$, the two assembled factors are

$$Q_1 = \tau^{-1/2} + \frac{s \operatorname{erf}(s/(2\sqrt{\tau})) + 2\sqrt{\tau/\pi} e^{-s^2/(4\tau)}}{2\tau}, \quad Q_2 = \frac{2\tau + s^2}{4\tau^2} + \tau^{-3/2} [\text{same bracket}] + \frac{1}{\tau},$$

the prefactor $\sqrt{\pi/\tau}$ of Lemma 72 cancelling against $\sqrt{4\pi\tau}$ in every term. On the whole range, $\sup \sqrt{\tau} Q_1 = 1.8273$ and $\sup \tau Q_2 = 3.6596$, both attained at $\tau = 0.0198$, $\xi = Z$. With the weights of Lemma 72 this assembles, for $\tau \leq 0.0198$ (so $r \leq 0.1981 < \frac{1}{2}$, $K_r \leq 0.631$, $e^{\tau w_r^2/4} \leq e^{0.506}$), to

$$|C'(y)| \leq e^{K_r - \tau + \tau w_r^2/4} (1.83 \tau^{-1/2}) r^{-2}, \quad |C''(y)| \leq e^{K_r - 2\tau + \tau w_r^2/4} (3.670/\tau) r^{-4}.$$

With $4\xi^2/r^4 \leq (2/\tau) e^{3\tau} (1 + 2\tau)$ the total $M_2 \tau^2$ is at most 25.04 at $\tau = 0.0198$, decreasing to 20.2 at $\tau = 1.29 \cdot 10^{-4}$, in both cases below the target $7w^4 \tau^2 = 28$. Those two numbers come from pairing

the largest K_r and the largest $\tau w_r^2/4$, both attained at $\xi = Z$, with the largest r^{-2} , attained at $\xi = qZ$; taking instead the supremum over $\xi \in [qZ, Z]$ of the assembled bound at a single ξ gives 23.33 at $\tau = 0.0198$, 20.06 at $\tau = 1.29 \cdot 10^{-4}$ and 19.84 as $\tau \rightarrow 0$, so the true margin against 28 is 16.7% rather than the 10.6% the factored form reports. Every constant in this proof, and the two ξ -suprema, are regenerated by `annulus_certificate.py` at two precisions; see Remark 74. $\square$

Remark 74 (what the certificate checks, and what it found). `annulus_certificate.py` recomputes, at 30 and again at 60 digits with zero disagreement: the window geometry ($\sup r = 0.19802$, $\sup K_r = 0.62963$, $\sup \tau w_r^2/4 = 0.50497$ against the quoted $\frac{1}{2}$, 0.631, 0.506); the excess constant $2/(\pi e) = 0.234199$ of Lemma 72; Lemma 71 against the true $|(x; q)_\infty^{-1}|$ on a (τ, ξ, θ) grid; the unfolding identity of this proof against numerical quadrature; the two moment suprema above; the assembled $M_2\tau^2$ on a 400-point τ grid with the ξ -supremum taken by search rather than at an endpoint; and the chain of Proposition 69 below. It also checks that the bound dominates the true $\max |f''|$ where both are computable, the ratio $\max |f''|/(\text{bound})$ coming out between 0.023 and 0.045 at $\tau = 0.0198$, 0.0127, $5 \cdot 10^{-3}$, 10^{-3} , consistent with Remark 75. Two constants of an earlier version of this proof did not survive it and are corrected above: $\sup \sqrt{\tau} Q_1 = 1.82733$ is not bounded by 1.825, and $2/(\pi e) = 0.234199$ is not bounded by 0.234. Both corrections move $M_2\tau^2$ by less than 0.01 and neither affects any conclusion. The arithmetic model is `mpmath mpf` at the two stated precisions, not interval arithmetic.

Remark 75 (how lossy Lemma 73 is, and why one sample cannot say). The ratio M_2/w^4 does not converge: f is the q -cosine, so $f''(z) \approx -(W/z_0)^2 \cos(zW/z_0)$ with $(W/z_0)^2 = e^{-\tau}/(\tau(1-q)) \sim w^4/4$, while the window $[qZ, Z]$ spans a phase of only $(1-q)Z/\tau \sim \sqrt{2\tau}$ radians. As τ varies the window slides through the cosine and M_2/w^4 sweeps $[0, \frac{1}{4}]$; a single value of τ therefore reports an arbitrary point of that sweep and says nothing about the maximum. The maximum is attained where the window covers an extremum of the phase, i.e. along the extreme-phase points $w = m\pi$, and there $M_2/w^4 \uparrow \frac{1}{4}$: measured 0.2499993 at $w = 120\pi$, against 0.22599 at the top of the range $\tau = 0.0198$ and 0.02337 at $\tau = 0.0163$. Over the 165 values of τ scanned, $M_2 < \frac{1}{4}w^4$ (`m2_certificate.py`, `mpmath` at two precisions, agreeing to ten digits at every point; a scan, not a supremum proof). Lemma 73 is therefore lossy by a factor of about 28 at the top of the range, and its target $7w^4$ clears the measured values by about 31.

Proof of Proposition 69. At a pole $f(Z) = 0$, so the Koornwinder–Swarttouw rules give the exact facts $f(qZ) = qZ s(Z)$ and $s(qZ) = s(Z)$; the mean value theorem on $[qZ, Z]$ then pins $f'(\zeta) = -q s(Z)/(1-q)$ at some $\zeta \in (qZ, Z)$, and a second mean value on $[z_0, Z] \subset [qZ, Z]$ gives

$$|S_e| = (Z - z_0) |f'(\tilde{\xi})| \geq \frac{q(1 - \sqrt{q})}{1 - q} Z |s(Z)| - (1 - \sqrt{q})(1 - q)Z^2 M_2.$$

The sine factor is bounded below by an invariant of Casoratian (q -Wronskian) type for the Hahn–Exton pair; the closed-form q -Wronskian is due to Swarttouw [43] (eq. (4.3.8)); see also [33, eq. (4.21)]. Concretely, with $c_n = \cos(Zq^n; q^2)$, $s_n = \sin(Zq^n; q^2)$ and $F_n = c_n^2 - Zq^{n+1}c_n s_n + q s_n^2$, the rules give the exact telescoping $F_n - F_{n+1} = -Zq^{n+1}(1-q)c_{n+1}s_{n+1}$ with $F_n \rightarrow 1$; the form has smallest eigenvalue $\lambda_n \geq q - Z/2$ (by $\sqrt{a^2 + b^2} \leq a + b$), $\sum_{n \geq 0} Zq^{n+1}(1-q) = qZ$, and the discrete Gronwall gives $F_0 \geq e^{-0.61Z} \geq 1 - 0.61Z$ for $Z \leq \min(0.2, q/3)$; at the pole $F_0 = q s(Z)^2$, so $|s(Z)| \geq \sqrt{1 - 0.61Z}$. (The Gronwall sum is $qZ/(2q - Z)$, whose supremum over the range is 0.5568 Z , so 0.61 has room; $Z \leq \min(0.2, q/3)$ holds throughout, $Z = 0.19999$ at $\tau = 0.0198$ and $Z = 0.016063$ at $\tau = 1.29 \cdot 10^{-4}$.) Inserting Lemma 73, the error fraction is $28\sqrt{2}(1.02)\sqrt{\tau} \leq 0.46$ for $\tau \leq 1.29 \cdot 10^{-4}$, a range containing every pole with $m \geq 41$, whence $|S_e(q_m)| \geq (1 - 0.02 - 0.46)\sqrt{\tau_m/2} \geq 0.35\sqrt{\tau_m}$ there. Evaluated rather than rounded, the same

chain gives $0.3856\sqrt{\tau_m}$ at the threshold and more below it, the exact error fraction being 0.4520 against the quoted 0.46 (`annulus_certificate.py`). The tabulated poles are verified numerically at 60+ digits (`qtrig_verify.py`, `mpmath`, no interval arithmetic: $|S_e|/\sqrt{\tau} \in [0.698, 0.708]$ as asserted by the script; the tabulated set is poles $2, \dots, 60$ in the indexing of §2 (the dominant pole is not among them, its value 0.581 coming from `star_gate_certificate.py`) and the table omits poles 53 and 60 (rows 52, 59), whose $\tau < 7.7 \cdot 10^{-5}$ lies inside the proved range), and the two ranges overlap on every tabulated pole with $m \geq 41$; only poles $2, \dots, 40$ (table rows $1, \dots, 39$) are load-bearing. The polynomial identities underlying the telescoping and the pole reduction are machine-checked (`GramCasoratian.lean`: `F_shift`, `F_drift`, `pole_qsine`, `sigma1_term`; each audits inside the standard three, the first two on `propext`, `Quot.sound` alone; no `sorry`). $\square$

On the intermediate window $1.29 \cdot 10^{-4} < \tau_m \leq 5 \cdot 10^{-3}$ the bound also follows, with margin, from Lemma 83 ($0.63 > 0.35$), independently of any enumeration of the poles there.

B Proof of the amplitude estimate: the gate holds at every travel pole

This appendix proves the estimate $(\star)$ of [7, Theorem thm:U] (at every travel pole, $|P_{12}(q_m)| w_m^3 < 2$, $w_m = \sqrt{2/\tau_m}$; quantitatively $|P_{12}| \leq \frac{3.5\sqrt{2}}{8} \tau^{3/2} < 0.619 \tau^{3/2}$ for $\tau \leq 5 \cdot 10^{-3}$, and $|P_{12}| w^3 \leq 0.75$ at the six poles above that threshold) and then closes the full gate (31): the same certificate yields the denominator bound $|S_e| \geq 0.63\sqrt{\tau}$ (Lemma 83), hence $s < 1$ at every travel pole, and, granting the transfer model (M) for the closed form of b_0 (Corollary 35), also $b_0 > 0$ there (Corollary 84). Nothing in this appendix depends on (M) except the statement just named as depending on it. Throughout, $h = \tau/2$, $\sigma^2 = \tau/2$, $\psi(t) = \cos(Ze^t; q^2)$; by Proposition 77, $P_{12} = -\frac{1}{2}[\psi(h) + \psi(-h)]$ and $\psi(0) = 0$, so

$$|P_{12}| \leq \frac{h^2}{2} \max_{|t| \leq h} |\psi''(t)|. \quad (36)$$

The gate block is a second difference at a zero

The following exact identity is not needed for [7, Theorem thm:V], but it is the sharpest available description of the object the U -gate must bound, and it makes the size $\tau^{3/2}$ structural rather than numerical.

Lemma 76 (the gate block in closed form). *Identically in $0 < q < 1$, with $c(u) = \cos(u; q^2)$ and $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$:*

$$P_{12} = \frac{qZ}{2} \mathfrak{s}(Z) - c(z_0), \quad S_o := \sum_{j \geq 0} \frac{(-1)^j [2(1-q)]^j q^{j^2+2j} (1-q)}{(q; q)_{2j+1}} = \frac{z_0}{2} \mathfrak{s}(z_0),$$

S_o being the odd bulk block of the closed form $b_0 = \frac{2q}{1-q} S_o/S_e$ of Corollary 35.

Proof. From (34), $P_{12} = \frac{2q^3}{1-q^3} \sum_k d_k$ with $d_k = (-2)^k (1-q)^k q^{k^2+3k} / [(q^2; q^2)_k (q^5; q^2)_k]$; since $(q; q)_{2k+3} = (q; q^2)_{k+2} (q^2; q^2)_{k+1}$ and $(q; q^2)_{k+2} = (1-q)(1-q^3)(q^5; q^2)_k$, this is

$$P_{12} = \sum_{k \geq 1} (-1)^{k-1} \frac{[2(1-q)]^k q^{k^2+k+1} (1-q^{2k})}{(q; q)_{2k+1}}. \quad (37)$$

Since $q^{1/2}Z = z_0$ and $z_0^2 = 2(1 - q)$, the j -th summand of $\frac{qZ}{2}\mathfrak{s}(Z) = \frac{z_0}{2}\sin(z_0; q^2)$ is $(-1)^j(1 - q)[2(1 - q)]^j q^{j^2+j}/(q; q)_{2j+1}$, and that of $c(z_0)$ is $(-1)^j[2(1 - q)]^j q^{j^2+j}/(q; q)_{2j}$. Their difference carries the factor $(1 - q) - (1 - q^{2j+1}) = q^{2j+1} - q$: the $j = 0$ terms cancel, and what remains is (37) term by term. The S_o identity is the same computation one half-step down: the j -th summand of $\frac{z_0}{2}\mathfrak{s}(z_0)$ is $(-1)^j(1 - q)[2(1 - q)]^j q^{j^2+2j}/(q; q)_{2j+1}$, which is S_o 's summand verbatim. All three matchings are exponent bookkeeping of the kind machine-checked in `QTrigIdentities.lean`; the identities are verified off-pole (hence as identities) and at the six exceptional poles to 10^{-100} (`star_gate_certificate.py`); the collapse identities behind Proposition 77 are further verified at the 58 tabulated poles (`halfstep_verify.py`). $\square$

Proposition 77. *At every travel pole, with $Z = z_0/\sqrt{q}$ the corresponding zero of $\cos(\cdot; q^2)$,*

$$P_{12} = -\frac{1}{2}[\cos(z_0; q^2) + \cos(z_0/q; q^2)].$$

Since $z_0 = Ze^{-\tau/2}$ and $z_0/q = Ze^{+\tau/2}$, and since $\cos(Z; q^2) = 0$, the right-hand side is exactly the symmetric second difference of $\cos(\cdot; q^2)$ at its own zero, taken in the variable $\log u$ with step $\tau/2$.

Proof. Write $c(u) = \cos(u; q^2)$ and $\mathfrak{s}(u) = q^{-1/2}\sin(q^{1/2}u; q^2)$. The shift rules and the pole invariant are [7, Lemma “half-step rules and the pole invariant”, `lem:halfstep`]: $c(qu) = c(u) + q^{3/2}u\mathfrak{s}(\sqrt{q}u)$, $\mathfrak{s}(qu) = \mathfrak{s}(u) - uc(\sqrt{q}u)$, and $q^{3/2}\mathfrak{s}(z_0)\mathfrak{s}(Z) = 1$ at every travel pole. Taking $u = Z/\sqrt{q}$ in the first shift rule gives $c(z_0) - c(z_0/q) = qZ\mathfrak{s}(Z)$. Taking $u = Z$ in the Hahn–Exton q -Wronskian $\mathfrak{s}(u)c(qu) - c(u)\mathfrak{s}(qu) = u$ and in the first shift rule, and using $c(Z) = 0$ twice, gives $\mathfrak{s}(Z)c(qZ) = Z$ and $c(qZ) = q^{3/2}Z\mathfrak{s}(z_0)$, whence again the pole invariant and $c(qZ) = Z/\mathfrak{s}(Z)$. Substituting the closed form $P_{12} = \frac{qZ}{2}\mathfrak{s}(Z) - c(z_0)$ of Lemma 76 into $c(z_0) + c(z_0/q) = 2c(z_0) - qZ\mathfrak{s}(Z)$ yields the boxed identity. $\square$

The two algebraic steps of the proof are machine-checked (`SelowerSkeleton.lean`): `shift_at_pole` is the instantiation of the half-step rule at $u = z_0/q$, written so that the index bookkeeping is explicit, and `P12_second_difference` is the elimination; both are stated in doubled form over an arbitrary commutative ring and depend only on `propext`. The identity is *verified* at all 58 tabulated poles to a worst relative deviation of $2 \cdot 10^{-44}$ (`halfstep_verify.py`). The consequence, (36) above, is that the gate is a bound on a second derivative on a window of width τ , not an asymptotic evaluation.

Lemma 78. *For $0 < \rho < 1$, with $\theta \sim N(\pi/2, \tau/2)$: $\mathbb{E}[(\rho e^{i\theta}; q)_\infty^{-1}] = \sum_{M \geq 0} q^{M^2/4} (i\rho)^M / (q; q)_M =: \Phi(\rho)$, and $\text{Re } \Phi(\rho) = \cos(\rho/\sqrt{q}; q^2)$. The sum is the q -Airy function of Kajiwara–Masuda–Noumi–Ohta–Yamada [20] at base $\sqrt{q}$, and the representation is Zhang’s q -Laplace transform of the second kind at level $\frac{1}{2}$ [45]; only the probabilistic packaging is ours.*

Proof. Euler’s identity $(x; q)_\infty^{-1} = \sum_M x^M / (q; q)_M$ integrated termwise (absolute uniform convergence on $|x| = \rho < 1$), with $\mathbb{E}[e^{iM\theta}] = i^M q^{M^2/4}$, splitting M by parity and matching exponents ($(2j+1)^2 = 4(j^2+j) + 1$; Lean `parity_exponent`) identifies $\text{Re } \Phi$ with the cosine series of [7, Proposition `prop:qtrig`]. $\square$

Lemma 79. *Write $A(\xi) = (\rho e^{i(\pi/2+\xi)}; q)_\infty^{-1}$. Then $\log A(\xi) = \sum_{k \geq 1} (i\rho)^k e^{ik\xi} / (k(1 - q^k))$; the cumulants $\Lambda_n = i^n \sum_k k^{n-1} (i\rho)^k / (1 - q^k)$ satisfy $|\Lambda_n| \leq (n-1)! \rho / ((1 - q)(1 - \rho)^n)$; Gaussian integration by parts gives, at radius $\rho(t) = z_0 e^t$,*

$$\psi'(t) = \frac{2}{\tau} \text{Im } \mathbb{E}[\xi A], \quad \psi''(t) = \frac{4}{\tau^2} \text{Re } \mathbb{E}[(\frac{\tau}{2} - \xi^2) A];$$

and for complex a, b with $\text{Re}(1 - b\sigma^2) > 0$, $\mathbb{E}[\xi^m e^{a\xi + b\xi^2/2}] = \beta^{-1/2} e^{a^2\sigma^2/(2\beta)} M_m$, $\beta = 1 - b\sigma^2$, $M_0 = 1$, $M_1 = \mu = a\sigma^2/\beta$, $M_m = \mu M_{m-1} + (m-1)(\sigma^2/\beta) M_{m-2}$.

Proof. $-\log(1-u) = \sum u^m/m$ summed over $u = \rho e^{i\theta} q^j$; $1-q^k \geq 1-q$ and a binomial estimate; $\rho \partial_\rho A = -i \partial_\theta A$ with $\mathbb{E}[f'] = \mathbb{E}[f\xi/\sigma^2]$ once and twice; completion of the square and analytic continuation in (a, b) . The exponent bookkeeping is machine-checked (`QTrigIdentities.lean`). $\square$

Lemma 80. *At every travel pole with $\tau \leq 5 \cdot 10^{-3}$ and every $t \in [-\tau, h]$: $|\psi'(t)| \leq 1.10 w$.*

Lemma 81. *At every travel pole with $\tau \leq 5 \cdot 10^{-3}$, writing $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$: $\cos(qZ; q^2) = Z/\mathfrak{s}(Z)$, and $|\mathfrak{s}(Z)| \geq Z/(\tau \max_{-\tau \leq t \leq 0} |\psi'(t)|) \geq 0.90$. Moreover, for $t \in [-\tau, 0]$, $|\Phi(\rho(t))| \geq q^{3/4}(|\mathfrak{s}(Z)| - 0.49\tau)$.*

Proof of Lemmas 80 and 81. The identity is the pole reduction underlying Proposition 77 (the Hahn–Exton q -Wronskian at the zero, cited there). The bound in Lemma 81 is the mean value theorem on $[-\tau, 0]$, where $qZ = Ze^{-\tau}$ exactly and all radii satisfy $\rho(t) \leq z_0$. For the last bound: $\text{Im } \Phi(\rho) = q^{3/4} \mathfrak{s}(\rho/\sqrt{q})$ (the odd half of the parity split in Lemma 78), and the drift of $\mathfrak{s}(\rho(t)/\sqrt{q})$ over $t \in [-\tau, 0]$ is the even/odd G_4 term of the certificate, $\leq 0.49\tau$ (`budget.py`, `G4_coef`). For Lemma 80: the principal term is the $m = 1$ tilted moment of Lemma 79 at $a = \Lambda_1$, $b = \Lambda_2$, at most $e^{c_1\sqrt{\tau}}|\beta|^{-1/2}L_1(\text{Re } \beta)^{-1} \cdot w \leq 1.088w$, where L_1 is the interval constant bounding $|\Lambda_1|/w$ (Lemma 79, $n = 1$) and $c_1 \leq 0.177$ bounds the exponent residual per $\sqrt{\tau}$ (the two leading halves of the exponent cancel exactly; every residual carries a geometric tail); the tail (linear orders 3..10 with ξ -weighted moments, the product convolution, and the degree- ≥ 11 remainder) is at most 0.131 *absolute*, i.e. $\leq 0.0066w$ at the top endpoint $w = 20$. All constants are produced by the committed certificate `code/zeta_probe/budget.py` (function `uniform_stack`), which works in 30-digit floating point with the interval constants realised as endpoint evaluations. That evaluation is *re-certified in directed-rounding interval arithmetic* by `budget_iv.py`: the same stack is evaluated with τ entering as an *interval*, over 160 cells covering $[1.2 \cdot 10^{-4}, 5 \cdot 10^{-3}]$ and 160 covering $[10^{-8}, 1.2 \cdot 10^{-4}]$ (contiguous, with both outer endpoints clamped exactly), giving enclosures valid for *every* τ in each cell; so on those ranges neither the floating-point arithmetic nor the monotonicity of any atom is assumed. The certified maxima are 3.349 and 2.650, against the budget 3.5. On the remaining range $(0, 10^{-8}]$, which no finite set of cells can cover, the power-counting argument below (every term a positive power of τ) is what proves the bound. The companion `star_gate_certificate.py` works in the floating-point model at 45–260 digits with explicit truncation control. $\square$

Theorem 82 (the amplitude estimate $(\star)$). *At every travel pole, $|P_{12}(q_m)| w_m^3 < 2$. Quantitatively: for $\tau \leq 5 \cdot 10^{-3}$, $\max_{|t| \leq h} |\psi''(t)| \leq 3.5w$, so (36) gives $|P_{12}| \leq \frac{3.5\sqrt{2}}{8}\tau^{3/2} < 0.619\tau^{3/2} < \tau^{3/2}/\sqrt{2}$; and the six poles with $\tau > 5 \cdot 10^{-3}$,*

$$\tau_1 = 0.79972, \quad \tau_2 = 9.049 \cdot 10^{-2}, \quad \tau_3 = 3.248 \cdot 10^{-2}, \quad \tau_4 = 1.656 \cdot 10^{-2}, \quad \tau_5 = 1.001 \cdot 10^{-2}, \quad \tau_6 = 6.701 \cdot 10^{-3},$$

satisfy $|P_{12}|w^3 = 0.745, 0.530, 0.510, 0.505, 0.503, 0.502$ by direct evaluation at 120-digit precision with explicit truncation control (`star_gate_certificate.py`; the underlying identities are independently verified by `halfstep_verify.py` at the 58 tabulated poles, 47–54 digits on the five it shares with this list: the dominant pole is covered by `star_gate_certificate.py` alone). The first is the dominant pole $q_1 = 1/\beta_2^2$ of [7, Corollary cor:beta]. Enumeration of the exceptional set: every travel pole has $q_m > \frac{1}{10}$ and the set is finite ([7, Lemma “The travel poles are locally finite”, lem:polefinite]); the count of six is established by a derivative-majorant subdivision for $\tau \geq 0.03$ (three poles; in the arithmetic model disclosed above) and needs no enumeration for $\tau \leq 5 \cdot 10^{-3}$, where the uniform certificate covers every pole; on the window $5 \cdot 10^{-3} < \tau < 0.03$ the three listed poles are located by a $\Delta w \approx 5 \cdot 10^{-4}$ scan (they sit on the $(k + \frac{1}{2})\pi$ ladder to three decimals, and $|\cos(Z; q^2)|$ swings to 1.00 between them with minimum 0.019 outside the root cells). The scan is the weakest step of the enumeration and is stated as such; transcendence in [7, Theorem thm:U] does not consume it.

Proof. By Lemma 79, $\psi'' = \frac{4}{\tau^2} \operatorname{Re}[-e^{\Lambda_0} G_Q(\Lambda_2 \sigma^4 / \beta + \mu^2) + e^{\Lambda_0} \mathbb{E}[(\frac{\tau}{2} - \xi^2) e^Q (T - 1)]]$, three terms, with $Q = \Lambda_1 \xi + \Lambda_2 \xi^2 / 2$, $G_Q = \mathbb{E}[e^Q] = \beta^{-1/2} e^{\Lambda_1^2 \sigma^2 / (2\beta)}$, $\beta = 1 - \Lambda_2 \sigma^2$, $\mu = \Lambda_1 \sigma^2 / \beta$ (Lemma 79 at $m = 0, 1$), and $T = e^\Omega$, $\Omega = \sum_{n \geq 3} \Lambda_n \xi^n / n!$, the exponential of the higher cumulants. The μ^2 term's phase is pinned by the pole: $|\cos \arg \Phi(\rho(t))| = |\psi(t)| / |\Phi(\rho(t))| \leq \max |\psi'| |t| / |\Phi|$ with $\psi(0) = 0$ (Lemma 80) and $|\Phi| \geq q^{3/4} (|\mathfrak{s}(Z)| - 0.49\tau)$ (Lemma 81); its argument's sine obeys $|\sin 2 \arg \mu| \leq 13.3 \tau^{3/2}$, the two $O(1/w)$ tilts of Λ_1 and β cancelling to $O(\tau)$ because they are the same Lambert data shifted by one index. Every remaining quantity is bounded by an interval constant in scaled variables, each resting on a one-line monotonicity ($(1 - e^{-\tau})/\tau$ decreasing; $e^{c\tau}$, $\rho^2 = 2(1 - q)e^\tau$ increasing; w, q decreasing), with tails via $1 - q^k \geq (1 - q)kq^{k-1}$ and the signed cancellations via the alternating-series bound $\tau^2/6 - \tau^3/12 \leq 1 + q - 2(1 - q)/\tau \leq \tau^2/6$. The degree- ≥ 11 remainder is handled on two intervals: on $[1.2 \cdot 10^{-4}, 5 \cdot 10^{-3}]$ by an exponential majorant whose profile $e^{0.097/\sqrt{\tau}} \tau^{4.5}$ is increasing exactly there (threshold $1.16 \cdot 10^{-4}$), total $\leq 3.03w$; on $(0, 1.2 \cdot 10^{-4}]$ with no exponential device (linear remainder geometric with ratio $\frac{1}{2}$ on the cutoff region; all products by $|e^\Omega - 1 - \Omega| \leq \frac{1}{2} |\Omega|^2 e^{|\Omega|}$ with $|\Omega| \leq 0.45$ on $|\xi| \leq t_0 \kappa^{-1/3}$; the intermediate zone suppressed by the Gaussian; here $t_0 = 1 - \rho$ and $\kappa = \rho/(1 - q)$), total $\leq 2.10w$. Both are endpoint evaluations of quantities proved monotone; on $[10^{-8}, 5 \cdot 10^{-3}]$ the same conclusion is obtained without any monotonicity assumption by the interval-arithmetic certification `budget_iv.py` described above, and the monotonicity argument is what carries the remaining range $(0, 10^{-8}]$. The assembly is the committed script `budget.py`. $\square$

Lemma 83 (the denominator at the pole). *At every travel pole, exactly: $S_e = \frac{qZ}{2} \mathfrak{s}(Z) - P_{12}$. Consequently, for $\tau \leq 5 \cdot 10^{-3}$,*

$$|S_e| \geq 0.90 \cdot \frac{qZ}{2} - |P_{12}| \geq \left[\frac{0.90}{\sqrt{2}} \sqrt{q} v - 0.619 \tau \right] \sqrt{\tau} \geq 0.63 \sqrt{\tau}, \quad v = \frac{1-q}{\tau},$$

and S_e has the sign of $\mathfrak{s}(Z)$.

Proof. The identity is Lemma 76 rearranged, since $S_e = \cos(z_0; q^2)$ ([7, Proposition prop:qtrig]). For the bound: $|\mathfrak{s}(Z)| \geq 0.90$ (Lemma 81), $\frac{qZ}{2} = \sqrt{q(1-q)/2} = \sqrt{\tau/2} \sqrt{q} v$, $|P_{12}| \leq 0.619 \tau^{3/2}$ (Theorem 82), and $\sqrt{q} v \geq \sqrt{(1-\tau)(1-\tau/2)} \geq 1 - \frac{4}{5}\tau$ for $\tau \leq \frac{1}{2}$ (since $(1 - \frac{4}{5}\tau)^2 \leq (1-\tau)(1 - \frac{\tau}{2})$ there); at $\tau = 5 \cdot 10^{-3}$ the bracket is ≥ 0.630 (exact endpoint value 0.6309) and it increases as τ decreases. The sign statement holds because $0.90 \cdot \frac{qZ}{2} > |P_{12}|$ with the same margin. $\square$

Corollary 84 (the gate closes). *At every travel pole the gate (31) holds; with [7, Theorem “closed form of the junction pairing”, thm:RJ] this gives $\Pi_1(q_m), \Pi_q(q_m) > 0$ for $\tau_m \leq 5 \cdot 10^{-3}$ (the enumeration standing of the six-pole list is that of Theorem 82). The half $|s| < 1$ is proved here outright; the half $b_0 > 0$ consumes the closed form $b_0 = \frac{2q}{1-q} S_o / S_e$, which is Corollary 35 and therefore rests on the transfer model (M), [7, Theorem “the transfer model (M)”, thm:model]. Quantitatively, for $\tau \leq 5 \cdot 10^{-3}$:*

$$|s| = \frac{q}{1-q} \frac{|P_{12}|}{|S_e|} \leq \frac{\tau}{e^\tau - 1} \cdot \frac{0.619}{0.63} \leq 0.983 < 1,$$

and $b_0 > 0$: with the closed form $b_0 = \frac{2q}{1-q} S_o / S_e$ of Corollary 35 and $S_o = \frac{z_0}{2} \mathfrak{s}(z_0)$ (Lemma 76), the pole invariant $q^{3/2} \mathfrak{s}(z_0) \mathfrak{s}(Z) = 1$ (Proposition 77's proof) gives $\operatorname{sign} S_o = \operatorname{sign} \mathfrak{s}(Z) = \operatorname{sign} S_e$ (Lemma 83), so $S_o / S_e > 0$. At the six poles with $\tau > 5 \cdot 10^{-3}$ the gate quantities are evaluated directly (`star_gate_certificate.py`): $|s| = 0.296, 0.256, 0.252, 0.251, 0.251, 0.250$ and $b_0 \tau = 2.174, 2.001, 2.000, 2.000, 2.000, 2.000$; moreover $|S_e|/\sqrt{\tau} \geq 0.58$ at all six, so the bound $0.35\sqrt{\tau}$ of Proposition 69 holds at the dominant pole, as its statement records.

Machine verification of the q -trigonometric layer

Every series identity of the q -trigonometric sections and of this appendix is proved by matching the j -th summand; the matching is exponent bookkeeping, which is where such arguments fail in practice. That bookkeeping, and the Casoratian constancy behind the half-step invariant, are machine-checked in Lean 4 (`QTrigIdentities.lean`; the last four rows of the table below live in `SelowerSkeleton.lean`); convergence and the interchange of summation are analytic and are not formalised.

Paper step	Lean declaration	axioms
half-step rule for c , exponent match	<code>cshift_exponent</code>	<code>propext</code>
half-step rule for s , exponent match	<code>sshift_exponent</code>	<code>propext</code>
parity split of Φ	<code>parity_exponent</code>	<code>propext</code>
functional equation of F	<code>qairy_exponent</code>	<code>propext</code>
P_{12} k -sum split, low branch	<code>P12_split_lo</code>	<code>propext</code>
P_{12} k -sum split, high branch	<code>P12_split_hi</code>	<code>propext</code>
Casoratian constancy	<code>casoratian_re_const</code>	standard
constancy $\Rightarrow$ every rung	<code>casoratian_re_eq_zero</code>	standard
shift rule at the pole	<code>shift_at_pole</code>	<code>propext</code>
P_{12} as a second difference	<code>P12_second_difference</code>	<code>propext</code>
scaling $\tau^2 w = \sqrt{2} \tau^{3/2}$	<code>tau_sq_mul_w_sq</code>	standard
gate threshold $2/w^3 = \tau^{3/2}/\sqrt{2}$	<code>gate_threshold</code>	standard

The algebraic skeleton of Proposition 69 is likewise machine-checked (`SelowerSkeleton.lean`): `pole_facts` derives the two exact facts $s(qZ) = s(Z)$ and $c(qZ) = qZ s(Z)$ from the two shift rules together with $c(Z) = 0$; `F_at_pole` collapses the invariant to $q s(Z)^2$ at a pole; `form_lower_bound` is the quadratic-form estimate behind $\lambda_n \geq q - Z/2$; and `prod_one_sub_ge` is the Weierstrass bound $\prod(1 - \delta_i) \geq 1 - \sum \delta_i$ used in the discrete Gronwall step. The analytic inputs (the annulus representation, the Poisson envelope and the mean value theorem) are hypotheses there, not formalised.

“standard” means `propext`, `Classical.choice`, `Quot.sound`. Every file named in this paper lives in the Mathlib-based package of the accompanying development; none of its 62 files contains `sorry`, each ends with a `#print axioms` audit, and every declaration cited anywhere in this paper appears in one of those audits. (The audits are selective in the other direction: a file may prove auxiliary lemmas that its audit block does not list.) Some declarations report “does not depend on any axioms”, which is the legitimate output for a goal closed by `decide` or `rfl` (for instance `OverDetermination.linear_order_19_ok`) and is also, character for character, what an audit prints for a declaration that failed to elaborate; the audits quoted here were therefore read from a cold elaboration of each file one at a time (`lake env lean`, exit status checked), not from a warm `lake build`, which does not replay the output for an up-to-date library. Three declarations cited in this paper are proved by `native_decide` and carry, besides the standard axioms, the compiler-trust axiom named at their point of use (Remark 5; §8). The realness of the ladder coefficient is what makes the added term in `casoratian_re_const` purely imaginary, and is carried as an explicit hypothesis.

The boundary of the formalisation: the analytic layer

The combinatorial layer of this paper is formalised (§6.6), the analytic layer is not, and the obstruction is one and the same for every statement in it: Mathlib carries no q -Pochhammer symbol $(a; q)_n$ or $(a; q)_\infty$, no product form of a theta function, no Bessel or q -Bessel function, no q -difference equations, and no steepest-descent, stationary-phase or van der Corput machinery. It does carry the

two-variable Jacobi theta series with its summability, analyticity and modular functional equation, and the Mellin transform; it does not carry the Jacobi triple product, which is the identity that connects the two. The list below records, for each analytic statement of this paper, the object whose absence blocks a formalisation. The statements are at PROVED, not at VERIFIED, and this table is the reason.

Statement	missing prerequisite
Lemma 71	$(x; q)_\infty$ and its logarithm
Lemma 76	$(q; q)_n$; the q -cosine and q -sine
Proposition 77	$(q; q)_n$; the q -cosine
Lemma 83	$(q; q)_\infty$; the q -sine
Lemma 70	the Jacobi triple product; $(q^2; q^2)_\infty$
[7, Lemma “The travel poles are locally finite”, lem:polefinite]	analyticity of a q -series and its zero set
Lemma 11	the same, plus a limit at $q \rightarrow 1$
Lemma 12	the same
Proposition 69	$(q; q)_\infty$; quantitative q -trigonometric bounds
Lemma 72	a Gaussian majorant for a theta series
Lemma 73	second derivatives of a q -series
Lemma 80	derivative bounds for the q -sine
Lemma 81	the q -sine and q -cosine at a zero
Theorem 82	a quantitative Taylor bound on a q -phase
Lemma 78	the q -Airy function; a Gaussian expectation
Lemma 79	cumulant bounds; Gaussian integration by parts
Lemma 6	$\log \Gamma$ with an explicit error term; $\log(q; q)_\infty$
Lemma 9	Mellin–Barnes contour shifting; a verified contour-integral
Lemma 65, confluence	Bessel J_ν at half-integer order

The first five rows would be reached by a q -Pochhammer layer alone, together with the Jacobi triple product; the next four would be reached by that layer together with Mathlib’s existing complex analysis; the remainder need asymptotic machinery that is a larger absence than the q -series one. The half of Lemma 65 that is an explicit rational inequality *is* verified (`AtomN.lean`); the confluence half is the row above (Remark 66).

C The number β_2 : value transcendence

This appendix concerns the *number* $\beta_2 = 1/\sqrt{q^*}$ of [7, Corollary cor:beta], not the function V . Nothing in §§3–9 or in Appendices A–B depends on any statement made here; the material is recorded because it locates the obstruction to value transcendence precisely, at the arithmetic ratio $\rho = \frac{1}{2}$, one notch beyond the proven $\rho \geq 1$ theory.

Remark 85 (the number β_2 : a value-transcendence question). Whether the *number* β_2 is transcendental is open, but the question has a clean reduction. Since $\beta_2 = 1/\sqrt{q^*}$ with $\Sigma_1^T(q^*) = 1$, the number β_2 is algebraic iff q^* is; if q^* were algebraic, the series Σ_1^T would take the *rational* value 1 at an algebraic interior point of its disc. For the Jacobi-theta class this is impossible: by Nesterenko’s theorem on Eisenstein values [31], $\sum_n \alpha^{n^2}$ is transcendental at *every* algebraic α with $0 < |\alpha| < 1$ [14]. The blocks here are *theta-like*; their coefficients ride the squares $q^{(j+1)^2}$, which is the source of the natural boundary of Theorem 1; but their Euler-product denominators $(q; q)_{2k+1}$ spread the support densely, and no analogue of [31] is known off the modular class.

Concretely the reduction localizes on a single confluent basic hypergeometric function. Writing

$$\Sigma_1^T(q) = 1 - S(q),$$

$$S(q) = \sum_{k \geq 0} \frac{(-2(1-q))^k q^{k^2}}{(q; q)_{2k}} = {}_1\phi_1(0; q; q^2, 2q(1-q)),$$

a q -Bessel-class function (using $(q; q)_{2k} = (q; q^2)_k (q^2; q^2)_k$); β_2 is algebraic iff the least positive zero $q^* = 0.449453630\dots$ of S is. What is proved about low-degree algebraic relations is the certified finite exclusion of Proposition 110: neither q^* nor β_2 is a root of a nonzero integer polynomial of degree ≤ 8 with height below 10^{228} , or of degree ≤ 16 with height below 10^{118} (for q^* alone: 10^{230} , 10^{120} , and also degree ≤ 24 below 10^{80} , degree ≤ 64 below 10^{28}). That proposition is a computer-assisted lattice certificate on a 2090-digit certified bracket. By contrast, a PSLQ run that returns “no relation” is heuristic, not a proof; so the PSLQ observation that no bilinear relation of height $\leq 10^{12}$ ties q^* to the fold point q_c of the zero curve (430 digits) is evidence only. The continued fraction of q^* (383 certified partial quotients, two-precision guard) is Gauss–Kuzmin generic: digit frequencies match, Khinchin mean 2.63 (2.685), Lévy exponent 3.21 (3.276), largest quotient $a_{170} = 8250$ (probability $\approx 6\%$ over 383 draws, consistent with chance). The number offers no arithmetic foothold of any tested kind. We record what the natural tools give here, since it locates the difficulty precisely. First, S is *neither* q -holonomic *nor* a Mahler function in q (verified to $O(q^{48})$): the diagonal specialization $z = 2q(1-q)$ ties the argument to the base and destroys the single-variable q -difference structure, so the methods of Mahler–Nishioka [32] and of holonomy do not apply. Second, two elementary refinements fail *decisively*, not for want of effort: an ultrametric argument is void; the v -adic and archimedean sums of $S(\alpha)$ are unrelated numbers, so $|S(\alpha)|_v = 1$ does not bound the real value; and a Liouville/height bound is beaten by the q -Pochhammer denominators, whose growth over the conjugates of modulus > 1 is cubic ($\sim K^3$) against the merely quadratic ($\sim K^2$) tail, Kronecker-sharply (equality would need Mahler measure 1, i.e. a root of unity). A more promising reformulation uses two variables. Set $G(q, z) = {}_1\phi_1(0; q; q^2, z) = \sum_k (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$, so $S(q) = G(q, 2q(1-q))$; unlike S , the function G is q -holonomic in z , satisfying the second-order q -difference equation

$$q G(q, z) - (q + 1 - qz) G(q, q^2 z) + G(q, q^4 z) = 0.$$

One checks that $z^* := 2q^*(1-q^*) = 0.4948901\dots$ is the *smallest positive zero* of $G(q^*, \cdot)$. Hence β_2 algebraic would force an *algebraic zero of a q -Bessel-class function at an algebraic base*; and the classical analogue is false: the zeros of the Bessel function J_ν are transcendental (Siegel [40]). A q -analogue of Siegel’s theorem for the zeros of ${}_1\phi_1$ would therefore settle β_2 . The right framework is arithmetic. Writing the equation as a system $Y(q^2 z) = A(z)Y(z)$ with $Y = (G(z), G(q^2 z))^\top$, the companion matrix

$$A(z) = \begin{pmatrix} 0 & 1 \\ -q & q + 1 - qz \end{pmatrix}$$

is *polynomial* in z with $\mathbb{Z}[q]$ entries and $\det A = q$, so the iterated matrices carry no denominators and the Casoratian obeys $W(q^2 z) = qW(z)$, i.e. $W(z) = cz^{1/2}$: the two solutions form a q -Bessel pair with Frobenius exponents $0, \frac{1}{2}$. The second solution is explicit, $\tilde{G}(z) = z^{1/2}H(z)$ with $H(z) = \sum_k (-1)^k q^{k^2} (1-q)z^k / (q; q)_{2k+1}$, and the pair satisfies the exact q -Wronskian identity

$$q G(z) H(q^2 z) - G(q^2 z) H(z) = q - 1,$$

so at the zero z^* one has the clean relation $G(q^2 z^*)H(z^*) = 1 - q$. Moreover $G(q, \cdot)$ is *entire of order zero*; a q - E -function, not a q - G -function. Hence β_2 transcendental would follow from a

q -analogue of the Siegel–Shidlovskii theorem for this q - E -function: such a theorem gives $G(q, \alpha)$ transcendental at every algebraic $\alpha \neq 0$ off the singular locus, whereas $G(q^*, z^*) = 0$ is algebraic; forcing q^* , and with it β_2 , transcendental (and by the shared q -connection core, likewise U). All the structural hypotheses hold here (polynomial companion matrix, constant determinant, entire order-zero solution); the sole open point is whether the available q -Siegel–Shidlovskii results (André [3]; Di Vizio–Hardouin [12]; and the Siegel-method theory of [2]) extend to a ${}_1\phi_1$ that is *irregular* at $z = \infty$; a precise, expert-checkable question, not an absence of theory. (Note the contrast with the *function-level* results of this paper, which say nothing about individual interior values.)

For *irrationality* (weaker than transcendence) the classical architecture transports verbatim. G and H are Hahn–Exton (third Jackson) q -Bessel functions: $G(q, z) \propto J_{-1/2}(\sqrt{z}/q; q^2)$ and $H(q, z) \propto J_{+1/2}(\sqrt{z}/q; q^2)$ (verified; the zeros correspond exactly), i.e. the q -analogues of $\cos$ and $\sin$; and z^* is the q -analogue of $(\pi/2)^2$. Niven’s irrationality proof for π runs on the Lommel-polynomial ladder $J_{m+1/2}(r) = R_{m,1/2}(r)J_{1/2}(r) - R_{m-1,3/2}(r)J_{-1/2}(r)$ evaluated at the zero of $J_{-1/2} = \sqrt{2/\pi r} \cos$; the exact q -analogue of every ingredient exists (Koelink–Swarttouw [22]): the ladder $J_{\nu+1}(x; q^2) = ((1 - q^{2\nu})/x + x)J_\nu - J_{\nu-1}$, the q -Lommel connection $J_{\nu+m} = R_{m,\nu}J_\nu - R_{m-1,\nu+1}J_{\nu-1}$ with $x^m R_{m,\nu} \in \mathbb{Z}[q, x^2]$, and the second-kind functions carrying $q^{m^2/2}$ prefactors; all verified numerically for our pair. Moreover, since a rational zero would make $G(q^*, z^*) = 0$ a *rational value* at rational base and argument, β_2 is irrational as soon as this Hahn–Exton function omits the value 0 at rational points: a plain *value-irrationality* statement, strictly weaker than every transcendence target above. What blocks even this is now a single quantified invariant: the *arithmetic ratio* ρ of the series (b -adic decay exponent of the terms over b -adic growth exponent of the cleared denominators, $q = s/t$ in lowest terms). Every proven q -irrationality theorem (Tschakaloff, q -exponential, the Bézivin–Bundschuh–Väänänen classes) lives at $\rho \geq 1$, where the Pochhammer denominator itself supplies the decay; our G has an independent q^{k^2} numerator against the double Pochhammer $(q; q)_{2k}$, giving $\rho = \frac{1}{2}$. All three classical constructions fail by exactly this margin, measured: the z -Padé remainder $q^{3.5n^2}$ against height q -degree $\approx 9n^2$; the z -direction continued fraction (convergents in $\mathbb{Z}[q, z]$, Pincherle limit the H -ratio) converges only geometrically against t^{n^2} heights; and the ν -ladder at the zero has $J_{m+1/2}(x_0) = R_{m,1/2}(x_0)J_{1/2}(x_0) \sim Cx_0^m$; geometric, because the q -factorial is geometric where Niven’s $m!$ is super-exponential; against t^{m^2} . The ν -ladder is in fact fully costed: the q -Lommel coefficients satisfy $\deg_q p_m = m^2$ *exactly* (immediate from the recurrence, verified symbolically), so their denominators at $q = s/t$ are the pure power t^{m^2} : carrying *no* cyclotomic structure, so the lcm-compression that powers the little- q -Legendre proofs has nothing to act on; while by Poincaré–Perron [36, 35] the m -recurrence, whose coefficient tends to $x_0 + 1/x_0$ with *distinct* characteristic roots $x_0, 1/x_0$, admits *only geometric* solutions: no theta-type q^{m^2} -decaying solution exists in the ν -direction for either kind (the second kind’s $q^{\nu^2/2}$ prefactor is exactly cancelled by the $(q^{-\nu+1}; q)_\infty$ blow-up of its shifted argument). Hence the cleared ν -ladder forms grow like $t^{m^2}x_0^{-m} \rightarrow \infty$ for *every* rational $q = s/t$, including $s = 1$. The $b \leftrightarrow z$ transformation ${}_1\phi_1(0; q; q^2, z) = \frac{(z; q^2)_\infty}{(q; q^2)_\infty} {}_1\phi_1(0; z; q^2, q)$ redistributes but conserves ρ . The structural summary: the theta-type decay lives only in the z -direction, where the equation is *irregular* at $z = \infty$ (Newton polygon slopes $\{0, 1\}$) and every integral structure carries the double-Pochhammer weight $\rho = \frac{1}{2}$; the ν -direction is *regular* (Poincaré class) with clean $\mathbb{Z}[q, X]$ arithmetic but only geometric decay. Decay and arithmetic live in different directions and never meet; the same irregularity that obstructs q -Siegel–Shidlovskii creates the arithmetic deficit. The precise open door remains a q -irrationality criterion at $\rho = \frac{1}{2}$; one notch beyond the proven $\rho \geq 1$ theory.

Remark 86 (the β_2 family). q^* is the first member of an infinite family. On $(0, 1)$ the parabola $w(q) = 2q(1 - q)$ falls like $1 - q$ while every zero branch falls like $(1 - q)^2$ (the confluent limit): the

branches are $z_j(q) = 2q(1 - q)\lambda_j(q)$ with λ_j the eigenvalue branches of [7, Theorem “the travel determinant”, thm:fredholm], and $w = z_j$ means $\lambda_j = 1$. So w overtakes each branch at least once, distinct branches at distinct points, and S has infinitely many zeros $q_1^* < q_2^* < \dots \rightarrow 1$, which are the travel poles q_j ([7, Proposition “infinitely many travel poles, without T_2 ”, prop:infpolesalt]). That w overtakes each branch *exactly* once, so that the j -th zero is the crossing of the j -th branch, $w(q_j^*) = z_j(q_j^*)$, is Proposition 20(i), at the standing of Theorem 19 (also verified at $j = 2$ to 43 digits: exactly one zero of $G(q_2^*, \cdot)$ lies below w). The accumulation obeys the classical cosine-zero law

$$1 - q_j^* = \frac{8}{\pi^2(2j - 1)^2}(1 + \theta_j), \quad |\theta_j| \leq \frac{3}{j}, \quad \text{measured } (1 - q_j^*) \frac{(2j - 1)^2 \pi^2}{8} = 0.679, 0.961, 0.986, 0.993, 0.9956, 0.9991, \dots$$

This law is now *proved*: it is Theorem 25. That theorem gives explicit two-sided bounds for every $j \geq 1$, from the eigenvalue sandwich of Lemma 24 and the index identification of Proposition 20(i), at the standing of Theorem 19. Every measured value above lies inside those bounds, and so do those of the first 41 poles. Only the first order is proved. The second-order corrections below are derived by a formal calculus and checked numerically; they are not proved. The second member is $q_2^* = 0.913486638731047141696\dots$, i.e. $\beta_2^{(2)} := 1/\sqrt{q_2^*} = 1.046282352687417855\dots$ PSLQ at 100 digits finds no integer polynomial of degree ≤ 8 and height $\leq 10^{10}$ vanishing at either number, and no bilinear relation between q_1^* and q_2^* . This is heuristic evidence, not a proof: a PSLQ “no relation” output certifies nothing, and unlike q_1^* (Proposition 110) no certified exclusion has been computed for q_2^* . The family interpolates between the open head ($j = 1$, the wild β_2) and the classical tail (the π^2 -law): the roots become “provably transcendental in the limit” while the first, the one of combinatorial interest, remains untouched; the confluent boundary of the step-ratio mechanism made visible inside a single object. (That the higher roots are genuine singularities of U itself is the non-vanishing $\Pi_q(q_m) \neq 0$, which is [7, Theorem “closed form of the junction pairing”, thm:RJ].)

Three refinements (all computations at 115 digits, 30 cosine and 10 sine members). First, the sine side has its own family, $H(q_j^s, 2q(1 - q)) = 0$ with $1 - q_j^s \sim 2/(j\pi)^2$ (law ratios $0.913, \dots, 0.9991$ for $j \leq 10$), and the two families *interlace*: $q_j^* < q_j^s < q_{j+1}^*$ (verified $j \leq 9$); its head is $q_1^s = 0.815032640290\dots$ Since $H(q, 2q(1 - q)) = (1 - q)\Sigma_0(q)/(2q)$, the sine family is exactly the zero set of Σ_0 in $(0, 1)$, that is, of $q \mapsto J_{1/2}^{(3)}(z_0(q); q^2)$ with $z_0 = \sqrt{2(1 - q)}$ ([7, Lemma “the numerators in closed form”, lem:qsinenum] and the normalisation after [7, Proposition prop:qtrig]). Half of the interlacing, at least one sine member in each (q_j^*, q_{j+1}^*) , is Proposition 22(b), at the standing of Theorem 19; that there is exactly one is verified only. Second, the accumulation law has an exact-looking rational second order, identical for both families:

$$1 - q_j^* = \frac{2}{\mu_j} \left(1 - \frac{8}{9\mu_j} + \frac{B}{\mu_j^2} + O(\mu_j^{-3}) \right), \quad \mu_j = \left(\frac{(2j-1)\pi}{2} \right)^2 \text{ resp. } (j\pi)^2,$$

with $A = -8/9$ matched to 12 digits (fit residual $3 \cdot 10^{-18}$ out of sample) and B derived below. The constant A is derived, and the derivation separates two regimes that must not be conflated. Writing $\varepsilon = 1 - q$, the logarithm of the ratio of the k -th term of $G(1 - \varepsilon, \varepsilon^2 y)$ to $(-1)^k y^k / (2k)!$ equals $\frac{k}{2}\varepsilon + c_2(k)\varepsilon^2 + O(\varepsilon^3)$ with

$$c_2(k) = -\frac{k^3}{9} - \frac{k^2}{12} + \frac{23k}{72};$$

to first order G is $\cos\sqrt{y e^{\varepsilon/2}}$, so at *fixed* j the zero branch obeys $z_j = \varepsilon^2 \mu_j (1 - \frac{1}{2}\varepsilon + O(\varepsilon^2))$: the fixed- j confluent correction is $-\frac{1}{2}$ (measured -0.5004 at $\varepsilon = 0.002$, converging linearly). The

crossing, however, sits in the *uniform* regime $\mu\varepsilon \approx 2$, where the k^3 -term acts as the operator $-\frac{\varepsilon^2}{9}D^3$, $D = (u/2)d/du$ on $\cos u$, $u = \sqrt{X}$; at the zeros $D^3 \cos u = \frac{u^3}{8} \sin u + O(u^2)$, while even powers of D are cos-weighted and drop out (parity protects the order). With $m = \varepsilon\mu$ the crossing equation $2(1 - \varepsilon)e^{\varepsilon/2} = \varepsilon\mu(1 - \varepsilon^2\mu/36)$ becomes $m = 2 - \varepsilon + \varepsilon m^2/36 + O(\varepsilon^2)$, whence $m = 2 - \frac{8}{9}\varepsilon$:

$$A = -1 + \frac{1}{9} = -\frac{8}{9},$$

identically for the sine family (its ε^2 -coefficient carries the same leading $-k^3/9$). The two regimes carry different constants ($-\frac{1}{2}$ versus the crossing's $-\frac{8}{9}$). One more order of the same calculus derives B exactly. The ingredients: the exact prefactor $2(1 - \varepsilon)e^{\varepsilon/2} = 2 - \varepsilon - \frac{3}{4}\varepsilon^2 - \dots$; the cubic coefficient $c_3(k) = -\frac{k^3}{9} - \frac{k^2}{12} + \frac{17k}{72}$ (the m^3 -terms cancel in its defining sum, so c_3 shares the leading $-k^3/9$); the quartic $c_4(k) = \frac{k^5}{450} + O(k^4)$ together with $\frac{1}{2}c_2(D)^2 = \frac{D^6}{162} + \frac{D^5}{108} + \dots$; and one subtlety: at order $\varepsilon^6 u^9$ the cube $\frac{1}{6}(c_2(D)\varepsilon^2)^3$ contributes $+\frac{u^9}{2239488} \sin u$ through the leading term of D^9 , cancelling the $\delta^3/6$ of the cosine expansion *identically*; truncating the exponential before the cube gives a spurious m^5 -term. With that cancellation the crossing equation closes as $m = 2 - \frac{8}{9}\varepsilon - \frac{1891}{8100}\varepsilon^2 + O(\varepsilon^3)$, and inversion gives

$$B = 2m_2 + \frac{64}{81} = \frac{1309}{4050} = 0.3232098765432\dots,$$

confirmed by a five-order free fit to the computed roots: $B = 0.3232098765429\dots$ agrees to 13 digits (with $A = -8/9$ simultaneously reproduced to 17), and the (A, B) -constrained fit has out-of-sample residuals below $4 \cdot 10^{-16}$. The accumulation law of the β_2 family therefore begins

$$1 - q_j^* = \frac{2}{\mu_j} \left(1 - \frac{8}{9\mu_j} + \frac{1309}{4050\mu_j^2} + O(\mu_j^{-3}) \right),$$

with both corrections rational and derived. Three closing observations. (a) *Universality*: the sine family obeys the *same* law; a constrained fit to 22 sine members gives $B_{\sin} \cdot 4050 = 1309.000000$ to the fit precision $2 \cdot 10^{-8}$; so through second order the two interlaced families share one accumulation series, with only μ_j distinguishing them. (b) Every ingredient of the crossing calculus is rational (the $c_n(k)$ are rational polynomials; $D^n \cos u$ has odd u -powers on sin and even on cos, an invariant of $D = (u/2)d/du$ preserved by the one-line induction; the δ -elimination therefore stays in odd powers and the m -equation in rational coefficients), so we conjecture the expansion is rational to all orders. The third constant resists identification: $C = -0.4745357072870\dots$ is stable to 10^{-14} across model orders and members through $j = 55$, excludes every denominator up to $\sim 10^6$ (PSLQ), and the fitted higher coefficients grow ($F \approx 4.3$), marking the series as asymptotic-divergent; which caps the extractable precision and leaves C 's (conjecturally rational, large-denominator) value to the third-order calculus. (c) At $j = 1$ the same series becomes a rational-coefficient expansion in $4/\pi^2$ evaluated at the wild head: divergent, hence a resummation question. If a Borel-type summation of the rational series could be shown to reproduce q^* , the arithmetic of π^2 would enter the problem for the first time through the family; we record this as a direction and have tested its carrying capacity twice. A Borel–Padé resummation of the seven known coefficients reproduces the four accessible samples (both family heads and both second members) to residuals $7.9 \cdot 10^{-3}$, $1.7 \cdot 10^{-6}$, $4.6 \cdot 10^{-9}$, $7.4 \cdot 10^{-11}$ at $\mu = 2.47, 9.87, 22.2, 39.5$. An attempted transseries extraction from these residuals then *refuted its own premise*: the log-slopes between consecutive samples are 1.14, 0.48, 0.24, matching the truncation signature $7/\mu = 1.17, 0.47, 0.24$ exactly (three points), not an exponential $e^{-\beta\mu}$; the residuals are the x^7 -tail of the divergent series, not a nonperturbative signal. At the head this is the standard resurgence degeneracy: seven coefficients place the evaluation at optimal truncation, where truncation floor and first transseries term coincide in scale ($e^{-t_0\mu_1} \approx 0.01$,

with $t_0 \approx 0.75 \pm 1.70i$ the Padé–Borel singularities), so the head residual $-0.008(4)$ *cannot* be attributed to a nonperturbative sector on present data. That cost has now been paid: the crossing calculus has been automated in exact rational arithmetic (weight-graded formal Newton for the zero shift; parity and residual controls exact), producing the law’s coefficients through order thirty; *all rational*, e.g. $A_3 = -1695089/3572100$ (whose denominator $2^2 3^6 5^2 7^2$ explains the failed PSLQ) and $A_4 = 10199641/535815000$. The coefficients are computed by running the calculus over $\mathbb{F}_p$ and recovering each A_n by rational reconstruction from twenty 62-bit primes; the values are independently confirmed by extracting A_n from the accumulation law applied to the roots themselves, computed to 105 digits.

The Borel geometry is only partly determined by these coefficients. The signs of A_n follow a three-positive, three-negative pattern from $n \approx 12$, indicating a conjugate pair of singularities near 60° ; the pattern is not exact over the full range. This period-6 oscillation dominates the naive growth diagnostics, and must be removed before the Gevrey order can be read off. Setting

$$G_n := \frac{1}{6}(\log |A_n| - \log |A_{n-6}|),$$

which is phase-matched and therefore free of the oscillation, the model $|A_n| \sim C^n (n!)^s$ gives $G_n \sim \log C + s \log n$, so s is the slope of G_n against $\log n$. Least squares yields $s = 0.844$ on $16 \leq n \leq 30$, 0.897 on $23 \leq n \leq 30$ and 1.033 on $26 \leq n \leq 30$: the estimate rises towards 1 as the fitting window advances, and the data are consistent with Gevrey-1. (The unnormalised ratio $\log |A_n|/(n \log n)$, which equals $s + (\log C - s)/\log n$, is 0.313 at $n = 14$ and 0.411 at $n = 30$; its corrections are of order $1/\log n$ and it is therefore useless for extrapolation at these lengths.)

For the singularity modulus three independent estimates are available and agree only in order of magnitude: the phase-matched growth constant gives $|t_0| = 1/C = 3.59\text{--}3.98$ over $24 \leq n \leq 30$; a three-term recurrence fit, appropriate to a conjugate pair, gives $|t_0| = 3.285, 3.394, 3.493, 3.558$ at $n_0 = 22, 25, 28, 30$, extrapolating in $1/n_0$ to 4.30 , with $\arg t_0 = 60.8^\circ, 61.7^\circ, 62.4^\circ, 62.3^\circ$; and the Padé–Borel approximants give $|t| = 2.61, 2.56, 2.96, 3.09, 3.33$ for $[6/6]$ through $[14/14]$. Thus $|t_0|$ lies between 2.9 and 4.3 and is not pinned. Gevrey-1 growth, and with it the $n!$ -Borel framework, is supported but not established; the location of the singularities is not established; and the Borel representation below remains a conjecture whose analytic premise is open. One structural constraint is available: by the weight lemma, only Bernoulli numbers of index $\lesssim n$ reach A_n (a B_j occurring in c_m sits at k -degree $\deg c_m - j$ and reaches A_n only if $j \leq 2n - m + 1$), which excludes the Gevrey-2 growth that B_{2n} would otherwise force, but does not pin the class. Where testable the resummation succeeds: at the sine head the Padé–Borel ladder converges $1.7 \cdot 10^{-6} \rightarrow 3.7 \cdot 10^{-11}$ (any transseries prefactor $\lesssim 10^{-6}$ unless phase-suppressed), while at the cosine head the plain ladder reaches its approximation wall at $\sim 10^{-3}$ with residuals oscillating about zero. We therefore pose the *Borel representation conjecture*: $1 - q_j^* = (2/\mu_j) \mathcal{S}(1/\mu_j)$ for both families and all j , where $\mathcal{S}$ is the Borel–Laplace sum of the explicit rational series $\sum A_n x^n$; in particular

$$q^* = 1 - \frac{8}{\pi^2} \mathcal{S}\left(\frac{4}{\pi^2}\right),$$

the first exact (conjectural) bridge between q^* and π^2 -arithmetic. Numerically the head representation is confirmed to $\sim 10^{-4}$ by conformal-map Borel resummation (the residual crosses zero smoothly with no plateau above the estimator scatter), and the *same* resummation machinery reproduces the three lower- x heads to $10^{-13}\text{--}10^{-26}$; the coefficients are now computed through A_{22} , still rational (e.g. A_{21}, A_{22} have denominators $\sim 10^{93}$), consistent with the rationality of the expansion at every order.

Two honest boundaries frame the bridge. *Analytically*, its proof is *not* a cite-the-theorems exercise: it reduces to a single hard lemma; a q -uniform, turning-point (Olver/Airy-grade) factorial

remainder bound $|A_n| \leq C\sigma^n n!$ for the Hahn–Exton crossing as $q \rightarrow 1$; which has no citable form in the literature and is plausibly the program’s recurring degenerate-saddle obstruction in Borel-summability costume. *Arithmetically*, the Borel transform $b(t) = \sum (A_n/n!)t^n$ carries a sharp, exceptionless prime law: for every odd prime $p \leq 43$, $p \mid \text{den}(b_n)$ exactly when $2n+1 \geq p$, with p -adic valuation exactly 2 at the onset $n = (p-1)/2$, and thereafter non-vanishing with valuation growing linearly *on average* (least-squares slopes 2.38, 1.17, 0.73, ... for $p = 3, 5, 7$; the valuation is non-monotone locally, e.g. v_5 dips $6 \rightarrow 5$ and v_{17} dips $2 \rightarrow 1$). The clean-before-onset half is a theorem for all p . (i) The polynomials $c_n(k)$ built from the calculus are p -integral for $n \leq p-2$ (von Staudt–Clausen [19] at the Faulhaber layer). (ii) A_n depends only on the calculus data through order $2n$; this is now proved, in two steps. First, by series reversion A_n is a function of $m_1, \dots, m_n$ alone (the order- n inverse coefficient uses only the first n forward coefficients; verified by perturbing $m_{>n}$). Second, m_i depends only on $c_1, \dots, c_{2i}$: in the turning-point grading, where $k \sim \varepsilon^{-1/2}$ assigns $\varepsilon^a u^b$ the weight $2a - b$ (physical order = weight/2) and every operation is additive on weights, the polynomial c_m enters at minimum weight $2m - \deg_k c_m$, attained by its top-degree term. Here $\deg_k c_m = m + [m \text{ even}]$: the degree- m coefficient of the $\varepsilon^m \log$ -ratio $\ell_m(i)$ equals $(-1)^m [x^m] \log((e^x - 1)/x)$, which vanishes for every odd $m \geq 3$ because $\log((e^x - 1)/x) = \frac{x}{2} + \log(2 \sinh(x/2)/x)$ and the last term is *even* in x ; the Faulhaber sum then raises the k -degree by one. Hence c_m has minimum weight $m - [m \text{ even}]$ and first reaches physical order i only when $m \leq 2i$, so m_i (and therefore A_n) sees no c_m with $m > 2n$. (iii) The assembly introduces no prime p at output order $n < p$: the Bernoulli denominators of the Faulhaber sums acquire p only when $(p-1) \mid 2m$, i.e. $2m \geq p-1$, and all factorial and division indices stay below p . For $n < (p-1)/2$ one has $2n \leq p-3$, so every ingredient feeding A_n is p -integral and $p \nmid \text{den}(A_n)$.

The matching lower direction; that p *does* appear at $n = (p-1)/2$, with valuation exactly 2; reduces to a unit. At $n = (p-1)/2$ the sole prime- p source is c_{p-1} (all lower c are p -integral), and it enters A_n *linearly*, since c_{p-1}^2 has weight $2(p-2) > 2n$. Only the top three coefficients of c_{p-1} (degrees $\geq p-2$) survive the weight cutoff; of the two that carry p^2 ; the k^1 and k^p monomials (von Staudt–Clausen at the Faulhaber layer, $\deg_k c_{p-1} = p$); only k^p reaches A_n , as k^1 sits at weight $2p-3 > 2n+1$. The coefficient of that top monomial in $A_{(p-1)/2}$ is a *unit*, $\partial A_{(p-1)/2} / \partial c_{p-1}[k^p] = \pm 1$, and this now has a leading-symbol proof. The monomial perturbs the confluent expansion of G by $\varepsilon^{p-1} k^p$; the turning-point map (the D^p -action, $D = (u/2) d/du$) sends k^p to $a_p u^p \sin u + b_p u^{p-1} \cos u$ with $a_p = (-1)^{n-1} 2^{-p}$ the leading sine coefficient and $b_p = (-1)^n n(2n+1) 2^{-p}$ the leading cosine coefficient ($p = 2n+1$; both closed forms follow from the recursion $(A, B) \mapsto (\frac{u}{2}(A' - B), \frac{u}{2}(B' + A))$). At a simple zero u_* of $\cos$ the cosine term vanishes and every sine power below u^p contributes only at higher order in ε , so the zero shift is $\delta = \varepsilon^{p-1} a_p u_*^p$ (the factor $\sin u_* = \pm 1$ cancels against $G' = -\sin$). With $m = \varepsilon u^2$ and the leading balance $u_*^2 = 2/\varepsilon$ (from $\varepsilon u^2 = 2(1 - \varepsilon)e^{\varepsilon/2} \rightarrow 2$),

$$\delta m = 2\varepsilon u_* \delta = 2a_p \varepsilon^p u_*^{p+1} = a_p 2^{n+2} \varepsilon^n, \quad \text{so} \quad \frac{\partial m_n}{\partial c_{p-1}[k^p]} = a_p 2^{n+2} = \pm 2^{1-n},$$

and with the reversion Jacobian $\partial A_n / \partial m_n = 2^{n-1}$ (from $\varepsilon \sim 2/\mu$) the product is ± 1 . Only the top sine power u^p reaches order ε^n , so the magnitude is exact; the sign is $(-1)^n$ (verified for all $n \leq 10$, independently of p). Since $c_{p-1}[k^p]$ has p -adic valuation -2 while the other reaching monomials (k^{p-1}, k^{p-2}) have valuation ≥ -1 against p -integral multipliers, $v_p(\text{den } A_{(p-1)/2}) = 2$ exactly. Thus the onset law; $p \mid \text{den}(A_n) \iff 2n+1 \geq p$, with valuation 2 at $n = (p-1)/2$; is *proved for all odd* p . The average-linear (Legendre/factorial) growth; not the logarithmic growth an $\text{lcm}(1, \dots, 2n)$ or ζ -type denominator would give; makes $\text{den}(A_n)$ super-geometric ($\log \text{den}(A_n)/n$ rises $5.4 \rightarrow 10.2$ over $n \leq 20$, unbounded). Consequently $b(t)$ is *not* a G -function (super-geometric denominators alone suffice, independent of holonomy), so $\sum A_n x^n$ is not of André arithmetic-Gevrey type [3] and lies outside the G -function, E -function, and E -operator (Fischler–Rivoal [16]) classes entirely. The

closest existing frame, the E -operator theory, has exactly one blocking gap; the hypothesis that b is a G -function; which the prime law refutes rather than supplies. The missing tool is therefore genuinely new, not an incremental hypothesis-removal: a value theory for Borel–Laplace sums whose Borel transform is non-holonomic with factorial denominators, evaluated at a point $(4/\pi^2)$ transcendental over $\mathbb{Q}$. This constant is a canonical first target for such a theory. Third, the family sum converges to

$$T := \sum_{j \geq 1} (1 - q_j^*) = 0.7357490877157094 \dots (\pm 3 \cdot 10^{-17}, \text{ tail summed via the fitted law and Hurwitz zeta}),$$

for which no relation with $\{1, \pi^{-2}, \pi^{-4}, \log 2, \gamma, \pi^2/6\}$ exists at height 10^4 within the stated uncertainty (and $T \neq 2/e$): the classical tail is π^2 -rational piece by piece, but the wild head shifts the sum off every tested constant; a sum rule, if one exists, must come from global structure not yet visible.

The failure pattern above is not an artifact of particular constructions; it is forced by exact arithmetic facts, which we record as a theorem-level ledger.

Proposition 87 (Arithmetic no-go ledger at $\rho = \frac{1}{2}$). *Let $q = s/t \in (0, 1)$ with $\gcd(s, t) = 1$, and let $g_k = (-1)^k q^{k(k-1)} / (q; q)_{2k}$ be the Taylor coefficients of $G(q, \cdot)$. Then:*

- (i) (exact denominators) *in lowest terms $g_k = (-1)^k s^{k^2-k} t^{k^2+2k} / \prod_{i=1}^{2k} (t^i - s^i)$; in particular $\text{den}(g_k) = \prod_{i=1}^{2k} (t^i - s^i) = t^{k(2k+1)(1+o(1))}$ exactly, while $|g_k| = (s/t)^{k^2(1+o(1))}$, so the arithmetic ratio is $\rho = \log(t/s) / (2 \log t) \leq \frac{1}{2}$, with equality iff $s = 1$.*
- (ii) (ν -ladder arithmetic) *the q -Lommel polynomials of Koelink–Swarttouw [22] satisfy $\deg_q p_m = m^2$ exactly, with leading q -coefficient $(-1)^m$; hence at $q = s/t$ their denominators are the pure power t^{m^2} , carrying no cyclotomic structure.*
- (iii) (ν -direction regularity) *the ladder recurrence $u_{m+1} = ((1 - q^{2m+1})/x_0 + x_0)u_m - u_{m-1}$ has coefficients converging exponentially to $x_0 + 1/x_0$ with distinct characteristic roots $x_0^{\pm 1}$; by Poincaré–Perron [36, 35] every nonzero solution satisfies $|u_m|^{1/m} \rightarrow x_0^{\pm 1}$. No theta-type (q^{cm^2} -decaying) solution exists in the ν -direction, for either kind. Consequently the cleared ν -ladder forms are $\gg t^{m^2} x_0^{-m} \rightarrow \infty$ for every rational q , including $s = 1$.*
- (iv) (cyclotomic floor) *any linear form built over the coefficients $g_0, \dots, g_M$ has decay at best $q^{M^2(1+o(1))}$, while its cyclotomic support is $\{\Phi_d(t, s) : d \leq 2M\}$; even if every Φ_d appeared at most once; the idealized best case of any lcm/acceleration argument; the denominator exceeds $t^{\sum_{d \leq 2M} \varphi(d)} = t^{(12/\pi^2 + o(1))M^2}$, and $12/\pi^2 = 1.2159 \dots > 1$. Hence no cyclotomic-compression (Rhin–Viola-type [39]) refinement of the Padé, continued-fraction, or ladder families can produce integer forms tending to 0. Moreover the true solved Padé denominators contain growing non-cyclotomic irreducible factors (at $n = 3$, one of degree 26 with coefficients up to 4; no integer d has $\varphi(d) = 26$), which no lattice-of-cyclotomes argument addresses.*

Proof. (i) $q^{k(k-1)} = s^{k(k-1)} / t^{k(k-1)}$ and $1/(q; q)_{2k} = t^{k(2k+1)} / \prod_{i \leq 2k} (t^i - s^i)$; the numerator $s^{k^2-k} t^{k^2+2k}$ is coprime to each factor since $\gcd(st, t^i - s^i) = 1$. (ii) Induction: $p_{m+1} = (X + 1 - q^{2m+1})p_m - Xp_{m-1}$, and the top q -degree term of p_{m+1} is $-q^{2m+1} \cdot (\text{top of } p_m)$, of degree $m^2 + 2m + 1 = (m+1)^2$, which cannot cancel against Xp_{m-1} (degree $(m-1)^2$). (iii) The perturbation $(1 - q^{2m+1})/x_0 - 1/x_0$ decays geometrically, so the Poincaré–Perron theorem [36, 35] applies with the limit equation $u_{m+1} = (x_0 + 1/x_0)u_m - u_{m-1}$, whose roots $x_0 \neq 1/x_0$ are simple. (iv) A form using $g_0, \dots, g_M$ vanishes to order at most M at $z = 0$, so its value at a fixed z_0 inside the disc is $\gg |g_{M+1} z_0^{M+1}| = q^{M^2(1+o(1))}$

in the truncation-based families; its denominators involve $(t^i - s^i)$ for $i \leq 2M$, i.e. $\Phi_d(t, s)$ for $d \leq 2M$, and $\sum_{d \leq N} \varphi(d) = \frac{3}{\pi^2} N^2 + O(N \log N)$ gives the floor with $N = 2M$. The degree-26 factor is exhibited by the exact $n = 3$ computation (`code/zeta_probe/beta2_qniven_ladder.py`). $\square$

Remark 88 (scope). Proposition 87 closes every construction whose coefficients live in the natural $\mathbb{Z}[q^{\pm 1}, z]$ -lattices of the q -difference module; Taylor Padé and Hermite–Padé, the argument-direction continued fraction, the ν -ladders of both kinds; together with all cyclotomic-compression refinements of them. It does *not* exclude a fundamentally different family of forms; producing one is exactly what a $\rho = \frac{1}{2}$ criterion would mean. The ledger thereby locates β_2 's arithmetic at a precise frontier of q -Diophantine approximation.

The remaining freedom can be narrowed further: the finite places of $\mathbb{Q}$ are provably useless here. The series $S(q) = \sum_k g_k z^{*k}$, $z^* = 2q(1 - q)$, has rational terms, so besides its real sum it possesses s -adic and t -adic *avatars*; the same series summed in $\mathbb{Q}_s$ and $\mathbb{Q}_t$, where it also converges (the valuations of the terms tend to $+\infty$).

Lemma 89 (adelic units). *Let $q = s/t \in (0, 1)$, $\gcd(s, t) = 1$, and $z^* = 2q(1 - q)$. For every $k \geq 1$ and every prime p : if $p \mid t$ with $e = v_p(t)$, then*

$$v_p(g_k z^{*k}) = k^2 e + k[p = 2] \geq k^2,$$

(exact, verified symbolically), and if $p \mid s$ then $v_p(g_k z^{*k}) \geq k^2$. Consequently the series converges in $\mathbb{Q}_p$ for every $p \mid st$, and the finite-place avatars satisfy $S_p \equiv 1 \pmod{p^{v_p(st)}}$; in particular $S_t \equiv 1 \pmod{t}$ and $S_s \equiv 1 \pmod{s}$: they are units, hence nonzero; regardless of whether the real avatar vanishes.

Proof. Each factor $t^i - s^i$ of $\text{den}(g_k)$ is a p -unit for every $p \mid st$ (it is $\equiv -s^i$ or $\equiv t^i \pmod{p}$), so by Proposition 87(i), $v_p(g_k) = (k^2 + 2k)e$ for $p \mid t$ and $v_p(g_k) = (k^2 - k)v_p(s)$ for $p \mid s$. From $z^* = 2s(t - s)/t^2$: for $p \mid t$, $v_p(t - s) = 0$ and $v_p(z^*) = [p = 2] - 2e$, giving $v_p(g_k z^{*k}) = (k^2 + 2k)e + k([p = 2] - 2e) = k^2 e + k[p = 2]$; for $p \mid s$, $v_p(z^*) = v_p(2s) \geq v_p(s) \geq 1$, giving $v_p(g_k z^{*k}) \geq (k^2 - k) + k = k^2$. All are ≥ 1 for $k \geq 1$, so each avatar is 1 plus terms of positive valuation. $\square$

Remark 90 (exact adelic balance, and the shape of any future criterion). Lemma 89 has two consequences. First, it extends the no-go ledger from the module's coefficient lattices to the *values*: no divisibility or smallness can be sourced from the finite places, since the finite-place avatars are units; a hypothetical rationality of β_2 would be a maximally place-asymmetric event, the real avatar vanishing while every finite avatar equals a unit $\equiv 1$. Second, it makes the $\rho = \frac{1}{2}$ balance exact in the sense of the product formula: the term $g_k z^{*k}$ carries archimedean decay $(s/t)^{k^2(1+o(1))}$, s -adic and t -adic numerator content $s^{k^2} t^{k^2}$, and denominator $t^{2k^2(1+o(1))}$; the books balance with *zero margin* at the k^2 scale, so any successful construction must be lossless at all three places simultaneously and win at the geometric scale. Every mechanism surveyed above loses a factor t^{ck^2} somewhere; a $\rho = \frac{1}{2}$ criterion is therefore forced to be a purely *archimedean cancellation* phenomenon. This suggests the precise principle that would settle β_2 , which we record as a conjecture: for the irregular rank-two q -difference module with polynomial companion matrix and $\det A = q$, the vanishing locus is *adelically rigid*: the entire function $G_v(q, \cdot)$ cannot vanish at a rational point z_0 at which all finite-place avatars are units (as at $z_0 = z^*$, by Lemma 89) at one place v of $\mathbb{Q}$ while the avatars at all other convergent places are units. The finite-place half of this rigidity is exactly Lemma 89; its archimedean half implies $\beta_2 \notin \mathbb{Q}$. This is a q -analogue, for irregular q -difference modules, of the place-uniformity enjoyed by E - and G -function values, where the exceptional sets of Siegel–Shidlovskii–André theory are independent of the place.

Further analysis of the archimedean obstruction yields several exact structural facts and a faithful reduction. All are verified numerically to 40–60 digits (`code/zeta_probe/beta2_*.py`).

Proposition 91 (structure of the zero curve). *Write $Q = q^2$ and let $0 < z_1(q) < z_2(q) < \dots$ be the positive zeros of $G(q, \cdot)$.*

- (i) (base change) $G(q, z) = \frac{(q^2; q^2)_\infty}{\sqrt{q} (q; q^2)_\infty} z^{1/4} J_{-1/2}^{(3)}(\sqrt{z}/q; q^2)$, where $J_\nu^{(3)}$ is the Hahn–Exton (third Jackson) q -Bessel function; thus G is, up to an explicit prefactor, the Hahn–Exton q -cosine, and its zeros are those of a classical, well-studied object.
- (ii) (zero lattice) $z_k(q) Q^{k-1} \rightarrow 1$ as $k \rightarrow \infty$, with theta-suppressed dressing $\delta_k := z_{k+1} Q^k - 1 = O(c^k Q^{k^2})$ (numerically $\delta_0 = -0.5051$, $\delta_1 = -1.17 \cdot 10^{-2}$, $\delta_2 = -1.03 \cdot 10^{-5}, \dots$); $z^* = z_1(q^*)$ is the maximally dressed member, displaced from the bare lattice point $Q^0 = 1$ by an $O(1)$ correction; it is not a torsion/lattice point of the connection curve.
- (iii) (power sums are rational) the symmetric functions $\sigma_p(q) := \sum_{n \geq 1} z_n(q)^{-p}$ lie in $\mathbb{Q}(q)$ for every $p \geq 1$ (Newton’s identities on the $\mathbb{Q}(q)$ -coefficients g_k); e.g. $\sigma_1 = 1/((1-q)(1-q^2))$. Consequently $z^*(q) = \lim_{p \rightarrow \infty} \sigma_p(q)/\sigma_{p+1}(q)$ with geometric rate $(z_1/z_2)^p$, giving an all-orders $\mathbb{Q}(q)$ -approximation of the zero curve.

Two of the no-go’s ingredients also sharpen. First, the *cause* of $\rho = \frac{1}{2}$ is exactly the double Pochhammer: $\rho = \log(t/s)/(2 \log t)$, and the factor 2 in the denominator is the $2k$ in $(q; q)_{2k}$; a single Pochhammer would give $\rho > 1$ (whence Liouville) when $t > s^2$. The intractability is intrinsic to the base- q^2 Hahn–Exton structure, not to the choice of construction. Second, the finite-place primes are *disjoint* from st : $\prod_{i \leq 2N} (t^i - s^i)$ is coprime to st (only the prime 2, via lifting-the-exponent when s, t are both odd, contributes an $O(N)$ term), so the arithmetic of every partial-sum denominator is supported off $\{s, t\}$; orthogonal to the unit residues of Lemma 89. The Hankel/determinantal escape is likewise closed exactly: $\det(g_{i+j})_{0 \leq i, j \leq n}$ has denominator of *cubic* archimedean size ($\log |\text{den}| \sim c n^3$), one power worse than the quadratic remainder.

Finally, the Nesterenko route is blocked more precisely than by non-holonomicity alone. Nesterenko’s method needs a finitely generated δ -closed ($\delta = q d/dq$) $\mathbb{C}(q)$ -algebra containing S , i.e. that S be *differentially algebraic*; non-holonomicity does not suffice (the quasimodular E_2 is non-holonomic yet δ -algebraic). One computes $\delta S = \sum_k a_k (k^2 - \frac{kq}{1-q} + L_k)$ with $L_k = \sum_{i=1}^{2k} i q^i / (1 - q^i) \rightarrow (1 - E_2)/24$: S carries *every* finite Eisenstein truncation L_k , not merely the limit, which is precisely why it escapes the modular differential ring. Whether S is differentially algebraic is itself open, but the exclusion is now proof-grade over a wide window: S satisfies no algebraic differential equation in the δ -jets at any of the profiles (order, degree, $\deg_q$) = (2, 12, 3), (3, 8, 3), (4, 5, 6), (5, 4, 8), (6, 3, 15), (8, 2, 35), (12, 1, 120); exact linear algebra on 2000 coefficients modulo a prime, every matrix of full column rank; since a primitive integer relation survives reduction, full rank modulo one prime excludes a relation over $\mathbb{Q}$ at that profile outright. (The pipeline is validated on E_2 , for which it recovers the q -Chazy equation $\delta^3 E_2 = E_2 \delta^2 E_2 - \frac{3}{2} (\delta E_2)^2$ at its exact profile.) The same exclusion holds verbatim for the zero branch $z_1(q)$. The Nesterenko route is blocked *modulo the differential transcendence of S* , which is itself open; and, with no functional equation in q available for S (no shift, q -difference, or Mahler equation exists), no current method can even attack it.

Remark 92 (reduction to a theta non-resonance). All routes converge on the same faithful reformulation: β_2 is irrational iff $G(q^*, z^*) \neq 0$ is forced by the arithmetic, and (Ramis–Sauloy–Zhang [38]) this is a *theta non-resonance* statement; the connection ratio $R_q(z) = [P_{21} y_{\text{irr}}]/[P_{11} y_{\text{reg}}]$, an explicit θ_{q^2} -quotient, must satisfy $R_q(2q(1-q)) \neq -1$ at every algebraic q . The evidence points toward

truth: in the confluent limit $q \rightarrow 1$ this is exactly Lindemann's theorem that $\cos$ has no algebraic zero. But the reduction is not yet shown *easier*: $R_q(z) = -1$ is a functional-equation-consistent value that a nonconstant theta-quotient must attain somewhere, and on the diagonal $z = 2q(1 - q)$ the theta-arguments appear algebraically dependent, so no current independence theorem (Bertrand; Nesterenko [31]; Schneider–Lang [25]) forbids it. The archimedean obstruction is thus the exact mirror of the proven finite-place balance: units ($\equiv 1$) at every $p \mid st$, and a putative -1 at ∞ , with neither forbidden. The required connection-entry computation is carried out below.

Proposition 93 (the connection entries). *Write $Q = q^2$, $\theta_b(x) = \sum_{n \in \mathbb{Z}} b^{n(n-1)/2} x^n$. The equation for G has, at $z = \infty$, the formal solution basis $f_1(z) = \theta_Q(-z) \hat{g}(z)$ and $f_2(z) = h(z)/\theta_Q(-z/q)$, where $h(z) = \sum_k c_k z^{-k}$ is convergent (entire in $1/z$, $c_k = O(Q^{k^2/2})$) while $\hat{g}(z) = \sum_k b_k z^{-k}$ is q -Gevrey divergent, with the closed form*

$$b_k = \frac{q^{k-k(k-1)/2}}{(q; q)_k}$$

(equivalent to the recursion $b_k = [-(q+1)Q b_{k-1} - Q^{3-k} b_{k-2}]/(q(Q^k - 1))$; verified symbolically for $k \leq 8$). By Euler's theorem $\sum_j x^j/(q; q)_j = 1/(x; q)_\infty$, the q -Borel transform is therefore explicit:

$$\hat{B}(\xi) = \sum_{k \geq 0} b_k q^{k(k+1)/2} \xi^{-k} = \sum_{k \geq 0} \frac{(q^2/\xi)^k}{(q; q)_k} = \frac{1}{(q^2/\xi; q)_\infty},$$

meromorphic on $\mathbb{C}^*$ with poles exactly on $\xi \in q^{2+\mathbb{N}_0}$, and satisfying $\hat{B}(\xi) + \frac{q+1}{\xi} \hat{B}(q\xi) + (\xi^{-2} - 1) \hat{B}(q^2\xi) = 0$ identically (direct computation from the product form; in particular $\hat{B}(-1) = (q+1) \hat{B}(-q)$). The theta-kernel q -Laplace

$$g^{[\lambda]}(z) = \frac{\sum_{m \in \mathbb{Z}} q^{m(m-1)/2} (\lambda/z)^m \hat{B}(\lambda q^m)}{\theta_q(\lambda/z)}, \quad f_1^{[\lambda]} = \theta_Q(-z) g^{[\lambda]}(z),$$

is then singular only for $\lambda q^{\mathbb{Z}}$ meeting the pole set, i.e. for the single class $[\lambda] \in q^{\mathbb{Z}}$ of $E_q = \mathbb{C}^*/q^{\mathbb{Z}}$ (the positive-real lattice class, near which the zeros of G lie); every other class is regular. For $\lambda \notin q^{\mathbb{Z}}$ the bilateral sum converges absolutely on compacts of $\mathbb{C}^* \setminus (-\lambda q^{\mathbb{Z}})$ and $Lf_1^{[\lambda]} = 0$ exactly: the theta shifts $\theta_q(q^{-2}x) = q^{-3}x^2\theta_q(x)$ and $\theta_q(q^{-4}x) = q^{-10}x^4\theta_q(x)$ turn $f_1^{[\lambda]}(Qz)$ and $f_1^{[\lambda]}(Q^2z)$ into the same kernel sum with $\hat{B}(\lambda q^m)$ replaced by $\hat{B}(\lambda q^{m+2})$, $\hat{B}(\lambda q^{m+4})$; collecting the coefficient of $q^{m(m-1)/2} (\lambda/z)^m / \theta_q(\lambda/z)$ in $Lf_1^{[\lambda]}/\theta_Q(-z)$ then yields $q[\hat{B}(\xi) + \frac{q+1}{\xi} \hat{B}(q\xi) + (\xi^{-2} - 1) \hat{B}(q^2\xi)]$ at $\xi = \lambda q^m$, identically zero. (Each step was verified numerically to 40 digits; the kernel normalization was validated on convergent input to 34 digits, and $Lf_1^{[\lambda]} \approx 10^{-61}$ numerically.) Writing $G = C_1 f_1^{[\lambda]} + C_2^{[\lambda]} f_2$, numerically to 59–60 digits:

- (i) since $\det A = q$, every Casoratian $\text{Cas}(u, v)(z) := u(z)v(Qz) - u(Qz)v(z)$ of solutions satisfies $\text{Cas}(Qz) = q \text{Cas}(z)$ (immediate from $u(Q^2z) = a u(Qz) - q u(z)$), so $C_1 = \text{Cas}(G, f_2)/\text{Cas}(f_1^{[\lambda]}, f_2)$ and $C_2^{[\lambda]} = \text{Cas}(f_1^{[\lambda]}, G)/\text{Cas}(f_1^{[\lambda]}, f_2)$ are Q -elliptic; proven, and verified to 10^{-60} ;
- (ii) C_1 is a constant, independent of both z and the resummation direction λ : this is proven in Theorem 94 below (the limit argument applies along every orbit class $\neq [1]_Q$, and ellipticity extends the value); the slope-1 component of G is Stokes-invariant, and all q -Stokes ambiguity is carried by $C_2^{[\lambda]}$, nonconstant elliptic with λ^2 in its zero divisor;

(iii) (dressing-product formula, verified to 59 digits)

$$C_1 = \frac{1}{(Q; Q)_\infty \prod_{k \geq 1} z_k Q^{k-1}} :$$

the Stokes-invariant connection constant is the regularized product of the zero dressings;

(iv) at $q = q^*$ the entire apparatus evaluates: $C_1(q^*) = 2.69898382616391375\dots$, and the zero condition $G(q^*, z^*) = 0$ takes the pinned form

$$C_2^{[\lambda]}(z^*) = -C_1 \frac{f_1^{[\lambda]}(z^*)}{f_2(z^*)} = 0.011592994120624266\dots \quad (\lambda = \tfrac{3}{2}),$$

verified to 30 digits. This is the explicit instance of the theta non-resonance: every object in it is now constructed.

The constant C_1 is in fact *modular*, and this upgrades the dressing-product formula into an exact evaluation of the zero product.

Theorem 94 (the connection constant, and the zero product of the Hahn–Exton q -cosine). *For $0 < q < 1$,*

$$C_1 = \frac{1}{(q; q)_\infty}, \quad \text{and consequently} \quad \prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty,$$

where $z_1 < z_2 < \dots$ are the positive zeros of $G(q, \cdot)$. Likewise, for the second solution's underlying entire function H with zeros $w_1 < w_2 < \dots$, $\prod_{k \geq 1} w_k(q) q Q^{k-1} = (q^3; q^2)_\infty$. (Verified at three bases including q^* , to 41, 23, 26 digits for C_1 and 58 digits for the product.)

Proof. (1) *An exact identity.* From $(q; q)_{2k} = (q; q)_\infty / (q^{2k+1}; q)_\infty$ and Euler's expansion $(x; q)_\infty = \sum_{j \geq 0} (-1)^j q^{j(j-1)/2} x^j / (q; q)_j$ at $x = q^{2k+1}$, together with $k(k-1) + 2kj = (k+j)(k+j-1) - j(j-1)$ and the substitution $m = k + j$, absolute convergence of the double sum permits the rearrangement

$$(q; q)_\infty G(z) = \sum_{m \geq 0} (-1)^m q^{m(m-1)} z^m P_m(1/z), \quad P_m(w) := \sum_{j=0}^m c_j w^j, \quad c_j = \frac{q^{j-j(j-1)/2}}{(q; q)_j}$$

(the c_j coincide with the b_j of Proposition 93; identity checked symbolically through z^8 and numerically to 41 digits).

(2) *The constant.* Fix $u < 0$ and put $z = uQ^{-N}$. The moduli $q^{m(m-1)}|z|^m$ peak at $m_0 = N + O(1)$ and decay like $q^{(m-m_0)^2 + O(1)}$ away from it. For $1 \leq j \leq m \leq 2N$, $2Nj - j(j-1)/2 \geq Nj$, so $|c_j z^{-j}| \leq Cq^{Nj}|u|^{-j}$ and $|P_m(1/z) - 1| \leq C'q^N$ uniformly; the terms with $m > 2N$ contribute a relative $O(q^{N^2})$. The negative- m part of $\theta_Q(-z)$ is $O(|z|^{-1})$, negligible against the peak. Hence $(q; q)_\infty G(z) / \theta_Q(-z) \rightarrow 1$ along $z = uQ^{-N}$, for each fixed $u < 0$.

(3) *Geometric zero dressing.* On the circle $|z - Q^{1-k}| = \varepsilon q^k Q^{1-k}$ the identity of step (1) and the bounds of step (2) give $|(q; q)_\infty G - \theta_Q(-z)| \leq Cq^k |\theta_Q(-z)|_{\text{peak}}$, while $\theta_Q(-z)$ has a simple zero at Q^{1-k} with derivative bounded below there; Rouché gives $z_k = Q^{1-k}(1 + O(q^k))$, i.e. $\log(z_k Q^{k-1}) = O(q^k)$, absolutely summable.

(4) *The product.* $G(z) = \prod_k (1 - z/z_k)$ (order 0, genus 0, $G(0) = 1$) and $\theta_Q(-z) = (Q; Q)_\infty(z; Q)_\infty(Q/z; Q)_\infty$ (Jacobi). Along $z = -R$, $R \rightarrow \infty$,

$$\frac{G(-R)}{\theta_Q(R)} = \frac{1}{(Q; Q)_\infty(-Q/R; Q)_\infty} \prod_{k \geq 1} \frac{1 + R/z_k}{1 + RQ^{k-1}},$$

and for $x, y > 0$ the function $R \mapsto \log \frac{1+Rx}{1+Ry}$ is monotone from 0 to $\log(x/y)$, so each factor is dominated by $|\log(z_k Q^{k-1})|$, summable by step (3). Dominated convergence gives $G(-R)/\theta_Q(R) \rightarrow 1/[(Q; Q)_\infty \prod_k z_k Q^{k-1}]$ as $R \rightarrow \infty$; evaluating along the subsequence $R = |u|Q^{-N}$ by step (2), the limit equals $1/(q; q)_\infty$, whence $\prod_k z_k Q^{k-1} = (q; q)_\infty / (Q; Q)_\infty = (q; q^2)_\infty$ by Euler's $(q; q)_\infty = (q; q^2)_\infty (q^2; q^2)_\infty$. Finally, in the basis of Proposition 93: along the negative axis (a regular class, $[-1] \notin q^\mathbb{Z}$) one has $g^{[\lambda]}(z) \rightarrow 1$ and $f_2(z)/\theta_Q(-z) = O(1/(\theta_Q(R)\theta_Q(R/q))) \rightarrow 0$, so the limit just computed is the entry C_1 ; hence $C_1 = 1/(q; q)_\infty$. The H -statement is identical with comparison function $\theta_Q(-qz)$ and ratio $(1-q)/(q; q)_\infty$. Finally, steps (1)–(2) apply verbatim along $z = uQ^{-N}$ for any $u \in \mathbb{C}^*$ whose orbit avoids the zero class $[1]_Q$ of $\theta_Q(-\cdot)$, using the standard theta lower bound $|\theta_Q(-z)| \geq c([u]) \cdot (\text{peak term})$ off that class; along each such orbit $C_1(z)$ is constant (ellipticity) and $f_2/f_1 \rightarrow 0$, $g^{[\lambda]} \rightarrow 1$, so $C_1([u]) = 1/(q; q)_\infty$ on every class $\neq [1]_Q$, and by meromorphy $C_1 \equiv 1/(q; q)_\infty$: for every regular λ . In particular C_1 is z - and λ -independent. $\square$

Remark 95 (Diophantine consequences at z^*). At $q = q^*$ the identity splits off the candidate rational zero:

$$\prod_{k \geq 2} z_k(q^*) Q^{k-1} = \frac{(q; q^2)_\infty}{z^*} = 0.988299884628746 \dots \quad (\text{verified to 30 digits}).$$

Were β_2 rational, the *modular* value $(q; q^2)_\infty$ (transcendental at algebraic q by Nesterenko's theorem applied to the Euler products) would equal the rational z^* times the tail dressing product $\prod_{k \geq 2} z_k Q^{k-1}$. Thus the entire modular sector of the problem is now explicit; the Stokes-invariant connection constant is $1/(q; q)_\infty$, the zero product is $(q; q^2)_\infty$; and *all* remaining unknown arithmetic is provably confined to the tail dressing product, equivalently to the q -Stokes cocycle $C_2^{[\lambda]}$: the non-modularity of the β_2 problem *is* its q -Stokes data. A transcendence statement for that tail (an algebraic independence of the higher-zero dressings from the modular values, or a rigidity theorem for the q -Stokes cocycle) would settle β_2 ; no current theorem provides it.

The cocycle itself is now explicit. Writing $g = g^{[\lambda]}$ for the resummed series and h for f_2 's convergent one,

$$C_2^{[\lambda]}(z) = \theta_Q(-z/q) \frac{g(z) G(Qz) + z^{-1} g(Qz) G(z)}{-(z/q) g(z) h(Qz) + z^{-1} g(Qz) h(z)}$$

This is an identity: it is the ratio of the Casoratian evaluations $\text{Cas}(f_1, G) = \theta_Q(-z)[g(z)G(Qz) + z^{-1}g(Qz)G(z)]$ and $\text{Cas}(f_1, f_2) = \frac{\theta_Q(-z)}{\theta_Q(-z/q)}[-(z/q)g(z)h(Qz) + z^{-1}g(Qz)h(z)]$, both instances of $\theta_Q(Qw) = w^{-1}\theta_Q(w)$ (and verified numerically to 10^{-61}); with divisor now fully described: an *explicit* zero lattice $z \in q^{-1}Q^\mathbb{Z}$ from the theta factor (the computed zero at $z = 1/q$, exact to 10^{-63}), a direction-dependent zero at $z = \lambda^2$ arising from the exact *resummation resonance* $g^{[\lambda]}(\lambda^2)G(Q\lambda^2) + \lambda^{-2}g^{[\lambda]}(Q\lambda^2)G(\lambda^2) = 0$ (verified to 10^{-62} at $\lambda \in \{\frac{3}{2}, 2, \frac{13}{5}\}$; a proof for general regular λ is open), and poles at the zeros of the h -bracket. Substituting into the pinned zero condition at z^* and cancelling the common theta factor reduces the entire β_2 question to *one* relation between the two modular values $\theta_Q(-z^*)$, $(q; q)_\infty$ and five values of the non-modular pair $(g^{[\lambda]}, h)$ on the orbit of z^* . The remaining input is a transcendence/independence statement for those five values; supplying it is one of three research programs (a non-modular extension of Nesterenko's method via the deformation of this connection, a $\rho < \frac{1}{2}$ -regime irrationality criterion, or adelic rigidity of the vanishing locus); none is currently available.

The first of those programs now has its foundation: the deformation of the module in q closes finitely.

Proposition 96 (the deformation module). *Let $L = q - a\sigma + \sigma^2$ with $a = q + 1 - qz$, $\sigma f(z) = f(Qz)$, and let $\theta = z\partial_z$, $\delta = q\partial_q$ (acting on all q -dependence, including the base $Q = q^2$, so that $\delta\sigma = \sigma\delta + 2\sigma\theta$). Then*

$$L(\theta G) = -qz\sigma G, \quad L(\delta G) = -qG + q(1 + 3z)\sigma G + 4q\theta G - 2a\sigma\theta G$$

(two commutator computations; verified to 51 digits). Consequently the $\mathbb{Q}(q, z)\langle\sigma\rangle$ -module generated by $\{G, \theta G, \delta G\}$ has finite rank ≤ 6 , with explicit σ -companion relations $\sigma^2 G = a\sigma G - qG$, $\sigma^2 \theta G = a\sigma\theta G - q\theta G - qz\sigma G$, and $\sigma^2 \delta G = a\sigma\delta G - q\delta G - qG + q(1 + 3z)\sigma G + 4q\theta G - 2a\sigma\theta G$.

This finiteness is of the first jet only: the module is not closed under a second application of θ or δ . Both $\theta^2 G$ and $\delta^2 G$ escape $\text{span}_{\mathbb{Q}(q, z)}\{G, \sigma G, \theta G, \sigma\theta G, \delta G, \sigma\delta G\}$ (verified by exact modular linear algebra at $q = \frac{1}{3}$, z -series to 70 terms, coefficient degrees ≤ 6 : the augmented rank strictly exceeds the plain rank). The two derivations are moreover linked by the exact identity

$$\delta G = (\theta^2 - \theta)G + C, \quad C = \sum_{k \geq 0} (-1)^k \left(\sum_{i=1}^{2k} \frac{iq^i}{1-q^i} \right) \frac{q^{k(k-1)} z^k}{(q; q)_{2k}}$$

($= z^2 G_{zz} + C$; immediate from g_k 's logarithmic q -derivative), whose weight C carries the truncated Eisenstein sums $\sum_{i \leq 2k} iq^i/(1 - q^i)$ term by term; the same off-modular data that obstructs the differential closure. In particular, differentiating the pinned relation of Proposition 93 in q does not close: the second q -derivative leaves the module, consistent with Proposition 104. The value of the rank-6 structure is at the function level; it is the module whose difference Galois group is computed in Theorem 101; not as a Nesterenko precondition in the descent variable, which Proposition 104 shows does not exist.

Corollary 97 (equation of the zero curve). *Implicit differentiation of $G(q, z_1(q)) = 0$ gives*

$$z_1'(q) = - \frac{z_1 (\delta G)(q, z_1)}{q (\theta G)(q, z_1)},$$

with $\theta G, \delta G$ the first-jet generators of Proposition 96 (formula checked against a finite-difference derivative to 20 digits at $q = 0.3$). The right-hand side is not a closed vector field; iterating d/dq leaves the rank-6 module (Proposition 96); so this is one exact relation, not an autonomous system. The number q^* is the crossing of the zero curve with the diagonal: $z_1(q^*) = 2q^*(1 - q^*)$ (residual $< 10^{-27}$).

Lemma 98 (irreducibility of the σ -module). *For $0 < q < 1$ the rank-2 σ -module of G (in the sense of [37]) is irreducible over $\mathbb{C}(z)$: the Riccati equation*

$$u(z)u(Qz) - a(z)u(z) + q = 0, \quad a = q + 1 - qz,$$

has no solution $u \in \mathbb{C}(z)$.

Proof. Suppose $u = P/R$ with $P, R \in \mathbb{C}[z]$ coprime, of degrees p, r , leading coefficients α, β . Clearing denominators,

$$F(z) := P(Qz)P(z) - a(z)P(z)R(Qz) + qR(z)R(Qz) \equiv 0.$$

At $z = 0$: $F(0) = P(0)^2 - (q + 1)P(0)R(0) + qR(0)^2 = 0$; if $R(0) = 0$ then $P(0) = 0$, contradicting coprimality, and symmetrically; so $P(0)R(0) \neq 0$ and $u(0) \in \{1, q\}$. *Top degree*: the three terms have degrees $2p, p + r + 1, 2r$. If $p = r$, the top degree $2p + 1$ is attained by the middle term alone, with coefficient $q\alpha\beta Q^r \neq 0$; if $|p - r| \geq 2$, the top degree is attained by the first or third term

alone, with coefficient $\alpha^2 Q^p$ or $q\beta^2 Q^r$. Both are impossible, so $|p - r| = 1$. *Root pairing:* let $\zeta \neq 0$ be a root of R of multiplicity m , so $P(\zeta) \neq 0$. At $z = \zeta/Q$ the three terms of F have orders k , $k + m + [a(\zeta/Q) = 0]$, $m_0 + m$, where $k = \text{ord}_{\zeta/Q} P$ and $m_0 = \text{ord}_{\zeta/Q} R$; since $F \equiv 0$ the minimum order cannot be attained by a single term, forcing $k \geq m_0 + m \geq m$. Symmetrically, a root $\xi \neq 0$ of P of multiplicity k forces $\text{ord}_{Q\xi} R \geq k$ (coprimality gives $\text{ord}_{Q\xi} P = 0$, so the first term has order k at ξ , the third $\text{ord}_{Q\xi} R$, and the middle at least $k + \text{ord}_{Q\xi} R$; again no single minimum). The two maps $\zeta \mapsto \zeta/Q$, $\xi \mapsto Q\xi$ are mutually inverse on the nonzero root sets, and chains close at length two by coprimality. Summing multiplicities, $p \leq r \leq p$, so $p = r$; contradicting $|p - r| = 1$. $\square$

Proposition 99 (the symmetric square). *Products $w = y\tilde{y}$ of solutions of L satisfy the third-order equation*

$$w(Q^3 z) = \frac{A_1(A_0 A_1 - q)}{A_0} w(Q^2 z) + q(q - A_0 A_1) w(Qz) + \frac{q^3 A_1}{A_0} w(z),$$

with $A_0 = a(z)$, $A_1 = a(Qz)$, obtained by eliminating the mixed product $m = y\tilde{y}(Q\cdot) + \tilde{y}y(Q\cdot)$ through $m(Qz) = 2A_0 w(Qz) - qm(z)$. Its characteristic roots at $z = 0$ are $\{1, q, q^2\}$; verified on the solutions G^2 , $z^{1/2}GH$, zH^2 to 41–43 digits.

Lemma 100 (divisor orientation). *Let $r \in \mathbb{C}(z)^\times$ satisfy the rational-quotient relation of Proposition 99, $r_0 r_1 r_2 = c_2 r_0 r_1 + c_1 r_0 + c_0$ with $r_k = r(Q^k z)$, $c_2 = A_1(A_0 A_1 - q)/A_0$, $c_1 = q(q - A_0 A_1)$, $c_0 = q^3 A_1/A_0$. On every Q -orbit in $\mathbb{C}^*$ disjoint from the singular orbit $z_a Q^\mathbb{Z}$ ($z_a = (q + 1)/q$, the zero of a), the maximal-modulus zero-or-pole of r cannot be a pole, and the minimal-modulus zero-or-pole cannot be a zero.*

Proof. Off $z_a Q^\mathbb{Z}$ the c_i are finite and $c_0 \neq 0$. If the maximal-modulus zero-or-pole ζ on such an orbit were a pole, evaluate at $z = Q^{-2}\zeta$: there r_0, r_1 are finite and nonzero (all higher-modulus orbit points carry no zero or pole), r_2 has a pole, so the left side has a pole while the right side is finite. If the minimal-modulus zero-or-pole ζ' were a zero, rewrite the relation for $s = 1/r$ as $c_0 s_0 s_1 s_2 + c_1 s_1 s_2 + c_2 s_2 = 1$ and evaluate at $z = \zeta'$: s_0 has a pole, s_1, s_2 are finite and nonzero, $c_0(\zeta') \neq 0$, so the left side has a pole while the right side is 1. $\square$

Theorem 101 (the difference Galois group contains SL_2). *For every $0 < q < 1$, the difference Galois group over $\mathbb{C}(z)$ of the module of G contains SL_2 .*

Proof. For $N \geq 1$ suppose $\text{Sym}^N M$ has a rank-1 submodule, i.e. there is a nonzero solution w of the N -th symmetric-power equation with $r := w(Q\cdot)/w \in \mathbb{C}(z)$. The products $f_1^{[\lambda]N-k} f_2^k$ ($0 \leq k \leq N$) form a solution basis (their Casoratian is a power of $\text{Cas}(f_1, f_2) \neq 0$; checked numerically for $N = 2$), so $w = \sum_k c_k f_1^{N-k} f_2^k$ with constants c_k , and the case split is λ -robust since Stokes jumps preserve $k_0 := \min\{k : c_k \neq 0\}$.

Case $k_0 < N$. Along a ray in a regular class, f_2/f_1 decays super-polynomially, so the rational function r has the Poincaré asymptotic expansion of the quotient of the dominant component:

$$r \sim (-1/z)^{N-k_0} (-z/q)^{k_0} \left(\frac{\hat{g}(Qz)}{\hat{g}(z)} \right)^{N-k_0} \left(\frac{h(Qz)}{h(z)} \right)^{k_0}.$$

The order- j coefficient d_j of $\hat{g}(Qz)/\hat{g}(z)$ satisfies $d_j \sim -b_j$ with $b_j = q^{(3j-j^2)/2}/(q; q)_j$ (verified: $d_j/(-b_j) = 0.99986, 0.99998, 0.999998$ at $j = 10, 12, 14$, $q = \frac{1}{3}$, converging to 1). Since this factor enters r to the power $N - k_0 \geq 1$ against rational, hence geometrically bounded, prefactors, the coefficients of r inherit the growth $b_j \sim q^{-j^2/2}$, which is super-geometric, whereas a rational function's expansion at ∞ has geometrically bounded coefficients. Since matching asymptotic expansions are unique, no rational r exists.

Case $k_0 = N$. Then $r = (-z/q)^N \rho^N$ with $\rho := h(Q \cdot)/h$ meromorphic on $\mathbb{C}^*$; $\rho^N \in \mathbb{C}(z)$ and single-valuedness force $\rho \in \mathbb{C}(z)$ (an N -th root of a rational function is single-valued on $\mathbb{C}^*$ only if every divisor multiplicity, including at 0 and ∞ , is divisible by N). Then $f_2(Qz)/f_2(z) = -(z/q)\rho(z)$ is rational, giving a rational solution of the rank-2 Riccati equation; contradicting Lemma 98.

Hence no $\text{Sym}^N M$ has a rank-1 submodule. $N = 1$ is irreducibility; $N = 2$ excludes the imprimitive (dihedral) case; and a projectively finite irreducible subgroup (A_4, S_4, A_5) fixes a line in some Sym^N with $N \leq 12$ (the classical binary polyhedral invariants), which is likewise excluded. By the classification of algebraic subgroups of GL_2 , the group contains SL_2 . $\square$

Proposition 102 (no $\sigma\delta$ -integrability). *For every $0 < q < 1$ there is no $B \in \text{gl}_2(\mathbb{C}(z))$ and no $c \in \mathbb{C}(z)$ with $\theta(A) = \sigma(B)A - AB + cA$, where $A = \begin{pmatrix} 0 & 1 \\ -q & a \end{pmatrix}$ and $\theta = z\partial_z$.*

Proof. Writing $B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, the entries give $\gamma = -q\beta_1$, $\delta = \alpha_1 + a\beta_1 - c$, and eliminating α (the central terms c cancel identically; verified symbolically) leaves the necessary condition

$$qa\beta_3 = a_1(aa_1 - q)\beta_2 - a(aa_1 - q)\beta_1 + qa_1\beta + qz(a_1 + q^2a),$$

$\beta_k = \beta(Q^k z)$, $a = a(z)$, $a_1 = a(Qz)$. A polynomial part of degree $m \geq 0$ contributes top coefficient $\propto \beta_m(1 - q^{2+2m}) \neq 0$ at degree $m + 3$, above the inhomogeneity: so β has no polynomial part. If β has a pole off the orbit $z_a Q^{\mathbb{Z}}$, the maximal-modulus pole appears in exactly one term (the slot $z = \zeta Q^{-3}$, coefficient $qa \neq 0$ there), and the minimal-modulus pole likewise (slot $z = \zeta'$, coefficient qa_1): both impossible; so the poles lie in the window $\{z_a Q^{-1}, \dots, z_a Q^3\}$ ($a(z_a Q^k) = (q+1)(1 - Q^k)$ vanishes only at $k = 0$). Sweeping the slots $z = z_a Q^{-4}, \dots, z_a Q^{-1}$ kills the orders at $z_a Q^{-1}, \dots, z_a Q^2$ one by one (each pole appears in a single term with nonvanishing coefficient), and the order at $z_a Q^3$ is killed at the slot $z = z_a Q$; coefficient $a_1(aa_1 - q)$ with $(aa_1 - q)(z_a Q) = (q+1)^2(1 - q^2)(1 - q^4) - q$; or, where that vanishes (one root $q \approx 0.78027$), at the slot $z = z_a Q^2$; coefficient $a(aa_1 - q)$ with $(aa_1 - q)(z_a Q^2) = (q+1)^2(1 - q^4)(1 - q^6) - q$ (one root $q \approx 0.86753$); the two polynomials are coprime, so for every $q \in (0, 1)$ at least one slot applies. Hence $\beta \equiv 0$, contradicting $z(a_1 + q^2a) = z[(q+1)(1 + q^2) - 2q^3z] \not\equiv 0$. (Cross-check: at $q = \frac{1}{3}$ the exact linear system for poles of order ≤ 3 in the window has $\text{rank}(A) = 15$, $\text{rank}[A|b] = 16$.) $\square$

Corollary 103 (differential transcendence of G). *For every $0 < q < 1$ the function $G(q, \cdot)$ satisfies no algebraic differential equation over $\mathbb{C}(z)$: the Hahn–Exton q -cosine is differentially transcendental.*

Proof, via parametrized difference-Galois theory. We apply the parametrized (σ, θ) -difference Galois theory with derivation $\theta = z\partial_z$, over the difference-Galois formalism of [37] and in the form of [13, 4]. The determinant equation $\sigma(y) = \det(A)y = qy$ has the solution $y = z^{1/2}$ (indeed $\sigma(z^{1/2}) = (Qz)^{1/2} = qz^{1/2}$, as $Q = q^2$), which is algebraic over $\mathbb{C}(z)$; hence the parametrized difference Galois group of $\sigma(y) = \det(A)y$ lies in $\mathbb{C}^\times$. This is exactly the hypothesis of Dreyfus–Hardouin–Roques [13, Prop. 2.9], which for a rank-2 system whose difference Galois group contains SL_2 (here Theorem 101) gives the dichotomy: the parametrized group Gal^θ either is conjugate into $\text{GL}_2(\mathbb{C})$ (all solutions differentially algebraic) or contains $\text{SL}_2(\tilde{\mathbb{C}})$ (all nonzero solutions differentially transcendental), and the *first* alternative holds if and only if there exists $B \in \text{gl}_2(\mathbb{C}(z))$ with $\theta(A) = \sigma(B)A - AB$ (their integrability condition (7); $\theta(\det A) = 0$ since $\det A = q$ is constant in z , so no trace term appears). By Proposition 102 no such B exists; indeed none exists even allowing a scalar twist cA . Therefore the second alternative holds: $\text{Gal}^\theta \supseteq \text{SL}_2(\tilde{\mathbb{C}})$, and every nonzero solution, in particular $G(q, \cdot)$, is differentially transcendental over $\mathbb{C}(z)$. $\square$

The transcendence of the *number* q^* is a value-level statement, and the natural route (Nesterenko’s multiplicity method fed by Corollary 103) meets a precise structural obstruction, which we isolate.

Proposition 104 (where the differential closure fails in the descent variable). Write $\delta = q\partial_q$, $\theta = z\partial_z$, and let $u_j(q) := G(q, Q^j z^*(q))$ denote the orbit values along the diagonal $z^*(q) = 2q(1 - q)$.

- (i) (orbit collapse) *The σ -orbit is not an obstruction to finiteness: the z -recurrence evaluated on the diagonal has coefficients $a(Q^j z^*) = q + 1 - qQ^j z^* \in \mathbb{Q}(q)$, so $u_{j+2} = a(Q^j z^*) u_{j+1} - q u_j$ and every orbit value lies in $\text{span}_{\mathbb{Q}(q)}\{u_0, u_1\}$ (verified for $j \leq 3$ to 32 digits).*
- (ii) (the E_2 -coupled jet structure) *The q -derivative of G obeys the exact identity*

$$\delta G = (\theta^2 - \theta)G + \varphi(q)G - T, \quad \varphi(q) = \sum_{i \geq 1} \frac{iq^i}{1 - q^i}, \quad T = \sum_{k \geq 0} (-1)^k \left(\sum_{i > 2k} \frac{iq^i}{1 - q^i} \right) \frac{q^{k(k-1)} z^k}{(q; q)_{2k}}$$

(verified to 41 digits), where φ is the quasimodular Eisenstein weight ($24\varphi = 1 - E_2$ up to normalization) and the correction T has coefficients smaller than G 's by the factor $\sum_{i > 2k} iq^i / (1 - q^i) = O(kq^{2k})$. Expanding these weights as Lambert series and interchanging the (absolutely convergent) sums exhibits T as an Eisenstein-weighted trace of the module over the forward σ -orbit:

$$T(q, z) = \sum_{d \geq 1} \frac{2q^d}{1 - q^d} (\theta G)(q, q^{2d}z) + \sum_{d \geq 1} \frac{q^d}{(1 - q^d)^2} G(q, q^{2d}z)$$

(verified to 41 digits): the entire non-modular content of δG is an orbit convolution of $(G, \theta G)$ against E_2 -type weights. On the diagonal the orbit values collapse by (i) into four generators u_0, u_1, θ -analogues v_0, v_1 (the v 's obey the inhomogeneous transported recurrence $v_{j+2} = a_j v_{j+1} - q v_j - qQ^j z^* u_{j+1}$), but the collapse coefficients accumulate into transfer-product series $A, B, C, D \in \mathbb{Z}[[q]]$ (integrality is immediate: transfer data and Lambert weights lie in $\mathbb{Z}[[q]]$), which are not of any classical type: they satisfy no relation over $\text{span}\{1, \varphi, \sum \sigma_3(n)q^n\}$ with rational-function coefficients of degree ≤ 8 , and no linear q -ODE of order ≤ 3 and degree ≤ 6 (exact linear algebra on 45 coefficients). The identity relocates, and does not remove, the obstruction. Since δ and θ commute, each application of δ raises the z -jet order by two: $\delta^n S$ involves $\theta^{2n} G|_{z=z^*}$ with unit leading coefficient.

- (iii) (the actual obstruction) *The z -jets do not close: $\theta^2 G$ and $\delta^2 G$ escape the rank-6 module of Proposition 96, and by Corollary 103 the tower $\{\theta^m G\}_m$ satisfies no algebraic relation over $\mathbb{C}(z)$. A finitely generated δ -closed $\mathbb{C}(q)$ -algebra containing S would force an algebraic relation over $\mathbb{C}(q)$ among the diagonal restrictions $\{\theta^m G(q, z^*(q))\}_{m \leq M}$ for some finite M ; a statement equivalent to S being differentially algebraic in q , for which there is no supporting structure and proof-grade negative evidence across a wide window: no algebraic differential equation at profiles up to $(2, 12, 3)$, $(3, 8, 3)$, $(4, 5, 6)$, $(5, 4, 8)$, $(6, 3, 15)$, $(8, 2, 35)$, $(12, 1, 120)$ in (order, degree, $\deg_q$), by full column rank on 2000 coefficients modulo a prime (`beta2_diffalg_search.py`, `beta2_ade_sweep.py`; E_2/q -Chazy control recovered).*

Thus the Nesterenko route closes if and only if S is differentially algebraic in q ; the obstruction is carried by the z -jet tower; whose function-level independence is exactly Corollary 103; and not by the σ -orbit, which collapses to rank two.

Remark 105 (the value-level descent as a joint multiplicity estimate). Nesterenko's method derives transcendence of a value from three ingredients: a finite ring closed under the relevant operator, a *multiplicity estimate* bounding $\text{ord } P(\text{generators}) \leq c(\deg P)^\kappa$, and routine analytic size/Siegel-lemma constructions. Theorem 101 and Proposition 102 furnish precisely the nondegeneracy a

multiplicity estimate consumes (no invariant subvariety; no first integral). But Proposition 104 shows the first ingredient fails in the descent variable q : the finiteness of the β_2 module is intrinsically *joint* in (σ_z, δ_q) , and σ_z is a non-infinitesimal q -lattice shift, outside the Pfaffian/differential multiplicity theories of Nesterenko and Binyamini. Thus the transcendence of $q^* = \beta_2^{-2}$ reduces to a *joint q -difference-differential multiplicity estimate* for the rank- ≤ 6 module at (q^*, z^*) : a bound $\text{ord}_{(q^*, z^*)} P(G, \theta G, \delta G) \leq c N^\kappa$ ($N = \deg P$) in the local ring, uniform along the σ -orbit. Such an estimate for finite (σ, δ) -modules is not available in the literature; it is the sole non-routine input remaining, and the precise open problem on which the (ir)rationality of β_2 now turns.

Remark 106 (the obstruction inside q -Siegel–Shidlovskii theory). The same wall appears, quantified, in the value-transcendence theory built on Siegel’s method. The approach of Amou–Matala-aho–Väänänen [2] (and the Matala-aho-school Baker-type refinements) proves non-vanishing of values of Poincaré-type q -difference solutions at algebraic points under two hypotheses: the system is Poincaré-type at 0 (met here: $A(0)$ is invertible, $\det A = q$), and a *Siegel margin* $\gamma < \Gamma$ holds, where γ is the ratio of the logarithmic height of the cleared denominators to that of the analytic decay. For the Hahn–Exton class that ratio is exactly $\gamma = \rho = \log(t/s)/(2 \log t) \leq \frac{1}{2}$ of Proposition 87: the double Pochhammer $(q; q)_{2k}$ is the factor 2; so the margin *fails*, and the theory does not apply. Thus Conjecture Z (transcendence of the Hahn–Exton zeros at algebraic base, which by $z^* = z_1(q^*)$ implies β_2 transcendental) reduces to extending q -Siegel–Shidlovskii past the margin for a class that is additionally irregular at ∞ . The connection theory of this paper supplies the irregular- ∞ data explicitly; the margin itself is the value-theoretic face of the multiplicity estimate of Remark 105, and is equal in strength to β_2 -transcendence, not weaker.

One further mechanism deserves record, because it isolates which half of the Siegel scheme is actually blocked. Under the rationality hypothesis all tail power sums $\tau_p = \sum_{k \geq 2} z_k^{-p} = \sigma_p - z^{*-p}$ are rational, and they are the moments of a positive measure with infinite support (unit atoms at $1/z_k$); hence every Hankel determinant $\det[\tau_{i+j+2}]_{i,j \leq n}$ is *strictly positive for free*; no auxiliary nonvanishing argument is needed; and, being a positive rational, is bounded below by the reciprocal of its denominator ($\leq t^{8n^3/3+O(n^2)}$). Analytically these determinants decay like $Q^{n^3/3(1+o(1))}$ (atoms at $\approx Q^{k-1}$; verified: strictly positive with super-quadratic exponent growth at q^*). The comparison $2 \log(t/s) > 8 \log t$ fails for every rational q : the positivity route conserves the ledger at a factor 4. The instructive point is that *nonvanishing was never the bottleneck* of any construction in this ledger; positivity supplies it gratis here; the entire difficulty of the problem resides in the size-versus-height comparison. Nor is there an Apéry-type denominator miracle hiding in that comparison: the *measured* reduced denominators of the partial sums $S_N(s/t)$ equal 96–100% of the naive clearing $\prod_{i \leq 2N} (t^i - s^i)$ in logarithmic size, with the ratio tending to 1 as N grows (exact rational arithmetic, $N \leq 12$, at $q = \frac{1}{2}$ and $\frac{2}{5}$); the cancellation is an $O(N)$ boundary effect in an $O(N^2)$ exponent. The arithmetic of the ledger is exactly its worst-case costing.

Remark 107 (backward-orbit normal form of the zero condition). The zero condition admits a compression into a single explicit rational sequence. If $G(q, z_1) = 0$, the three-term recurrence along the backward σ -orbit, $u_{-m} = (a(Q^{-m} z_1) u_{-m+1} - u_{-m+2})/q$ with $u_n = G(q, Q^n z_1)$, loses one solution constant: $u_{-m} = u_1 r_m(q)$ with $r_0 = 0$, $r_1 = -1/q$, and $r_m \in \mathbb{Q}(q, z_1)$ computable by the same recurrence. The asymptotics of Theorem 94 then give the exact evaluation

$$\lim_{m \rightarrow \infty} r_m(q) Q^{m(m+1)/2} (-z_1)^{-m} = \frac{\theta_Q(-z_1)}{(q; q)_\infty G(q, Q z_1)},$$

with geometric rate Q^m and q -Gevrey correction series $\sum_j b_j z_1^{-j} Q^{mj}$ (verified to 17 digits at q^* by $m = 24$). Under the hypothesis $q^* = s/t$ the normalized terms are rationals of height $(st)^{m^2+O(m)}$ approximating the limit to $O(Q^m)$; accelerated forms that cancel J correction terms gain decay

Q^{mJ} but pay the quadratic heights of the b_j , and the arithmetic ratio stays $\leq \frac{1}{2}$; the balance of Proposition 87 once more. The normal form is recorded as the sharpest single-sequence statement of what any future method must exploit: the entire rationality hypothesis is equivalent to the arithmetic of this one explicit sequence. The θ -jet admits the same normal form one level up: along the backward orbit,

$$\frac{(\theta G)(q, Q^{-m} z_1)}{G(q, Q^{-m} z_1)} - m \longrightarrow -\frac{z_1 \theta'_Q(-z_1)}{\theta_Q(-z_1)} \quad (\text{rate } Q^m; \text{ verified to 10 digits at } q = 0.3),$$

an exact theta log-derivative: each backward step increments $\theta \log \theta_Q(-z)$ by exactly one (quasi-periodicity), and the $\hat{g}$ -correction vanishes in the limit.

The *forward* orbit admits the complementary normal form, in the language of rational dynamics. The Riccati relation gives the explicit continued fraction $q/(a(z) - q/(a(Qz) - \cdots))$, which by Pincherle's theorem [26] converges to the minimal-solution ratio $q H(Qz)/H(z)$; the zero condition $G(q, z_1) = 0$ forces the *dominant* ratios $w_n := G(q, Q^{n+1} z_1)/G(q, Q^n z_1)$ to obey $w_1 = a(z_1)$ exactly and thereafter the rational map

$$w_{n+1} = a(Q^n z_1) - \frac{q}{w_n},$$

so that under the hypothesis $q, z_1 \in \mathbb{Q}$ the entire forward orbit is an explicit $\mathbb{Q}$ -orbit converging to the attracting fixed point 1 with $w_n - 1 = \Theta(Q^n)$ (verified to 25 digits at q^*). Its heights compound quadratically ($t^{O(n^2)}$), so the Liouville comparison Q^n vs. t^{-n^2} is again consistent: both normal forms; backward theta-growth and forward rational dynamics; conserve the balance of Proposition 87, and together they exhaust the one-orbit reformulations of the zero condition.

Combining the two orbits with the q -Wronskian makes the relation-count of the rationality hypothesis explicit. Under $G(q, z_1) = 0$ the unknown transcendental data reduces to three numbers ($g_1 := G(q, Qz_1)$, $H(q, z_1)$, and the backward limit ρ_∞), bound by exactly two relations: $g_1 H(q, z_1) = 1 - q$ (the q -Wronskian at the zero) and $\rho_\infty g_1 = \theta_Q(-z_1)/(q; q)_\infty$ (the backward normal form). Eliminating g_1 ,

$$\frac{\rho_\infty}{H(q, z_1)} = \frac{\theta_Q(-z_1)}{(1 - q)(q; q)_\infty}$$

(verified to 32 digits at q^*), with modular right-hand side. Three unknowns, two relations: the rationality hypothesis is consistent at this level, and a proof requires precisely one further independent relation; the multiplicity statement of Remark 105, in yet another equivalent form.

Remark 108 (truncation roots, and the exact deficit). A scheme not covered by the lattice ledger prices the wall in its simplest form. Clearing the truncation of S gives integer polynomials $P_N = (q; q)_{2N} \sum_{k \leq N} (-2)^k q^{k^2} (1 - q)^k / (q; q)_{2k} \in \mathbb{Z}[q]$ of degree $2N^2 + N$, whose heights are only $e^{O(N)}$ (the pentagonal sparsity of q -Pochhammer coefficients; measured $\log_2 H = 2.6, 4.6, \dots, 12.1$ for $N = 2, \dots, 7$), and whose root nearest q^* satisfies $|q_N - q^*| \asymp q^{*(N+1)^2}$ (verified $N \leq 7$). If $q^* = s/t$ then $|P_N(q^*)| \geq t^{-(2N^2+N)}$, while analytically $|P_N(q^*)| = e^{O(N)} q^{*(N+1)^2}$; contradiction would require $q^* < t^{-2}$, impossible for $q^* = s/t \geq 1/t$. The deficit is exactly the factor 2 in the clearing degree; the double Pochhammer once more, now in s -independent form. The same computation shows that an equivalent representation of the root condition with *single*-Pochhammer denominators (clearing degree $\sim N^2/2$) would yield $q^* \neq s/t$ for all $s < \sqrt{t}$. No such representation exists within the Rogers–Ramanujan/Bailey framework [17]: those transformations require the argument to be a q -power, and the argument here is the catalytic point $z = 2q(1 - q)$. In this precise sense the obstruction to β_2 's irrationality is the non- q -power (catalytic) argument itself.

The catalytic locus is moreover *seat-independent*. Writing $S(q) = {}_1\phi_1(0; q; Q, z^*)$ ($Q = q^2$), the parameter–argument transformation ${}_1\phi_1(0; b; Q, x) = \frac{(x; Q)_\infty}{(b; Q)_\infty} {}_1\phi_1(0; x; Q, b)$ gives the equivalent zero

condition

$$T(q) := {}_1\phi_1(0; 2q(1-q); q^2, q) = \sum_{k \geq 0} \frac{(-1)^k q^{k^2}}{(q^2; q^2)_k (2q(1-q); q^2)_k} = 0$$

(verified to 51 digits; the prefactor cannot vanish since $z^* < 1$), in which the argument is the q -power q and the catalytic value sits in the *parameter*; equivalently, q^* is the point where the Hahn–Exton q -Bessel of q -dependent order $\nu(q) = \log_{q^2}(2q(1-q)) - 1$ vanishes at the fixed argument $q^{-1/2}$. The clearing degree of T ’s truncations is again $\sim 2N^2$: the deficit is invariant under the swap, and in either seat it is the non-special value $2q(1-q)$ that lies outside every Rogers–Ramanujan/Bailey lattice. The well-poised generalization does not help: a WP-Bailey specialization producing a single-Pochhammer form would require $2q(1-q)$ to be a q -power (the only special-value type such summations admit), whereas at $q = q^*$ one has $q^* < 2q^*(1-q^*) < 1$, strictly between consecutive q -powers (the two loci where $2q(1-q)$ is a q -power, $q = \frac{1}{2}$ and $q = \frac{2}{3}$, both miss q^*). The catalytic argument is thus q -power-free in every seat, which is the transformation-theoretic content of the $\rho = \frac{1}{2}$ wall.

Proposition 109 (effective non-rationality of q^*). *The least positive root q^* of S admits no rational representation s/t with $t \leq 10^{1044}$. Consequently, if $\beta_2 = u/v$ in lowest terms then $u > 10^{522}$.*

Proof. All steps are certified interval arithmetic (outward rounding; `mpmath.iv`; script `beta2_effective_irrational` with the series tail bounded by $(q; q)_{2k} \geq (1-2q)/(1-q) > 0$ for $q < \frac{1}{2}$ and term ratios $< \frac{1}{2}$ beyond the truncation. (i) $S > 0$ on $[0, 0.40]$ (400 certified chunks). (ii) $S' \leq -1$ on $[0.40, 0.46]$ (600 certified chunks; S' evaluated term-by-term with the logarithmic-derivative weights), so S has at most one zero in $[0.40, 0.46]$. (iii) with c the 2090-digit truncation of q^* , $a = c - 2 \cdot 10^{-2090}$ and $b = c + 2 \cdot 10^{-2090}$, certified evaluation gives $S(a) > 0 > S(b)$ (working precision 2130 digits, series tail enclosed in $\pm 10^{-2125}$); by (i)–(ii), $q^* \in [a, b]$. (iv) the interval $[a, b]$ contains the rational s_0/t_0 (Stern–Brocot construction) with $9.5 \cdot 10^{1044} < t_0 < 10^{1045}$, verified by exact rational comparison, and certified $S(s_0/t_0) \neq 0$. If $m/n \in [a, b]$ with $n \leq 10^{1044}$ and $m/n \neq s_0/t_0$, then $|m/n - s_0/t_0| \geq 1/(nt_0) > 10^{-2089} > b - a = 4 \cdot 10^{-2090}$, impossible; so $m/n = s_0/t_0$, whence $n = t_0 > 10^{1044}$, a contradiction. Thus no rational with denominator $\leq 10^{1044}$ lies in $[a, b]$ at all, in particular none equals q^* . For $\beta_2 = u/v$ in lowest terms, $q^* = v^2/u^2$ in lowest terms, so $u^2 > 10^{1044}$. $\square$

Proposition 109 is the degree-one case of a finite algebraic exclusion. The same certified bracket excludes low-degree algebraic relations. For $P = \sum_{i \leq d} p_i x^i \in \mathbb{Z}[x]$ write $H(P) = \max_i |p_i|$ for the height.

Proposition 110 (certified finite algebraic exclusion; computer-assisted). *Let q^* be the least positive zero of S and $\beta_2 = 1/\sqrt{q^*}$, and let $P \in \mathbb{Z}[x] \setminus \{0\}$.*

- (i) *If $P(q^*) = 0$, then $H(P) \geq 10^{230}$ when $\deg P \leq 8$, $H(P) \geq 10^{120}$ when $\deg P \leq 16$, $H(P) \geq 10^{80}$ when $\deg P \leq 24$, and $H(P) \geq 10^{28}$ when $\deg P \leq 64$.*
- (ii) *If $P(\beta_2) = 0$, then $H(P) \geq 10^{228}$ when $\deg P \leq 8$, and $H(P) \geq 10^{118}$ when $\deg P \leq 16$.*

Proof. The proof is a finite computation in two steps. The scripts are `step1_bracket.py` and `step2_lattice.py`, in the folder `code/zeta_probe/beta2_certificate/`. The trust base is outward-rounded interval arithmetic (`mpmath.iv`) in step 1, and exact integer and rational arithmetic (`python-flint` and Python integers) in step 2.

Step 1: certified bracket. Steps (i)–(iii) of the proof of Proposition 109 give $q^* \in [a, b]$ with $a = (A - 2) \cdot 10^{-2090}$, $b = (A + 2) \cdot 10^{-2090}$ and $A \in \mathbb{Z}$ the 2090-digit truncation of q^* . Then

$\beta_2 \in [1/\sqrt{b}, 1/\sqrt{a}] \subset [\beta_{\text{lo}}, \beta_{\text{hi}}]$ with $\beta_{\text{lo}} = m_{\text{lo}} \cdot 10^{-2080}$, $\beta_{\text{hi}} = m_{\text{hi}} \cdot 10^{-2080}$ and $m_{\text{hi}} - m_{\text{lo}} = 5$. The two inclusions are checked exactly, as the integer inequalities $m_{\text{lo}}^2(A+2) \leq 10^{2 \cdot 2080 + 2090} \leq m_{\text{hi}}^2(A-2)$.

Step 2: lattice. Let $x^* \in [c, c+w]$ be one of the two numbers, with c rational and w the width of its bracket. Let R bound $|x|$ on the bracket ($R = 1$ for q^* , $R = \frac{3}{2}$ for β_2), and put $M = \lfloor 1/w \rfloor$. Let $L_d \subset \mathbb{Z}^{d+2}$ be the lattice with basis rows $e_i + \lfloor Mc^i \rfloor e_{d+1}$, $0 \leq i \leq d$. To P with $\deg P \leq d$ assign $v_P = (p_0, \dots, p_d, \sum_i p_i \lfloor Mc^i \rfloor) \in L_d$. It is nonzero when $P \neq 0$. Its last coordinate differs from $MP(c)$ by at most $(d+1)H(P)$. If $P(x^*) = 0$, then $|P(c)| \leq w \max_{[c, c+w]} |P'| \leq wH(P)s_d$ with $s_d = \sum_{i=1}^d iR^{i-1}$, and $Mw \leq 1$. Hence

$$\|v_P\| \leq H(P)K_d, \quad K_d^2 = (d+1) + (s_d + d+1)^2.$$

For any basis $b_0, \dots, b_d$ of L_d with Gram–Schmidt vectors b_k^* , every nonzero lattice vector has norm at least $\min_k \|b_k^*\|$. We take the basis returned by LLL reduction ($\delta = 0.99$). LLL serves only to find a good basis: the script checks exactly that the output is TB with B the basis above and $|\det T| = 1$, so it is a basis of L_d . It then computes $\|b_k^*\|^2 = \Delta_k/\Delta_{k-1}$ exactly, where Δ_k are the leading principal minors of the integer Gram matrix. Any P with $P(x^*) = 0$ and $H(P)K_d < \min_k \|b_k^*\|$ would give a nonzero lattice vector shorter than this bound, which is impossible. The certified values of $\log_{10}(\min_k \|b_k^*\|/K_d)$ are 230.45, 120.63, 80.90 and 28.29 for q^* at $d = 8, 16, 24, 64$, and 228.66 and 118.15 for β_2 at $d = 8, 16$. These give the stated bounds. $\square$

Controls and consistency. As a negative control, the same procedure applied to $x^* = \sqrt{2} - 1$, a root of $x^2 + 2x - 1$ of height 2, certifies at $d = 2$ only $H < 10^{-0.30}$, which excludes nothing, as it must. At $d = 1$ the procedure gives $H(P) \geq 10^{1043.99}$ for q^* . This agrees with Proposition 109, whose Stern–Brocot argument on the same bracket gives the marginally sharper $t > 10^{1044}$. Both statements are finite. Neither says anything about irrationality or transcendence. A PSLQ search that returns “no relation” is weaker still: it is heuristic, and it is not a proof of any exclusion.

Remark 111 (the obstruction to transcendence and to irrationality of β_2). No available theory turns Propositions 109 and 110 into an irrationality or transcendence proof. Three features of the problem are responsible.

- (i) *A zero in the base variable.* q^* is a zero of S in the base q itself, not a value at a fixed base. Base-variable transcendence results exist for modular forms (Nesterenko [31]) and for Mahler functions (Nishioka [32]). S is neither modular nor Mahler in any available sense. No modular transformation law is known for it. A search to $O(q^{48})$ finds no Mahler equation for S (Remark 85), and for the travel resolvent $G_0^T = \Sigma_0/S$ [7, Proposition prop:noqdiff] excludes Mahler equations unconditionally.
- (ii) *The fixed-base ratio is $\rho = \frac{1}{2}$.* Freezing the base and working in the argument, as the q -Siegel–Shidlovskii and q -irrationality methods do, gives the arithmetic ratio $\rho = \log(t/s)/(2 \log t) \leq \frac{1}{2}$ of Proposition 87, with equality at the most favourable bases $q = 1/t$. The factor 2 is the double Pochhammer $(q; q)_{2k}$. The fixed-base value theorems of Amou–Katsurada–Väänänen [1] and of Krattenthaler–Rochev–Väänänen–Zudilin [24] therefore do not apply: their constructions need the decay of the terms to outrun the growth of the cleared denominators, and at $\rho = \frac{1}{2}$ it does not. The same margin fails in Remark 106.
- (iii) *The Hahn–Exton identification is analytic only.* [7, Corollary “the travel poles are Hahn–Exton zeros”, cor:qbessel] identifies the zeros of S in $(0, 1)$ with the zeros of $J_{-1/2}^{(3)}(Z(q); q^2)$. This is an identity of analytic functions. It transports zeros, asymptotics and q -Wronskians, but no arithmetic. No irrationality or transcendence theorem is known for zeros of Hahn–Exton functions, and here the base is itself the unknown.

D Proof of the block assembly (M3')

Status. This appendix proves the assembly step (M3) of [7, Theorem “the transfer model (M)”, thm:model] in the corrected form [7, (eq:assembly)]. It is a *hand proof*. Two independent adversarial reviews reconstructed it and found it correct; the second also compared an independent tuple enumerator against the model, with no mismatch. The proof is *not* formalised. The Lean development formalises the objects it is about: the site costs, the cut predicate and $\ell_R^{\text{true}}, c^{\text{true}}$ (`Realisation.lean`, `CorrectedSpan.lean`), the junction-cut characterisation (`RJLemmaJ.lean`), and the metric identity $|g| = \ell_R^{\text{true}}(g) + 2c^{\text{true}}(g)$ (`metricAll`). The proof is supported by the computational checks recorded in §7.1: with y symbolic, all 816 coefficients with $\ell_R + 2c \leq 31$ agree with exhaustive enumeration in all three sectors (5.03×10^6 elements), and a deliberately corrupted model is detected; at $y = q$ the model agrees to x^{26} and at $y = q^2$ to x^{22} . The checks are evidence, not part of the proof.

In this appendix we write k, ε, δ for the marker data $k^*, \varepsilon^*, \delta^*$ of the main text, and `Elt`, `SameElt` for the element type of the Lean development and its equality of the four normal-form fields [5, Theorem “Normal form and lamp model”, thm:normal-form].

D.1 Setting and statement

Elements. Edges and sites are indexed by $\mathbb{Z}$. Site s lies between edge $s - 1$ and edge s . For $k \in \mathbb{Z}$, the *travel indicator* is

$$f_j^{(k)} = \begin{cases} +1, & 0 \leq j < k, \\ -1, & k \leq j < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Let $\mathcal{G}$ be the set of tuples $(k, \varepsilon, \delta, d)$ such that

- $k \in \mathbb{Z}$, $\varepsilon \in \{\pm 1\}$ and $\delta \in \{0, 1\}$;
- $d : \mathbb{Z} \rightarrow \mathbb{Z}$ has finite support;
- $d_j \equiv f_j^{(k)} \pmod{2}$ for all j .

Lemma 112 (classes). *The map $g \mapsto (g.\text{kstar}, g.\text{eps}, g.\text{delta}, g.\text{d})$ induces a bijection from SameElt-classes of `Elt` onto $\mathcal{G}$. The functions ℓ_R^{true} and c^{true} of `CorrectedSpan.lean` are constant on classes, so they are functions on $\mathcal{G}$.*

Proof. `SameElt` is by definition equality of the four fields, so the induced map is well defined and injective. The fields of an `Elt` satisfy the conditions defining $\mathcal{G}$: `hpar` gives the parity condition, and `hsupp` gives finite support.

For surjectivity, take $(k, \varepsilon, \delta, d) \in \mathcal{G}$. Put $\text{supp} = \text{supp } d \cup \{j : f_j^{(k)} \neq 0\}$. This set is finite, and it gives an `Elt` with the prescribed fields.

The only field not determined by the class is `supp`. It enters ℓ_R^{true} and c^{true} only through $\text{occTrue} = \text{supp} \cap \{j : d_j \neq 0 \vee f_j \neq 0\}$. By `hsupp`, this intersection equals $\{j : d_j \neq 0 \vee f_j \neq 0\}$, which is independent of `supp`. All the other ingredients are functions of $(k, \varepsilon, \delta, d)$: `mu`, `siteCost`, `cut` and `ShieldFires`. $\square$

From now on we write $g = (k, \varepsilon, \delta, d) \in \mathcal{G}$ and $f = f^{(k)}$.

Local data. These are transcribed from `Realisation.lean`. Write $[\cdot]$ for the Iverson bracket.

$$\begin{aligned}\alpha_s &= d_{s-1} - [s = 0] + \varepsilon[s = k][\delta = 0], & \beta_s &= d_s - \varepsilon[s = k][\delta = 1], \\ \Phi_s &= f_{s-1} + [s = 0] - [s = k][\delta = 0], & \text{site}(s) &= \max(|\alpha_s|, |\beta_s|), \\ \mu_j &= \begin{cases} 2, & d_j = f_j = 0, \\ \max(|d_j|, |f_j|), & \text{otherwise,} \end{cases} & \text{cut}(s) &\iff \alpha_s = \beta_s = \Phi_s = 0.\end{aligned}$$

The term $-[s = 0]$ is the virtual arrival **vArr**. It is the “ ± 1 correction” at the junction.

Corrected span. These are transcribed from `CorrectedSpan.lean`. Let $\text{occ} = \{j : d_j \neq 0 \vee f_j \neq 0\}$. Then

$$\begin{aligned}A &= \begin{cases} \min(0, \min \text{occ}), & \text{occ} \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases} & B &= \begin{cases} \max(-1, \max \text{occ}), & \text{occ} \neq \emptyset, \\ -1, & \text{otherwise,} \end{cases} \\ \ell_R^{\text{true}}(g) &= \sum_{j=A}^B \mu_j + \sum_{s=A}^{B+1} \text{site}(s), & c^{\text{true}}(g) &= \#\{s : A < s < B + 1, \text{cut}(s)\} + [\text{SF}(g) \wedge \text{cut}(0)],\end{aligned}$$

where the *boundary shield* is

$$\text{SF}(g) \iff k = 0 \wedge \delta = 0 \wedge (\forall j < 0 : d_j = 0) \wedge (\exists j \geq 0 : d_j \neq 0).$$

Series. Put $q = x^2$ and

$$\mathfrak{g} = \sum_{L \geq 1} q^L y^{L-1}.$$

The bulk vector $P = (P_b)_{b \geq 1}$ is the solution of

$$P_b = 2q^b \left(q^b + \sum_{a \geq 1} K_{\mathfrak{g}}(a, b) P_a \right), \quad K_{\mathfrak{g}}(a, b) = q^{\max(a, b)} + q^{a+b} \mathfrak{g}, \quad (38)$$

which exists and is unique by Lemma 115. From it we form

$$S = \sum_{b \geq 1} q^b P_b, \quad B_0 = 1 + \mathfrak{g}S, \quad B_{0z} = 1 + y \mathfrak{g}S,$$

and, for $s \geq 0$,

$$\mu_s^{(y)} = B_0 x^{2s+1} + \sum_{b \geq 1} \frac{P_b}{2} (x^{\max(2b-1, 2s+1)} + x^{\max(2b+1, 2s+1)}), \quad (39)$$

$$E_s^{(y)} = B_0^{[s]} x^{2s} + B_0 x^{2s+2} + \sum_{b \geq 1} P_b (x^{\max(2s, 2b)} + x^{\max(2s+2, 2b)}), \quad B_0^{[0]} = B_{0z}, \quad B_0^{[s]} = B_0 \quad (s \geq 1), \quad (40)$$

$$\lambda_s^{(y)} = E_s^{(y)} + 2\mu_s^{(y)}. \quad (41)$$

Let $D_e = \text{diag}(e_s)_{s \geq 0}$ with $e_s = 2q^{1+s}$, let $K(s, t) = q^{\max(s, t)}$, and let $\mathcal{T} = D_e K$. Define

$$\mathcal{W}_+ = x^{-1} \langle \mu^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \quad \mathcal{W}_- = x^{-1} \langle \tfrac{1}{2} E^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle,$$

and let $\mathcal{W}_0$ be the branch sum (48) of §D.8, a double sum over the left and right chain kinds (summable, not finite).

Theorem 113 ((M3')). *The following identity holds in $\mathbb{Z}[y][[x]]$:*

$$\sum_{[g] \in \text{Elt}/\text{SameElt}} x^{\ell_R^{\text{true}}(g)} y^{c^{\text{true}}(g)} = \sum_{g \in \mathcal{G}} x^{\ell_R^{\text{true}}(g)} y^{c^{\text{true}}(g)} = \mathcal{W}_+ + \mathcal{W}_- + \mathcal{W}_0.$$

The three terms are the contributions of the sectors $k > 0$, $k < 0$ and $k = 0$. Moreover

$$\mathcal{W}_+ + \mathcal{W}_- = \frac{1}{2x} \langle \lambda^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle,$$

and each of $\mathcal{W}_+$, $\mathcal{W}_-$, $\mathcal{W}_0$ lies in $\mathbb{Z}[y][[x]]$.

Corollary 114. *By metricAll (Lean: $|g| = \ell_R^{\text{true}}(g) + 2c^{\text{true}}(g)$ for every Elt , all of which are reachable), the specialisation $y = x^2$ of Theorem 113 is the growth series $\sum_{[g]} x^{|g|}$ of the group modelled by Elt . This group is W_{univ} . For a particular triangle group W_T , the growth series agrees with this one only up to the depth at which ρ_T stops being faithful on W_{univ} [5, Theorem “Effective universality”, thm:effective-universality]. For the (1, 2) triangle the first deviation is at depth 33 [5, thm:universality-sharp].*

The first equality of Theorem 113 is Lemma 112. The rest of this appendix proves the second.

D.2 Convergence and integrality

We use the x -adic topology on $\mathbb{Z}[y][[x]]$. A family of series is *summable* if only finitely many of its members have a nonzero coefficient of any given x^n . We write ord for the x -adic order.

Lemma 115. (a) *For each n , only finitely many $g \in \mathcal{G}$ have $\ell_R^{\text{true}}(g) = n$, and for these $c^{\text{true}}(g) \leq n + 1$. So the left side of Theorem 113 is a well-defined element of $\mathbb{Z}_{\geq 0}[y][[x]]$.*

(b) *Equation (38) has a unique solution with $P_b \in \mathbb{Z}[y][[x]]$ and $\text{ord } P_b \geq 4b$. Every P_b is divisible by 2.*

(c) *$(I - \mathcal{T})^{-1} = \sum_{n \geq 0} \mathcal{T}^n$ converges entrywise, and every pairing in the definitions of $\mathcal{W}_{\pm}$ is a summable sum.*

(d) *$\mu^{(y)}$, $E^{(y)}$, $\lambda^{(y)}$, $\mathcal{W}_+$, $\mathcal{W}_-$ and $\mathcal{W}_0$ all have coefficients in $\mathbb{Z}[y]$.*

Proof. (a) Every edge of $[A, B]$ has $\mu_j \geq 1$, so $B - A + 1 \leq n$. Moreover $|d_j| \leq \mu_j \leq n$ and $|k| \leq B - A + 1$. This leaves finitely many choices of $(k, \varepsilon, \delta, d)$. Also $c^{\text{true}} \leq \#\{s : A < s < B + 1\} + 1 = (B - A + 1) + 1 \leq n + 1$.

(b) Let $\Psi(P)_b = 2q^b(q^b + \sum_a K_{\mathfrak{g}}(a, b)P_a)$. The entry $K_{\mathfrak{g}}(a, b)$ has order $\geq 2 \max(a, b) \geq 2a$, so $\sum_a$ is summable whenever $\text{ord } P_a \geq 0$. If two inputs satisfy $\text{ord}(P_a - P'_a) \geq m$ for all a , then $\text{ord}(\Psi(P)_b - \Psi(P')_b) \geq 2b + 2 + m \geq m + 4$. So Ψ is a contraction. Starting from $P = 0$, the iterates therefore converge to the unique fixed point. Every iterate has $\text{ord } P_b \geq 4b$ and carries the factor 2. The coefficients lie in $\mathbb{Z}[y]$, since the coefficient of x^n in $\mathfrak{g}$ is 0 or $y^{n/2-1}$.

(c) $\mathcal{T}(s, t) = 2x^{2s+2+2\max(s, t)}$ has order $\geq 2s + 2t + 2$. So $(\mathcal{T}^n)(s, t)$ has order $\geq 2n + 2t$, the Neumann series converges, and for fixed s the sums over t are summable. By (b), $\text{ord } \mu_s^{(y)}, \text{ord } E_s^{(y)} \geq 2s$, so the outer pairings are summable as well.

(d) $\mu_s^{(y)}$ is integral because $P_b/2$ is integral, by (b). For $\mathcal{W}_{\pm}$, factor $D_e = 2D'$ with $D' = \text{diag}(q^{1+s})$. The push-through identity gives

$$(I - D_e K)^{-1} D_e = D_e (I - K D_e)^{-1} = 2D' (I - K D_e)^{-1}.$$

Hence

$$\mathcal{W}_- = x^{-1} \langle \frac{1}{2} E^{(y)}, 2D'(I - KD_e)^{-1} \lambda^{(y)} \rangle = \langle E^{(y)}, (x^{-1} D')(I - KD_e)^{-1} \lambda^{(y)} \rangle,$$

and $x^{-1} D' = \text{diag}(x^{2s+1})$ is integral. **So $\mathcal{W}_-$ is integral because D_e carries the factor 2.** The $\frac{1}{2}$ on $E^{(y)}$ is cancelled by that factor, and $E^{(y)}$ itself is not divisible by 2: its $s = 0$ entry has constant term 1. The same factorisation shows $\mathcal{W}_+ \in \mathbb{Z}[y][[x]]$, with the factor 2 to spare. $\mathcal{W}_0$ is a summable double sum of products $x^j w_L w_R$ with weights from 1, $\mathfrak{g}S$, $y\mathfrak{g}S$ and $P_b/2$, as (48) shows. $\square$

All rearrangements below are of families of monomials with nonnegative integer coefficients, and each family is summable by Lemma 115(a). So they are legitimate in $\mathbb{Z}[y][[x]]$.

D.3 Decomposition of an element: chains and the clamp analysis

Chains. A *chain* is a finitely supported $c : \mathbb{Z}_{\geq 1} \rightarrow 2\mathbb{Z}$. Its *length* is $N(c) = \max\{i : c_i \neq 0\}$, with $N(0) = 0$. There are three kinds of chain:

- *empty* (kind E): $N = 0$;
- *gap-opening* (kind Z): $N \geq 1$ and $c_1 = 0$;
- *deposit-opening* (kind $N_{\sigma,b}$): $c_1 = 2\sigma b$ with $\sigma = \pm 1$ and $b \geq 1$.

Given $g = (k, \varepsilon, \delta, d)$, put $k_- = \min(0, k)$ and $k_+ = \max(0, k)$. The *left chain* and the *right chain* of g are

$$c_i^L = d_{k_- - i}, \quad c_i^R = d_{k_+ - 1 + i} \quad (i \geq 1).$$

Both consist of even deposits, since $f = 0$ outside $[k_-, k_+)$. The *travel deposits* are d_j for $k_- \leq j < k_+$. There are $|k|$ of them, all odd. Write $\ell = N(c^L)$ and $r = N(c^R)$.

Lemma 116 (bijection). *The map*

$$g \mapsto (k, \varepsilon, \delta, (d_j)_{k_- \leq j < k_+}, c^L, c^R)$$

is a bijection from $\mathcal{G}$ onto the set of tuples consisting of $k \in \mathbb{Z}$, ε, δ , a word of $|k|$ odd integers, and two chains.

Proof. The three index sets $\{j < k_-\}$, $[k_-, k_+)$ and $\{j \geq k_+\}$ partition $\mathbb{Z}$, and the parity condition says exactly that the deposits are odd on the middle set and even off it. $\square$

Lemma 117 (span; clamp analysis). *For every $g \in \mathcal{G}$,*

$$A = k_- - \ell, \quad B = k_+ - 1 + r.$$

Hence:

(i) *The edges of $[A, B]$ are the ℓ left-chain edges, the $|k|$ travel edges and the r right-chain edges.*

(ii) *The site range $[A, B + 1]$ consists of the following sites:*

- *the junction sites k_- and k_+ , which coincide when $k = 0$;*
- *the $|k| - 1$ travel-interior sites $k_- < s < k_+$;*
- *the chain-interior sites: $A < s < k_-$ on the left and $k_+ < s < B + 1$ on the right;*
- *the outer sites A and $B + 1$, but only when $\ell > 0$, resp. $r > 0$.*

(iii) The junction site k_- lies in the open interval $(A, B + 1)$ if and only if $\ell > 0$ or $k < 0$, i.e. $k_- < k_+$. Symmetrically, k_+ lies in it if and only if $r > 0$ or $k > 0$. The exception is $k = 0$: there site 0 lies in the open interval if and only if $\ell > 0$ and $r > 0$.

Proof. occ consists of the travel range $[k_-, k_+)$, the edges $k_- - i$ with $c_i^L \neq 0$, and the edges $k_+ - 1 + i$ with $c_i^R \neq 0$.

Case $k > 0$. Here $0 \in \text{occ}$. So $\min \text{occ} = -\ell \leq 0$ (it is 0 when $\ell = 0$), and the clamp $\min(0, \cdot)$ is inactive. Likewise $\max \text{occ} = k - 1 + r \geq 0 > -1$, and the clamp $\max(-1, \cdot)$ is inactive.

Case $k < 0$. The argument is symmetric, using $-1 \in \text{occ}$.

Case $k = 0$. Now $\text{occ} = \{-i : c_i^L \neq 0\} \cup \{i - 1 : c_i^R \neq 0\}$ and the clamps are active:

- If $\ell = 0$, then $\text{occ} \subseteq [0, \infty)$ or $\text{occ} = \emptyset$, and either way $A = 0 = -\ell$.
- If $r = 0$, then $B = -1 = r - 1$.
- If $\ell > 0$, resp. $r > 0$, the unclamped values $-\ell$, resp. $r - 1$, already lie on the correct side of the clamp.

So the formula for A and B holds in all cases.

Items (i) and (ii) now follow by listing $[A, B]$ and $[A, B + 1]$. When $\ell = 0$, the outer site A coincides with the junction site k_- , and similarly on the right.

For (iii), $A < k_-$ if and only if $\ell > 0$, and $k_- < B + 1 = k_+ + r$ if and only if $k_- < k_+$ or $r > 0$. The symmetric statement for k_+ is proved the same way. $\square$

So the clamps in **ATrue** and **BTrue** change nothing when $k \neq 0$. When $k = 0$ they do exactly two things. First, they keep site 0 in the site range even when both chains are empty. Second, they put site 0 *outside* the open interval unless both chains are nonempty. That second effect is what SF partly undoes (§D.8).

Lemma 118 (locality). *Let s be a site of $[A, B + 1]$ other than a junction site. Then $\alpha_s = d_{s-1}$, $\beta_s = d_s$ and $\text{site}(s) = \max(|d_{s-1}|, |d_s|)$. Moreover:*

- A travel-interior site has odd cost $\max(|d_{s-1}|, |d_s|)$, and it is never a cut, because $\Phi_s = f_{s-1} = \pm 1$.*
- A chain-interior site is a cut if and only if $d_{s-1} = d_s = 0$. Every chain-interior site lies in $(A, B + 1)$.*
- An outer site has cost $|c_N|$, where c is the corresponding chain and N its length. It is not a cut, and it does not lie in $(A, B + 1)$.*

In particular ℓ_R^{true} splits as a sum of the following parts:

- left-chain edges and left chain-interior and outer sites;
- travel edges and travel-interior sites;
- right-chain edges and right chain-interior and outer sites;
- the junction sites.

The cut count c^{true} splits the same way, except that the junction sites are governed by Lemma 117(iii) and by SF.

Proof. For $s \notin \{0, k\}$ all the brackets vanish, so $\alpha_s = d_{s-1}$, $\beta_s = d_s$ and $\Phi_s = f_{s-1}$. For the cut criterion see also `RJLemmaJ.bulk_cut`.

For (a), if $k_- < s < k_+$ then $s-1 \in [k_-, k_+)$, so d_{s-1} and d_s are odd and $f_{s-1} = \pm 1$.

For (b), if s is chain-interior then $s-1$ and s are chain edges with $f = 0$. Such s lie strictly between A and $B+1$ by Lemma 117.

For (c), at the outer site $B+1$ (when $r > 0$) we have $\alpha = d_B = c_r^R \neq 0$ and $\beta = 0$. The site A is symmetric.

The cost and cut conditions of every non-junction site involve only the deposits of one block, which gives the splitting. $\square$

D.4 Chain generating functions

For a chain c of length $N \geq 1$, define its *weight* $\omega(c)$ as $x^{\ell(c)}y^{\gamma(c)}$, where

$$\begin{aligned} \ell(c) &= \underbrace{\sum_{i=1}^N (|c_i| + 2[c_i = 0])}_{\text{edges}} + \underbrace{\sum_{i=2}^N \max(|c_{i-1}|, |c_i|)}_{\text{interior sites}} + \underbrace{|c_N|}_{\text{outer site}}, \\ \gamma(c) &= \#\{2 \leq i \leq N : c_{i-1} = c_i = 0\}. \end{aligned}$$

The empty chain has weight 1. By Lemmas 117 and 118, the left and right chains of g contribute exactly $\omega(c^L)$ and $\omega(c^R)$ to $x^{\ell^{\text{true}}}y^{\gamma^{\text{true}}}$. Everything else comes from the junction sites and the travel block. Note that the junction site is *not* part of ω .

Lemma 119 (bulk). *For $b \geq 1$,*

$$\sum_{c: |c_1|=2b} \omega(c) = P_b, \quad \sum_{c: c_1=2\sigma b} \omega(c) = \tfrac{1}{2}P_b \quad (\sigma = \pm 1), \quad \sum_{c \text{ of kind } Z} \omega(c) = \mathfrak{g}S.$$

Consequently

$$\sum_{c \text{ of kind } E \text{ or } Z} \omega(c) = B_0.$$

Proof. Let $Q_b = \sum_{|c_1|=2b} \omega(c)$. We split according to what follows edge 1.

- $N = 1$. The weight is x^{2b} for the edge times x^{2b} for the outer site, i.e. q^{2b} , for each of the two signs.
- *Next deposit* $c_2 = \pm 2a \neq 0$. Site 2 costs $x^{2\max(a,b)}$.
- *A gap run* $c_2 = \dots = c_{L+1} = 0$ with $L \geq 1$, followed by $c_{L+2} = \pm 2a$. Site 2 costs x^{2b} . The L gap edges give q^L . The $L-1$ sites strictly inside the gap cost 1 and are cuts, giving y^{L-1} . The site before c_{L+2} costs x^{2a} .

The tail $(c_{i+1})_{i \geq 1}$ in the second case, and $(c_{L+1+i})_{i \geq 1}$ in the third, is again a chain with first entry of magnitude $2a$. Its interior sites, outer site and cuts are exactly those counted in its own ω . So

$$Q_b = 2q^b \left(q^b + \sum_a (q^{\max(a,b)} + q^b \mathfrak{g} q^a) Q_a \right),$$

which is (38). We have $\text{ord } Q_b \geq 4b$, so $Q = P$ by the uniqueness in Lemma 115(b).

The negation $c \mapsto -c$ preserves ω , which gives the signed statement.

A chain of kind Z is a gap run of length $L \geq 1$, followed by a chain with $|c_1| = 2a$. Its weight is $q^L y^{L-1} \cdot x^{2a} \cdot \omega(\text{tail})$. Here the sites strictly inside the gap are the $L-1$ cuts, and x^{2a} is the site between the gap and the deposit. Summing gives $\mathfrak{g} \sum_a q^a P_a = \mathfrak{g}S$. $\square$

D.5 Travel block

Lemma 120 (travel). *Let $k \neq 0$. Fix everything except the signs of the travel deposits, and write $|d_j| = 2s_j + 1$. The travel edges and travel-interior sites contribute*

$$\prod_j x^{2s_j+1} \prod_{\text{interior}} x^{2\max(s_j, s_{j+1})+1} = x^{-1} \prod_j (x^{2s_j+2}) \prod_{\text{interior}} x^{2\max(s_j, s_{j+1})}.$$

This does not depend on the signs.

Proof. The run has $|k|$ edges and $|k| - 1$ interior sites. Moving one x from each interior site to the edge that follows it leaves one extra x , which is where the factor x^{-1} comes from. Sign independence follows from Lemma 118(a). $\square$

Summing over the travel words with the two end magnitudes fixed, and counting the $2^{|k|}$ sign choices as the factor 2 in each e_s , gives

$$\sum_{|k| \geq 1} \sum_{\text{words}} (\cdots) = x^{-1} \sum_{n \geq 0} \langle v, (D_e K)^n D_e w \rangle = x^{-1} \langle v, (I - \mathcal{T})^{-1} D_e w \rangle. \quad (42)$$

This formula needs a condition on the ends: the end weights $v_{s_{\text{near}}}$ and $w_{s_{\text{far}}}$ must be the *sign-averaged* end weights. That is, the ends together must contribute the factor $\frac{1}{2} \cdot 2$ per end sign, which the next lemma makes precise. Here $n = |k| - 1$, and K is symmetric, so the reading direction does not matter.

Lemma 121 (sign decoupling). *Let $k \neq 0$. Let σ_n and σ_f be the signs of the travel deposits adjacent to the near junction (site 0) and the far junction (site k). Write the contribution of the two junction sites, together with the adjacent chains, as $N(\sigma_n) \cdot F(\sigma_f, \varepsilon, \delta)$. Here N is summed over the chain at site 0 and F over the chain at site k . Suppose that*

$$\Lambda := \sum_{\varepsilon, \delta} F(\sigma, \varepsilon, \delta) \quad \text{does not depend on } \sigma. \quad (43)$$

Then the sum of $N \cdot F$ over ε, δ and the end signs equals $2 \cdot \bar{N} \cdot \Lambda$ when $|k| = 1$ (so $\sigma_n = \sigma_f$), and $4 \cdot \bar{N} \cdot \Lambda$ when $|k| \geq 2$. Here $\bar{N} = \frac{1}{2} \sum_{\sigma} N(\sigma)$. In both cases this is what (42) produces with $v = \bar{N}$ and $w = \Lambda$.

Proof. If $|k| \geq 2$ the two signs are independent, and $\sum_{\sigma_n, \sigma_f, \varepsilon, \delta} N F = (\sum_{\sigma} N)(\sum_{\sigma_f} \Lambda) = 2\bar{N} \cdot 2\Lambda$.

If $|k| = 1$ there is one travel edge and one sign σ , so $\sum_{\sigma, \varepsilon, \delta} N(\sigma) F(\sigma, \varepsilon, \delta) = \sum_{\sigma} N(\sigma) \Lambda = 2\bar{N} \Lambda$, using (43).

In both cases these are the edge factors 2 carried by D_e at the end edges. $\square$

Remark 122 (decoupling holds only after the sum over ε). Hypothesis (43) is essential when $|k| = 1$, and it fails before the sum over ε .

Take $k = -1$, so that $d_{-1} = \sigma(2s + 1)$ is both the near and the far travel deposit. Then $N(\sigma)$ involves $|d_{-1} - 1|$, which is $2s$ or $2s + 2$ according to σ . For fixed ε and $\delta = 1$, the far site involves $|d_{-1} - \varepsilon|$, which is also $2s$ or $2s + 2$ according to $\sigma\varepsilon$. So for fixed ε the sum $\sum_{\sigma} N(\sigma) F(\sigma, \varepsilon, 1)$ pairs the cheap near cost with the cheap or the expensive far cost depending on ε . It is *not* equal to $\bar{N} \sum_{\sigma} F(\sigma, \varepsilon, 1)$.

Only after summing over ε does the far weight lose its σ -dependence, because the substitution $\varepsilon \mapsto -\varepsilon$ exchanges the two values. Then the product factorises.

For $k = +1$, the near weight depends only on $|d_0|$ (see (44)), so there the factorisation is harmless. The model nevertheless uses the ε -summed far vector $\lambda^{(y)}$ in both sectors, and that is legitimate only because of (43).

Condition (43) is verified in (45) and (47) below. In both, the σ -independence comes from the involution $\varepsilon \mapsto -\varepsilon$, combined with negating the far chain where necessary. Negating the far chain preserves ω and all cut conditions.

D.6 Sector $k > 0$

The near junction is site 0. There $\alpha_0 = d_{-1} - 1$, $\beta_0 = d_0$ and $\Phi_0 = f_{-1} + 1 = 1$. So **site 0 is never a cut** (RJLemmaJ.J_site0_pos). Its chain is c^L , with $u := c_1^L = d_{-1}$. Let $d_0 = \sigma(2s + 1)$. The cost is $\max(|u - 1|, 2s + 1)$, which does not depend on σ . Summing over c^L with Lemma 119:

- $u = 0$ (kinds E, Z): the cost is $2s + 1$ and the weight is B_0 ;
- $u = +2b$: the cost is $\max(2b - 1, 2s + 1)$ and the weight is $\frac{1}{2}P_b$;
- $u = -2b$: the cost is $\max(2b + 1, 2s + 1)$ and the weight is $\frac{1}{2}P_b$.

Hence

$$N(\sigma) = \overline{N} = \mu_s^{(y)}. \quad (44)$$

No y occurs here: site 0 is never a cut, and in any case it lies in $(A, B + 1)$ only when $\ell > 0$.

The far junction is site k . Put $d_{k-1} = \sigma(2s + 1)$ and $v := c_1^R = d_k$. Then

$$\alpha_k = \sigma(2s + 1) + \varepsilon[\delta = 0], \quad \beta_k = v - \varepsilon[\delta = 1], \quad \Phi_k = f_{k-1} - [\delta = 0] = [\delta = 1].$$

By Lemma 117(iii), site k lies in $(A, B + 1)$ if and only if $r > 0$.

$\delta = 1$. Here $\Phi_k = 1$, so there is no cut. The cost is $\max(2s + 1, |v - \varepsilon|)$.

- $v = 0$: the cost is $2s + 1$ for both ε , contributing $2B_0x^{2s+1}$.
- $v = \pm 2b$: as ε runs over ± 1 , $|v - \varepsilon|$ runs over $\{2b - 1, 2b + 1\}$. Each sign of v has weight $\frac{1}{2}P_b$, so the contribution is $P_b(x^{\max(2b-1, 2s+1)} + x^{\max(2b+1, 2s+1)})$.

The total is $2\mu_s^{(y)}$.

$\delta = 0$. Here $\Phi_k = 0$ and $\beta_k = v$. As ε runs over ± 1 , $|\alpha_k|$ takes the value $2s + 2$ (for $\varepsilon = \sigma$) and $2s$ (for $\varepsilon = -\sigma$).

- $\varepsilon = \sigma$: the cost is $\max(2s + 2, |v|)$ and there is no cut. The contribution is $B_0x^{2s+2} + \sum_b P_b x^{\max(2s+2, 2b)}$.
- $\varepsilon = -\sigma$: the cost is $\max(2s, |v|)$. Site k is a cut if and only if $s = 0$ and $v = 0$ (RJLemmaJ.J_sitek_pos: $\delta = 0$, $d_{k-1} = -\varepsilon$, $d_k = 0$). The cut is *counted* if and only if moreover $r > 0$, i.e. the right chain has kind Z; for kind E, site $k = B + 1$ is the outer end. So for $v = 0$ the weight is $1 + y^{[s=0]} \mathbf{g}S = B_0^{[s]}$, and the contribution is $B_0^{[s]}x^{2s} + \sum_b P_b x^{\max(2s, 2b)}$.

The total is $E_s^{(y)}$.

Therefore

$$\sum_{\varepsilon, \delta} F(\sigma, \varepsilon, \delta) = E_s^{(y)} + 2\mu_s^{(y)} = \lambda_s^{(y)} \quad \text{for both } \sigma, \quad (45)$$

which verifies (43).

Combining Lemmas 118, 119, 120 and 121 with (42):

$$\sum_{g: k > 0} x^{\ell_R^{\text{true}}} y^{c^{\text{true}}} = x^{-1} \langle \mu^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle = \mathcal{W}_+.$$

There is no shield term, because SF requires $k = 0$.

D.7 Sector $k < 0$, and the factor $\frac{1}{2}$ on E

The near junction is site 0. Its chain is now c^R , with $v = c_1^R = d_0$, and the adjacent travel deposit is $d_{-1} = \sigma(2s + 1)$. Then

$$\alpha_0 = \sigma(2s + 1) - 1, \quad \beta_0 = v, \quad \Phi_0 = f_{-1} + 1 = 0.$$

So $|\alpha_0| = 2s$ for $\sigma = +1$, and $|\alpha_0| = 2s + 2$ for $\sigma = -1$. Site 0 is a cut if and only if $d_{-1} = 1$ and $d_0 = 0$ (RJLemmaJ.J_site0_neg). It lies in $(A, B + 1)$ if and only if $r > 0$, by Lemma 117(iii). So:

- $\sigma = +1$: the contribution is $B_0^{[s]} x^{2s} + \sum_b P_b x^{\max(2s, 2b)}$. The y appears exactly for $s = 0$ with c^R of kind Z.
- $\sigma = -1$: the contribution is $B_0 x^{2s+2} + \sum_b P_b x^{\max(2s+2, 2b)}$, with no cut.

Thus $N(+1) + N(-1) = E_s^{(y)}$, and

$$\bar{N} = \frac{1}{2} E_s^{(y)}. \quad (46)$$

The $E/2$ argument. Here, unlike the case $k > 0$, the near weight depends on σ . The sum over σ of the near end produces the whole of $E_s^{(y)}$. But the edge carrying σ is a travel edge, and its two signs are already counted by the factor 2 in e_s . Pairing $E^{(y)}$ with D_e would therefore count that sign twice. The correct near vector is the sign average, $\frac{1}{2} E^{(y)}$. This is exactly what Lemma 121 says. Its integrality was recorded in Lemma 115(d).

The far junction is site k . Its chain is c^L , with $u = c_1^L = d_{k-1}$, and $d_k = \sigma(2s + 1)$. Then

$$\alpha_k = u + \varepsilon[\delta = 0], \quad \beta_k = \sigma(2s + 1) - \varepsilon[\delta = 1], \quad \Phi_k = f_{k-1} - [\delta = 0] = -[\delta = 0].$$

By Lemma 117(iii), site k lies in $(A, B + 1)$ if and only if $\ell > 0$.

$\delta = 0$. Here $\Phi_k = -1$, so there is no cut. The cost is $\max(|u + \varepsilon|, 2s + 1)$. By the same computation as for $\delta = 1$ in §D.6, with the chains mirrored, the sum over ε and c^L is $2\mu_s^{(y)}$.

$\delta = 1$. Here $\Phi_k = 0$ and $\alpha_k = u$, while $|\beta_k|$ is $2s$ for $\varepsilon = \sigma$ and $2s + 2$ for $\varepsilon = -\sigma$. Site k is a cut if and only if $u = 0$, $s = 0$ and $\varepsilon = \sigma$ (RJLemmaJ.J_sitek_neg). The cut is counted if and only if $\ell > 0$, i.e. the left chain has kind Z. The sum over ε and c^L is $E_s^{(y)}$, with the slot B_{0z} at $s = 0$.

Hence

$$\sum_{\varepsilon, \delta} F(\sigma, \varepsilon, \delta) = \lambda_s^{(y)} \quad \text{for both } \sigma. \quad (47)$$

By Lemma 121, which is now needed in earnest at $k = -1$ (Remark 122),

$$\sum_{g: k < 0} x^{\ell_R^{\text{true}}} y^{c^{\text{true}}} = x^{-1} \langle \frac{1}{2} E^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle = \mathcal{W}_-.$$

Finally, $\mu^{(y)} + \frac{1}{2} E^{(y)} = \frac{1}{2} \lambda^{(y)}$ by (41), which gives $\mathcal{W}_+ + \mathcal{W}_- = (2x)^{-1} \langle \lambda^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle$.

D.8 Sector $k = 0$: the complete case analysis

When $k = 0$ there is no travel, every d_j is even, and site 0 is the only junction site. Put $u = c_1^L = d_{-1}$ and $v = c_1^R = d_0$. Then

$$\alpha_0 = u - 1 + \varepsilon[\delta = 0], \quad \beta_0 = v - \varepsilon[\delta = 1], \quad \Phi_0 = 1 - [\delta = 0] = [\delta = 1].$$

The term -1 is **vArr**. By Lemmas 117, 118 and 119,

$$\mathcal{W}_0 = \sum_{\varepsilon=\pm 1} \sum_{\delta \in \{0,1\}} \sum_{(u, w_L)} \sum_{(v, w_R)} x^{j(\varepsilon, \delta, u, v)} w_L w_R y^\chi, \quad j = \max(|u - 1 + \varepsilon[\delta = 0]|, |v - \varepsilon[\delta = 1]|), \quad (48)$$

where:

- (u, w_L) ranges over the left kinds: **E** : $(0, 1)$, **Z** : $(0, \mathfrak{g}S)$ and $\mathbf{N}_{\pm, b} : (\pm 2b, \frac{1}{2}P_b)$ for $b \geq 1$;
- (v, w_R) ranges over the same list for the right chain;
- $\chi \in \{0, 1\}$ records whether site 0 is counted in c^{true} .

Only w_R of kind **Z** can carry the extra y : $y^\chi w_R$ is $y\mathfrak{g}S$ when $\chi = 1$. The chain interiors are accounted for inside w_L and w_R .

Site-0 cost, by branch.

δ	ε	$j(\varepsilon, \delta, u, v)$
0	+1	$\max(u , v)$
0	-1	$\max(u - 2 , v)$
1	+1	$\max(u - 1 , v - 1)$
1	-1	$\max(u - 1 , v + 1)$

For example, $u = v = 0$ gives 0, 2, 1 and 1 in the four rows. These are the identity and three elements of length ≤ 2 .

When is site 0 a cut? $\text{cut}(0)$ holds if and only if all three of the following hold:

- $\delta = 0$, since $\Phi_0 = [\delta = 1]$;
- $v = 0$;
- $u = 1 - \varepsilon$, that is, $u = 0$ if $\varepsilon = +1$ and $u = 2$ if $\varepsilon = -1$.

When is site 0 counted? Site 0 is counted if and only if $\text{cut}(0)$ holds and one of the following holds:

- (I) $0 \in (A, B + 1)$, which by Lemma 117(iii) means $\ell > 0$ and $r > 0$;
- (S) $\text{SF}(g)$ holds. This means $\delta = 0$, $\ell = 0$ (the left chain has kind **E**; recall $d_j = 0$ for all $j < 0$ says exactly that), and $r > 0$.

Since $v = 0$ and $r > 0$ force the right chain to have kind **Z**, we get $\chi = 1$ only if $\delta = 0$ and c^R has kind **Z**. The full table follows. Rows are the left kind, and the **N** kinds are split by u .

δ	ε	left kind (u)	right kind	cut(0)	(I)	(S)	χ
0	+1	E ($u = 0$)	Z	yes	no ($\ell = 0$)	yes	1
0	+1	Z ($u = 0$)	Z	yes	yes	no	1
0	+1	N ($u \neq 0$)	Z	no ($u \neq 0$)	yes	no	0
0	-1	E ($u = 0$)	Z	no ($u \neq 2$)	no	yes	0
0	-1	Z ($u = 0$)	Z	no ($u \neq 2$)	yes	no	0
0	-1	N _{+,1} ($u = 2$)	Z	yes	yes	no	1
0	-1	N, $u \neq 2$	Z	no	yes	no	0
0	± 1	E	N	no ($v \neq 0$)	no	yes	0
0	± 1	Z, N	N	no ($v \neq 0$)	yes	no	0
0	± 1	E	E	$[\varepsilon = +1]$	no ($\ell = r = 0$)	no ($r = 0$)	0
0	± 1	Z, N	E	possible	no ($r = 0$)	no ($\ell > 0, r = 0$)	0
1	± 1	any	any	no ($\Phi_0 = 1$)	—	no ($\delta = 1$)	0

The table exhausts all branches. There are 2×2 marker choices, the left kinds E, Z, N _{$\pm$, b} , and the right kinds E, Z, N _{$\pm$, b} . Every marker choice with $\delta = 0$ appears in the first eleven rows, split by right kind, and $\delta = 1$ is the last row. The rows marked (S)=yes with $\chi = 0$ are the cases in which the shield *fires but site 0 is not a cut*. They contribute nothing, which is why **cTrue** conjoins SF with cut(0). The first row is the only branch in which the shield changes the count. There site 0 is a genuine cut, but it lies at the clamped left end $A = 0$ of the span and so falls outside the open interval.

Summary. The right Z weight is **ygS** exactly in the following cases:

- $\delta = 0, \varepsilon = +1$ and the left chain has kind E or Z;
- $\delta = 0, \varepsilon = -1$ and $u = +2$.

In every other case it is **gS**. This is branch for branch the rule used by the bivariate enumeration check of §7.1. This proves

$$\sum_{g: k=0} x^{\ell_R^{\text{true}}} y^{c^{\text{true}}} = \mathcal{W}_0.$$

Integrality is clear from (48): $P_b/2 \in \mathbb{Z}[y][[x]]$, and the sum is summable because $\text{ord } P_b \geq 4b$.

Remark 123 (the regrouping of $\mathcal{W}_0$). Let $\Psi = (\Psi_b)_{b \geq 1}$ be the ungapped bulk vector, the solution of (38) with **g** = 0, which depends on q alone. By the Sherman–Morrison formula (27), $P_b = B_0 \Psi_b$, and **gS** = $B_0 - 1$. So every weight in (48) is affine in B_0 : $w \in \{1, B_0 - 1, \frac{1}{2} B_0 \Psi_b\}$, and $y^\chi w_R = y(B_0 - 1)$ when $\chi = 1$. Each term $x^j w_L w_R y^\chi$ is therefore a polynomial of degree at most 2 in B_0 , and collecting powers gives

$$\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2,$$

with A_0, A_1, A_2 absolutely convergent sums of monomials in x and y times at most two entries of Ψ . The A_i contain no travel resolvent. As functions of q they are singular only where Ψ is, i.e. where $I - M_0$ is not invertible ($S_e = 0$, Corollary 38), and $B_0 = 1/(1 - \mathbf{g}t_1)$ only where $1 - \mathbf{g}t_1 = 0$. Neither occurs at a travel pole (Proposition 69, Corollary 84), so $\mathcal{W}_0$ has no travel pole.

D.9 Conclusion of the proof

The sectors $k > 0$, $k < 0$ and $k = 0$ partition $\mathcal{G}$. Lemma 116 identifies each sector with the product set of its blocks, and Lemma 118 shows that $x^{\ell_R^{\text{true}}} y^{c^{\text{true}}}$ is the product of the block weights and the junction-site factors. The three sector computations above then give Theorem 113. The rearrangements are justified by Lemma 115. $\square$

What is and is not established here. The following are established:

- the identity of Theorem 113, as a hand proof;
- integrality;
- x -adic convergence.

The following are not established here:

- any statement about a specific rational triangle beyond the universality depth (Corollary 114);
- a Lean formalisation of the assembly.

The inputs taken from Lean are the definitions of `siteCost`, `cut`, `ATrue`, `BTrue`, `lRTrue`, `cTrue` and `ShieldFires`, the junction lemmas of `RJLemmaJ.lean`, and `metricAll` (for the corollary only).

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